

DevOrSysAdminScripts

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Abstract

Some “small” useful scripts for enhancing development or system-administrators tasks

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Code lines: 14921 total lines, 13643 not empty lines, 1278 empty lines.

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1 Files trees

```
.
├── archive_libre_office_documents.exec.php
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.lib.php
│   ├── build_file_from_printable_file__standard.exec.sh
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.lib.sh
│   ├── check_shell_scripts_indentation.lib.sh
│   ├── check_URLs.lib.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.lib.sh
│   ├── comparisons.lib.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.lib.sh
│   ├── get_common_text_glob_patterns.lib.sh
│   ├── grammar_and_spelling_check.lib.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.lib.sh
│   ├── lines_filters.lib.sh
│   ├── lines_maps.lib.sh
│   ├── overwrite_if_not_equal.lib.sh
│   ├── phpm_d_baseline.xslt
│   ├── python_black_complement.lib.sh
│   ├── python_isort_complement.lib.sh
│   ├── repository_data.lib.sh
│   ├── shell_checks_complement.lib.sh
│   ├── split_score.exec.php
│   ├── split_score.lib.php
│   ├── strings_functions.lib.sh
│   ├── too_long_code_lines.lib.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   └── black.toml
```

- dependencies_data.json
- dependencies_notes.txt
- DevOrSysAdminScripts.html.tpl
- DevOrSysAdminScripts.tex
- DevOrSysAdminScripts.tex.tpl
- files_names_listing.txt
- .gitignore
- licenses_notes.txt
- phpcs_ruleset.xml
- phpmd_baseline.xml
- phpstan_baseline.neon
- phpstan_baseline.neon.tpl
- phpstan_config.neon
- phpstan_config.neon.tpl
- pylintrc.toml
- repository_data.txt
- repository_data.txt.tpl
- ruff.toml
- shellcheck.ini
- shellcheck_notes.txt
- COPYING
- COPYING.LESSER
- DevOrSysAdminScripts.html
- DevOrSysAdminScripts.pdf
- files.lib.php
- git_helpers
 - git_checks.lib.sh
 - post-commit
 - post_commit_renaming_comment.lib.sh
 - pre-commit
- .gitignore
- IMAP_transfert.exec.py
- LiensUtiles.txt
- miscellaneous
 - a_long_long_long_long_long_long_long_long_filename_so_that_you_c
 - an_see_that_it_works.txt
 - funny.sh
 - grep_benchmark.sh
 - spampty_data.txt
 - string_escaping.html
 - string_escaping.xhtml
 - yet_another_long_long_long_long_long_long_long_long_filename_so_
 - that_you_can_see_that_it_works.txt
- README.html

```

├── README.md
├── README.md.tpl
├── string_escaping.lib.php
├── unicode_exceptions.lib.php
├── unicode.lib.php
├── upgrade_build_and_checks.exec.sh
├── wget_sha512.lib.sh

```

6 directories, 87 files

```

[ 12K 2026-09-18 22:17] ./
├── [1.8K 2026-08-14 17:47] archive_libre_office_documents.exec.php*
├── [4.0K 2026-09-18 22:17] build_and_checks_dependencies/
│   ├── [4.4K 2026-05-12 19:33] build_and_checks.exec.sh.tpl*
│   ├── [1.5K 2026-08-14 17:46] build_dependencies_notes.exec.php*
│   ├── [ 10K 2026-08-14 15:46] build_dependencies_notes.lib.php
│   ├── [2.6K 2026-08-13 12:28]
│   │   └── build_file_from_printable_file__standard.exec.sh*
│   ├── [4.2K 2026-08-12 19:31] build_md_from_printable_md.exec.sh*
│   ├── [3.1K 2026-05-12 19:34] check_shell_scripts_beginnings.lib.sh
│   ├── [5.3K 2026-05-12 19:37] check_shell_scripts_indentation.lib.sh
│   ├── [4.0K 2026-08-12 19:14] check_URLs.lib.sh
│   ├── [ 43K 2026-09-18 22:17] common_build_and_checks.exec.sh*
│   ├── [ 38K 2026-09-18 22:14] common_build_and_checks.exec.sh.tpl
│   ├── [ 19K 2026-05-12 19:37] common_options.lib.sh
│   ├── [6.2K 2026-05-12 19:37] comparisons.lib.sh
│   ├── [ 22K 2026-08-12 18:12] create_PDF.exec.sh*
│   ├── [3.7K 2026-05-12 19:32] generate_from_template.lib.sh
│   ├── [2.1K 2026-08-12 19:16] get_common_text_glob_patterns.lib.sh
│   ├── [3.3K 2026-05-12 19:37] grammar_and_spelling_check.lib.sh
│   ├── [4.0K 2026-08-13 13:00] licenses_templates/
│   │   ├── [ 16K 2026-08-13 13:00] build_licenses_templates.exec.sh*
│   │   ├── [ 720 2026-05-12 19:16] license_file_header_AGPLv3+.tpl
│   │   ├── [ 699 2026-05-12 19:17] license_file_header_GPLv3+.tpl
│   │   └── [ 720 2026-05-12 19:18] license_file_header_LGPLv3+.tpl
│   ├── [6.0K 2026-05-13 17:48] lines_counts.lib.sh
│   ├── [ 15K 2026-05-13 15:25] lines_filters.lib.sh
│   ├── [3.0K 2026-05-12 19:37] lines_maps.lib.sh
│   ├── [3.3K 2026-05-12 19:37] overwrite_if_not_equal.lib.sh
│   ├── [1.6K 2026-08-14 14:40] phpmd_baseline.xslt
│   ├── [2.4K 2026-05-12 19:37] python_black_complement.lib.sh
│   ├── [3.1K 2026-05-13 15:29] python_isort_complement.lib.sh
│   ├── [2.9K 2026-05-13 15:08] repository_data.lib.sh
│   ├── [3.9K 2026-05-12 19:37] shell_checks_complement.lib.sh
│   ├── [2.8K 2026-08-14 17:45] split_score.exec.php*
│   └── [5.9K 2026-08-14 14:56] split_score.lib.php

```

```

|— [ 47K 2026-05-13 16:32] strings_functions.lib.sh
|— [4.8K 2026-08-12 19:25] too_long_code_lines.lib.sh
|— [4.5K 2026-08-14 14:33] update_common_build_and_checks.exec.sh*
|— [6.0K 2026-05-12 19:37]
|   update_or_check_files_names_listing.exec.sh*
|— [4.6K 2026-09-18 22:17] build_and_checks.exec.sh*
|— [4.0K 2026-09-18 22:18] build_and_checks_variables/
|   |— [ 15 2026-01-05 20:00] bandit_baseline.json
|   |— [ 25 2026-01-05 18:49] bandit.ini
|   |— [ 30 2026-02-28 20:30] black.toml
|   |— [5.1K 2026-04-17 16:50] dependencies_data.json
|   |— [4.3K 2026-04-17 16:57] dependencies_notes.txt
|   |— [5.7K 2026-05-12 19:31] DevOrSysAdminScripts.html.tpl
|   |— [ 37K 2026-08-14 18:11] DevOrSysAdminScripts.tex
|   |— [4.2K 2026-05-12 19:38] DevOrSysAdminScripts.tex.tpl
|   |— [5.3K 2026-08-14 17:42] files_names_listing.txt
|   |— [ 23 2026-03-24 19:33] .gitignore
|   |— [1.4K 2026-04-17 16:49] licenses_notes.txt
|   |— [ 16K 2026-08-14 17:54] phpcs_ruleset.xml
|   |— [5.2K 2026-08-14 14:53] phpmd_baseline.xml
|   |— [1.6K 2026-08-14 18:06] phpstan_baseline.neon
|   |— [1.7K 2026-08-14 18:06] phpstan_baseline.neon.tpl
|   |— [ 236 2026-08-13 12:29] phpstan_config.neon
|   |— [ 244 2026-08-12 19:23] phpstan_config.neon.tpl
|   |— [ 678 2026-02-24 22:38] pylintrc.toml
|   |— [5.5K 2026-05-13 17:14] repository_data.txt
|   |— [ 932 2026-05-13 13:46] repository_data.txt.tpl
|   |— [1.5K 2026-08-13 16:02] ruff.toml
|   |— [ 166 2026-03-24 19:33] shellcheck.ini
|   |— [3.6K 2026-05-13 17:18] shellcheck_notes.txt
|— [ 34K 2024-05-06 00:49] COPYING
|— [7.5K 2025-12-27 21:48] COPYING.LESSER
|— [569K 2026-08-14 18:11] DevOrSysAdminScripts.html
|— [465K 2026-08-14 18:11] DevOrSysAdminScripts.pdf
|— [ 21K 2026-08-14 18:01] files.lib.php
|— [4.0K 2026-05-13 17:28] git_helpers/
|   |— [3.8K 2026-05-13 17:22] git_checks.lib.sh
|   |— [2.1K 2026-05-12 19:31] post-commit*
|   |— [ 15K 2026-05-13 17:28] post_commit_renaming_comment.lib.sh
|   |— [1.8K 2026-05-12 19:31] pre-commit*
|— [ 8 2026-03-26 10:37] .gitignore
|— [8.4K 2026-05-12 19:31] IMAP_transfert.exec.py
|— [1.1K 2026-03-31 19:59] LiensUtiles.txt
|— [ 24K 2026-08-13 18:22] miscellaneous/
|   |— [ 0 2026-03-04 09:13]
|   |a_long_long_long_long_long_long_long_long_filename_so_that_you_c

```

```

| an_see_that_it_works.txt
|— [1.3K 2026-05-12 19:37] funny.sh
|— [2.4K 2026-08-13 18:22] grep_benchmark.sh
|— [ 94 2026-03-03 19:56] spampty_data.txt
|— [ 461 2026-05-13 18:15] string_escaping.html
|— [ 653 2026-05-12 14:38] string_escaping.xhtml
|— [ 0 2026-03-04 09:13]
| yet_another_long_long_long_long_long_long_long_long_filename_so_
| that_you_can_see_that_it_works.txt
|— [8.1K 2026-05-12 12:17] README.html
|— [4.3K 2026-03-22 21:29] README.md
|— [4.3K 2026-03-22 21:29] README.md.tpl
|— [7.5K 2026-08-14 17:03] string_escaping.lib.php
|— [2.5K 2026-08-13 18:54] unicode_exceptions.lib.php
|— [ 31K 2026-08-14 17:04] unicode.lib.php
|— [6.4K 2026-05-13 18:46] upgrade_build_and_checks.exec.sh*
|— [2.5K 2026-05-12 19:37] wget_sha512.lib.sh

```

6 directories, 87 files

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: <https://www.gnu.org/licenses/>. The git repository of this source code is also available at: <https://github.com/LLyaudet/DevOrSysAdminScripts/>. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff, xmllint;

- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmid;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, utils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:

<https://www.gnu.org/software/bash/>,
<https://tiswww.case.edu/php/chet/bash/bashtop.html>,
<https://savannah.gnu.org/projects/bash/>.

GNU coreutils can be found at the following URLs:

<https://www.gnu.org/software/coreutils/>,
<https://savannah.gnu.org/projects/coreutils/>.

utils coreutils can be found at the following URLs:

<https://utils.github.io/coreutils/>,
<https://github.com/utils/coreutils>.

diff (GNU diffutils) can be found at the following URLs:

<https://www.gnu.org/software/diffutils/>,
<https://savannah.gnu.org/projects/diffutils/>.

find, xargs (GNU findutils) can be found at the following URLs:

<https://www.gnu.org/software/findutils/>,
<https://savannah.gnu.org/projects/findutils/>.

grep (GNU grep) can be found at the following URLs:

<https://www.gnu.org/software/grep/>,
<https://savannah.gnu.org/projects/grep/>.

sed (GNU sed) can be found at the following URLs:

<https://www.gnu.org/software/sed/>,
<https://savannah.gnu.org/projects/sed/>.

ShellCheck can be found at the following URLs:

<https://www.shellcheck.net/>,
<https://github.com/koalaman/shellcheck>.

GNU Wget can be found at the following URLs:

<https://www.gnu.org/software/wget/>,
<https://savannah.gnu.org/projects/wget/>.

more of util-linux can be found at the following URLs:

<https://www.kernel.org/pub/linux/utils/util-linux/>,
<https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/>,
<https://github.com/util-linux/util-linux> official?.

pandoc can be found at the following URLs:

<https://pandoc.org/>,
<https://github.com/jgm/pandoc>.

pdftex/pdflatex can be found at the following URLs:

<https://www.tug.org/applications/pdftex/>,
<https://tug.org/texlive/svn/>.

tree can be found at the following URLs:

<https://oldmanprogrammer.net/source.php?dir=projects/tree>,
<https://gitlab.com/OldManProgrammer/unix-tree>,
<https://github.com/Old-Man-Programmer/tree>.

git can be found at the following URLs:

<https://git-scm.com/about>,
<https://github.com/git/git>.

pylint can be found at the following URLs:

<https://www.pylint.org/>,
<https://pylint.readthedocs.io/en/stable/>,
<https://github.com/pylint-dev/pylint>,
<https://pypi.org/project/pylint/>.

notify-send can be found at the following URLs:

<https://gitlab.gnome.org/GNOME/libnotify/>,
<https://gnome.pages.gitlab.gnome.org/libnotify/>.

black can be found at the following URLs:

<https://github.com/psf/black>,
<https://black.readthedocs.io/en/stable/>,
<https://pypi.org/project/black/>.

isort can be found at the following URLs:

<https://pycqa.github.io/isort/>,
<https://github.com/pycqa/isort/>,
<https://pypi.org/project/isort/>.

mypy can be found at the following URLs:

<https://mypy-lang.org/>,
<https://mypy.readthedocs.io/en/stable/>,
<https://github.com/python/mypy>,
<https://pypi.org/project/mypy/>.

phpdoc (phpDocumentor) can be found at the following URLs:

<https://phpdoc.org/>,
<https://docs.phpdoc.org/>,
<https://github.com/phpdocumentor/phpdocumentor>.

pip can be found at the following URLs:

<https://pip.pypa.io/en/stable/>,
<https://github.com/pypa/pip>,
<https://pypi.org/project/pip/> (egg and chicken ;)).

ruff can be found at the following URLs:
<https://docs.astral.sh/ruff/>,
<https://github.com/astrol-sh/ruff>,
<https://pypi.org/project/ruff/>.

xmllint (libxml2) can be found at the following URLs:
<https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home>,
<https://gitlab.gnome.org/GNOME/libxml2>.

bandit can be found at the following URLs:
<https://bandit.readthedocs.io/en/latest/>,
<https://github.com/PyCQA/bandit>,
<https://pypi.org/project/bandit/>.

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
<https://phpmd.org/>,
<https://github.com/phpmd/phpmd>,
<https://packagist.org/packages/phpmd/phpmd>.

php can be found at the following URLs:
<https://www.php.net/>,
<https://github.com/php/php-src>.

tex can be found at the following URL:
<https://ctan.org/tex-archive/systems/knuth/dist/tex>.

pcgrep, pcre2grep can be found at the following URLs:
<https://www.pcre.org/>,
<https://pcre2project.github.io/pcre2/>,
<https://github.com/PCRE2Project/pcre1>,
<https://github.com/PCRE2Project/pcre2>,
<https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c>.

python3 can be found at the following URLs:
<https://www.python.org/>,
<https://docs.python.org/3/>.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 archive_libre_office_documents.exec.php

```

1  #!/usr/bin/php
2  <?php
3
4  /**
5   This file is part of DevOrSysAdminScripts.
6

```

```

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23 along with DevOrSysAdminScripts.
24 If not, see <https://www.gnu.org/licenses/>.
25
26 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27
28 @category Executable
29 @package DevOrSysAdminScripts
30 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33 */
34
35 declare(strict_types=1);
36 declare(encoding='UTF-8');
37
38 require_once 'files.lib.php';
39
40 if(PHP_SAPI !== 'cli'){
41     fwrite(
42         STDERR,
43         "archive_libre_office_documents.exec.php must be run in a shell.\n",
44     );
45     die(1);
46 }
47
48 if(count($argv) < 3 || count($argv) > 4){
49     fwrite(
50         STDERR,
51         "split_score.exec.php must be given 2 or 3 arguments.\n",
52     );

```

```

53     die(1);
54 }
55
56 DOSASfiles\archive_directory_into_another(
57     '/home/' . $argv[1] . '/.config/libreoffice/4/user/backup/',
58     '/home/' . $argv[2] . '/Documents/LibreOfficeArchives/',
59     true,
60     true,
61     // Following argument is if you want same relative path and same content
62     // at the same time.
63     // This is in case you want to track when the file goes back to some
64     // old state.
65     isset($argv[3]) ? (bool) (int) $argv[3] : false,
66     true,
67     true,
68     '\\.~\\d+~',
69 );
70 ?>

```

3.2 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
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8  # either version 3 of the License,
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10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
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14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "build_and_checks.exec.sh.tpl" was renamed from
25 # "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".

```

```

26 # The file "build_and_checks.exec.sh" generated from the file
27 # "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
28 # was renamed from
29 # "build_and_checks.sh" to "build_and_checks.exec.sh".
30
31 build_and_checks() {
32     # $1=work_directory
33     # Options:
34     #   --check-download=y|keep (=keep to keep the downloaded files)
35     #   --verbose
36     local LFBFL_work_directory="."
37     if [[ -n "$1" ]]; then
38         LFBFL_work_directory="$1"
39     fi
40     readonly LFBFL_work_directory
41
42     local LFBFL_verbose=""
43     if [[ "$*" == *--verbose* ]]; then
44         printf "%s %s\n" "$0" "$*"
45         LFBFL_verbose="--verbose"
46     fi
47     readonly LFBFL_verbose
48
49     local LFBFL_check_download=""
50     if [[ "$*" == *--check-download=keep* ]]; then
51         LFBFL_check_download="--check-download=keep"
52     elif [[ "$*" == *--check-download=y* ]]; then
53         LFBFL_check_download="--check-download=y"
54     fi
55     readonly LFBFL_check_download
56
57     source ./wget_sha512.lib.sh
58
59     LFBFL_subdir="build_and_checks_dependencies"
60     mkdir --parents "${LFBFL_subdir}/licenses_templates"
61     local LFBFL_variables_directory
62     LFBFL_variables_directory="${LFBFL_work_directory}/"
63     LFBFL_variables_directory+="build_and_checks_variables"
64     readonly LFBFL_variables_directory
65     mkdir --parents -- "${LFBFL_variables_directory}/temp"
66     if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67         printf "temp/\nupgrade_venvs_ts\n" \
68             > "${LFBFL_variables_directory}/.gitignore"
69     fi
70
71     local LFBFL_dependencies_URL

```

```

72 LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73 LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74 readonly LFBFL_dependencies_URL
75
76 declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77 local LFBFL_script_download_URL
78 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79 LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80 readonly LFBFL_script_download_URL
81 declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82 @sha512-common_build_and_checks.exec.sh@
83 wget_sha512 "${LFBFL_file_path}" \
84     "${LFBFL_script_download_URL}" \
85     "${LFBFL_correct_sha512}" \
86     "${LFBFL_verbose}" \
87     "${LFBFL_check_download}"
88 chmod +x "${LFBFL_file_path}"
89
90 "${LFBFL_file_path}" "${LFBFL_work_directory}" \
91     "${LFBFL_dependencies_URL}" \
92     "${LFBFL_verbose}" \
93     "${LFBFL_check_download}"
94 }
95
96 if [[ "$*" == *--fixed-point-build* ]]; then
97     printf -- "--fixed-point-build\n"
98     source "./build_and_checks_dependencies/lines_filters.lib.sh"
99     LFBFL_work_directory="."
100     if [[ -n "$1" ]]; then
101         LFBFL_work_directory="$1"
102     fi
103     LFBFL_variables_directory="${LFBFL_work_directory}/"
104     LFBFL_variables_directory+="build_and_checks_variables"
105     LFBFL_data_file_name="${LFBFL_variables_directory}/"
106     LFBFL_data_file_name+="repository_data.txt"
107     LFBFL_repository_name=""
108     grep_variable "${LFBFL_data_file_name}" repository_name \
109         --result-variable-prefix="LFBFL_" \
110         --replace-line-returns-by=""
111     # Touching the 3 following files first let us iterate
112     # upgrade_build_and_checks 2 times instead of 3 to reach
113     # a fixed point.
114     touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
115     touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
116     touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
117     # The two iterations of build_and_checks needs to complete

```

```

118     # during the same minute to have a fixed point.
119     build_and_checks "$@"
120     # build_and_checks "$@" second iteration below
121     # Then calling this script without --fixed-point-build should not
122     # yield any new superficial modifications.
123 fi
124
125 build_and_checks "$@"

```

3.3 build_and_checks_dependencies/build_dependencies_notes.exec.php

```

1  #!/usr/bin/php
2  <?php
3
4  /**
5   This file is part of DevOrSysAdminScripts.
6
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25
26  ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
27
28  @category Executable
29  @package DevOrSysAdminScripts
30  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31  @copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
32  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33  */
34
35  declare(strict_types=1);

```



```

36 declare(encoding='UTF-8');
37
38 require_once 'string_escaping.lib.php';
39 require_once 'build_dependencies_notes.lib.php';
40
41 if(PHP_SAPI !== 'cli'){
42     fwrite(
43         STDERR,
44         "build_dependencies_notes.exec.php must be run in a shell.\n",
45     );
46     die(1);
47 }
48
49 if(count($argv) !== 2){
50     fwrite(
51         STDERR,
52         "split_score.exec.php must be given exactly 1 argument.\n",
53     );
54     die(1);
55 }
56
57 \DOSAS_dependencies_notes\build_dependencies_notes($argv[1]);
58 ?>

```

3.4 build_and_checks_dependencies/build_dependencies_notes.lib.php

```

1  <?php
2
3  /**
4   This file is part of DevOrSysAdminScripts.
5
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8   under the terms of the GNU Lesser General Public License
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```

```

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23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 2026-03-26T11:59+01:00: This file was renamed from
27 "build_dependencies_notes.exec.php"
28 to
29 "build_dependencies_notes.lib.php".
30
31 @category Library
32 @package DevOrSysAdminScripts
33 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
34 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
35 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
36 */
37
38 declare(strict_types=1);
39 declare(encoding='UTF-8');
40
41 // require_once 'string_escaping.lib.php';
42 namespace DOSAS_dependencies_notes;
43
44 use Exception;
45 use string_escaping;
46
47 /**
48 This function looks at the dependencies data found in a work directory,
49 and generate three files: a txt file, a TeX file (.sub),
50 and an HTML file (.sub) with the data in a nicer format.
51
52 @param string $s_work_directory The directory where the work must be done.
53
54 @throws \Exception When the content of one of the files needed as input
55 cannot be loaded into a string.
56 Or that some part of the structure from the JSON
57 isn't valid.
58
59 @return void
60 */
61 function build_dependencies_notes(string $s_work_directory) : void {
62     $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
63     $s_subdir2 = $s_subdir1.'/temp';
64     $s_data_file_path = $s_subdir1.'/dependencies_data.json';
65     $s_data = file_get_contents($s_data_file_path);
66     if($s_data === false){

```

```

67     throw new Exception(
68         'Unexpected failure when fetching content of dependencies_data.json'
69     );
70 }
71 $arr_data = json_decode($s_data, true);
72 if(!is_array($arr_data)){
73     throw new Exception(
74         "dependencies_data.json doesn't contain an array."
75     );
76 }
77 $s_data_file_path2 = $s_subdir1.'/repository_data.txt';
78 $s_data2 = file_get_contents($s_data_file_path2);
79 if($s_data2 === false){
80     throw new Exception(
81         'Unexpected failure when fetching content of repository_data.txt'
82     );
83 }
84 $i_max_line_length = 70;
85 foreach(explode("\n", $s_data2) as $s_line){
86     if(str_starts_with($s_line, 'max_line_length=')){
87         $i_max_line_length = (
88             (int) substr($s_line, strlen('max_line_length='))
89         );
90         break;
91     }
92 }
93
94 $s_result_content1 = '';
95 $s_result_content2 = '';
96 $s_result_content3 = '';
97
98 foreach($arr_data as $arr_dependency){
99     if(!is_array($arr_dependency)){
100         throw new Exception("Dependency isn't an array.");
101     }
102     if(!isset($arr_dependency['URLs'])){
103         throw new Exception("Dependency doesn't have an URLs array.");
104     }
105     if(!is_array($arr_dependency['URLs'])){
106         throw new Exception("URLs isn't an array.");
107     }
108     if(!isset($arr_dependency['dependency_name'])){
109         throw new Exception("Dependency doesn't have a name.");
110     }
111     if(!is_string($arr_dependency['dependency_name'])){
112         throw new Exception("Dependency name isn't a string.");

```

```

113     }
114
115     $s_URL_word = 'URL';
116     if (count($arr_dependency['URLs']) > 1) {
117         $s_URL_word .= 's';
118     }
119
120     $s_result_content3 .= "<p>\n";
121
122     $s_dependency1 = $arr_dependency['dependency_name'];
123     $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
124     $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);
125
126     $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
127     $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
128     $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
129     $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";
130
131     for(
132         $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
133         $i <= $i_max;
134         ++$i
135     ){
136         $mixed_URL = $arr_dependency['URLs'][$i];
137         $s_comment = '';
138         $s_URL = '';
139         if(is_array($mixed_URL)){
140             if(!isset($mixed_URL['URL'])){
141                 throw new Exception("Dependency URL doesn't have an URL.");
142             }
143             if(!isset($mixed_URL['comment'])){
144                 throw new Exception("Dependency URL doesn't have a comment.");
145             }
146             $s_URL = $mixed_URL['URL'];
147             $s_comment = $mixed_URL['comment'];
148         }
149         else{
150             $s_URL = $mixed_URL;
151         }
152         if(!is_string($s_URL)){
153             throw new Exception("URL isn't a string.");
154         }
155         if(!is_string($s_comment)){
156             throw new Exception("Comment isn't a string.");
157         }
158     }

```

```

159     $i_URL_length = strlen($s_URL);
160     $i_comment_length = strlen($s_comment);
161
162     $s_result_content1 .= $s_URL;
163     $i_previous_line_length = $i_URL_length;
164     if($i_comment_length > 0){
165         if(
166             $i_previous_line_length + $i_comment_length + 1
167             > $i_max_line_length
168         ){
169             $s_result_content1 .= "\n";
170         }
171         else{
172             $s_result_content1 .= ' ';
173         }
174         $s_result_content1 .= $s_comment;
175     }
176     $s_result_content1 .= "\n";
177
178     if($i_URL_length > 60){
179         $s_result_content2 .= (
180             "{\\small\n"
181             .'\url{' . $s_URL . "}'\n"
182             .'}'
183         );
184         $i_previous_line_length = 1;
185     }
186     else{
187         $s_result_content2 .= '\url{' . $s_URL . '}' ;
188         $i_previous_line_length = $i_URL_length + 6;
189     }
190     if($i === $i_max){
191         ++$i_previous_line_length;
192     }
193     else{
194         $i_previous_line_length += 9;
195     }
196     $s_comment2 = string_escaping\TeX\escape_text($s_comment);
197     $i_comment_length2 = strlen($s_comment2);
198     if($i_comment_length2 > 0){
199         if(
200             $i_previous_line_length + $i_comment_length2 + 1
201             > $i_max_line_length
202         ){
203             $s_result_content2 .= "\n";
204         }

```

```

205         else{
206             $s_result_content2 .= ' ';
207         }
208         $s_result_content2 .= $s_comment2;
209     }
210     if($i === $i_max){
211         $s_result_content2 .= ".\n";
212     }
213     else{
214         $s_result_content2 .= ",\\newline\n";
215     }
216
217     $s_result_content3 .= (
218         "<a\n"
219         . 'href="'.string_escaping\HTML\escape_attribute($s_URL)."\".\n"
220         . "><span style=\"font-family:monospace\".\n"
221         . '>' .string_escaping\HTML\escape_text($s_URL)."</span\n"
222         . '></a>'
223     );
224     if($s_comment !== ''){
225         $s_result_content3 .= (
226             ' '.string_escaping\HTML\escape_text($s_comment)
227         );
228     }
229     if($i === $i_max){
230         $s_result_content3 .= ".\n";
231     }
232     else{
233         $s_result_content3 .= ",<br>\n";
234     }
235 }//end for urls
236
237 $s_result_content1 .= "\n";
238 $s_result_content2 .= "\n";
239 $s_result_content3 .= "</p>\n\n";
240 }//end foreach($arr_data as $arr_dependency)
241
242 // Add a "footer".
243 $s_closing_sentence = (
244     "If you see something that I missed regarding my dependencies,\n"
245     ."please tell me/email me :).\n"
246 );
247 $s_result_content1 .= $s_closing_sentence;
248 $s_result_content2 .= $s_closing_sentence;
249 $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
250

```

```

251 // We overwrite results if needed.
252 $$_result_path1 = $_subdir1.'/dependencies_notes.txt';
253 $_existing_result_content1 = '';
254 if(file_exists($_result_path1)){
255     $_existing_result_content1 = file_get_contents($_result_path1);
256 }
257 if($_result_content1 != $_existing_result_content1){
258     file_put_contents($_result_path1, $_result_content1);
259 }
260 /*
261 Example of generated content of dependencies_notes.txt:
262 -----
263 more of util-linux can be found at the following URLs:
264 https://www.kernel.org/pub/linux/utils/util-linux/
265 https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
266 https://github.com/util-linux/util-linux official?
267 -----
268 */
269 $_result_path2 = $_subdir2.'/dependencies_notes.tex.sub';
270 $_existing_result_content2 = '';
271 if(file_exists($_result_path2)){
272     $_existing_result_content2 = file_get_contents($_result_path2);
273 }
274 if($_result_content2 != $_existing_result_content2){
275     file_put_contents($_result_path2, $_result_content2);
276 }
277 /*
278 Example of generated content of dependencies_notes.tex.sub:
279 -----
280 more of util-linux can be found at the following URLs:\newline
281 \url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
282 {\small
283     \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
284 },\newline
285 \url{https://github.com/util-linux/util-linux} official?.
286 -----
287 */
288 $_result_path3 = $_subdir2.'/dependencies_notes.html.sub';
289 $_existing_result_content3 = '';
290 if(file_exists($_result_path3)){
291     $_existing_result_content3 = file_get_contents($_result_path3);
292 }
293 if($_result_content3 != $_existing_result_content3){
294     file_put_contents($_result_path3, $_result_content3);
295 }
296 /*

```

```

297 Example of generated content of dependencies_notes.html.sub:
298 <p>
299 more of util-linux can be found at the following URLs:<br>
300 <a
301 href="https://www.kernel.org/pub/linux/utils/util-linux/"
302 ><span style="font-family:monospace"
303 >https://www.kernel.org/pub/linux/utils/util-linux/</span
304 ></a>, <br>
305 <a
306 href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
307 ><span style="font-family:monospace"
308 >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
309 ></a>, <br>
310 <a
311 href="https://github.com/util-linux/util-linux"
312 ><span style="font-family:monospace"
313 >https://github.com/util-linux/util-linux</span
314 ></a> official?.
315 </p>
316 */
317 }//end build_dependencies_notes()
318 ?>

```

3.5 build_and_checks_dependencies/build_file_from_printable_file__standard.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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```



```

22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.lib.sh
27 source "${LFBFL_subdir}/common_options.lib.sh"
28 # shellcheck source=overwrite_if_not_equal.lib.sh
29 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
30
31 build_file_from_printable_file__standard() {
32     # $1=base_name
33     # Options:
34     #   --verbose
35     #   --work-directory=""
36     #   --remove-leading-whitespaces-after-escaped-line-return
37     declare -i LFBFL_i_verbose=0
38     get_verbose_option "$@"
39     local LFBFL_work_directory=""
40     get_work_directory_option "$@"
41     declare -i LFBFL_remove_leading_whitespaces_after_escaped_line_return=0
42     get_some_flag LFBFL_remove_leading_whitespaces_after_escaped_line_return\
43         --remove-leading-whitespaces-after-escaped-line-return 1 "$@"
44     declare -a LFBFL_return_traps_stack
45     init_return_trap
46     pushd_to_work_directory --trap-popd
47     can_continue_after_enhanced_pushd || return 1
48
49     local LFBFL_base_name
50     get_some_option LFBFL_base_name --base-name '' '' 1 "$@"
51     if [[ LFBFL_i_verbose -eq 1 ]]; then
52         if [[ "${LFBFL_base_name}" == "" ]]; then
53             return 2
54         fi
55         printf "Searching file to build from printable file: %s.\n"\
56             "${LFBFL_base_name}"
57     fi
58
59     local LFBFL_s_expression='s/\\n/Mg'
60     if [[
61         LFBFL_remove_leading_whitespaces_after_escaped_line_return -eq 1
62     ]]; then
63         LFBFL_s_expression='s/\\n */Mg'
64     fi
65     if [[ -f "${LFBFL_base_name}.tpl" ]]; then
66         sed --regexp-extended --null-data --expression="${LFBFL_s_expression}"\
67             -- "${LFBFL_base_name}.tpl"

```

```

68         > "${LFBFL_base_name}.temp"
69         overwrite_if_not_equal "${LFBFL_base_name}" "${LFBFL_base_name}.temp"
70     else
71         printf "No file %s.tpl\n" "${LFBFL_base_name}"
72         return 3
73     fi
74
75     if [[ ! -f "${LFBFL_base_name}" ]]; then
76         printf "No file %s\n" "${LFBFL_base_name}"
77         return 4
78     fi
79 }
80
81 build_file_from_printable_file__standard "$@"

```

3.6 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "build_md_from_printable_md.sh"
25 # to "build_md_from_printable_md.exec.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"

```

```

30 # shellcheck source=overwrite_if_not_equal.lib.sh
31 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
32
33 build_md_from_printable_md() {
34     # Options:
35     #     --verbose
36     #     --work-directory=""
37     #     --base-name="README"
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     local LFBFL_work_directory=""
41     get_work_directory_option "$@"
42     declare -a LFBFL_return_traps_stack
43     init_return_trap
44     pushd_to_work_directory --trap-popd
45     can_continue_after_enhanced_pushd || return 1
46
47     local LFBFL_base_name
48     get_some_option LFBFL_base_name --base-name README README 1 "$@"
49     if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
50         printf "Searching md file: %s.\n" "${LFBFL_base_name}"
51     fi
52
53     # Remove line returns here to keep lines short.
54     local LFBFL_sed_expression
55     LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]): [^\n\\]*\\\\n/\\1/Mg'
56     LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*\\\\n/\\1/Mg'
57     LFBFL_sed_expression+=';s/(\n[ ]*-<http[^\n\\]*\\\\n/\\1/Mg'
58     readonly LFBFL_sed_expression
59     # Since Markdown can contain code, it is limited to Markdown URLs.
60     # Hence, it is way more complicated than 's/\\\\n/\\1/Mg' expression
61     # in other files.
62     # Another downside is that more than one pass is needed.
63
64     if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
65         sed --regext-extended --null-data\
66             --expression="${LFBFL_sed_expression}"\
67             --expression="${LFBFL_sed_expression}"\
68             --expression="${LFBFL_sed_expression}"\
69             --expression="${LFBFL_sed_expression}"\
70             -- "${LFBFL_base_name}.md.tpl"\
71             > "${LFBFL_base_name}.md.temp"
72         overwrite_if_not_equal "${LFBFL_base_name}.md"\
73             "${LFBFL_base_name}.md.temp"
74     else
75         printf "No file %s.md.tpl\n" "${LFBFL_base_name}"

```

```

76     # Unfortunately, we cannot return early, since we want to propose
77     # HTML file generation with pandoc from .md file,
78     # if the file .md.tpl doesn't exist.
79 fi
80
81 if [[ ! -f "${LFBFL_base_name}.md" ]]; then
82     printf "No file %s.md\n" "${LFBFL_base_name}"
83     return 2
84 fi
85
86 pandoc --from=markdown --to=html --standalone\
87     --shift-heading-level-by=-1 --output="${LFBFL_base_name}.html.temp"\
88     -- "${LFBFL_base_name}.md"
89 # Correcting pandoc HTML "by hand".
90 # https://github.com/jgm/pandoc/issues/10957
91 # With the most open-minded answer found on Internet ;) XD.
92 # This second sed expression is what I need,
93 # but it may be unsuitable for you.
94 # Other problems:
95 # https://github.com/jgm/pandoc/issues/11484
96 local LFBFL_prewrap="s~(code \{(\n|[^}])*) ( \}) "
97 LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
98 readonly LFBFL_prewrap
99 sed --in-place --regexp-extended\
100     --expression='s~( <meta .*) />~\1>~g'\
101     --expression='s~<html .*>~<html lang="en">~'\
102     --expression='/      code\{white-space: pre-wrap;\}/d'\
103     -- "${LFBFL_base_name}.html.temp"
104 sed --in-place --regexp-extended --null-data\
105     --expression='s~(<img(\n|^[a-z0-9])(\n|^[>])*) />~\1>~Mg'\
106     --expression="${LFBFL_prewrap}"\
107     -- "${LFBFL_base_name}.html.temp"
108 overwrite_if_not_equal "${LFBFL_base_name}.html"\
109     "${LFBFL_base_name}.html.temp"
110
111 # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
112 #     --verbose "${LFBFL_base_name}.html"
113 }
114
115 build_md_from_printable_md "$@"

```

3.7 build_and_checks_dependencies/check_shell_scripts_beginnings.lib.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #

```

```

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23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "check_shell_scripts_beginnings.sh"
25 # to "check_shell_scripts_beginnings.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=lines_filters.lib.sh
31 source "${LFBFL_subdir}/lines_filters.lib.sh"
32
33 declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"
34
35 check_one_shell_script_beginning() {
36     # $1=file_path
37     #
38     # declare -r LFBFL_file_path=$(basename -- "$1")
39     # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
40     #     return 0
41     # fi
42     # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
43     # -----
44     # is of course slower than
45     # -----
46     # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
47     # -----
48     # is of course slower than
49     # -----

```

```

50 declare -r LFBFL_head=$(head --lines=1 -- "$1")
51 [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
52 || printf "%s:File has wrong shell script beginning.\n" "$1"
53 }
54
55 check_shell_scripts_beginnings() {
56     # Options:
57     #   --verbose
58     #   --work-directory=""
59     declare -i LFBFL_i_verbose=0
60     get_verbose_option "$@"
61     local LFBFL_work_directory=""
62     get_work_directory_option "$@"
63     declare -a LFBFL_return_traps_stack
64     init_return_trap
65     pushd_to_work_directory --trap-popd
66     can_continue_after_enhanced_pushd || return 1
67
68     # shopt -s globstar
69     # for LFBFL_file_path in **/*.sh; do
70     local LFBFL_file_path
71     local LFBFL_head
72     declare -r LFBFL_s_files_paths=$(
73         find . -type f -name "*.sh" -printf '%P\n'\
74         | relevant_find
75     )
76     if [[ -z "${LFBFL_s_files_paths}" ]]; then
77         [[ LFBFL_i_verbose -eq 1 ]]\
78         && printf "check_shell_scripts_beginnings: No file found.\n"
79         return
80     fi
81     declare -a LFBFL_arr_files_paths
82     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
83     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
84         [[ LFBFL_i_verbose -eq 1 ]]\
85         && printf "%s:Checking shell script beginning.\n"\
86             "${LFBFL_file_path}"
87         check_one_shell_script_beginning "${LFBFL_file_path}"
88     done
89 }

```

3.8 build_and_checks_dependencies/check_shell_scripts_indentation.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #

```

```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.lib.sh
27 source "${LFBFL_subdir}/common_options.lib.sh"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30
31 check_one_shell_script_indentation() {
32     # $1=file_path
33     declare -r LFBFL_NULL_PREVIOUS_LINE="--"
34     declare -i LFBFL_i_file_with_error=0
35     local LFBFL_some_line
36     local LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
37     local LFBFL_line_end
38     local LFBFL_previous_line_end
39     declare -i LFBFL_i_line_length
40     declare -i LFBFL_i_line_end_length
41     declare -i LFBFL_i_line_start_length
42     local LFBFL_spacing
43     declare -i LFBFL_i_offset
44     local LFBFL_substring1
45     local LFBFL_substring2
46     declare -r LFBFL_pattern='^ * -e '
47
48     # This grep would have been enough without some sed commands.
49     declare -r LFBFL_s_lines=$(

```

```

50     grep --before-context=1 --extended-regexp --line-number\
51         --regexp='^ ( ) * ([^ ]|$)' --with-filename -- "$1"
52 )
53 if [[ -z "${LFBFL_s_lines}" ]]; then
54     return
55 fi
56
57 declare -a LFBFL_arr_lines
58 mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
59 readonly LFBFL_arr_lines
60
61 for LFBFL_some_line in "${LFBFL_arr_lines[@]"; do
62     if [[ "${LFBFL_some_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]]; then
63         # When switching between code fragments, we reinit.
64         # It cannot happen for other grep output lines with line-number.
65         LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
66         continue
67     fi
68     # Note how it's funny that this loop always work because we check
69     # shell scripts beginnings in another script and thus we always
70     # have a line before.
71     if [[ "${LFBFL_previous_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]];
72     then
73         LFBFL_previous_line="${LFBFL_some_line}"
74         continue
75     fi
76     # Remove the name of the file and then the line number.
77     LFBFL_line_end=${LFBFL_some_line#*:}
78     LFBFL_line_end=${LFBFL_line_end#*:}
79     # But for the previous line '-' or ':' is used instead of ':'
80     # and filenames with '-' are much more frequent than with ':'...
81     LFBFL_i_line_length=${#LFBFL_some_line}
82     LFBFL_i_line_end_length=${#LFBFL_line_end}
83     LFBFL_i_line_start_length=$((
84         LFBFL_i_line_length - LFBFL_i_line_end_length
85     ))
86     LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
87     # Checking if increasing number of digits in the line numbers
88     # (99 -> 100).
89     LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
90     # Nice puzzle below :) Look for comments above :)
91     case ${LFBFL_substring1} in
92         -) ((LFBFL_i_offset += 1));;
93         :) ((LFBFL_i_offset += 1));;
94         [0-9]) ((LFBFL_i_offset += 2));;
95         *) printf ":-(\n";;

```



```

96     esac
97     LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
98     if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
99         # Line starts with spacing and -e
100        # Keep max spaces
101        LFBFL_spacing=${LFBFL_line_end%%-*}
102        LFBFL_i_offset=${#LFBFL_spacing}
103        LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
104        LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
105        if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
106            printf "%s\n" "${LFBFL_some_line}"
107            LFBFL_i_file_with_error=1
108        fi
109    else
110        printf "%s\n" "${LFBFL_some_line}"
111        LFBFL_i_file_with_error=1
112    fi
113    LFBFL_previous_line="${LFBFL_some_line}"
114 done
115
116 if [[ LFBFL_i_file_with_error -eq 1 ]]; then
117     printf "%s:File has some lines with odd number of spaces.\n" "$1"
118 fi
119 }
120
121 check_shell_scripts_indentation(){
122     # Options:
123     #   --verbose
124     #   --work-directory=""
125     declare -i LFBFL_i_verbose=0
126     get_verbose_option "$@"
127     local LFBFL_work_directory=""
128     get_work_directory_option "$@"
129     declare -a LFBFL_return_traps_stack
130     init_return_trap
131     pushd_to_work_directory --trap-popd
132     can_continue_after_enhanced_pushd || return 1
133
134     # shopt -s globstar
135     # for LFBFL_file_path in **/*.sh; do
136     local LFBFL_file_path
137     declare -r LFBFL_s_files_paths=$(
138         find . -type f -name "*.sh" -printf '%P\n'\
139         | relevant_find
140     )
141     if [[ -z "${LFBFL_s_files_paths}" ]]; then

```

```

142     [[ LFBFL_i_verbose -eq 1 ]]\
143     && printf "check_shell_scripts_indentation: No file found.\n"
144     return
145 fi
146 declare -a LFBFL_arr_files_paths
147 mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
148 readonly LFBFL_arr_files_paths
149 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
150     [[ LFBFL_i_verbose -eq 1 ]]\
151     && printf "%s:Checking shell script indentation.\n"\
152         "${LFBFL_file_path}"
153     check_one_shell_script_indentation "${LFBFL_file_path}"
154 done
155 }

```

3.9 build_and_checks_dependencies/check_URLs.lib.sh

```

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2  # This file is part of DevOrSysAdminScripts.
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23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "check_URLs.sh" to "check_URLs.lib.sh".
25
26 LFBFL_subdir="build_and_checks_dependencies"
27 # shellcheck source=common_options.lib.sh
28 source "${LFBFL_subdir}/common_options.lib.sh"
29 # shellcheck source=get_common_text_glob_patterns.lib.sh

```

```

30 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
31 # shellcheck source=lines_filters.lib.sh
32 source "${LFBFL_subdir}/lines_filters.lib.sh"
33
34 check_URLs() {
35     # Options:
36     #   --verbose
37     #   --work-directory=""
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     local LFBFL_work_directory=""
41     get_work_directory_option "$@"
42     declare -a LFBFL_return_traps_stack
43     init_return_trap
44     pushd_to_work_directory --trap-popd
45     can_continue_after_enhanced_pushd || return 1
46
47     enhanced_set_shell_option pipefail --trap-unset
48
49     get_COMMON_TEXT_FILES_GLOB_PATTERNS
50
51     declare -Ar LFBFL_substitutions=(
52         ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
53     )
54
55     local LFBFL_pattern
56     local LFBFL_base_file_name
57     local LFBFL_substitution
58     local LFBFL_substitution2
59     local LFBFL_file_path
60     local LFBFL_s_files_paths
61     declare -a LFBFL_arr_files_paths
62     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}; do
63         if [[ LFBFL_i_verbose -eq 1 ]]; then
64             printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
65         fi
66         find . -type f -name "${LFBFL_pattern}" -printf '%P\0'\
67             | xargs -0 grep --with-filename -- 'http:'\
68             | grep --invert-match " | xargs -0 grep --with-filename -- 'htt"\
69             | grep --invert-match --perl-regexp\
70             "(['\"]http(?:[^'\"]*)\\1.*(['\"]https\\2\\3"\
71             | grep --invert-match\
72             --regexp='http://www.w3.org/1999/xhtml'\
73             --regexp="http://www.w3.org/1999/XSL/Transform"\
74             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd'\
75             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'\

```

```

76         --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd'\
77         --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd'
78     # Second grep after xargs grep just above will remove false positives
79     # from substitutions that fit on one line.
80
81     LFBFL_s_files_paths=$(
82         find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
83         | relevant_find\
84         | not_main_html_find
85     )
86     if [[ -z "${LFBFL_s_files_paths}" ]]; then
87         [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
88         continue
89     fi
90     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
91     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
92         LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
93         [[ "${LFBFL_base_file_name}" != "check_URLs.lib.sh" ]]\
94         || continue
95         if [[ LFBFL_i_verbose -eq 1 ]]; then
96             printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
97         fi
98         for LFBFL_substitution in "${!LFBFL_substitutions[@]"; do
99             LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
100             if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
101             then
102                 sed --in-place\
103                 --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
104                 -- "${LFBFL_file_path}"
105             fi
106         done
107     done
108 done
109 }

```

3.10 build_and_checks_dependencies/common_build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "common_build_and_checks.exec.sh.tpl" was renamed from
25 # "common_build_and_checks.sh.tpl" to
26 # "common_build_and_checks.exec.sh.tpl".
27 # The file "common_build_and_checks.exec.sh" generated from the file
28 # "common_build_and_checks.sh.tpl"
29 # or "common_build_and_checks.exec.sh.tpl"
30 # was renamed from
31 # "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".
32
33 common_build_and_checks() {
34     # $1 LFBFL_work_directory
35     # $2 LFBFL_dependencies_URL
36     # Options:
37     # --check-download=y|keep (=keep to keep the downloaded files)
38     # --verbose
39     local LFBFL_work_directory="${1:-}"
40     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41     declare -r LFBFL_dependencies_URL="$2"
42     local LFBFL_verbose=""
43     declare -i LFBFL_i_verbose=0
44     if [[ "$*" == *--verbose* ]]; then
45         printf "%s %s\n" "$0" "$*"
46         LFBFL_verbose="--verbose"
47         LFBFL_i_verbose=1
48     fi
49     readonly LFBFL_verbose
50     readonly LFBFL_i_verbose
51     local LFBFL_check_download=""
52     if [[ "$*" == *--check-download=keep* ]]; then
53         LFBFL_check_download="--check-download=keep"
54     elif [[ "$*" == *--check-download=y* ]]; then
55         LFBFL_check_download="--check-download=y"

```

```

56 fi
57 readonly LFBFL_check_download
58
59 declare -a LFBFL_some_common_options=(
60     "${LFBFL_verbose}"
61     "--work-directory=${LFBFL_work_directory}"
62 )
63
64 source ./wget_sha512.lib.sh
65
66 local LFBFL_subdir="build_and_checks_dependencies"
67 declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68 declare -r LFBFL_subdir3="build_and_checks_variables"
69 local LFBFL_file_name
70 local LFBFL_script_download_URL
71 local LFBFL_file_path
72 local LFBFL_correct_sha512
73
74 wrapped_wget_sha512() {
75     wget_sha512 "${LFBFL_file_path}" \
76         "${LFBFL_script_download_URL}" \
77         "${LFBFL_correct_sha512}" \
78         "${LFBFL_verbose}" \
79         "${LFBFL_check_download}"
80 }
81
82 LFBFL_file_name="build_dependencies_notes.exec.php"
83 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85 LFBFL_correct_sha512='3d881a757c69fbcc02cbc5a568c508cfa6249c1330ea6'
86 LFBFL_correct_sha512+='7eeca6d284950b3ca2ced7b4acce4142cdf90c5aaa88'
87 LFBFL_correct_sha512+='5bb3ee77f3a32ae0d4b412173cec0d944278ab0'
88 wrapped_wget_sha512
89 chmod +x "${LFBFL_file_path}"
90
91 LFBFL_file_name="build_dependencies_notes.lib.php"
92 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
93 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
94 LFBFL_correct_sha512='6203cb1c31b656004ec8467d4f84288f342fcc7f5c236'
95 LFBFL_correct_sha512+='270e819b986fe967aa6abe101188a7fcl1a55e2f4c9b5'
96 LFBFL_correct_sha512+='9a6dde6c9aa125b385d0ce85e02b1de69ce1084'
97 wrapped_wget_sha512
98
99 LFBFL_file_name="build_file_from_printable_file_standard.exec.sh"
100 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
101 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"

```

```

102 LFBFL_correct_sha512='9c5b11280019b4bbc679bddb8eb5c66d9ef715d04c8c1'
103 LFBFL_correct_sha512+='9e2cbd55e08ad38169b82aa2de5fe59f46053552a358'
104 LFBFL_correct_sha512+='d92f3554cfc8ce7d930055b400ec1e6ee5e3069'
105 wrapped_wget_sha512
106 chmod +x "./${LFBFL_file_path}"
107
108 LFBFL_file_name="build_md_from_printable_md.exec.sh"
109 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
110 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
111 LFBFL_correct_sha512='a39101cb7c0a7a743386b0c8745071cc15f8fd93be3ac'
112 LFBFL_correct_sha512+='94bd5d69860836e4ae94b08a3219c2542f9d6501f0b4'
113 LFBFL_correct_sha512+='d4b1bc40700afcb1b555bbbe25de6485f95c361'
114 wrapped_wget_sha512
115 chmod +x "./${LFBFL_file_path}"
116
117 LFBFL_file_name="check_shell_scripts_beginnings.lib.sh"
118 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
119 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
120 LFBFL_correct_sha512='0b3e861820e09148a141142c8bb5549e7fa29d906dda9'
121 LFBFL_correct_sha512+='5e0fe44132610910687e6f767f1050646abelea262b8'
122 LFBFL_correct_sha512+='37b3a5d5e79497d0cdc42e43d3c74967f266abb'
123 wrapped_wget_sha512
124
125 LFBFL_file_name="check_shell_scripts_indentation.lib.sh"
126 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
127 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
128 LFBFL_correct_sha512='3596fddc4b20ce353facac3e355e38fadb5593caee3c8'
129 LFBFL_correct_sha512+='05464aedd513d34860a79b296efa648dae18717ae4c0'
130 LFBFL_correct_sha512+='24a6df11eb0549c4797643427c49c4f80947840'
131 wrapped_wget_sha512
132
133 LFBFL_file_name="check_URLs.lib.sh"
134 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
135 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
136 LFBFL_correct_sha512='d3ddcd56520d1786cd5afb8fd0f7ee278ed7307a6f5d2'
137 LFBFL_correct_sha512+='1f948d0f2f6f2ed5210072d923f2036643c7181012fc'
138 LFBFL_correct_sha512+='15d7921f5617105012a14b8adc9acc76d2b50cb'
139 wrapped_wget_sha512
140
141 LFBFL_file_name="common_options.lib.sh"
142 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
143 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
144 LFBFL_correct_sha512='3cfaf6cea99f67ade931cf21e9f49658c421e182dc4ac'
145 LFBFL_correct_sha512+='211be526f9d99b2e57a09844991cf0d2e6d5e49c19e2'
146 LFBFL_correct_sha512+='a63bf9e8f1e3fe999542e9a4e45e5196b375fa9'
147 wrapped_wget_sha512

```

```

148
149 LFBFL_file_name="comparisons.lib.sh"
150 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
151 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
152 LFBFL_correct_sha512='f146beca0d7b4046c140f3b2292cec4c31515cf8c7dbe'
153 LFBFL_correct_sha512+='7e9b07fc6dae32513aff6692c6286fa31055caff1a20'
154 LFBFL_correct_sha512+='9a6a7384568fea0b50b3cbb2608a3ed9e6e5125'
155 wrapped_wget_sha512
156
157 LFBFL_file_name="create_PDF.exec.sh"
158 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
159 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
160 LFBFL_correct_sha512='664cb7d2f3f312796f3651bc2c552bbc3ffa2b6d0ff80'
161 LFBFL_correct_sha512+='d5d9acbad4a874b6f3db6e9d1e446a2e69477c5cda69'
162 LFBFL_correct_sha512+='b84fccbdd3f112361850b93c84bbf4cab6b2892'
163 wrapped_wget_sha512
164 chmod +x "${LFBFL_file_path}"
165
166 LFBFL_file_name="generate_from_template.lib.sh"
167 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
168 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
169 LFBFL_correct_sha512='5f322f27bd20d7042c729f4cb5d8f14f587b5b415c77e'
170 LFBFL_correct_sha512+='4a224902d8d4cdeb1483639ae85d81873c9f20c175bb'
171 LFBFL_correct_sha512+='4de76a6eaa3dbcdf8a300368e65bbd5303afd3d'
172 wrapped_wget_sha512
173
174 LFBFL_file_name="get_common_text_glob_patterns.lib.sh"
175 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
176 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
177 LFBFL_correct_sha512='0945c9896ce9d26637a8ba5c08a1fdad102b725de26ce'
178 LFBFL_correct_sha512+='088e519a3696925465007ad0b460ae7433174d8af848'
179 LFBFL_correct_sha512+='b77dfdf0da78534453e90303d3bd458bf378e9'
180 wrapped_wget_sha512
181
182 LFBFL_file_name="grammar_and_spelling_check.lib.sh"
183 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
184 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
185 LFBFL_correct_sha512='88cc4f031c05d576a74ea7b14e79077f06c53b3b8a294'
186 LFBFL_correct_sha512+='e9b8ddcfe884db5e2cb5cebde09a8d9aea7980fb595c'
187 LFBFL_correct_sha512+='85c562462fb67e10503a471ee46da89c8f73b47'
188 wrapped_wget_sha512
189
190 # /licenses_templates/ -----
191 local LFBFL_dependencies_URL2
192 LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
193 readonly LFBFL_dependencies_URL2

```



```

194 LFBFL_file_name="build_licenses_templates.exec.sh"
195 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
196 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
197 LFBFL_correct_sha512='9b06e0631a64496c0e412ef412fb676bff0ace10cdbf7'
198 LFBFL_correct_sha512+='a6b92b7cb087049f8d2ae38261b316f74eedb5a9d2e3'
199 LFBFL_correct_sha512+='4825396438729e70ce82e38df766df7fb416b5b'
200 wrapped_wget_sha512
201 chmod +x "${LFBFL_file_path}"
202
203 LFBFL_file_name="license_file_header_AGPLv3+.tpl"
204 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
205 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
206 LFBFL_correct_sha512='5dc0dc20914c57feee2e26c2d329f020f221357b11b0d'
207 LFBFL_correct_sha512+='0ff6add69941ad953f8356b68be32cd60df8c56e91a2'
208 LFBFL_correct_sha512+='716c9cbf70ab8698e74e6aacc9c5439a93cc2bc'
209 wrapped_wget_sha512
210
211 LFBFL_file_name="license_file_header_GPLv3+.tpl"
212 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
213 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
214 LFBFL_correct_sha512='2d4b31697e78ba36ee0e3bd2b5393a57e15b5f5e62b2e'
215 LFBFL_correct_sha512+='802b812dcb087588f63cb92a851b21180be24bf5dee4'
216 LFBFL_correct_sha512+='eff7927244cde817e964512eaa1fb51d9064c01'
217 wrapped_wget_sha512
218
219 LFBFL_file_name="license_file_header_LGPLv3+.tpl"
220 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
221 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
222 LFBFL_correct_sha512='43976b6dc9a54fcea0440cfaf87530dade73efe22b5ae'
223 LFBFL_correct_sha512+='a968453c9c5ac1fdc4d3798fdd445fb38a53f997da82'
224 LFBFL_correct_sha512+='7c30be6b7ad42d5e56ae90aac5095b9526b203d'
225 wrapped_wget_sha512
226 # -----
227
228 LFBFL_file_name="lines_counts.lib.sh"
229 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
230 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
231 LFBFL_correct_sha512='d131e24ad596988a6953b9362b13e5d17330f3bd8013b'
232 LFBFL_correct_sha512+='03fc2acf74a6b931660a619149cce59d1fa3b075c11c'
233 LFBFL_correct_sha512+='0a5e9c85e847b437d7bc3b58f034c00df3c5702'
234 wrapped_wget_sha512
235
236 LFBFL_file_name="lines_filters.lib.sh"
237 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
238 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
239 LFBFL_correct_sha512='d7b9d7456d18965b7910516cf80f96b5a89b76c59f315'

```

```

240 LFBFL_correct_sha512+= 'f7558487f5f37afc8d0af13d4f11928f7c2bec8a03a1'
241 LFBFL_correct_sha512+= '7b49a9a5f71945d2a6fff955ce98518f5305234'
242 wrapped_wget_sha512
243
244 LFBFL_file_name="lines_maps.lib.sh"
245 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
246 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
247 LFBFL_correct_sha512='ccb9115be582a5b39790725dc96a628beb21b0c9951e7'
248 LFBFL_correct_sha512+= '03a7a7bc8ab2736e3940ed3df6a2fcb04e75350c10d7'
249 LFBFL_correct_sha512+= '3e92bcfada424b4049cc6141d41bf7901c0525b'
250 wrapped_wget_sha512
251
252 LFBFL_file_name="overwrite_if_not_equal.lib.sh"
253 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
254 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
255 LFBFL_correct_sha512='378f03b8ba77430e05db4ef5d5926803b466a87e1a2ec'
256 LFBFL_correct_sha512+= 'eef0b51036f25460a93a018bd86d3f90146b4dfec9ab'
257 LFBFL_correct_sha512+= '15c5d1d57a526296bbdc4a0c03650088f3e9653'
258 wrapped_wget_sha512
259
260 LFBFL_file_name="phpmd_baseline.xslt"
261 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
262 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
263 LFBFL_correct_sha512='88f39ea6952acebdad60c267e0f49a0489b486f7ead53'
264 LFBFL_correct_sha512+= '4c6ccf32e9f2bfa86f92d772c2fa59fecef00d6c9e6c'
265 LFBFL_correct_sha512+= '4d31cf074dda4943070db069c64581a18bfe724'
266 wrapped_wget_sha512
267
268 LFBFL_file_name="python_black_complement.lib.sh"
269 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
270 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
271 LFBFL_correct_sha512='4f08461329f1b78b58e54f675fb433aec363de2c96a91'
272 LFBFL_correct_sha512+= '34ba5e96f3c77b93c9d7d6e28771a0b886501c286457'
273 LFBFL_correct_sha512+= 'a4a410389556f180f2dc8f66d7956bf5dd5acf2'
274 wrapped_wget_sha512
275
276 LFBFL_file_name="python_isort_complement.lib.sh"
277 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
278 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
279 LFBFL_correct_sha512='f91f3236ff9720412b31ec9fec2a5a5b3ef2424931815'
280 LFBFL_correct_sha512+= 'c0e6d1bd9821e86ef80a13a27a3f6d831327dc4b9698'
281 LFBFL_correct_sha512+= '5935a75793c87bdf735c4ae2a94327289d1c2ed'
282 wrapped_wget_sha512
283
284 LFBFL_file_name="repository_data.lib.sh"
285 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"

```

```

286 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
287 LFBFL_correct_sha512='8148fae16b5853b5ed6ca141b73b32f6d8695d8d34a54'
288 LFBFL_correct_sha512+='73ad8e8cc0dab61d92e87fd925b6f25c0632f70b2fee'
289 LFBFL_correct_sha512+='78ee53b0e3af6aa25b3cc58d8769292c3e4defd'
290 wrapped_wget_sha512
291
292 LFBFL_file_name="shell_checks_complement.lib.sh"
293 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
294 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
295 LFBFL_correct_sha512='143575d86b5fc0fd78e575b6e46054b0c8facaf2c6df6'
296 LFBFL_correct_sha512+='07dc596b6db0494c2b88211d5e0e011dee9abe70a2fc'
297 LFBFL_correct_sha512+='b8f2bc04eb5cd50b248b0afe423103ee1cd568b'
298 wrapped_wget_sha512
299
300 LFBFL_file_name="split_score.exec.php"
301 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
302 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
303 LFBFL_correct_sha512='2845905b970c2dae5cf4a072df3e00670607f576d88db'
304 LFBFL_correct_sha512+='22176e57f47081d487eaf0d5d006950550932bae2686'
305 LFBFL_correct_sha512+='9ad1933bb2adcaf2719469fad83ea55e53e154e'
306 wrapped_wget_sha512
307 chmod +x "${LFBFL_file_path}"
308
309 LFBFL_file_name="split_score.lib.php"
310 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
311 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
312 LFBFL_correct_sha512='a66cf904ba51bf9180c887d67f81b8d64eb0ac9b95a3c'
313 LFBFL_correct_sha512+='696e51e050743956029d2d247e2252e2763ceffbd201'
314 LFBFL_correct_sha512+='459ac09ed5ed53e2af2436ed3610fe56a88d384'
315 wrapped_wget_sha512
316
317 LFBFL_file_name="strings_functions.lib.sh"
318 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
319 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
320 LFBFL_correct_sha512='526bdaeca9ce0732ad429c16410909f9f8014eef9e031'
321 LFBFL_correct_sha512+='837c428786b19bc31fea3eff6c8d93eea260b9076b67'
322 LFBFL_correct_sha512+='71d21220528e70aa08a1a2fb7579926db662ff2'
323 wrapped_wget_sha512
324
325 LFBFL_file_name="too_long_code_lines.lib.sh"
326 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
327 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
328 LFBFL_correct_sha512='0571332f70cd5700f35ba5eb804bbe20a58d01ca33635'
329 LFBFL_correct_sha512+='c36f40a8aa24ddc7926ee66d6372191b91e6b2ffcac3'
330 LFBFL_correct_sha512+='9d4e89ef8a1e0c32cbae6c2e1e69437a65d3efc'
331 wrapped_wget_sha512

```

```

332
333 LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
334 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
335 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
336 LFBFL_correct_sha512='188fa5dbcab63be8e0f5b3c8997e099c7259f7c804433'
337 LFBFL_correct_sha512+='581c2c05a7d9c7bd3d82bc424e2021afa90cb9e9d6a5'
338 LFBFL_correct_sha512+='f84f549ed692c18b6351983c31b875846d7e1af'
339 wrapped_wget_sha512
340 chmod +x "${LFBFL_file_path}"
341
342 # shellcheck source=check_shell_scripts_beginnings.lib.sh
343 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
344 # shellcheck source=check_shell_scripts_indentation.lib.sh
345 source "${LFBFL_subdir}/check_shell_scripts_indentation.lib.sh"
346 # shellcheck source=check_URLs.lib.sh
347 source "${LFBFL_subdir}/check_URLs.lib.sh"
348 # shellcheck source=common_options.lib.sh
349 source "${LFBFL_subdir}/common_options.lib.sh"
350 # shellcheck source=comparisons.lib.sh
351 source "${LFBFL_subdir}/comparisons.lib.sh"
352 # shellcheck source=generate_from_template.lib.sh
353 source "${LFBFL_subdir}/generate_from_template.lib.sh"
354 # shellcheck source=get_common_text_glob_patterns.lib.sh
355 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
356 # shellcheck source=grammar_and_spelling_check.lib.sh
357 source "${LFBFL_subdir}/grammar_and_spelling_check.lib.sh"
358 # shellcheck source=lines_counts.lib.sh
359 source "${LFBFL_subdir}/lines_counts.lib.sh"
360 # shellcheck source=lines_filters.lib.sh
361 source "${LFBFL_subdir}/lines_filters.lib.sh"
362 # shellcheck source=lines_maps.lib.sh
363 source "${LFBFL_subdir}/lines_maps.lib.sh"
364 # shellcheck source=overwrite_if_not_equal.lib.sh
365 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
366 # shellcheck source=python_black_complement.lib.sh
367 source "${LFBFL_subdir}/python_black_complement.lib.sh"
368 # shellcheck source=python_isort_complement.lib.sh
369 source "${LFBFL_subdir}/python_isort_complement.lib.sh"
370 # shellcheck source=shell_checks_complement.lib.sh
371 source "${LFBFL_subdir}/shell_checks_complement.lib.sh"
372 # shellcheck source=strings_functions.lib.sh
373 source "${LFBFL_subdir}/strings_functions.lib.sh"
374 # shellcheck source=too_long_code_lines.lib.sh
375 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
376
377 enhanced_set_bash_option extglob

```

```

378 # shellcheck source=repository_data.lib.sh
379 source "${LFBFL_subdir}/repository_data.lib.sh"
380 # shellcheck disable=SC2154,SC2309
381 [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
382   && enhanced_unset_bash_option_extglob
383
384 declare -a LFBFL_return_traps_stack
385 init_return_trap
386
387 enhanced_set_shell_option pipefail --trap-unset
388
389 declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
390 keep_options LFBFL_some_common_options2 --verbose
391
392 printf "Building license headers\n"
393 "${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
394   "${LFBFL_some_common_options[@]}"
395
396 printf "Building README.md\n"
397 "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
398   --work-directory="${LFBFL_work_directory}"\
399   --base-name=README\
400   "${LFBFL_verbose}"
401
402 printf "Building other MarkDown files\n"
403 local LFBFL_some_directory
404 declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
405 local LFBFL_s_files_paths
406 LFBFL_s_files_paths=(
407   find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
408   | relevant_find
409 )
410 declare -a LFBFL_arr_files_paths
411 if [[ -n "${LFBFL_s_files_paths}" ]]; then
412   mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
413   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
414     [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
415     printf "Found template %s.\n" "${LFBFL_file_path}"
416     LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
417     LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
418     LFBFL_file_name=${LFBFL_file_name%.md.tpl}
419     "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
420       --work-directory="${LFBFL_some_directory}"\
421       --base-name="${LFBFL_file_name}"\
422       "${LFBFL_verbose}"
423   done

```

```

424 fi
425
426 printf "Building other standard files\n"
427 LFBFL_s_files_paths=$(
428     find "${LFBFL_work_directory}" -type f -name "*.neon.tpl"\
429     | relevant_find
430 )
431 if [[ -n "${LFBFL_s_files_paths}" ]]; then
432     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
433     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
434         printf "Found template %s.\n" "${LFBFL_file_path}"
435         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
436         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
437         LFBFL_file_name=${LFBFL_file_name%.tpl}
438         "${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh"\
439         --work-directory="${LFBFL_some_directory}"\
440         --base-name="${LFBFL_file_name}"\
441         "${LFBFL_verbose}"\
442         --remove-leading-whitespaces-after-escaped-line-return
443     done
444 fi
445
446 printf "Building dependencies_notes\n"
447 # https://github.com/php/php-src/issues/21538
448 local LFBFL_annoying_warning
449 LFBFL_annoying_warning="PHP Warning: declare(encoding=...) ignored"
450 LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
451 LFBFL_annoying_warning=" by settings in"
452 readonly LFBFL_annoying_warning
453 if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
454     php -f="${LFBFL_subdir}/build_dependencies_notes.exec.php"\
455     -- "${LFBFL_work_directory}"\
456     2>&1\
457     | grep --invert-match "${LFBFL_annoying_warning}"
458 elif [[ LFBFL_i_verbose -eq 1 ]]; then
459     printf "No file dependencies_data.json.\n"
460 fi
461
462 declare -i LFBFL_i_directory_changed
463 pushd_to_work_directory
464 LFBFL_i_directory_changed=$?
465 can_continue_after_enhanced_pushd || return 1
466
467 local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
468
469 declare -i LFBFL_i_max_line_length

```

```

470 grep_variable "${LFBFL_data_file_path}" max_line_length\
471     --result-variable-prefix=LFBFL_i_
472
473 LFBFL_some_common_options+=(
474     "--max-line-length=${LFBFL_i_max_line_length}"
475 )
476 LFBFL_some_common_options2+=(
477     "--max-line-length=${LFBFL_i_max_line_length}"
478 )
479
480 declare -i LFBFL_i_upgrade_venvs=0
481 local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
482 readonly LFBFL_upgrade_venvs_ts_file
483 declare -i LFBFL_i_upgrade_venvs_ts
484 declare -i LFBFL_i_current_ts
485 local LFBFL_upgrade_venvs_answer
486
487 local LFBFL_upgrade_venvs_time_interval_in_seconds=""
488 get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
489     "${LFBFL_some_common_options2[@]}"
490
491 if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
492     LFBFL_i_upgrade_venvs_ts=$(
493         stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
494     )
495     LFBFL_i_current_ts=$(date +%s)
496     ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
497     ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
498     if [[ LFBFL_i_current_ts -gt 0 ]]; then
499         LFBFL_i_upgrade_venvs=1
500     fi
501 else
502     LFBFL_i_upgrade_venvs=1
503 fi
504 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
505     local LFBFL_upgrade_venvs=""
506     grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
507         --result-variable-prefix=LFBFL_
508     if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
509         read -r -n 1 -t 10\
510             -p "Upgrade venvs and composer global? [Y/n]"\
511             LFBFL_upgrade_venvs_answer
512         if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
513             LFBFL_i_upgrade_venvs=0
514         fi
515     fi

```

```

516     fi
517
518     printf "Analyzing too long lines\n"
519     too_long_code_lines "${LFBFL_some_common_options2[@]}"
520
521     printf "Analyzing URLs\n"
522     check_URLs "${LFBFL_some_common_options2[@]}" \
523         | relevant_grep
524
525     printf "Analyzing strange characters: hover over in doubt\n"
526     # shellcheck disable=SC1111
527     LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç@'\"`|_|«»..."
528     grep --binary-files=without-match --color=always --exclude-dir .git\
529         --line-number --perl-regexp --recursive --\
530         "[^${LFBFL_usual_characters}]" .\
531         | relevant_grep
532
533     printf "Grammar and spelling check\n"
534     grammar_and_spelling_check "${LFBFL_data_file_path}" \
535         "${LFBFL_some_common_options2[@]}"
536
537     printf "Running shellcheck\n"
538     if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
539         LFBFL_s_files_paths=(
540             find . -type f -name "*.sh" \
541             | relevant_find
542         )
543         if [[ -n "${LFBFL_s_files_paths}" ]]; then
544             mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
545             for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
546                 shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini" \
547                     -- "${LFBFL_file_path}"
548             done
549         fi
550     elif [[ LFBFL_i_verbose -eq 1 ]]; then
551         printf "No file shellcheck.ini.\n"
552     fi
553     printf "Running shell_checks_complement\n"
554     shell_checks_complement "${LFBFL_some_common_options2[@]}"
555
556     printf "Analyzing shell scripts beginnings\n"
557     check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
558         | relevant_grep
559
560     printf "Analyzing shell scripts indentation\n"
561     check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \

```



```

562         | relevant_grep
563
564 printf -- "---Python---\n"
565
566 maybe_deactivate() {
567     if command -v deactivate
568     then
569         deactivate
570     fi
571 }
572
573 source_without_return_trap() {
574     declare -i LFBFL_i_no_return_traps=1
575     # shellcheck disable=SC1090,SC1091
576     source -- "$1"
577 }
578
579 printf "Running isort\n"
580 local LFBFL_isort_venv=""
581 grep_variable "${LFBFL_data_file_path}" isort_venv\
582     --result-variable-prefix=LFBFL_\
583     --replace-line-returns-by=""
584 if [[ -n "${LFBFL_isort_venv}" ]]; then
585     maybe_deactivate
586     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
587 fi
588 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
589     pip install --upgrade isort
590 fi
591 isort --profile=black .
592 if [[ -n "${LFBFL_isort_venv}" ]]; then
593     deactivate
594 fi
595 printf "Running python_isort_complement\n"
596 python_isort_complement "${LFBFL_some_common_options2[@]}"
597
598 printf "Running black\n"
599 # First, we update the configuration file with max_line_length.
600 if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
601     local LFBFL_update_max_length="s/^line-length = [0-9]*$/
602     LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
603     sed --regext-extended --expression="${LFBFL_update_max_length}"\
604         "${LFBFL_subdir3}/black.toml"\
605         > "${LFBFL_subdir3}/black.toml.temp"
606     overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
607         "${LFBFL_subdir3}/black.toml.temp"

```

```

608 elif [[ LFBFL_i_verbose -eq 1 ]]; then
609     printf "No file black.toml.\n"
610 fi
611
612 local LFBFL_black_venv=""
613 grep_variable "${LFBFL_data_file_path}" black_venv\
614     --result-variable-prefix=LFBFL_\
615     --replace-line-returns-by=""
616 if [[ -n "${LFBFL_black_venv}" ]]; then
617     maybe_deactivate
618     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
619 fi
620 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
621     pip install --upgrade black
622 fi
623 if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
624     black --config="${LFBFL_subdir3}/black.toml" .
625 fi
626 if [[ -n "${LFBFL_black_venv}" ]]; then
627     deactivate
628 fi
629 printf "Running python_black_complement\n"
630 python_black_complement "${LFBFL_some_common_options2[@]}"
631
632 printf "Probing if mypy should be runned\n"
633 local LFBFL_mypy_venv=""
634 grep_variable "${LFBFL_data_file_path}" mypy_venv\
635     --result-variable-prefix=LFBFL_\
636     --replace-line-returns-by=""
637 if [[ -n "${LFBFL_mypy_venv}" ]]; then
638     maybe_deactivate
639     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
640 fi
641 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
642     pip install --upgrade mypy
643 fi
644 local LFBFL_directory_path
645 declare -i LFBFL_i_no_toml=1
646 LFBFL_s_files_paths=$(
647     find . -type f -name "pyproject.toml"\
648     | relevant_find
649 )
650 if [[ -n "${LFBFL_s_files_paths}" ]]; then
651     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
652     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
653         if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then

```

```

654         printf "Running mypy\n"
655         LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
656         mypy -- "${LFBFL_directory_path}"
657     fi
658     LFBFL_i_no_toml=0
659 done
660 fi
661 if [[ LFBFL_i_no_toml -eq 1 ]]; then
662     printf "Running mypy\n"
663     mypy .
664 fi
665 if [[ -n "${LFBFL_mypy_venv}" ]]; then
666     deactivate
667 fi
668
669 printf "Running bandit\n"
670 LFBFL_bandit_venv=""
671 grep_variable "${LFBFL_data_file_path}" bandit_venv\
672     --result-variable-prefix=LFBFL\
673     --replace-line-returns-by=""
674 if [[ -n "${LFBFL_bandit_venv}" ]]; then
675     maybe_deactivate
676     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
677 fi
678 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
679     pip install --upgrade bandit
680 fi
681 if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
682     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
683         --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
684         --recursive .
685     # Saving new baseline in temp if necessary.
686     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
687         --format=json\
688         --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
689         --recursive .
690 elif [[ LFBFL_i_verbose -eq 1 ]]; then
691     printf "No file bandit.ini.\n"
692 fi
693 if [[ -n "${LFBFL_bandit_venv}" ]]; then
694     deactivate
695 fi
696
697 printf "Running pylint\n"
698 LFBFL_pylint_venv=""
699 grep_variable "${LFBFL_data_file_path}" pylint_venv\

```

```

700     --result-variable-prefix=LFBFL_\
701     --replace-line-returns-by=""
702 if [[ -n "${LFBFL_pylint_venv}" ]]; then
703     maybe_deactivate
704     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
705 fi
706 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
707     pip install --upgrade pylint
708 fi
709 if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
710     pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
711 elif [[ LFBFL_i_verbose -eq 1 ]]; then
712     printf "No file pylintrc.toml.\n"
713 fi
714 if [[ -n "${LFBFL_pylint_venv}" ]]; then
715     deactivate
716 fi
717
718 printf "Running ruff\n"
719 LFBFL_ruff_venv=""
720 grep_variable "${LFBFL_data_file_path}" ruff_venv\
721     --result-variable-prefix=LFBFL_\
722     --replace-line-returns-by=""
723 if [[ -n "${LFBFL_ruff_venv}" ]]; then
724     maybe_deactivate
725     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
726 fi
727 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
728     pip install --upgrade ruff
729 fi
730 if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
731     ruff check --config="${LFBFL_subdir3}/ruff.toml"
732 elif [[ LFBFL_i_verbose -eq 1 ]]; then
733     printf "No file ruff.toml.\n"
734 fi
735 if [[ -n "${LFBFL_ruff_venv}" ]]; then
736     deactivate
737 fi
738 printf -- "----Python end---\n"
739
740 printf -- "----PHP---\n"
741 printf "Running PHP native linter\n"
742 LFBFL_s_files_paths=$(
743     find . -type f -name "*.php"\
744     | relevant_find
745 )

```

```

746 if [[ -n "${LFBFL_s_files_paths}" ]]; then
747     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
748     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"}; do
749         php --syntax-check -f="${LFBFL_file_path}"\
750             2>&1\
751             | grep --invert-match "${LFBFL_annoying_warning}"
752     done
753 fi
754
755 printf "Running PHP_CodeSniffer\n"
756 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
757     composer global require squizlabs/php_codesniffer
758 fi
759 if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
760     phpcs --report=code\
761         --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
762 elif [[ LFBFL_i_verbose -eq 1 ]]; then
763     printf "No file phpcs_ruleset.xml.\n"
764 fi
765
766 printf "Running PHPMD\n"
767 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
768     composer global require phpmd/phpmd
769 fi
770
771 declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
772 local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
773 LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
774 readonly LFBFL_temp_phpmd_baseline
775
776 local LFBFL_phpmd_rulesets="cleancode, codesize, controversial, "
777 LFBFL_phpmd_rulesets+="design, naming, unusedcode"
778 readonly LFBFL_phpmd_rulesets
779
780 phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
781     . text "${LFBFL_phpmd_rulesets}"
782 # Saving new baseline in temp if necessary.
783 rm "${LFBFL_temp_phpmd_baseline}"
784 phpmd --color --generate-baseline\
785     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
786     . text "${LFBFL_phpmd_rulesets}"
787 # Remove all superfluous whitespaces.
788 # sed --in-place --regexp-extended --null-data\
789 #     --expression='s/\n +//Mg'\
790 #     "${LFBFL_temp_phpmd_baseline}"
791 xsltproc --output "${LFBFL_temp_phpmd_baseline}"\

```

```

792     build_and_checks_dependencies/phpmd_baseline.xslt\
793     "${LFBFL_temp_phpmd_baseline}"
794 # sed --in-place --expression='s/" file="/"\n    file="/g'\
795 #     "${LFBFL_temp_phpmd_baseline}"
796 # sed --in-place --regexp-extended\
797 #     --expression='s~(file=".*)/>~\1\n    />~g'\
798 #     "${LFBFL_temp_phpmd_baseline}"
799 sed --in-place\
800     --expression='s/\([^"]*\)(("[^"]*"*)*) rule="/\1\n    rule="/'\
801     --expression='s/\([^"]*\)(("[^"]*"*)*) file="/\1\n    file="/'\
802     --expression='s/\([^"]*\)(("[^"]*"*)*) method="/\1\n    method="/'\
803     --expression='s~/>~\n    />~'\
804     "${LFBFL_temp_phpmd_baseline}"
805 diff --suppress-common-lines\
806     "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
807     | grep --invert-match\
808     --regexp='\<(!-- \|^ [0-9]\+ \+ (, [0-9]\+ \+ ) \? d[0-9]\+ $)\>'
809 # See miscellaneous/grep_benchmark.
810
811 printf "Running PHPSTAN\n"
812 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
813     composer global require phpstan/phpstan
814     composer global require phpstan/phpstan-strict-rules
815 fi
816
817 local LFBFL_temp_phpstan_baseline="${LFBFL_subdir3}/temp/"
818 LFBFL_temp_phpstan_baseline+="phpstan_baseline.neon"
819 readonly LFBFL_temp_phpstan_baseline
820 if [[ -e "${LFBFL_subdir3}/phpstan_config.neon" ]]; then
821     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"
822     # Saving new baseline in temp if necessary.
823     rm "${LFBFL_temp_phpstan_baseline}"
824     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"\
825         --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
826         --allow-empty-baseline
827     sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
828     cat "${LFBFL_temp_phpstan_baseline}"
829 elif [[ LFBFL_i_verbose -eq 1 ]]; then
830     printf "No file phpstan_config.neon.\n"
831 fi
832
833 printf "Running phpDocumentor\n"
834 # shellcheck disable=SC2312
835 docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
836 printf -- "----PHP end----\n"
837

```

```

838 printf -- "----JS---\n"
839 local LFBFL_npm_lint_directories=""
840 grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
841 --result-variable-prefix=LFBFL_
842 if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
843     printf "Running ESLint\n"
844     local LFBFL_JS_directory
845     declare -a LFBFL_arr_paths
846     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
847     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]"; do
848         (cd -- "${LFBFL_JS_directory}" && npm run lint)
849     done
850 fi
851 printf -- "----JS end---\n"
852
853 [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
854
855 printf "Checking listed files\n"
856 "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
857 "${LFBFL_some_common_options[@]}"
858
859 printf "Creating the PDF file of the listing of the source code\n"
860 "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
861
862 pushd_to_work_directory --trap-popd
863 can_continue_after_enhanced_pushd || return 1
864
865 if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
866     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
867 fi
868
869 printf "Running xmllint\n"
870 # In some cases, we need to handle a cache of common DTDs by hand.
871 # XHTML DTDs validation can yield errors:
872 # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
873 # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
874 LFBFL_file_name="xhtml1-strict.dtd"
875 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
876 LFBFL_script_download_URL="${LFBFL_a}"
877 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
878 LFBFL_correct_sha512='fe1ldb06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
879 LFBFL_correct_sha512+='d8352bf19947168c9bcflae212eca035824293abb1f9'
880 LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
881 wrapped_wget_sha512
882
883 LFBFL_file_name="xhtml1-transitional.dtd"

```

```

884 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
885 LFBFL_script_download_URL="{LFBFL_a}"
886 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
887 LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
888 LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
889 LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
890 wrapped_wget_sha512
891
892 LFBFL_file_name="xhtml1-frameset.dtd"
893 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
894 LFBFL_script_download_URL="{LFBFL_a}"
895 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
896 LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
897 LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
898 LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
899 wrapped_wget_sha512
900
901 LFBFL_file_name="xhtml1.dtd"
902 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1.dtd"
903 LFBFL_script_download_URL="{LFBFL_a}"
904 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
905 LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
906 LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
907 LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
908 wrapped_wget_sha512
909
910 LFBFL_file_name="xhtml-lat1.ent"
911 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
912 LFBFL_script_download_URL="{LFBFL_a}"
913 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
914 LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
915 LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
916 LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
917 wrapped_wget_sha512
918
919 LFBFL_file_name="xhtml-special.ent"
920 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
921 LFBFL_script_download_URL="{LFBFL_a}"
922 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
923 LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
924 LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
925 LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
926 wrapped_wget_sha512
927
928 LFBFL_file_name="xhtml-symbol.ent"
929 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"

```



```

930 LFBFL_script_download_URL="${LFBFL_a}"
931 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
932 LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
933 LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
934 LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
935 wrapped_wget_sha512
936
937 declare -r LFBFL_s_files_for_xmllint=$(
938     find . -type f -iregex ".*\.(xml|xhtml)" "\
939     | relevant_find\
940     | sort
941 )
942 if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
943     LFBFL_xmllint_use_downloaded_DTDs=""
944     grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
945     --result-variable-prefix=LFBFL_
946
947     declare -a LFBFL_arr_files_for_xmllint
948     mapfile -t LFBFL_arr_files_for_xmllint\
949     <<< "${LFBFL_s_files_for_xmllint}"
950     readonly LFBFL_arr_files_for_xmllint
951     local LFBFL_xmllint_files_without_DTD=""
952     grep_variable_with_multiple_files "${LFBFL_data_file_path}" "\
953     xmllint_files_without_DTD\
954     --result-variable-prefix=LFBFL_
955     declare -a LFBFL_arr_xmllint_files_without_DTD
956     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
957         mapfile -t LFBFL_arr_xmllint_files_without_DTD\
958         <<< "${LFBFL_xmllint_files_without_DTD}"
959     fi
960     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
961         [[ LFBFL_i_verbose -eq 1 ]]\
962         && printf "xmllint on %s\n" "${LFBFL_file_path}"
963         # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
964         # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
965         # xmllint --pedantic --noout -- "${LFBFL_file_path}"
966         xmllint --pedantic --noout "${LFBFL_file_path}"
967     done
968
969     local LFBFL_file_path2
970     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
971         for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
972         do
973             if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
974                 LFBFL_file_path=""
975             fi

```

```

976     done
977     if [[ "${LFBFL_file_path}" == "" ]]; then
978         continue
979     fi
980     [[ LFBFL_i_verbose -eq 1 ]]\
981     && printf "xmllint on %s\n" "${LFBFL_file_path}"
982     if [[
983         "${LFBFL_file_path}" == *.xhtml
984         && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
985     ]]; then
986         if grep --fixed-strings\
987             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
988             "${LFBFL_file_path}"
989         then
990             xmllint --pedantic --noout\
991                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
992                 "${LFBFL_file_path}"
993         elif grep --fixed-strings\
994             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
995             "${LFBFL_file_path}"
996         then
997             xmllint --pedantic --noout\
998                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
999                 "${LFBFL_file_path}"
1000         elif grep --fixed-strings\
1001             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
1002             "${LFBFL_file_path}"
1003         then
1004             xmllint --pedantic --noout\
1005                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
1006                 "${LFBFL_file_path}"
1007         elif grep --fixed-strings\
1008             "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
1009             "${LFBFL_file_path}"
1010         then
1011             xmllint --pedantic --noout\
1012                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
1013                 "${LFBFL_file_path}"
1014         else
1015             printf "Unknown XHTML DTD.\n"
1016         fi
1017     else
1018         xmllint --pedantic --noout --valid "${LFBFL_file_path}"
1019     fi
1020 done
1021 fi

```

```

1022 # -----
1023
1024 # -----
1025 # It is at the end that all the HTML files are generated and that
1026 # we can check there are no mistakes.
1027 printf "Running Nu W3C HTML CSS and SVG validator\n"
1028 local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
1029 grep_variable "${LFBFL_data_file_path}" \
1030     upgrade_vnu_jar_time_interval_in_seconds \
1031     --result-variable-prefix=LFBFL_
1032
1033 if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
1034 then
1035     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
1036 fi
1037
1038 declare -i LFBFL_i_vnu_jar_ts
1039 local LFBFL_vnu_jar_path=""
1040 grep_variable "${LFBFL_data_file_path}" vnu_jar_path \
1041     --result-variable-prefix=LFBFL_ \
1042     --replace-line-returns-by=""
1043 if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
1044     LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
1045     LFBFL_i_current_ts=$(date +%s)
1046     ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
1047     ((
1048         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
1049     ))
1050     if [[ LFBFL_i_current_ts -gt 0 ]]; then
1051         printf $'/!\ATTENTION: Nu W3C validator is too old./!\n'
1052         printf "Go get the latest version here:\n"
1053         printf "https://github.com/validator/validator/releases\n"
1054     fi
1055     # vnu has false positives for xhtml.
1056     local LFBFL_use_vnu_for_xhtml=""
1057     grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml \
1058         --result-variable-prefix=LFBFL_
1059     local LFBFL_find_regexp
1060     if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
1061         LFBFL_find_regexp=".*\.(css|html|svg|xhtml)"
1062     else
1063         LFBFL_find_regexp=".*\.(css|html|svg)"
1064     fi
1065     declare -r LFBFL_s_files_for_Nu=$(
1066         find . -type f -iregex "${LFBFL_find_regexp}" \
1067         | relevant_find

```

```

1068     )
1069     if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
1070         declare -a LFBFL_arr_files_for_Nu
1071         mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
1072         readonly LFBFL_arr_files_for_Nu
1073         java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
1074             --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
1075     fi
1076 fi
1077 # -----
1078
1079 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
1080     touch -- "${LFBFL_upgrade_venvs_ts_file}"
1081 fi
1082 }
1083
1084 common_build_and_checks "$@"

```

3.11 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "common_build_and_checks.exec.sh.tpl" was renamed from
25 # "common_build_and_checks.sh.tpl" to
26 # "common_build_and_checks.exec.sh.tpl".

```

```

27 # The file "common_build_and_checks.exec.sh" generated from the file
28 # "common_build_and_checks.sh.tpl"
29 # or "common_build_and_checks.exec.sh.tpl"
30 # was renamed from
31 # "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".
32
33 common_build_and_checks() {
34     # $1 LFBFL_work_directory
35     # $2 LFBFL_dependencies_URL
36     # Options:
37     #   --check-download=y|keep (=keep to keep the downloaded files)
38     #   --verbose
39     local LFBFL_work_directory="${1:-.}"
40     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41     declare -r LFBFL_dependencies_URL="$2"
42     local LFBFL_verbose=""
43     declare -i LFBFL_i_verbose=0
44     if [[ "$*" == *--verbose* ]]; then
45         printf "%s %s\n" "$0" "$*"
46         LFBFL_verbose="--verbose"
47         LFBFL_i_verbose=1
48     fi
49     readonly LFBFL_verbose
50     readonly LFBFL_i_verbose
51     local LFBFL_check_download=""
52     if [[ "$*" == *--check-download=keep* ]]; then
53         LFBFL_check_download="--check-download=keep"
54     elif [[ "$*" == *--check-download=y* ]]; then
55         LFBFL_check_download="--check-download=y"
56     fi
57     readonly LFBFL_check_download
58
59     declare -a LFBFL_some_common_options=(
60         "${LFBFL_verbose}"
61         "--work-directory=${LFBFL_work_directory}"
62     )
63
64     source ./wget_sha512.lib.sh
65
66     local LFBFL_subdir="build_and_checks_dependencies"
67     declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68     declare -r LFBFL_subdir3="build_and_checks_variables"
69     local LFBFL_file_name
70     local LFBFL_script_download_URL
71     local LFBFL_file_path
72     local LFBFL_correct_sha512

```

```

73
74 wrapped_wget_sha512() {
75     wget_sha512 "${LFBFL_file_path}" \
76         "${LFBFL_script_download_URL}" \
77         "${LFBFL_correct_sha512}" \
78         "${LFBFL_verbose}" \
79         "${LFBFL_check_download}"
80 }
81
82 LFBFL_file_name="build_dependencies_notes.exec.php"
83 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85 @sha512_build_dependencies_notes.exec.php@
86 wrapped_wget_sha512
87 chmod +x "${LFBFL_file_path}"
88
89 LFBFL_file_name="build_dependencies_notes.lib.php"
90 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
91 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
92 @sha512_build_dependencies_notes.lib.php@
93 wrapped_wget_sha512
94
95 LFBFL_file_name="build_file_from_printable_file__standard.exec.sh"
96 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
97 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
98 @sha512_build_file_from_printable_file__standard.exec.sh@
99 wrapped_wget_sha512
100 chmod +x "${LFBFL_file_path}"
101
102 LFBFL_file_name="build_md_from_printable_md.exec.sh"
103 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
104 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
105 @sha512_build_md_from_printable_md.exec.sh@
106 wrapped_wget_sha512
107 chmod +x "${LFBFL_file_path}"
108
109 LFBFL_file_name="check_shell_scripts_beginnings.lib.sh"
110 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
111 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
112 @sha512_check_shell_scripts_beginnings.lib.sh@
113 wrapped_wget_sha512
114
115 LFBFL_file_name="check_shell_scripts_indentation.lib.sh"
116 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
117 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
118 @sha512_check_shell_scripts_indentation.lib.sh@

```

```

119 wrapped_wget_sha512
120
121 LFBFL_file_name="check_URLs.lib.sh"
122 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
123 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
124 @sha512_check_URLs.lib.sh@
125 wrapped_wget_sha512
126
127 LFBFL_file_name="common_options.lib.sh"
128 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
129 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
130 @sha512_common_options.lib.sh@
131 wrapped_wget_sha512
132
133 LFBFL_file_name="comparisons.lib.sh"
134 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
135 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
136 @sha512_comparisons.lib.sh@
137 wrapped_wget_sha512
138
139 LFBFL_file_name="create_PDF.exec.sh"
140 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
141 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
142 @sha512_create_PDF.exec.sh@
143 wrapped_wget_sha512
144 chmod +x "${LFBFL_file_path}"
145
146 LFBFL_file_name="generate_from_template.lib.sh"
147 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
148 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
149 @sha512_generate_from_template.lib.sh@
150 wrapped_wget_sha512
151
152 LFBFL_file_name="get_common_text_glob_patterns.lib.sh"
153 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
154 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
155 @sha512_get_common_text_glob_patterns.lib.sh@
156 wrapped_wget_sha512
157
158 LFBFL_file_name="grammar_and_spelling_check.lib.sh"
159 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
160 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
161 @sha512_grammar_and_spelling_check.lib.sh@
162 wrapped_wget_sha512
163
164 # /licenses_templates/ -----

```

```

165 local LFBFL_dependencies_URL2
166 LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
167 readonly LFBFL_dependencies_URL2
168 LFBFL_file_name="build_licenses_templates.exec.sh"
169 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
170 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
171 @sha512_build_licenses_templates.exec.sh@
172 wrapped_wget_sha512
173 chmod +x "${LFBFL_file_path}"
174
175 LFBFL_file_name="license_file_header_AGPLv3+.tpl"
176 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
177 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
178 @sha512_license_file_header_AGPLv3+.tpl@
179 wrapped_wget_sha512
180
181 LFBFL_file_name="license_file_header_GPLv3+.tpl"
182 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
183 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
184 @sha512_license_file_header_GPLv3+.tpl@
185 wrapped_wget_sha512
186
187 LFBFL_file_name="license_file_header_LGPLv3+.tpl"
188 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
189 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
190 @sha512_license_file_header_LGPLv3+.tpl@
191 wrapped_wget_sha512
192 # -----
193
194 LFBFL_file_name="lines_counts.lib.sh"
195 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
196 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
197 @sha512_lines_counts.lib.sh@
198 wrapped_wget_sha512
199
200 LFBFL_file_name="lines_filters.lib.sh"
201 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
202 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
203 @sha512_lines_filters.lib.sh@
204 wrapped_wget_sha512
205
206 LFBFL_file_name="lines_maps.lib.sh"
207 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
208 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
209 @sha512_lines_maps.lib.sh@
210 wrapped_wget_sha512

```



```

211
212 LFBFL_file_name="overwrite_if_not_equal.lib.sh"
213 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
214 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
215 @sha512_overwrite_if_not_equal.lib.sh@
216 wrapped_wget_sha512
217
218 LFBFL_file_name="phpmd_baseline.xslt"
219 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
220 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
221 @sha512_phpmd_baseline.xslt@
222 wrapped_wget_sha512
223
224 LFBFL_file_name="python_black_complement.lib.sh"
225 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
226 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
227 @sha512_python_black_complement.lib.sh@
228 wrapped_wget_sha512
229
230 LFBFL_file_name="python_isort_complement.lib.sh"
231 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
232 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
233 @sha512_python_isort_complement.lib.sh@
234 wrapped_wget_sha512
235
236 LFBFL_file_name="repository_data.lib.sh"
237 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
238 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
239 @sha512_repository_data.lib.sh@
240 wrapped_wget_sha512
241
242 LFBFL_file_name="shell_checks_complement.lib.sh"
243 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
244 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
245 @sha512_shell_checks_complement.lib.sh@
246 wrapped_wget_sha512
247
248 LFBFL_file_name="split_score.exec.php"
249 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
250 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
251 @sha512_split_score.exec.php@
252 wrapped_wget_sha512
253 chmod +x "${LFBFL_file_path}"
254
255 LFBFL_file_name="split_score.lib.php"
256 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"

```

```

257 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
258 @sha512_split_score.lib.php@
259 wrapped_wget_sha512
260
261 LFBFL_file_name="strings_functions.lib.sh"
262 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
263 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
264 @sha512_strings_functions.lib.sh@
265 wrapped_wget_sha512
266
267 LFBFL_file_name="too_long_code_lines.lib.sh"
268 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
269 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
270 @sha512_too_long_code_lines.lib.sh@
271 wrapped_wget_sha512
272
273 LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
274 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
275 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
276 @sha512_update_or_check_files_names_listing.exec.sh@
277 wrapped_wget_sha512
278 chmod +x "${LFBFL_file_path}"
279
280 # shellcheck source=check_shell_scripts_beginnings.lib.sh
281 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
282 # shellcheck source=check_shell_scripts_indentation.lib.sh
283 source "${LFBFL_subdir}/check_shell_scripts_indentation.lib.sh"
284 # shellcheck source=check_URLs.lib.sh
285 source "${LFBFL_subdir}/check_URLs.lib.sh"
286 # shellcheck source=common_options.lib.sh
287 source "${LFBFL_subdir}/common_options.lib.sh"
288 # shellcheck source=comparisons.lib.sh
289 source "${LFBFL_subdir}/comparisons.lib.sh"
290 # shellcheck source=generate_from_template.lib.sh
291 source "${LFBFL_subdir}/generate_from_template.lib.sh"
292 # shellcheck source=get_common_text_glob_patterns.lib.sh
293 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
294 # shellcheck source=grammar_and_spelling_check.lib.sh
295 source "${LFBFL_subdir}/grammar_and_spelling_check.lib.sh"
296 # shellcheck source=lines_counts.lib.sh
297 source "${LFBFL_subdir}/lines_counts.lib.sh"
298 # shellcheck source=lines_filters.lib.sh
299 source "${LFBFL_subdir}/lines_filters.lib.sh"
300 # shellcheck source=lines_maps.lib.sh
301 source "${LFBFL_subdir}/lines_maps.lib.sh"
302 # shellcheck source=overwrite_if_not_equal.lib.sh

```

```

303 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
304 # shellcheck source=python_black_complement.lib.sh
305 source "${LFBFL_subdir}/python_black_complement.lib.sh"
306 # shellcheck source=python_isort_complement.lib.sh
307 source "${LFBFL_subdir}/python_isort_complement.lib.sh"
308 # shellcheck source=shell_checks_complement.lib.sh
309 source "${LFBFL_subdir}/shell_checks_complement.lib.sh"
310 # shellcheck source=strings_functions.lib.sh
311 source "${LFBFL_subdir}/strings_functions.lib.sh"
312 # shellcheck source=too_long_code_lines.lib.sh
313 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
314
315 enhanced_set_bash_option extglob
316 # shellcheck source=repository_data.lib.sh
317 source "${LFBFL_subdir}/repository_data.lib.sh"
318 # shellcheck disable=SC2154,SC2309
319 [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
320     && enhanced_unset_bash_option extglob
321
322 declare -a LFBFL_return_traps_stack
323 init_return_trap
324
325 enhanced_set_shell_option pipefail --trap-unset
326
327 declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
328 keep_options LFBFL_some_common_options2 --verbose
329
330 printf "Building license headers\n"
331 "${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
332     "${LFBFL_some_common_options[@]}"
333
334 printf "Building README.md\n"
335 "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
336     --work-directory="${LFBFL_work_directory}"\
337     --base-name=README\
338     "${LFBFL_verbose}"
339
340 printf "Building other Markdown files\n"
341 local LFBFL_some_directory
342 declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
343 local LFBFL_s_files_paths
344 LFBFL_s_files_paths=$(
345     find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
346     | relevant_find
347 )
348 declare -a LFBFL_arr_files_paths

```

```

349 if [[ -n "${LFBFL_s_files_paths}" ]]; then
350     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
351     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
352         [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
353         printf "Found template %s.\n" "${LFBFL_file_path}"
354         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
355         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
356         LFBFL_file_name=${LFBFL_file_name%.md.tpl}
357         "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh" \
358             --work-directory="${LFBFL_some_directory}" \
359             --base-name="${LFBFL_file_name}" \
360             "${LFBFL_verbose}"
361     done
362 fi
363
364 printf "Building other standard files\n"
365 LFBFL_s_files_paths=$(
366     find "${LFBFL_work_directory}" -type f -name "*.neon.tpl" \
367     | relevant_find
368 )
369 if [[ -n "${LFBFL_s_files_paths}" ]]; then
370     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
371     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
372         printf "Found template %s.\n" "${LFBFL_file_path}"
373         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
374         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
375         LFBFL_file_name=${LFBFL_file_name%.tpl}
376         "./${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh" \
377             --work-directory="${LFBFL_some_directory}" \
378             --base-name="${LFBFL_file_name}" \
379             "${LFBFL_verbose}" \
380             --remove-leading-whitespaces-after-escaped-line-return
381     done
382 fi
383
384 printf "Building dependencies_notes\n"
385 # https://github.com/php/php-src/issues/21538
386 local LFBFL_annoying_warning
387 LFBFL_annoying_warning="PHP Warning: declare(encoding=...) ignored"
388 LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
389 LFBFL_annoying_warning=" by settings in"
390 readonly LFBFL_annoying_warning
391 if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
392     php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php" \
393         -- "${LFBFL_work_directory}" \
394         2>&1 \

```

```

395         | grep --invert-match "${LFBFL_annoying_warning}"
396     elif [[ LFBFL_i_verbose -eq 1 ]]; then
397         printf "No file dependencies_data.json.\n"
398     fi
399
400     declare -i LFBFL_i_directory_changed
401     pushd_to_work_directory
402     LFBFL_i_directory_changed=$?
403     can_continue_after_enhanced_pushd || return 1
404
405     local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
406
407     declare -i LFBFL_i_max_line_length
408     grep_variable "${LFBFL_data_file_path}" max_line_length\
409         --result-variable-prefix=LFBFL_i_
410
411     LFBFL_some_common_options+=(
412         "--max-line-length=${LFBFL_i_max_line_length}"
413     )
414     LFBFL_some_common_options2+=(
415         "--max-line-length=${LFBFL_i_max_line_length}"
416     )
417
418     declare -i LFBFL_i_upgrade_venvs=0
419     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
420     readonly LFBFL_upgrade_venvs_ts_file
421     declare -i LFBFL_i_upgrade_venvs_ts
422     declare -i LFBFL_i_current_ts
423     local LFBFL_upgrade_venvs_answer
424
425     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
426     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
427         "${LFBFL_some_common_options2[@]}"
428
429     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
430         LFBFL_i_upgrade_venvs_ts=$(
431             stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
432         )
433         LFBFL_i_current_ts=$(date +%s)
434         ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
435         ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
436         if [[ LFBFL_i_current_ts -gt 0 ]]; then
437             LFBFL_i_upgrade_venvs=1
438         fi
439     else
440         LFBFL_i_upgrade_venvs=1

```

```

441 fi
442 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
443     local LFBFL_upgrade_venvs=""
444     grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
445         --result-variable-prefix=LFBFL_
446     if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
447         read -r -n 1 -t 10\
448             -p "Upgrade venvs and composer global? [Y/n]"\
449             LFBFL_upgrade_venvs_answer
450         if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
451             LFBFL_i_upgrade_venvs=0
452         fi
453     fi
454 fi
455
456 printf "Analyzing too long lines\n"
457 too_long_code_lines "${LFBFL_some_common_options2[@]}"
458
459 printf "Analyzing URLs\n"
460 check_URLs "${LFBFL_some_common_options2[@]}" \
461     | relevant_grep
462
463 printf "Analyzing strange characters: hover over in doubt\n"
464 # shellcheck disable=SC1111
465 LFBFL_usual_characters="\x00-\x7Fàâéèëîïôûç@\"`|«»..."
466 grep --binary-files=without-match --color=always --exclude-dir .git\
467     --line-number --perl-regexp --recursive --\
468     "[^${LFBFL_usual_characters}]" .\
469     | relevant_grep
470
471 printf "Grammar and spelling check\n"
472 grammar_and_spelling_check "${LFBFL_data_file_path}"\
473     "${LFBFL_some_common_options2[@]}"
474
475 printf "Running shellcheck\n"
476 if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
477     LFBFL_s_files_paths=$(
478         find . -type f -name "*.sh"\
479         | relevant_find
480     )
481     if [[ -n "${LFBFL_s_files_paths}" ]]; then
482         mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
483         for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
484             shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
485                 -- "${LFBFL_file_path}"
486         done

```

```

487     fi
488 elif [[ LFBFL_i_verbose -eq 1 ]]; then
489     printf "No file shellcheck.ini.\n"
490 fi
491 printf "Running shell_checks_complement\n"
492 shell_checks_complement "${LFBFL_some_common_options2[@]}"
493
494 printf "Analyzing shell scripts beginnings\n"
495 check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
496     | relevant_grep
497
498 printf "Analyzing shell scripts indentation\n"
499 check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
500     | relevant_grep
501
502 printf -- "---Python---\n"
503
504 maybe_deactivate(){
505     if command -v deactivate
506     then
507         deactivate
508     fi
509 }
510
511 source_without_return_trap(){
512     declare -i LFBFL_i_no_return_traps=1
513     # shellcheck disable=SC1090,SC1091
514     source -- "$1"
515 }
516
517 printf "Running isort\n"
518 local LFBFL_isort_venv=""
519 grep_variable "${LFBFL_data_file_path}" isort_venv\
520     --result-variable-prefix=LFBFL_\
521     --replace-line-returns-by=""
522 if [[ -n "${LFBFL_isort_venv}" ]]; then
523     maybe_deactivate
524     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
525 fi
526 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
527     pip install --upgrade isort
528 fi
529 isort --profile=black .
530 if [[ -n "${LFBFL_isort_venv}" ]]; then
531     deactivate
532 fi

```

```

533 printf "Running python_isort_complement\n"
534 python_isort_complement "${LFBFL_some_common_options2[@]}"
535
536 printf "Running black\n"
537 # First, we update the configuration file with max_line_length.
538 if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
539     local LFBFL_update_max_length="s/^line-length = [0-9]*$"
540     LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
541     sed --regex-extended --expression="${LFBFL_update_max_length}"\
542         "${LFBFL_subdir3}/black.toml"\
543         > "${LFBFL_subdir3}/black.toml.temp"
544     overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
545         "${LFBFL_subdir3}/black.toml.temp"
546 elif [[ LFBFL_i_verbose -eq 1 ]]; then
547     printf "No file black.toml.\n"
548 fi
549
550 local LFBFL_black_venv=""
551 grep_variable "${LFBFL_data_file_path}" black_venv\
552     --result-variable-prefix=LFBFL_\
553     --replace-line-returns-by=""
554 if [[ -n "${LFBFL_black_venv}" ]]; then
555     maybe_deactivate
556     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
557 fi
558 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
559     pip install --upgrade black
560 fi
561 if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
562     black --config="${LFBFL_subdir3}/black.toml" .
563 fi
564 if [[ -n "${LFBFL_black_venv}" ]]; then
565     deactivate
566 fi
567 printf "Running python_black_complement\n"
568 python_black_complement "${LFBFL_some_common_options2[@]}"
569
570 printf "Probing if mypy should be runned\n"
571 local LFBFL_mypy_venv=""
572 grep_variable "${LFBFL_data_file_path}" mypy_venv\
573     --result-variable-prefix=LFBFL_\
574     --replace-line-returns-by=""
575 if [[ -n "${LFBFL_mypy_venv}" ]]; then
576     maybe_deactivate
577     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
578 fi

```



```

579 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
580     pip install --upgrade mypy
581 fi
582 local LFBFL_directory_path
583 declare -i LFBFL_i_no_toml=1
584 LFBFL_s_files_paths=$(
585     find . -type f -name "pyproject.toml" \
586     | relevant_find
587 )
588 if [[ -n "${LFBFL_s_files_paths}" ]]; then
589     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
590     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
591         if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
592             printf "Running mypy\n"
593             LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
594             mypy -- "${LFBFL_directory_path}"
595         fi
596         LFBFL_i_no_toml=0
597     done
598 fi
599 if [[ LFBFL_i_no_toml -eq 1 ]]; then
600     printf "Running mypy\n"
601     mypy .
602 fi
603 if [[ -n "${LFBFL_mypy_venv}" ]]; then
604     deactivate
605 fi
606
607 printf "Running bandit\n"
608 LFBFL_bandit_venv=""
609 grep_variable "${LFBFL_data_file_path}" bandit_venv \
610     --result-variable-prefix=LFBFL \
611     --replace-line-returns-by=""
612 if [[ -n "${LFBFL_bandit_venv}" ]]; then
613     maybe_deactivate
614     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
615 fi
616 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
617     pip install --upgrade bandit
618 fi
619 if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
620     bandit --ini="${LFBFL_subdir3}/bandit.ini" \
621         --baseline="${LFBFL_subdir3}/bandit_baseline.json" \
622         --recursive .
623     # Saving new baseline in temp if necessary.
624     bandit --ini="${LFBFL_subdir3}/bandit.ini" \

```

```

625         --format=json\
626         --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
627         --recursive .
628     elif [[ LFBFL_i_verbose -eq 1 ]]; then
629         printf "No file bandit.ini.\n"
630     fi
631     if [[ -n "${LFBFL_bandit_venv}" ]]; then
632         deactivate
633     fi
634
635     printf "Running pylint\n"
636     LFBFL_pylint_venv=""
637     grep_variable "${LFBFL_data_file_path}" pylint_venv\
638         --result-variable-prefix=LFBFL_\
639         --replace-line-returns-by=""
640     if [[ -n "${LFBFL_pylint_venv}" ]]; then
641         maybe_deactivate
642         source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
643     fi
644     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
645         pip install --upgrade pylint
646     fi
647     if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
648         pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
649     elif [[ LFBFL_i_verbose -eq 1 ]]; then
650         printf "No file pylintrc.toml.\n"
651     fi
652     if [[ -n "${LFBFL_pylint_venv}" ]]; then
653         deactivate
654     fi
655
656     printf "Running ruff\n"
657     LFBFL_ruff_venv=""
658     grep_variable "${LFBFL_data_file_path}" ruff_venv\
659         --result-variable-prefix=LFBFL_\
660         --replace-line-returns-by=""
661     if [[ -n "${LFBFL_ruff_venv}" ]]; then
662         maybe_deactivate
663         source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
664     fi
665     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
666         pip install --upgrade ruff
667     fi
668     if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
669         ruff check --config="${LFBFL_subdir3}/ruff.toml"
670     elif [[ LFBFL_i_verbose -eq 1 ]]; then

```

```

671     printf "No file ruff.toml.\n"
672 fi
673 if [[ -n "${LFBFL_ruff_venv}" ]]; then
674     deactivate
675 fi
676 printf -- "----Python end---\n"
677
678 printf -- "----PHP---\n"
679 printf "Running PHP native linter\n"
680 LFBFL_s_files_paths=$(
681     find . -type f -name "*.php" \
682     | relevant_find
683 )
684 if [[ -n "${LFBFL_s_files_paths}" ]]; then
685     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
686     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
687         php --syntax-check -f="${LFBFL_file_path}" \
688             2>&1 \
689             | grep --invert-match "${LFBFL_annoying_warning}"
690     done
691 fi
692
693 printf "Running PHP_CodeSniffer\n"
694 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
695     composer global require squizlabs/php_codesniffer
696 fi
697 if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
698     phpcs --report=code \
699         --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
700 elif [[ LFBFL_i_verbose -eq 1 ]]; then
701     printf "No file phpcs_ruleset.xml.\n"
702 fi
703
704 printf "Running PHPMD\n"
705 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
706     composer global require phpmd/phpmd
707 fi
708
709 declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
710 local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
711 LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
712 readonly LFBFL_temp_phpmd_baseline
713
714 local LFBFL_phpmd_rulesets="cleancode, codesize, controversial, "
715 LFBFL_phpmd_rulesets+="design, naming, unusedcode"
716 readonly LFBFL_phpmd_rulesets

```

```

718 phpmd --color --baseline-file="\${LFBFL_phpmd_baseline}"\
719 . text "\${LFBFL_phpmd_rulesets}"
720 # Saving new baseline in temp if necessary.
721 rm "\${LFBFL_temp_phpmd_baseline}"
722 phpmd --color --generate-baseline\
723 --baseline-file="\${LFBFL_temp_phpmd_baseline}"\
724 . text "\${LFBFL_phpmd_rulesets}"
725 # Remove all superfluous whitespaces.
726 # sed --in-place --regexp-extended --null-data\
727 # --expression='s/\n +//Mg'\
728 # "\${LFBFL_temp_phpmd_baseline}"
729 xsltproc --output "\${LFBFL_temp_phpmd_baseline}"\
730 build_and_checks_dependencies/phpmd_baseline.xslt\
731 "\${LFBFL_temp_phpmd_baseline}"
732 # sed --in-place --expression='s/" file="/"\n      file="/g'\
733 # "\${LFBFL_temp_phpmd_baseline}"
734 # sed --in-place --regexp-extended\
735 # --expression='s~(file=".*)/>~1\n />~g'\
736 # "\${LFBFL_temp_phpmd_baseline}"
737 sed --in-place\
738 --expression='s/\([^"]*\("[^"]*"*)*) rule="/\1\n      rule="/'\
739 --expression='s/\([^"]*\("[^"]*"*)*) file="/\1\n      file="/'\
740 --expression='s/\([^"]*\("[^"]*"*)*) method="/\1\n      method="/'\
741 --expression='s~/>~\n />~'\
742 "\${LFBFL_temp_phpmd_baseline}"
743 diff --suppress-common-lines\
744 "\${LFBFL_phpmd_baseline}" "\${LFBFL_temp_phpmd_baseline}"\
745 | grep --invert-match\
746 --regexp='\
```

```

763     --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
764     --allow-empty-baseline
765     sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
766     cat "${LFBFL_temp_phpstan_baseline}"
767 elif [[ LFBFL_i_verbose -eq 1 ]]; then
768     printf "No file phpstan_config.neon.\n"
769 fi
770
771 printf "Running phpDocumentor\n"
772 # shellcheck disable=SC2312
773 docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
774 printf -- "----PHP end---\n"
775
776 printf -- "----JS---\n"
777 local LFBFL_npm_lint_directories=""
778 grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
779 --result-variable-prefix=LFBFL_
780 if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
781     printf "Running ESLint\n"
782     local LFBFL_JS_directory
783     declare -a LFBFL_arr_paths
784     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
785     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]"; do
786         (cd -- "${LFBFL_JS_directory}" && npm run lint)
787     done
788 fi
789 printf -- "----JS end---\n"
790
791 [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
792
793 printf "Checking listed files\n"
794 "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
795 "${LFBFL_some_common_options[@]}"
796
797 printf "Creating the PDF file of the listing of the source code\n"
798 "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
799
800 pushd_to_work_directory --trap-popd
801 can_continue_after_enhanced_pushd || return 1
802
803 if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
804     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
805 fi
806
807 printf "Running xmllint\n"
808 # In some cases, we need to handle a cache of common DTDs by hand.

```

```

809 # XHTML DTDs validation can yield errors:
810 # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
811 # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
812 LFBFL_file_name="xhtml1-strict.dtd"
813 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
814 LFBFL_script_download_URL="${LFBFL_a}"
815 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
816 LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
817 LFBFL_correct_sha512+='d8352bf19947168c9bcflae212eca035824293abb1f9'
818 LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
819 wrapped_wget_sha512
820
821 LFBFL_file_name="xhtml1-transitional.dtd"
822 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
823 LFBFL_script_download_URL="${LFBFL_a}"
824 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
825 LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efaela9dfa447a5'
826 LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
827 LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
828 wrapped_wget_sha512
829
830 LFBFL_file_name="xhtml1-frameset.dtd"
831 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
832 LFBFL_script_download_URL="${LFBFL_a}"
833 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
834 LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
835 LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
836 LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
837 wrapped_wget_sha512
838
839 LFBFL_file_name="xhtml11.dtd"
840 LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
841 LFBFL_script_download_URL="${LFBFL_a}"
842 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
843 LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
844 LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
845 LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
846 wrapped_wget_sha512
847
848 LFBFL_file_name="xhtml-lat1.ent"
849 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
850 LFBFL_script_download_URL="${LFBFL_a}"
851 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
852 LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
853 LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
854 LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'

```

```

855 wrapped_wget_sha512
856
857 LFBFL_file_name="xhtml-special.ent"
858 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
859 LFBFL_script_download_URL="${LFBFL_a}"
860 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
861 LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
862 LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
863 LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
864 wrapped_wget_sha512
865
866 LFBFL_file_name="xhtml-symbol.ent"
867 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
868 LFBFL_script_download_URL="${LFBFL_a}"
869 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
870 LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
871 LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
872 LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
873 wrapped_wget_sha512
874
875 declare -r LFBFL_s_files_for_xmllint=$(
876     find . -type f -iregex ".*\\.\\(xml\\|xhtml\\)" "\
877     | relevant_find\
878     | sort
879 )
880 if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
881     LFBFL_xmllint_use_downloaded_DTDs=""
882     grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
883     --result-variable-prefix=LFBFL_
884
885     declare -a LFBFL_arr_files_for_xmllint
886     mapfile -t LFBFL_arr_files_for_xmllint\
887     <<< "${LFBFL_s_files_for_xmllint}"
888     readonly LFBFL_arr_files_for_xmllint
889     local LFBFL_xmllint_files_without_DTD=""
890     grep_variable_with_multiple_files "${LFBFL_data_file_path}" "\
891     xmllint_files_without_DTD\
892     --result-variable-prefix=LFBFL_
893     declare -a LFBFL_arr_xmllint_files_without_DTD
894     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
895         mapfile -t LFBFL_arr_xmllint_files_without_DTD\
896         <<< "${LFBFL_xmllint_files_without_DTD}"
897     fi
898     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
899         [[ LFBFL_i_verbose -eq 1 ]]\
900         && printf "xmllint on %s\n" "${LFBFL_file_path}"

```

```

901     # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
902     # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
903     # xmllint --pedantic --noout -- "${LFBFL_file_path}"
904     xmllint --pedantic --noout "${LFBFL_file_path}"
905 done
906
907 local LFBFL_file_path2
908 for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
909     for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
910     do
911         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
912             LFBFL_file_path=""
913         fi
914     done
915     if [[ "${LFBFL_file_path}" == "" ]]; then
916         continue
917     fi
918     [[ LFBFL_i_verbose -eq 1 ]]\
919     && printf "xmllint on %s\n" "${LFBFL_file_path}"
920     if [[
921         "${LFBFL_file_path}" == *.xhtml
922         && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
923     ]]; then
924         if grep --fixed-strings\
925             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
926             "${LFBFL_file_path}"
927         then
928             xmllint --pedantic --noout\
929                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
930                 "${LFBFL_file_path}"
931         elif grep --fixed-strings\
932             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
933             "${LFBFL_file_path}"
934         then
935             xmllint --pedantic --noout\
936                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
937                 "${LFBFL_file_path}"
938         elif grep --fixed-strings\
939             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
940             "${LFBFL_file_path}"
941         then
942             xmllint --pedantic --noout\
943                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
944                 "${LFBFL_file_path}"
945         elif grep --fixed-strings\
946             "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"

```



```

947         "${LFBFL_file_path}"
948     then
949         xmllint --pedantic --noout\
950             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
951             "${LFBFL_file_path}"
952     else
953         printf "Unknown XHTML DTD.\n"
954     fi
955 else
956     xmllint --pedantic --noout --valid "${LFBFL_file_path}"
957 fi
958 done
959 fi
960 # -----
961
962 # -----
963 # It is at the end that all the HTML files are generated and that
964 # we can check there are no mistakes.
965 printf "Running Nu W3C HTML CSS and SVG validator\n"
966 local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
967 grep_variable "${LFBFL_data_file_path}"\
968     upgrade_vnu_jar_time_interval_in_seconds\
969     --result-variable-prefix=LFBFL_
970
971 if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
972 then
973     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
974 fi
975
976 declare -i LFBFL_i_vnu_jar_ts
977 local LFBFL_vnu_jar_path=""
978 grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
979     --result-variable-prefix=LFBFL_\
980     --replace-line-returns-by=""
981 if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
982     LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
983     LFBFL_i_current_ts=$(date +%s)
984     ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
985     ((
986         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
987     ))
988     if [[ LFBFL_i_current_ts -gt 0 ]]; then
989         printf $'/!\ATENTION: Nu W3C validator is too old./!\n'
990         printf "Go get the latest version here:\n"
991         printf "https://github.com/validator/validator/releases\n"
992     fi

```

```

993     # vnu has false positives for xhtml.
994     local LFBFL_use_vnu_for_xhtml=""
995     grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
996         --result-variable-prefix=LFBFL_
997     local LFBFL_find_regexp
998     if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
999         LFBFL_find_regexp=".*\\.\\(css\\|html\\|svg\\|xhtml\\)"
1000     else
1001         LFBFL_find_regexp=".*\\.\\(css\\|html\\|svg\\)"
1002     fi
1003     declare -r LFBFL_s_files_for_Nu=$(
1004         find . -type f -iregex "${LFBFL_find_regexp}"\
1005             | relevant_find
1006     )
1007     if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
1008         declare -a LFBFL_arr_files_for_Nu
1009         mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
1010         readonly LFBFL_arr_files_for_Nu
1011         java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
1012             --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
1013     fi
1014 fi
1015 # -----
1016
1017 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
1018     touch -- "${LFBFL_upgrade_venvs_ts_file}"
1019 fi
1020 }
1021
1022 common_build_and_checks "$@"

```

3.12 build_and_checks_dependencies/common_options.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;

```

```

14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 get_where_was_i() {
26     # Auxiliary function needed below.
27     # See usages below if you want to reuse it.
28     # $1=offset
29     LFBFL_where_was_i="$0"
30     if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
31         LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
32     fi
33     if [[ "${FUNCNAME[$1]}" != "" ]]; then
34         LFBFL_where_was_i+=" ${FUNCNAME[$1]} () "
35     fi
36     readonly LFBFL_where_was_i
37     # Comment out line above in case someone would use distinct
38     # offsets in the same function.
39 }
40
41 # RETURN traps handling
42 # Add the following lines at the start of a function.
43 # declare -a LFBFL_return_traps_stack
44 # init_return_trap
45
46 init_return_trap() {
47     local LFBFL_previous_return_trap
48     local LFBFL_previous_return_trap2
49     LFBFL_previous_return_trap=$(trap -p RETURN)
50     # shellcheck disable=SC2154,SC2309
51     if [[ LFBFL_i_verbose -eq 1 ]]; then
52         local LFBFL_where_was_i
53         get_where_was_i 2
54         printf "%s previous_return_trap is: %s.\n" \
55             "${LFBFL_where_was_i}" \
56             "${LFBFL_previous_return_trap}"
57     fi
58     if [[ -z "${LFBFL_previous_return_trap}" ]]; then
59         LFBFL_previous_return_trap="trap - RETURN"

```

```

60     fi
61     readonly LFBFL_previous_return_trap
62     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
63     LFBFL_previous_return_trap2="${LFBFL_function_depth}:"
64     LFBFL_previous_return_trap2+="${LFBFL_previous_return_trap}"
65     readonly LFBFL_previous_return_trap2
66     LFBFL_return_traps_stack+="${LFBFL_previous_return_trap2}"
67     trap 'execute_return_traps' RETURN
68 }
69
70 execute_return_traps() {
71     if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
72         return
73     fi
74     # -----
75     # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
76     if [[ "${FUNCNAME[1]}" == "source" ]]; then
77         # Not Working
78         return
79     fi
80     if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
81         return
82     fi
83     # -----
84     local LFBFL_some_return_trap
85     local LFBFL_where_was_i
86     get_where_was_i 2
87     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
88     while [[ ${LFBFL_return_traps_stack[@]} -gt 0 ]]; do
89         LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
90         if [[ "${LFBFL_some_return_trap}" != "${LFBFL_function_depth}:" ]]; then
91             # We only treat the top of the stack that should correspond to the
92             # current function.
93             break
94         fi
95         LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
96         # shellcheck disable=SC2154,SC2309
97         if [[ LFBFL_i_verbose -eq 1 ]]; then
98             if [[ ${#LFBFL_return_traps_stack[@]} -gt 1 ]]; then
99                 printf "%s executing some return trap: %s.\n" \
100                     "${LFBFL_where_was_i}" \
101                     "${LFBFL_some_return_trap}"
102             else
103                 printf "%s setting back to previous_return_trap: %s.\n" \
104                     "${LFBFL_where_was_i}" \
105                     "${LFBFL_some_return_trap}"

```

```

106         fi
107     fi
108     eval -- "${LFBFL_some_return_trap}"
109     unset 'LFBFL_return_traps_stack[-1]'
110 done
111 }
112
113
114 is_top_dirstack_directory(){
115     # Although "${DIRSTACK[0]}" may seem to be already a realpath,
116     # possible use of symbolic links makes this function necessary.
117     # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
118     # $1=to_directory needed to be "realpathed" already
119     # but I removed this micro-optimization.
120     # A compiled language could have a "realpathed" property" on
121     # "path object/string" to optimize when possible.
122     if [[ -z "$1" ]]; then
123         return 0
124     fi
125     local LFBFL_real_path
126     LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
127     readonly LFBFL_real_path
128     local LFBFL_real_path2
129     LFBFL_real_path2=$(realpath -- "$1")
130     readonly LFBFL_real_path2
131     [[ "${LFBFL_real_path}" == "${LFBFL_real_path2}" ]]
132 }
133
134 enhanced_pushd(){
135     # Auxiliary function needed below.
136     # See usages below if you want to reuse it.
137     # $1=to_directory (needed to be sanitized with realpath before call)
138     # $2=offset_for_where_was_i
139     # $3=error_message_intermediate_complement
140     # returns an error code whenever no pushd happened
141     # $1 can be any path now, see my comment on the micro-optimization in
142     # is_top_dirstack_directory() just above.
143     # shellcheck disable=SC2154,SC2309
144     [[ LFBFL_i_verbose -eq 1 ]]\
145     && printf "enhanced_pushd requested: %s.\n" "$1"
146     # Value 0 will always be overwritten, it is superfluous.
147     declare -gi i_enhanced_pushd_result=0
148     if [[ -z "$1" ]]; then
149         i_enhanced_pushd_result=110
150         # shellcheck disable=SC2248
151         return ${i_enhanced_pushd_result}

```

```

152     fi
153     local LFBFL_real_path
154     LFBFL_real_path=$(realpath -- "$1")
155     readonly LFBFL_real_path
156     # shellcheck disable=SC2154,SC2309
157     [[ LFBFL_i_verbose -eq 1 ]]\
158         && printf "enhanced_pushd passed validation 1.\n"
159     is_top_dirstack_directory "${LFBFL_real_path}" && {
160         i_enhanced_pushd_result=111
161         # shellcheck disable=SC2248
162         return ${i_enhanced_pushd_result}
163     }
164     # shellcheck disable=SC2154,SC2309
165     [[ LFBFL_i_verbose -eq 1 ]]\
166         && printf "enhanced_pushd passed validation 2.\n"
167
168     pushd -- "${LFBFL_real_path}" || {
169         i_enhanced_pushd_result=$?
170         local LFBFL_where_was_i
171         get_where_was_i "$2"
172         printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3"\
173             "${LFBFL_real_path}"
174         # shellcheck disable=SC2248
175         return ${i_enhanced_pushd_result}
176     }
177     # shellcheck disable=SC2154,SC2309
178     [[ LFBFL_i_verbose -eq 1 ]]\
179         && printf "enhanced_pushd executed.\n"
180     # shellcheck disable=SC2248
181     return ${i_enhanced_pushd_result}
182 }
183
184 enhanced_popd() {
185     # Auxiliary function needed below.
186     # See usages below if you want to reuse it.
187     # $1=to_directory
188     # $2=offset_for_where_was_i
189     # shellcheck disable=SC2154,SC2309
190     [[ LFBFL_i_verbose -eq 1 ]]\
191         && printf "enhanced_popd requested: %s.\n" "$1"
192     if [[ -z "$1" ]]; then
193         return 0
194     fi
195
196     declare -i LFBFL_i_popd_result
197     popd || {
198         LFBFL_i_popd_result=$?

```

```

198     readonly LFBFL_i_popd_result
199     local LFBFL_where_was_i
200     get_where_was_i "$2"
201     printf "%s popd failed.\n" "${LFBFL_where_was_i}"
202     # shellcheck disable=SC2248
203     return ${LFBFL_i_popd_result}
204 }
205 # shellcheck disable=SC2154,SC2309
206 [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed.\n"
207 return 0
208 }
209
210 can_continue_after_enhanced_pushd() {
211     [[
212         i_enhanced_pushd_result -eq 0
213         || i_enhanced_pushd_result -eq 110
214         || i_enhanced_pushd_result -eq 111
215     ]]
216 }
217
218 # Main usage for the next functions relative to verbose mode:
219 # Add the two following lines at the start of a function.
220 #   declare -i LFBFL_i_verbose=0
221 #   get_verbose_option "$@"
222 # /\ You really need all of the two lines. /\
223 # Otherwise, the call to get_verbose_option may impact a function above;
224 # in case you didn't copy all of "$@" in nested function calls...
225 # /\ The same warning applies for other functions below. /\
226 # /\ I will not repeat it below. /\
227 # Use get_verbose_options_array, if needed.
228
229 get_verbose_options_array() {
230     # Use this constant with code like:
231     # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
232     # It could be extended if there is more than one level of verboseness.
233     if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
234         return
235     fi
236
237     declare -agr LFBFL_VERBOSE_OPTIONS=(
238         ""
239         "--verbose"
240     )
241 }
242
243 get_some_flag() {

```

```

244 # $1=variable_name
245 # $2=some_flag_long_name
246 # $3=read_only 0 or 1 to make the variable readonly
247 # This command is to be called in another one with same arguments on top
248 # of the three previous.
249 # This test:
250 # if [[ "$*" == *--verbose* ]]; then
251 # is faster than below.
252 # But it is not bug free in a calling function,
253 # and even incompatible with the current function where
254 # the flag_long_name is a mandatory argument.
255 printf -v "$1" "%d" 0
256 declare -i LFBFL_i
257 local LFBFL_argument
258 for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
259     LFBFL_argument="${!LFBFL_i}"
260     if [[ "${LFBFL_argument}" == "$2" ]]; then
261         printf -v "$1" "%d" 1
262         break
263     fi
264 done
265 if [[ $3 -eq 1 ]]; then
266     readonly "$1"
267 fi
268 }
269
270 get_verbose_option(){
271     # This command is to be called in another one with same arguments.
272     # Options:
273     # --verbose
274     get_some_flag LFBFL_i_verbose --verbose 1 "$@"
275     # shellcheck disable=SC2154,SC2309
276     if [[ LFBFL_i_verbose -eq 1 ]]; then
277         local LFBFL_where_was_i
278         get_where_was_i 2
279         printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
280     fi
281 }
282
283
284 get_some_option(){
285     # $1=variable_name
286     # $2=some_option_long_name
287     # $3=option_default_value_if_flag_without_option_value_or_empty_value
288     # $4=option_default_value_if_nothing
289     # $5=read_only 0 or 1 to make the variable readonly

```



```

290 # This command is to be called in another one with same arguments on top
291 # of the four previous.
292 # If you truly have a need to distinguish:
293 # option_default_value_if_flag_without_option_value --thing
294 # option_default_value_if_flag_with_empty_option_value --thing= or '='
295 # then think twice before modifying this function.
296 printf -v "$1" "%s" "$4"
297 declare -i LFBFL_i
298 local LFBFL_argument
299 for ((LFBFL_i = 6; LFBFL_i <= $#; ++LFBFL_i)); do
300     LFBFL_argument="${!LFBFL_i}"
301     if [[ "${LFBFL_argument}" == "$2" ]]; then
302         printf -v "$1" "%s" "$3"
303         break
304     fi
305     if [[ "${LFBFL_argument}" == "$2"=* ]]; then
306         printf -v "$1" "%s" "${LFBFL_argument#"${2}="}"
307         printf -v "$1" "%s" "${!1:-$3}"
308         break
309     fi
310 done
311 if [[ $5 -eq 1 ]]; then
312     readonly "$1"
313 fi
314 }
315
316 keep_options() {
317     # $1=variable_name of array of options
318     # Other arguments: the options to keep.
319     declare -a LFBFL_array_of_filtered_options
320     declare -n LFBFL_array_of_options=$1
321     local LFBFL_some_argument
322     local LFBFL_some_argument2
323     for LFBFL_some_argument in "${LFBFL_array_of_options[@]"; do
324         for LFBFL_some_argument2 in "$@"; do
325             if [[
326                 "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
327                 || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
328             ]]; then
329                 LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
330             fi
331         done
332     done
333     LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
334 }
335

```

```

336
337 # Main usage for the next functions relative to work_directory:
338 # Add the five following lines at the start of a function.
339 #   local LFBFL_work_directory=""
340 #   get_work_directory_option "$@"
341 #   pushd_to_work_directory\
342 #   && trap 'popd_from_work_directory' RETURN
343 #   can_continue_after_enhanced_pushd || return 1
344 # Or:
345 #   local LFBFL_work_directory=""
346 #   get_work_directory_option "$@"
347 #   pushd_to_work_directory --trap-popd
348 #   can_continue_after_enhanced_pushd || return 1
349
350 get_work_directory_option(){
351     # This command is to be called in another one with same arguments.
352     # Options:
353     #   --work-directory=""
354     get_some_option LFBFL_work_directory --work-directory "." "." 0 "$@"
355     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
356     readonly LFBFL_work_directory
357 }
358
359 pushd_to_work_directory(){
360     # Options:
361     #   --trap-popd
362     enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
363     if [[ i_enhanced_pushd_result -eq 0 ]]; then
364         declare -i LFBFL_i_trap_popd
365         get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
366         if [[ LFBFL_i_trap_popd -eq 1 ]]; then
367             declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
368             local LFBFL_some_return_trap
369             LFBFL_some_return_trap="${LFBFL_function_depth}:"
370             LFBFL_some_return_trap+="popd_from_work_directory 4"
371             readonly LFBFL_some_return_trap
372             LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
373         fi
374     fi
375     # shellcheck disable=SC2248
376     return ${i_enhanced_pushd_result}
377 }
378
379 popd_from_work_directory(){
380     # $1=offset for where_was_i
381     enhanced_popd "${LFBFL_work_directory}" "${1:-3}"

```

```

382 }
383
384 work_directory_is_top_dirstack_directory() {
385     is_top_dirstack_directory "${LFBFL_work_directory}"
386 }
387
388
389 get_offset_option() {
390     # This command is to be called in another one with same arguments on top
391     # of the default value.
392     # $1=option_default_value
393     # Options:
394     #     --offset=2
395     get_some_option LFBFL_i_offset --offset "$1" "$1" 1 "$@"
396 }
397
398 get_max_line_length_option() {
399     # This command is to be called in another one with same arguments.
400     # Its most useful feature is that it factorizes the default maximum line
401     # length of 70 characters.
402     # Options:
403     #     --max-line-length
404     get_some_option LFBFL_i_max_line_length --max-line-length 70 70 1 "$@"
405 }
406
407
408 # Functions relative to shell and bash options
409 # (See SHELLOPTS and BASHOPTS, set and shopt):
410 # - shell options usage, example with pipefail:
411 #     enhanced_set_shell_option pipefail\
412 #         && trap 'enhanced_unset_shell_option pipefail' RETURN
413 # or maybe:
414 #     enhanced_unset_shell_option pipefail\
415 #         && trap 'enhanced_set_shell_option pipefail' RETURN
416 # if you need to unset it in some function.
417 # or better if you use my RETURN traps handler:
418 #     enhanced_set_shell_option pipefail --trap-unset
419 #     enhanced_unset_shell_option pipefail --trap-set
420 # - bash options usage, example with globstar:
421 #     enhanced_set_bash_option globstar\
422 #         && trap 'enhanced_unset_bash_option globstar' RETURN
423 # or maybe:
424 #     enhanced_unset_bash_option globstar\
425 #         && trap 'enhanced_set_bash_option globstar' RETURN
426 # or better if you use my RETURN traps handler:
427 #     enhanced_set_bash_option globstar --trap-unset

```

```

428 #      enhanced_unset_bash_option globstar --trap-set
429
430 enhanced_set_shell_option(){
431     # $1=optname some shell option for set
432     # Options:
433     #     --offset=2
434     #     --trap-unset
435     declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
436     declare -gi "${LFBFL_result_name}"=0
437     if [[ -o "$1" ]]; then
438         printf -v "${LFBFL_result_name}" "%d" 1
439         # shellcheck disable=SC2086
440         return ${!LFBFL_result_name}
441     fi
442     set -o "$1"
443     declare -i LFBFL_i_offset=2
444     get_offset_option 2 "$@"
445     # shellcheck disable=SC2154,SC2309
446     if [[ LFBFL_i_verbose -eq 1 ]]; then
447         local LFBFL_where_was_i
448         # shellcheck disable=SC2248
449         get_where_was_i ${LFBFL_i_offset}
450         printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
451     fi
452     declare -i LFBFL_i_trap_unset
453     get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
454     if [[ LFBFL_i_trap_unset -eq 1 ]]; then
455         declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
456         local LFBFL_some_return_trap
457         LFBFL_some_return_trap="${LFBFL_function_depth}:"
458         LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
459         readonly LFBFL_some_return_trap
460         LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
461     fi
462     # shellcheck disable=SC2086
463     return ${!LFBFL_result_name}
464 }
465
466 enhanced_unset_shell_option(){
467     # $1=optname some shell option for set
468     # Options:
469     #     --offset=2
470     #     --trap-set
471     declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
472     declare -gi "${LFBFL_result_name}"=0
473     if [[ ! -o "$1" ]]; then

```

```

474     printf -v "${LFBFL_result_name}" "%d" 1
475     # shellcheck disable=SC2086
476     return ${!LFBFL_result_name}
477 fi
478 set +o "$1"
479 declare -i LFBFL_i_offset=2
480 get_offset_option 2 "$@"
481 # shellcheck disable=SC2154,SC2309
482 if [[ LFBFL_i_verbose -eq 1 ]]; then
483     local LFBFL_where_was_i
484     # shellcheck disable=SC2248
485     get_where_was_i ${LFBFL_i_offset}
486     printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
487 fi
488 declare -i LFBFL_i_trap_set
489 get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
490 if [[ LFBFL_i_trap_set -eq 1 ]]; then
491     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
492     local LFBFL_some_return_trap
493     LFBFL_some_return_trap="${LFBFL_function_depth}:"
494     LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
495     readonly LFBFL_some_return_trap
496     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
497 fi
498 # shellcheck disable=SC2086
499 return ${!LFBFL_result_name}
500 }
501
502 enhanced_set_bash_option() {
503     # $1=optname some bash option for shopt
504     # Options:
505     #   --offset=2
506     #   --trap-unset
507     declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
508     declare -gi "${LFBFL_result_name}"=0
509     # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
510     # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
511     if shopt -q "$1"; then
512         printf -v "${LFBFL_result_name}" "%d" 1
513         # shellcheck disable=SC2086
514         return ${!LFBFL_result_name}
515     fi
516     shopt -s "$1"
517     declare -i LFBFL_i_offset=2
518     get_offset_option 2 "$@"
519     # shellcheck disable=SC2154,SC2309

```

```

520 if [[ LFBFL_i_verbose -eq 1 ]]; then
521     local LFBFL_where_was_i
522     # shellcheck disable=SC2248
523     get_where_was_i ${LFBFL_i_offset}
524     printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
525 fi
526 declare -i LFBFL_i_trap_unset
527 get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
528 if [[ LFBFL_i_trap_unset -eq 1 ]]; then
529     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
530     local LFBFL_some_return_trap
531     LFBFL_some_return_trap="${LFBFL_function_depth}:"
532     LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
533     readonly LFBFL_some_return_trap
534     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
535 fi
536 # shellcheck disable=SC2086
537 return ${!LFBFL_result_name}
538 }
539
540 enhanced_unset_bash_option() {
541     # $1=optname some bash option for shopt
542     # Options:
543     #   --offset=2
544     #   --trap-set
545     declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
546     declare -gi "${LFBFL_result_name}"=0
547     if ! shopt -q "$1"; then
548         printf -v "${LFBFL_result_name}" "%d" 1
549         # shellcheck disable=SC2086
550         return ${!LFBFL_result_name}
551     fi
552     shopt -u "$1"
553     declare -i LFBFL_i_offset=2
554     get_offset_option 2 "$@"
555     # shellcheck disable=SC2154,SC2309
556     if [[ LFBFL_i_verbose -eq 1 ]]; then
557         local LFBFL_where_was_i
558         # shellcheck disable=SC2248
559         get_where_was_i ${LFBFL_i_offset}
560         printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
561     fi
562     declare -i LFBFL_i_trap_set
563     get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
564     if [[ LFBFL_i_trap_set -eq 1 ]]; then
565         declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))

```

```

566     local LFBFL_some_return_trap
567     LFBFL_some_return_trap="${LFBFL_function_depth}:"
568     LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
569     readonly LFBFL_some_return_trap
570     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
571 fi
572 # shellcheck disable=SC2086
573 return ${!LFBFL_result_name}
574 }

```

3.13 build_and_checks_dependencies/comparisons.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "comparisons.sh" was renamed to "comparisons.lib.sh".
25
26 # shellcheck source=common_options.lib.sh
27 source "./build_and_checks_dependencies/common_options.lib.sh"
28
29 equal(){
30     # Returns 1 if all the arguments strings are equal.
31     # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
32     # J'ai rerepensé au 3-way-merge
33     # et à mon article sur le principe de première différence
34     # appliqué à la largeur modulaire, de clique ou de rang avec

```

```

35 # l'idée d'un principe de première différence ternaire.
36 declare -i LFBFL_i_first=1
37 local LFBFL_current
38 local LFBFL_argument
39 for LFBFL_argument in "$@"; do
40     if [[ LFBFL_i_first -eq 1 ]]; then
41         LFBFL_current="{LFBFL_argument}"
42         LFBFL_i_first=0
43         continue
44     fi
45     if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
46         return 0
47     fi
48 done
49 return 1
50 }
51
52 some_distinct(){
53     ! equal "$@"
54 }
55
56 all_distinct(){
57     # Returns 1 if all the arguments strings are distinct.
58     declare -ir LFBFL_ij_max=$#
59     declare -i LFBFL_i
60     declare -i LFBFL_j
61     for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
62         for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
63             [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
64         done
65     done
66     return 1
67 }
68
69 all_distinct2(){
70     # Returns 1 if all the arguments strings are distinct.
71     local LFBFL_element
72     local LFBFL_previous_element
73     if [[ $# -le 1 ]]; then
74         return 1
75     fi
76     declare -r LFBFL_s_sorted_arguments=$(
77         printf "%s\n" "$@" \
78         | sort
79     )
80     declare -a LFBFL_arr_sorted_arguments

```



```

81 mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
82 declare -i LFBFL_i
83 declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
84 LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
85 for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
86     LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
87     if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
88         return 0
89     fi
90     LFBFL_previous_element="${LFBFL_element}"
91 done
92 return 1
93 }
94
95 all_distinct3(){
96     # Returns 1 if all the arguments strings are distinct.
97     # all_distinct3 is slower than old all_distinct2 since a call to sort is
98     # needed:
99     # the manual of uniq says it doesn't detect non-adjacent repeated lines.
100    # A few weeks ago, when testing without sort, I had no bug...
101    # I may have tested all_distinct3 a a b and not all_distinct3 a b a
102    # There is a sort also in all_distinct2, but uniq could have an option to
103    # sort before "applying -d", or sort could have an option for printing
104    # duplicates only. It would complement -u option of sort.
105    # Both solutions would avoid one pipe and one more shell command call.
106    # Note that old all_distinct2 was faster with pipes and read -r commands,
107    # but read -r has small drawbacks. Using mapfile is a cleaner way to do
108    # it, since it handles arguments with spaces from scratch.
109    # Arguments with line-returns are still not handled.
110    # Use C, PHP or Python for example for that.
111    # Somehow Bash was made to push people to stop being lazy and do actual C
112    # code, even if it needs compiling something, adding a makefile, etc. XD
113    # Note that C code would be faster with all_distinct2 blueprint,
114    # that's the reason I kept all 3 versions, see the (asymptotic)
115    # complexity and simple algorithms ideas on this toy problem.
116    if [[ $# -le 1 ]]; then
117        return 1
118    fi
119    declare -r LFBFL_s_duplicated_arguments=$(
120        printf "%s\n" "$@" \
121        | sort \
122        | uniq --repeated
123    )
124    [[ -n "${LFBFL_s_duplicated_arguments}" ]]
125 }
126

```

```

127 some_equal() {
128     ! all_distinct "$@"
129 }
130
131 some_equal2() {
132     ! all_distinct2 "$@"
133 }
134
135 some_equal3() {
136     ! all_distinct3 "$@"
137 }
138
139 max() {
140     # Returns the maximum string among the arguments.
141     # Renvoie la chaîne la plus grande parmi celles proposées.
142     # $1=sort_command with the flags you want to be applied.
143     # $ max sort a b c
144     # c
145     if [[ $# -eq 0 ]]; then
146         return 1
147     fi
148     enhanced_set_shell_option pipefail\
149     && trap 'enhanced_unset_shell_option pipefail' RETURN
150     printf "%s\n" "${@:2}" \
151         | eval -- "$1 --reverse"\
152         | head --lines=1
153 }
154
155 min() {
156     # Returns the minimum string among the arguments.
157     # Renvoie la chaîne la plus petite parmi celles proposées.
158     # $1=sort_command with the flags you want to be applied.
159     # $ min sort a b c
160     # a
161     if [[ $# -eq 0 ]]; then
162         return 1
163     fi
164     enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166     printf "%s\n" "${@:2}" \
167         | eval -- "$1"\
168         | head --lines=1
169 }
170
171 is_substring() {
172     # $1=string

```

```

173 # $2=substring
174 declare -r LFBFL_var_1=$(
175     printf "%s" "$1"\
176     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
177 )
178 declare -r LFBFL_var_2=$(
179     printf "%s" "$2"\
180     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
181 )
182 local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
183 LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
184 LFBFL_var_3+=" \$c = '\/'.addslashes(\$b, '\/').'\/';"
185 LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
186 readonly LFBFL_var_3
187 php --run="\${LFBFL_var_3}"
188 return $?
189 }
190
191 is_subfile(){
192     # $1=file_path
193     # $2=sub_file_path
194     local LFBFL_var_1="\$a = file_get_contents('$1');"
195     LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
196     LFBFL_var_1+=" \$c = '\/'.addslashes(\$b, '\/').'\/';"
197     LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
198     readonly LFBFL_var_1
199     php --run="\${LFBFL_var_1}"
200     return $?
201 }
202
203 alias min_from_int_sort="min 'sort --numeric-sort'"
204 alias max_from_int_sort="max 'sort --numeric-sort'"

```

3.14 build_and_checks_dependencies/create_PDF.exec.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #
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6 # under the terms of the GNU General Public License
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8 # either version 3 of the License,
9 # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed

```

```

12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".
25
26 LFBFL_subdir="build_and_checks_dependencies"
27 # shellcheck source=common_options.lib.sh
28 source "${LFBFL_subdir}/common_options.lib.sh"
29 # shellcheck source=generate_from_template.lib.sh
30 source "${LFBFL_subdir}/generate_from_template.lib.sh"
31 # shellcheck source=lines_counts.lib.sh
32 source "${LFBFL_subdir}/lines_counts.lib.sh"
33 # shellcheck source=lines_filters.lib.sh
34 source "${LFBFL_subdir}/lines_filters.lib.sh"
35 # shellcheck source=lines_maps.lib.sh
36 source "${LFBFL_subdir}/lines_maps.lib.sh"
37 # shellcheck source=overwrite_if_not_equal.lib.sh
38 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
39 # shellcheck source=strings_functions.lib.sh
40 source "${LFBFL_subdir}/strings_functions.lib.sh"
41
42 create_PDF() {
43     # Options:
44     #   --verbose
45     #   --work-directory=""
46     declare -i LFBFL_i_verbose=0
47     get_verbose_option "$@"
48     local LFBFL_work_directory=""
49     get_work_directory_option "$@"
50     declare -a LFBFL_return_traps_stack
51     init_return_trap
52     pushd_to_work_directory --trap-popd
53     can_continue_after_enhanced_pushd || return 1
54
55     enhanced_set_shell_option pipefail --trap-unset
56
57     declare -a LFBFL_common_options=("$@")

```

```

58 keep_options LFBFL_common_options --verbose
59
60 local LFBFL_subdir2="build_and_checks_variables"
61 local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
62 local LFBFL_repository_name=""
63 grep_variable "${LFBFL_data_file_path}" repository_name\
64     --result-variable-prefix="LFBFL_"\
65     --replace-line-returns-by=""
66
67 local LFBFL_repository_web_url=""
68 grep_variable "${LFBFL_data_file_path}" repository_web_url\
69     --result-variable-prefix="LFBFL_"\
70     --replace-line-returns-by=""
71
72 local LFBFL_repository_git_url=""
73 grep_variable "${LFBFL_data_file_path}" repository_git_url\
74     --result-variable-prefix="LFBFL_"\
75     --replace-line-returns-by=""
76
77 local LFBFL_abstract=""
78 grep_variable "${LFBFL_data_file_path}" abstract\
79     --result-variable-prefix="LFBFL_"
80
81 local LFBFL_acknowledgments=""
82 grep_variable "${LFBFL_data_file_path}" acknowledgments\
83     --result-variable-prefix="LFBFL_"
84
85 local LFBFL_author_email=""
86 grep_variable "${LFBFL_data_file_path}" author_email\
87     --result-variable-prefix="LFBFL_"\
88     --replace-line-returns-by=""
89
90 local LFBFL_author_full_name=""
91 grep_variable "${LFBFL_data_file_path}" author_full_name\
92     --result-variable-prefix="LFBFL_"\
93     --replace-line-returns-by=""
94
95 local LFBFL_author_website=""
96 grep_variable "${LFBFL_data_file_path}" author_website\
97     --result-variable-prefix="LFBFL_"\
98     --replace-line-returns-by=""
99
100 declare -i LFBFL_i_max_line_length
101 grep_variable "${LFBFL_data_file_path}" max_line_length\
102     --result-variable-prefix="LFBFL_i_"
103

```

```

104 declare -r LFBFL_current_date=$(date --iso-8601="date")
105
106 declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
107
108 declare -ir LFBFL_i_number_of_commits=$(
109     git shortlog\
110     | space_starting_lines\
111     | wc --lines
112 )
113
114 local LFBFL_redirect="/dev/stdout"
115 if [[ LFBFL_i_verbose -eq 0 ]]; then
116     LFBFL_redirect="/dev/null"
117 fi
118 {
119     code_lines_count_all "${LFBFL_common_options[@]}"
120     code_lines_count_not_empty "${LFBFL_common_options[@]}"
121     code_lines_count_empty "${LFBFL_common_options[@]}"
122 } &> "${LFBFL_redirect}"
123
124 local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
125 LFBFL_number_of_lines+=" total lines,"
126 LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
127 LFBFL_number_of_lines+=" not empty lines,"
128 LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
129 LFBFL_number_of_lines+=" empty lines."
130 readonly LFBFL_number_of_lines
131
132 declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"
133
134 # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
135 # ?show=eyJpaWQiOiI0MyIsImZlbGxhcGF0aCI6Ik9sZE1hblByb2dyYWlt
136 # ZXIvdW5peC10cmVlIiwiaWQiOiJlE4NTE0NTgzOH0%3D
137 tree -a --gitignore\
138     -I "node_modules/"\
139     -I "__pycache__/"\
140     -I ".mypy_cache/"\
141     -I ".ruff_cache/"\
142     -I ".git/"\
143     | replace_non_ascii_spaces\
144     > "${LFBFL_temp_path}/current_tree_light.txt"
145
146 tree -a -Fh --timefmt='%F %R' --gitignore\
147     -I "node_modules/"\
148     -I "__pycache__/"\
149     -I ".mypy_cache/"\

```

```

150     -I ".ruff_cache/"\
151     -I ".git/"\
152     | replace_non_ascii_spaces\
153     > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155 local LFBFL_s_files_paths
156 # Remove line returns that are here to keep lines short.
157 LFBFL_s_files_paths=$(
158     sed --regext-extended --null-data\
159         --expression='s/(.)\\n/\1/Mg'\
160         -- "${LFBFL_subdir2}/files_names_listing.txt"\
161         | grep --invert-match '^/'
162 )
163 readonly LFBFL_s_files_paths
164 declare -a LFBFL_arr_files_paths
165 if [[ -n "${LFBFL_s_files_paths}" ]]; then
166     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
167 fi
168 readonly LFBFL_arr_files_paths
169
170 local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
171 LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
172 readonly LFBFL_temp_files_listing
173 : > "${LFBFL_temp_files_listing}"
174 local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
175 LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
176 readonly LFBFL_temp_files_listing2
177 : > "${LFBFL_temp_files_listing2}"
178 # HTML <li> elements, hence "lis".
179 local LFBFL_temp_files_lis
180 LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
181 readonly LFBFL_temp_files_lis
182 : > "${LFBFL_temp_files_lis}"
183 # shellcheck disable=SC2248
184 get_split_score_simple 1 ${LFBFL_i_max_line_length} /
185 declare -r LFBFL_score_command="${get_split_score_result}"
186 declare -i LFBFL_i_score_command_properties
187 LFBFL_i_score_command_properties=${i_get_split_score_result2}
188 readonly LFBFL_i_score_command_properties
189 # shellcheck disable=SC2248
190 get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
191 declare -r LFBFL_score_command2="${get_split_score_result}"
192 declare -i LFBFL_i_score_command_properties2
193 LFBFL_i_score_command_properties2=${i_get_split_score_result2}
194 readonly LFBFL_i_score_command_properties2
195 declare -r LFBFL_suffix='%'
```

```

196 declare -i LFBFL_i=0
197 local LFBFL_cleaned_path1
198 local LFBFL_cleaned_path2
199 local LFBFL_new_lines
200 local LFBFL_new_lines2
201 local LFBFL_new_lines3
202 local LFBFL_file_path
203 local LFBFL_s_format
204 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
205     printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
206     # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
207     # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
208     LFBFL_cleaned_path1="${LFBFL_file_path//_/\_}"
209     LFBFL_cleaned_path2=$(
210         printf "%s" "${LFBFL_file_path}"\
211         | sed --expression='s/\:\/\/:/g;s/\./\/g'
212     )
213     LFBFL_new_lines=" ${LFBFL_cleaned_path1}"
214     if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
215         # shellcheck disable=SC2248
216         repeated_split_last_line "${LFBFL_new_lines}"\
217             "\
218             ${LFBFL_i_max_line_length}\
219             "${LFBFL_suffix}"\
220             "${LFBFL_score_command}"\
221             ${LFBFL_i_score_command_properties}\
222             "\
223         LFBFL_new_lines=${repeated_split_last_line_result}
224     fi
225     LFBFL_new_lines2=" ${LFBFL_cleaned_path2}"
226     if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
227         # shellcheck disable=SC2248
228         repeated_split_last_line "${LFBFL_new_lines2}"\
229             "\
230             ${LFBFL_i_max_line_length}\
231             "${LFBFL_suffix}"\
232             "${LFBFL_score_command2}"\
233             ${LFBFL_i_score_command_properties2}
234     LFBFL_new_lines2=${repeated_split_last_line_result}
235     fi
236     LFBFL_new_lines3="${LFBFL_file_path}"
237     if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
238         # shellcheck disable=SC2248
239         repeated_split_last_line "${LFBFL_new_lines3}"\
240             "\
241             ${LFBFL_i_max_line_length}\

```



```

242     "${LFBFL_suffix}"\
243     "${LFBFL_score_command}"\
244     "${LFBFL_i_score_command_properties}"
245     LFBFL_new_lines3=${repeated_split_last_line_result}
246 fi
247 LFBFL_s_format="\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
248 LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
249 LFBFL_s_format+="{\n%s\n}\n\n\n"
250 # shellcheck disable=SC2059
251 printf "${LFBFL_s_format}"\
252     "${LFBFL_new_lines}"\
253     "${LFBFL_new_lines2}"\
254     "${LFBFL_new_lines3}"\
255     >> "${LFBFL_temp_files_listing}"
256
257 # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
258 ((++LFBFL_i))
259 LFBFL_new_lines="${LFBFL_file_path}"
260 if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
261     # shellcheck disable=SC2248
262     repeated_split_last_line "${LFBFL_new_lines}"\
263         "-->\n\
264         ${LFBFL_i_max_line_length}\n\
265         "<!--\n\
266         "${LFBFL_score_command}"\
267         "${LFBFL_i_score_command_properties}"
268     LFBFL_new_lines=${repeated_split_last_line_result}
269 fi
270 {
271     LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
272     LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
273     # shellcheck disable=SC2059
274     printf "${LFBFL_s_format}"\
275         "${LFBFL_i}"\
276         "${LFBFL_i}"\
277         "${LFBFL_new_lines}"
278     sed --expression='s/&/\&/g;s/</\&lt;/g'\
279         < "${LFBFL_file_path}"
280     printf "</pre>\n\n\n"
281 } >> "${LFBFL_temp_files_listing2}"
282 LFBFL_s_format="<li><a href=\"\#subsection3.%s\">\n%s\n"
283 LFBFL_s_format+="</a></li>\n"
284 # shellcheck disable=SC2059
285 printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
286     >> "${LFBFL_temp_files_lis}"
287 done

```

```

288 overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
289 "${LFBFL_temp_files_listing}"
290 # Example of generated content of files_listing.tex.sub:
291 # -----
292 # \subsection{
293 #   build\_and\_checks\_dependencies/licenses/templates/%
294 # build\_licenses/templates.exec.sh
295 # }
296 # \label{
297 #   build_and_checks_dependencies:licenses_templates:%
298 # build_licenses_templatesexecsh
299 # }
300 #
301 # \VerbatimInput[numbers=left,xleftmargin=-5mm]{
302 #   build_and_checks_dependencies/licenses_templates/%
303 #   build_licenses_templates.exec.sh
304 # }
305 # -----
306 sed --in-place --expression="s|\\n</pre>|</pre>|g"\
307 -- "${LFBFL_temp_files_listing2}"
308 overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
309 "${LFBFL_temp_files_listing2}"
310 # Example of generated content of files_listing.html.sub:
311 # -----
312 # <h3 id="subsection3.15">3.15
313 #   build_and_checks_dependencies/licenses_templates/<!--
314 #   -->build_licenses_templates.exec.sh
315 # </h3>
316 # ...
317 # <h3 id="subsection3.33">3.33
318 #   build_and_checks_variables/.gitignore
319 # </h3>
320 # <pre class="numbered_lines">
321 #   temp/
322 #   upgrade_venvs_ts
323 # </pre>
324 # -----
325 overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
326 "${LFBFL_temp_files_lis}"
327 # Example of generated content of files_lis.html.sub:
328 # -----
329 #   <li><a href="#subsection3.15">
330 #   build_and_checks_dependencies/licenses_templates/<!--
331 #   -->build_licenses_templates.exec.sh
332 #   </a></li>
333 # -----

```

```

334
335 # shellcheck disable=SC2248
336 get_overlength_regex ${LFBFL_i_max_line_length}
337
338 # We verify if some lines are beyond max_line_length characters
339 # in current_tree_light.txt and current_tree.txt.
340 declare -ar LFBFL_trees=(
341     "current_tree_light.txt"
342     "current_tree.txt.temp"
343 )
344 local LFBFL_tree
345 local LFBFL_tree_path
346 local LFBFL_line
347 local LFBFL_prefix
348 declare -i LFBFL_i_prefix_last_position
349 local LFBFL_char
350 local LFBFL_line_start
351 local LFBFL_file_name
352 local LFBFL_s_lines
353 declare -a LFBFL_arr_lines
354 declare -i LFBFL_i_keep_length
355 for LFBFL_tree in "${LFBFL_trees[@]}"; do
356     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
357     # shellcheck disable=SC2248
358     LFBFL_s_lines=$(
359         grep --regex="${get_overlength_regex_result}" --\
360             "${LFBFL_tree_path}"
361     )
362     if [[ -z "${LFBFL_s_lines}" ]]; then
363         continue
364     fi
365     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
366     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
367         # printf "LFBFL_line: %s\n" "${LFBFL_line}"
368         LFBFL_prefix=$(
369             printf "%s" "${LFBFL_line}"\
370             | sed --regex-extended --expression='s/(.*)—[^-]+$/\1/g'
371         )
372         # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
373         LFBFL_i_prefix_last_position=$(( ${#LFBFL_prefix} - 1 ) )
374         LFBFL_char="${LFBFL_prefix}${LFBFL_i_prefix_last_position}:1}"
375         # printf "LFBFL_char: %s\n" "${LFBFL_char}"
376         LFBFL_prefix=$(
377             printf "%s" "${LFBFL_prefix}"\
378             | sed --regex-extended --expression='s/[^ ]+$//g'
379         )

```

```

380     if [[ "${LFBFL_char}" == "L" ]]; then
381         LFBFL_prefix+=" "
382     else
383         LFBFL_prefix+="| "
384     fi
385     # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
386     LFBFL_file_name=$(
387         printf "%s" "${LFBFL_line}"\
388         | sed --regex-extended --expression='s|.* ([^ ]+)$|\1|g'
389     )
390     # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
391     LFBFL_i_keep_length=$(( ${#LFBFL_line} - ${#LFBFL_file_name} ))
392     LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
393     LFBFL_line_start=$(
394         printf "%s" "${LFBFL_line_start}"\
395         | sed --expression='s/ *///g'
396     )
397     # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
398     LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
399     if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
400         # shellcheck disable=SC2248
401         repeated_split_last_line "${LFBFL_new_lines}"\
402             "${LFBFL_prefix}"\
403             ${LFBFL_i_max_line_length}\
404             ""\
405             ""\
406             ""
407         LFBFL_new_lines=${repeated_split_last_line_result}
408     fi
409     printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
410         > "${LFBFL_tree_path}.some_line.temp"
411     insert_file_at_token "${LFBFL_tree_path}"\
412         "${LFBFL_line}"\
413         "${LFBFL_tree_path}.some_line.temp"\
414         ""\
415         --quiet
416     done
417     rm --force -- "${LFBFL_tree_path}.some_line.temp"
418 done
419
420 overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
421     "${LFBFL_temp_path}/current_tree.txt.temp" --keep-temp --tree-mode
422
423 local LFBFL_tex_path_start
424 LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
425 readonly LFBFL_tex_path_start

```

```

426 local LFBFL_html_path_start
427 LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
428 readonly LFBFL_html_path_start
429
430 if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
431     # If there is a template, we init the process from it.
432     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
433         "${LFBFL_tex_path_start}.1"
434 elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
435     # Otherwise if there is a tex file, we init the process from it.
436     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
437         "${LFBFL_tex_path_start}.1"
438 else
439     printf "Neither .tex.tpl, nor .tex in ./s/.\n" "${LFBFL_subdir2}"
440 fi
441
442 # Same logic with repository HTML file.
443 if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
444     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
445         "${LFBFL_html_path_start}.1"
446 elif [[ -f "${LFBFL_repository_name}.html" ]]; then
447     cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
448 else
449     printf "Neither .html.tpl in ./s/, nor .html in ./.\n"\
450         "${LFBFL_subdir2}"
451 fi
452
453 # shellcheck disable=SC2001
454 printf "%s\n" "${LFBFL_abstract}"\
455     | sed --expression='s/\n/\n/g'\
456     > "${LFBFL_temp_path}/abstract_temp"
457
458 # shellcheck disable=SC2001
459 printf "%s\n" "${LFBFL_acknowledgments}"\
460     | sed --expression='s/\n/\n/g'\
461     > "${LFBFL_temp_path}/acknowledgments_temp"
462
463 # But the filling still occurs, in case the dev want to refill
464 # part of the tex/html file that he modified with @token@
465 # using some computed results.
466 # Tex filling:
467 if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
468     sed --in-place\
469         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
470         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
471         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"

```

```

472 --expression="s|@author_email@|${LFBFL_author_email}|g"\
473 --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
474 --expression="s|@author_website@|${LFBFL_author_website}|g"\
475 --expression="s|@current_date@|${LFBFL_current_date}|g"\
476 --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
477 --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
478 --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
479 -- "${LFBFL_tex_path_start}.1"
480
481 insert_file_at_token "${LFBFL_tex_path_start}.1"\
482 @abstract@\
483 "${LFBFL_temp_path}/abstract_temp"\
484 "${LFBFL_tex_path_start}.2"
485
486 insert_file_at_token "${LFBFL_tex_path_start}.2"\
487 @acknowledgments@\
488 "${LFBFL_temp_path}/acknowledgments_temp"\
489 "${LFBFL_tex_path_start}.3"
490
491 insert_file_at_token "${LFBFL_tex_path_start}.3"\
492 @current_tree_light@\
493 "${LFBFL_temp_path}/current_tree_light.txt"\
494 "${LFBFL_tex_path_start}.4"
495
496 insert_file_at_token "${LFBFL_tex_path_start}.4"\
497 @current_tree@\
498 "${LFBFL_temp_path}/current_tree.txt"\
499 "${LFBFL_tex_path_start}.5"
500
501 insert_file_at_token "${LFBFL_tex_path_start}.5"\
502 @dependencies_notes@\
503 "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
504 "${LFBFL_tex_path_start}.6"
505
506 insert_file_at_token "${LFBFL_tex_path_start}.6"\
507 @files_listing_VerbatimInput@\
508 "${LFBFL_temp_path}/files_listing.tex.sub"\
509 "${LFBFL_tex_path_start}.7"
510
511 overwrite_if_not_equal\
512 "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
513 "${LFBFL_tex_path_start}.7" --keep-temp
514 fi
515
516 # HTML filling:
517 declare -i LFBFL_i_HTML_updated=0

```

```

518 if [[ -f "${LFBFL_html_path_start}.1" ]]; then
519     sed --in-place\
520         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
521         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
522         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
523         --expression="s|@author_email@|${LFBFL_author_email}|g"\
524         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
525         --expression="s|@author_website@|${LFBFL_author_website}|g"\
526         --expression="s|@current_date@|${LFBFL_current_date}|g"\
527         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
528         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
529         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
530         -- "${LFBFL_html_path_start}.1"
531
532     insert_file_at_token "${LFBFL_html_path_start}.1"\
533         @abstract@\
534         "${LFBFL_temp_path}/abstract_temp"\
535         "${LFBFL_html_path_start}.2"
536
537     insert_file_at_token "${LFBFL_html_path_start}.2"\
538         @acknowledgments@\
539         "${LFBFL_temp_path}/acknowledgments_temp"\
540         "${LFBFL_html_path_start}.3"
541
542     insert_file_at_token "${LFBFL_html_path_start}.3"\
543         @files_lis@\
544         "${LFBFL_temp_path}/files_lis.html.sub"\
545         "${LFBFL_html_path_start}.4"
546
547     insert_file_at_token "${LFBFL_html_path_start}.4"\
548         @current_tree_light@\
549         "${LFBFL_temp_path}/current_tree_light.txt"\
550         "${LFBFL_html_path_start}.5"
551
552     insert_file_at_token "${LFBFL_html_path_start}.5"\
553         @current_tree@\
554         "${LFBFL_temp_path}/current_tree.txt"\
555         "${LFBFL_html_path_start}.6"
556
557     insert_file_at_token "${LFBFL_html_path_start}.6"\
558         @dependencies_notes@\
559         "${LFBFL_temp_path}/dependencies_notes.html.sub"\
560         "${LFBFL_html_path_start}.7"
561
562     insert_file_at_token "${LFBFL_html_path_start}.7"\
563         @files_listing_HTMLPreInput@

```

```

564     "${LFBFL_temp_path}/files_listing.html.sub"\
565     "${LFBFL_html_path_start}.8"
566
567     overwrite_if_not_equal "${LFBFL_repository_name}.html"\
568     "${LFBFL_html_path_start}.8" --keep-temp
569     LFBFL_i_HTML_updated=$?
570 fi
571
572 # The HTML contains everything including listings with source code.
573 # If it doesn't get updated,
574 # then it is useless to recompute the PDF.
575 if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
576     return
577 fi
578
579 local LFBFL_latex_command=""
580 grep_variable "${LFBFL_data_file_path}" latex_command\
581     --result-variable-prefix="LFBFL_"\
582     --replace-line-returns-by=""
583
584 case ${LFBFL_latex_command} in
585     pdflatex) : ;;
586     lualatex) : ;;
587     *) printf "Unsupported latex command" ; return 1 ;;
588 esac
589
590 local LFBFL_command
591 LFBFL_command="${LFBFL_latex_command}"
592 LFBFL_command+=" ' ${LFBFL_subdir2}/${LFBFL_repository_name}.tex' "
593 if [[ LFBFL_i_verbose -eq 0 ]]; then
594     LFBFL_command+=" > /dev/null"
595 fi
596 readonly LFBFL_command
597
598 for ((LFBFL_i = 0; LFBFL_i < 3; ++LFBFL_i)); do
599     eval -- "${LFBFL_command}"
600 done
601
602 LFBFL_files_to_temp=(\
603     "${LFBFL_repository_name}.aux"\
604     "${LFBFL_repository_name}.log"\
605     "${LFBFL_repository_name}.out"\
606     "${LFBFL_repository_name}.toc"
607 )
608 for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
609     mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/"

```



```

610     done
611 }
612
613 create_PDF "$@"

```

3.15 build_and_checks_dependencies/generate_from_template.lib.sh

```

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3  #
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10 #
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15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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17 #
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "generate_from_template.sh" to
25 # "generate_from_template.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30 # shellcheck source=overwrite_if_not_equal.lib.sh
31 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
32
33 generate_from_template_with_block_comments() {
34     # $1=base_file_path
35     # $2=target_file_path
36     # $3=enter_block_comment
37     # $4=exit_block_comment
38     local LFBFL_temp
39     LFBFL_temp="$2.generate_from_template_with_block_comments.temp"

```

```

40     readonly LFBFL_temp
41     {
42         printf "%s\n" "$3"
43         cat -- "$1"
44         printf "%s\n" "$4"
45     } > "${LFBFL_temp}"
46     overwrite_if_not_equal "$2" "${LFBFL_temp}"
47 }
48
49 generate_from_template_with_line_comments() {
50     # $1=base_file_path
51     # $2=target_file_path
52     # $3=line_comment_prefix
53     # $4=optional_post_processing
54     # $5=file_prefix
55     local LFBFL_temp
56     LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57     readonly LFBFL_temp
58     if [[ -n "$5" ]]; then
59         printf "%s\n" "$5" \
60             > "${LFBFL_temp}"
61     fi
62     sed --expression="s~^~$3~g" -- "$1" \
63         >> "${LFBFL_temp}"
64     if [[ -n "$4" ]]; then
65         eval -- "$4"
66     fi
67     overwrite_if_not_equal "$2" "${LFBFL_temp}"
68 }
69
70 split_file_in_two() {
71     # $1=file_path
72     # $2=token assume that token is the only thing on his line,
73     #     except for a possible indentation.
74     # $3=path_of_part1_of_file
75     # $4=path_of_part2_of_file
76     # We cannot add --line-regexp with --fixed-strings because of the
77     # possible indentation in file,
78     # instead we use grep_fixed_string_with_anchor.
79     declare -ir LFBFL_i_line_number=$(
80         grep_fixed_string_with_anchor "$1" "$2" \
81             --enforce-line-ends-with-fixed-string --line-number \
82             | head --lines=1 \
83             | cut --delimiter=':' --fields=1
84     )
85     declare -ir LFBFL_i_line_count=$((ll_wc -1 -n "$1"))

```

```

86 declare -ir LFBFL_i_lines_after=$((
87     LFBFL_i_line_count - LFBFL_i_line_number
88 ))
89 head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
90     > "$3"
91 tail --lines="${LFBFL_i_lines_after}" -- "$1"\
92     > "$4"
93 }
94
95 insert_file_at_token() {
96     # $1=file_path
97     # $2=token assume that token is the only thing on his line,
98     #     except for a possible indentation.
99     # $3=path_of_file_to_insert
100    # $4=path_of_result_file
101    # Options:
102    #     $5=--quiet to suppress diff output in overwrite_if_not_equal
103    declare -r LFBFL_start_file_path="$1.insert_file_at_token1.temp"
104    declare -r LFBFL_end_file_path="$1.insert_file_at_token2.temp"
105    local LFBFL_result_file_path
106    if [[ -n "$4" ]]; then
107        LFBFL_result_file_path="$4"
108    else
109        LFBFL_result_file_path="$1.insert_file_at_token3.temp"
110    fi
111    readonly LFBFL_result_file_path
112    split_file_in_two "$1" "$2" "${LFBFL_start_file_path}"\
113        "${LFBFL_end_file_path}"
114    cat -- "${LFBFL_start_file_path}" "$3" "${LFBFL_end_file_path}"\
115        > "${LFBFL_result_file_path}"
116    rm -- "${LFBFL_start_file_path}" "${LFBFL_end_file_path}"
117    if [[ -z "$4" ]]; then
118        overwrite_if_not_equal "$1" "${LFBFL_result_file_path}" "$5"
119    fi
120 }

```

3.16 build_and_checks_dependencies/get_common_text_glob_patterns.lib.sh

```

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```

```

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20 # along with DevOrSysAdminScripts.
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23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "get_common_text_glob_patterns.sh"
25 # to "get_common_text_glob_patterns.lib.sh".
26
27 get_COMMON_TEXT_PATHS_GLOB_PATTERNS() {
28     if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
29         return
30     fi
31
32     declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
33         '**/*.c'
34         '**/*.css'
35         '**/*.h'
36         '**/*.htm'
37         '**/*.html'
38         '**/*.ini'
39         '**/*.js'
40         '**/*.json'
41         '**/*.md'
42         '**/*.neon'
43         '**/*.php'
44         '**/*.py'
45         '**/*.rs'
46         '**/*.rlib'
47         '**/*.sh'
48         '**/*.sql'
49         '**/*.sub'
50         '**/*.tex'
51         '**/*.toml'
52         '**/*.tpl'
53         '**/*.ts'
54         '**/*.txt'

```

```

55     '**/*.xml'
56     '**/*.xslt'
57     '**/*.yaml'
58     '**/COPYING'
59     '**/COPYING.LESSER'
60     '**/pre-commit'
61     '**/post-commit'
62     '**/.gitignore'
63     '**/py.typed'
64 )
65 }
66
67
68 get_COMMON_TEXT_FILES_GLOB_PATTERNS() {
69     if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
70         return
71     fi
72
73     declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
74         '*.c'
75         '*.css'
76         '*.h'
77         '*.htm'
78         '*.html'
79         '*.ini'
80         '*.js'
81         '*.json'
82         '*.md'
83         '*.neon'
84         '*.php'
85         '*.py'
86         '*.rs'
87         '*.rlib'
88         '*.sh'
89         '*.sql'
90         '*.sub'
91         '*.tex'
92         '*.toml'
93         '*.tpl'
94         '*.ts'
95         '*.txt'
96         '*.xml'
97         '*.xslt'
98         '*.yaml'
99         'COPYING'
100        'COPYING.LESSER'

```

```

101     'pre-commit'
102     'post-commit'
103     '.gitignore'
104     'py.typed'
105 )
106 }

```

3.17 build_and_checks_dependencies/grammar_and_spelling_check.lib.sh

```

1  #!/usr/bin/env bash
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # 2026-03-22T15:30+01:00: This file was renamed from
25 # "grammar_and_spell_check.lib.sh"
26 # to
27 # "grammar_and_spelling_check.lib.sh".
28 # This file was renamed from "grammar_and_spell_check.sh"
29 # to "grammar_and_spell_check.lib.sh".
30
31 LFBFL_subdir="build_and_checks_dependencies"
32 # shellcheck source=common_options.lib.sh
33 source "${LFBFL_subdir}/common_options.lib.sh"
34 # shellcheck source=get_common_text_glob_patterns.lib.sh
35 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
36 # shellcheck source=lines_filters.lib.sh
37 source "${LFBFL_subdir}/lines_filters.lib.sh"

```

```

38
39 grammar_and_spelling_check() {
40     # $1=configuration_path build_and_checks_variables/repository_data.txt
41     # Options:
42     #     --verbose
43     #     --work-directory=""
44     declare -i LFBFL_i_verbose=0
45     get_verbose_option "$@"
46     local LFBFL_work_directory=""
47     get_work_directory_option "$@"
48     declare -a LFBFL_return_traps_stack
49     init_return_trap
50     pushd_to_work_directory --trap-popd
51     can_continue_after_enhanced_pushd || return 1
52
53     enhanced_set_shell_option pipefail --trap-unset
54
55     get_COMMON_TEXT_FILES_GLOB_PATTERNS
56     local LFBFL_grammar_and_spelling_checker_command=""
57     grep_variable "$1" grammar_and_spelling_checker_command\
58         --result-variable-prefix=LFBFL_\
59         --replace-line-returns-by=""
60     declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
61     if [[ -z "${LFBFL_command}" ]]; then
62         if [[ LFBFL_i_verbose -eq 1 ]]; then
63             printf "grammar_and_spelling_check: No command in configuration.\n"
64         fi
65         return
66     fi
67     if [[ LFBFL_i_verbose -eq 1 ]]; then
68         printf "grammar_and_spelling_check: Command found: %s.\n" \
69             "${LFBFL_command}"
70     fi
71     local LFBFL_file_path
72     local LFBFL_s_files_paths
73     declare -a LFBFL_arr_files_paths
74     local LFBFL_pattern
75     local LFBFL_eval_string
76     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]"}; do
77         if [[ LFBFL_i_verbose -eq 1 ]]; then
78             printf "grammar_and_spelling_check: Iterating on pattern: %s.\n" \
79                 "${LFBFL_pattern}"
80         fi
81         LFBFL_s_files_paths=$(
82             find . -type f -name "${LFBFL_pattern}" -printf '%P\n' \
83                 | relevant_find

```

```

84     )
85     if [[ -z "${LFBFL_s_files_paths}" ]]; then
86         [[ LFBFL_i_verbose -eq 1 ]]\
87         && printf "grammar_and_spelling_check: No file found.\n"
88         continue
89     fi
90     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
91     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
92         LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
93         eval -- "${LFBFL_eval_string}"
94     done
95 done
96 }

```

3.18 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```

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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "build_licenses_templates.sh"
25 # to "build_licenses_templates.exec.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=check_shell_scripts_beginnings.lib.sh
29 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
30 # shellcheck source=common_options.lib.sh

```



```

31 source "${LFBFL_subdir}/common_options.lib.sh"
32 # shellcheck source=comparisons.lib.sh
33 source "${LFBFL_subdir}/comparisons.lib.sh"
34 # shellcheck source=generate_from_template.lib.sh
35 source "${LFBFL_subdir}/generate_from_template.lib.sh"
36 # shellcheck source=lines_filters.lib.sh
37 source "${LFBFL_subdir}/lines_filters.lib.sh"
38 # shellcheck source=overwrite_if_not_equal.lib.sh
39 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
40
41 enhanced_set_bash_option extglob
42 # shellcheck source=repository_data.lib.sh
43 source "${LFBFL_subdir}/repository_data.lib.sh"
44 # shellcheck disable=SC2154,SC2309
45 [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
46   && enhanced_unset_bash_option extglob
47
48 build_licenses_templates() {
49   # Options:
50   #   --verbose
51   #   --work-directory=""
52   declare -i LFBFL_i_verbose=0
53   get_verbose_option "$@"
54   local LFBFL_work_directory=""
55   get_work_directory_option "$@"
56   declare -a LFBFL_return_traps_stack
57   init_return_trap
58
59   enhanced_set_shell_option pipefail --trap-unset
60
61   local LFBFL_s_format
62
63   # Source
64   local LFBFL_license_source_subdir
65   LFBFL_license_source_subdir="${LFBFL_subdir}/licenses_templates/"
66   readonly LFBFL_license_source_subdir
67   local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
68   LFBFL_source_license_prefix+="license_file_header_"
69   readonly LFBFL_source_license_prefix
70   # Target
71   local LFBFL_license_target_subdir
72   LFBFL_license_target_subdir="${LFBFL_work_directory}/"
73   LFBFL_license_target_subdir="build_and_checks_variables/temp/"
74   readonly LFBFL_license_target_subdir
75   local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
76   LFBFL_target_license_prefix+="license_file_header_"

```

```

77     readonly LFBFL_target_license_prefix
78
79     declare -ar LFBFL_licenses=(
80         "AGPLv3+"
81         "GPLv3+"
82         "LGPLv3+"
83     )
84
85     # -----
86     # Various language definitions for text file formats
87     declare -ar LFBFL_block_comment_languages=(
88         "c"
89         "py"
90         "xml"
91     )
92     declare -ar LFBFL_block_comment_enters=(
93         '/*'
94         '"""'
95         '<!--'
96     )
97     declare -ar LFBFL_block_comment_exits=(
98         '*/'
99         '"""'
100        '-->'
101    )
102    equal "${#LFBFL_block_comment_languages[@]}" "\
103        "${#LFBFL_block_comment_enters[@]}" "\
104        "${#LFBFL_block_comment_exits[@]}"
105    declare -i LFBFL_i_equal_result=$?
106    if [[ LFBFL_i_equal_result -eq 0 ]]; then
107        printf $'/!\\Problem with the definitions of languages arrays 1/!\\n'
108        return 1
109    fi
110
111    declare -ar LFBFL_line_comment_languages=(
112        "rs"
113        "sh"
114        "sql"
115        "tex"
116    )
117    declare -ar LFBFL_line_comment_prefixes=(
118        '// '
119        '# '
120        '-- '
121        '% '
122    )

```

```

123 equal "${#LFBFL_line_comment_languages[@]}"\
124      "${#LFBFL_line_comment_prefixes[@]}"
125 LFBFL_i_equal_result=$?
126 if [[ LFBFL_i_equal_result -eq 0 ]]; then
127     printf $'/!\n\Problem with the definitions of languages arrays 2/!\n\'
128     return 1
129 fi
130 # -----
131
132 # -----
133 # We generate the license header templates adapted to various text
134 # files formats.
135 # Kamoulox do endfor, Jacques Beauregard XD
136 # (C'est toujours mieux que l'inverse :) !)
137 # for do endfor -> for do done (Fort Doux Donne).
138 local LFBFL_some_license
139 local LFBFL_license_prefix
140 local LFBFL_license_prefix2
141 local LFBFL_extension
142 local LFBFL_license_file_path
143 local LFBFL_enter_string
144 local LFBFL_exit_string
145 local LFBFL_temp2
146 local LFBFL_prefix_string
147 local LFBFL_intermediate_file_path
148 local LFBFL_file_prefix
149 declare -i LFBFL_i_generate_from_template_result
150 declare -i LFBFL_i
151 for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
152     LFBFL_license_prefix="${LFBFL_source_license_prefix}"
153     LFBFL_license_prefix+="${LFBFL_some_license}"
154     LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
155     LFBFL_license_prefix2+="${LFBFL_some_license}"
156     for (
157         LFBFL_i = 0;
158         LFBFL_i < ${#LFBFL_block_comment_languages[@]};
159         ++LFBFL_i
160     ); do
161         LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
162         LFBFL_license_file_path="${LFBFL_license_prefix2}"
163         LFBFL_license_file_path+=".${LFBFL_extension}"
164         LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
165         LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
166         generate_from_template_with_block_comments\
167             "${LFBFL_license_prefix}.tpl"\
168             "${LFBFL_license_file_path}.tpl"

```

```

169     "${LFBFL_enter_string}" "${LFBFL_exit_string}"
170 LFBFL_i_generate_from_template_result=?
171 if [[ LFBFL_i_verbose -eq 1 ]]; then
172     if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
173         printf "License template %s.tpl updated.\n" \
174             "${LFBFL_license_file_path}"
175     elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
176         printf "License template %s.tpl created.\n" \
177             "${LFBFL_license_file_path}"
178     fi
179 fi
180 done
181 LFBFL_temp2=".generate_from_template_with_line_comments.temp"
182 for ( (
183     LFBFL_i = 0;
184     LFBFL_i < ${#LFBFL_line_comment_languages[@]};
185     ++LFBFL_i
186 ); do
187     LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
188     LFBFL_license_file_path="${LFBFL_license_prefix2}"
189     LFBFL_license_file_path+=".${LFBFL_extension}"
190     LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
191     LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
192     LFBFL_intermediate_file_path+="${LFBFL_temp2}"
193     LFBFL_file_prefix=""
194     if [[ "${LFBFL_extension}" == "sh" ]]; then
195         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
196     fi
197     # The sed command is here to remove trailing spaces of previously
198     # empty lines after comment prefix is added to the line.
199     generate_from_template_with_line_comments \
200         "${LFBFL_license_prefix}.tpl" \
201         "${LFBFL_license_file_path}.tpl" \
202         "${LFBFL_prefix_string}" \
203         "sed -Ei -e 's/\s*$/g' '${LFBFL_intermediate_file_path}'" \
204         "${LFBFL_file_prefix}"
205     LFBFL_i_generate_from_template_result=?
206     if [[ LFBFL_i_verbose -eq 1 ]]; then
207         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
208             printf "License template %s.tpl updated.\n" \
209                 "${LFBFL_license_file_path}"
210         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
211             printf "License template %s.tpl created.\n" \
212                 "${LFBFL_license_file_path}"
213         fi
214     fi

```

```

215     done
216 done
217 # -----
218
219 # Now that base license templates generated languages license
220 # templates in work directory, we can go there.
221 pushd_to_work_directory --trap-popd
222 can_continue_after_enhanced_pushd || return 1
223
224 local LFBFL_data_file_path="build_and_checks_variables/"
225 LFBFL_data_file_path+="repository_data.txt"
226 readonly LFBFL_data_file_path
227 local LFBFL_repository_name=""
228 grep_variable "${LFBFL_data_file_path}" repository_name\
229     --result-variable-prefix="LFBFL_"\
230     --replace-line-returns-by=""
231 local LFBFL_license=""
232 grep_variable "${LFBFL_data_file_path}" license\
233     --result-variable-prefix="LFBFL_"
234 local LFBFL_license2=""
235 grep_variable "${LFBFL_data_file_path}" license2\
236     --result-variable-prefix="LFBFL_"
237 local LFBFL_author_full_name=""
238 grep_variable "${LFBFL_data_file_path}" author_full_name\
239     --result-variable-prefix="LFBFL_"\
240     --replace-line-returns-by=""
241
242 LFBFL_license_prefix="build_and_checks_variables/temp/"
243 LFBFL_license_prefix+="license_file_header_"
244 readonly LFBFL_license_prefix
245 LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
246 readonly LFBFL_license_prefix2
247 local LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
248 readonly LFBFL_license_prefix3
249
250 # First year according to current state of git repository.
251 declare -ir LFBFL_i_first_year=$(
252     git log\
253     | grep 'Date:'\
254     | tail --lines=1\
255     | cut --delimiter=' ' --fields=8
256 )
257 # Last year according to current state of git repository.
258 declare -ir LFBFL_i_last_year=$(
259     git log\
260     | grep 'Date:'\

```

```

261 | head --lines=1\
262 | cut --delimiter=' ' --fields=8
263 )
264 local LFBFL_copyright_string
265 LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
266 LFBFL_copyright_string+="${LFBFL_author_full_name}"
267 readonly LFBFL_copyright_string
268
269 LFBFL_files_without_license_header=""
270 grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
271 files_without_license_header\
272 --result-variable-prefix="LFBFL_"
273 declare -a LFBFL_arr_files_without_license_header
274 if [[ -n "${LFBFL_files_without_license_header}" ]]; then
275 mapfile -t LFBFL_arr_files_without_license_header\
276 <<< "${LFBFL_files_without_license_header}"
277 fi
278
279 prepare_filled_license_file_block() {
280 # $1=license_file_path
281 sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
282 --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
283 -- "$1.tpl"\
284 > "$1.temp"
285 overwrite_if_not_equal "$1" "$1.temp"
286 declare -ir LFBFL_i_result=?
287 if [[ LFBFL_i_verbose -eq 1 ]]; then
288 if [[ LFBFL_i_result -eq 1 ]]; then
289 printf "License template %s updated.\n" "$1"
290 elif [[ LFBFL_i_result -eq 2 ]]; then
291 printf "License template %s created.\n" "$1"
292 fi
293 fi
294 }
295
296 prepare_filled_license_file_block_post() {
297 # $1=license_file_path
298 head --lines=-1 -- "$1"\
299 | tail --lines=+2\
300 > "$1.temp"
301 # Strange world: tail --lines=-1 was working
302 # to exclude first line.
303 # I tested it months ago, and saw with my eyes the start and end
304 # block comments removed with:
305 # head --lines=-1 "$1" | tail --lines=-1.
306 # Now, man page says to use --lines=+2 instead.

```

```

307     # Welcome in the world where AI change your environment and your
308     # documentation on the fly...
309 }
310
311 declare -Ar LFBFL_all_block_comment_languages=(
312     ["c"]="c"
313     ["css"]="c"
314     ["html"]="xml"
315     ["php"]="c"
316     ["py"]="py"
317     ["js"]="c"
318     ["ts"]="c"
319     ["xml"]="xml"
320 )
321
322 local LFBFL_key
323 local LFBFL_dest
324 local LFBFL_license_file_path2
325 declare -i LFBFL_i_not_subfile
326 declare -i LFBFL_i_not_subfile2
327 local LFBFL_file_path
328 local LFBFL_file_path2
329 local LFBFL_s_files_paths
330 declare -a LFBFL_arr_files_paths
331 declare -i LFBFL_i_is_whitelisted
332 for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]"; do
333     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
334     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
335     prepare_filled_license_file_block "${LFBFL_license_file_path}"
336     if [[ -n "${LFBFL_license2}" ]]; then
337         LFBFL_license_file_path2="${LFBFL_license_prefix3}"
338         LFBFL_license_file_path2+=".${LFBFL_dest}"
339         prepare_filled_license_file_block "${LFBFL_license_file_path2}"
340     fi
341
342     LFBFL_s_files_paths=(
343         find . -type f -iregex ".$${LFBFL_key}|${LFBFL_key}.tpl)" \
344         -printf '%P\n' \
345         | relevant_find
346     )
347     if [[ -z "${LFBFL_s_files_paths}" ]]; then
348         if [[ LFBFL_i_verbose -eq 1 ]]; then
349             LFBFL_s_format="build_licenses_templates: "
350             LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
351             # shellcheck disable=SC2059
352             printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"

```

```

353         fi
354         continue
355     fi
356     prepare_filled_license_file_block_post "${LFBFL_license_file_path}"
357     if [[ -n "${LFBFL_license2}" ]]; then
358         prepare_filled_license_file_block_post "${LFBFL_license_file_path2}"
359     fi
360     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
361     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"}; do
362         LFBFL_i_is_whitelisted=0
363         for LFBFL_file_path2 in \
364             "${LFBFL_arr_files_without_license_header[@]}";
365         do
366             if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
367                 LFBFL_i_is_whitelisted=1
368             fi
369         done
370         if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
371             continue
372         fi
373         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}.temp"
374         LFBFL_i_not_subfile=$?
375         LFBFL_i_not_subfile2=1
376         if [[ -n "${LFBFL_license2}" ]]; then
377             is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}.temp"
378             LFBFL_i_not_subfile2=$?
379         fi
380         if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
381         then
382             printf "File %s has no/wrong license header?\n"\
383                 "${LFBFL_file_path}"
384         fi
385     done
386     rm -- "${LFBFL_license_file_path}.temp"
387     if [[ -n "${LFBFL_license2}" ]]; then
388         rm -- "${LFBFL_license_file_path2}.temp"
389     fi
390 done
391
392 prepare_filled_license_file_line() {
393     # $1=LFBFL_license_file_path
394     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
395         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
396         -- "$1.tpl"\
397         > "$1.temp"
398     overwrite_if_not_equal "$1" "$1.temp"

```



```

399     declare -ir LFBFL_i_result=$?
400     if [[ LFBFL_i_verbose -eq 1 ]]; then
401         if [[ LFBFL_i_result -eq 1 ]]; then
402             printf "License template %s updated.\n" "$1"
403         elif [[ LFBFL_i_result -eq 2 ]]; then
404             printf "License template %s created.\n" "$1"
405         fi
406     fi
407 }
408
409 declare -Ar LFBFL_all_line_comment_languages=(
410     ["rlib"]="rs"
411     ["rs"]="rs"
412     ["sh"]="sh"
413     ["sql"]="sql"
414     ["tex"]="tex"
415     ["toml"]="sh"
416     ["yaml"]="sh"
417 )
418
419 for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}; do
420     LFBFL_dest=${LFBFL_all_line_comment_languages[$LFBFL_key]}
421     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
422     prepare_filled_license_file_line "${LFBFL_license_file_path}"
423     if [[ -n "${LFBFL_license2}" ]]; then
424         LFBFL_license_file_path2="${LFBFL_license_prefix3}"
425         LFBFL_license_file_path2+=".${LFBFL_dest}"
426         prepare_filled_license_file_line "${LFBFL_license_file_path2}"
427     fi
428
429     LFBFL_s_files_paths=$(
430         find . -type f -iregex ".$*\.\.(${LFBFL_key})\|${LFBFL_key}\.tpl\" \
431             -printf '%P\n' \
432             | relevant_find
433     )
434     if [[ -z "${LFBFL_s_files_paths}" ]]; then
435         if [[ LFBFL_i_verbose -eq 1 ]]; then
436             LFBFL_s_format="build_licenses_templates: "
437             LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
438             # shellcheck disable=SC2059
439             printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
440         fi
441         continue
442     fi
443     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
444     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}; do

```

```

445     LFBFL_i_is_whitelisted=0
446     for LFBFL_file_path2 in\
447         "${LFBFL_arr_files_without_license_header[@]}";
448     do
449         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
450             LFBFL_i_is_whitelisted=1
451         fi
452     done
453     if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
454         continue
455     fi
456     is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
457     LFBFL_i_not_subfile=$?
458     LFBFL_i_not_subfile2=1
459     if [[ -n "${LFBFL_license2}" ]]; then
460         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
461         LFBFL_i_not_subfile2=$?
462     fi
463     if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
464     then
465         printf "File %s has no/wrong license header?\n"\
466             "${LFBFL_file_path}"
467     fi
468 done
469 done
470 }
471
472 build_licenses_templates "$@"

```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```

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2
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```

```

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```

3.20 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

```

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8 or (at your option) any later version.
9
10 @repository_name@ is distributed
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12 but WITHOUT ANY WARRANTY;
13 without even the implied warranty of
14 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 See the GNU General Public License for more details.
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19 along with @repository_name@.
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21
22 ©Copyright @copyright_string@

```

3.21 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```

1 This file is part of @repository_name@.
2
3 @repository_name@ is free software:
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5 under the terms of the GNU Lesser General Public License
6 as published by the Free Software Foundation,
7 either version 3 of the License,
8 or (at your option) any later version.
9
10 @repository_name@ is distributed
11 in the hope that it will be useful,

```

```

12 but WITHOUT ANY WARRANTY;
13 without even the implied warranty of
14 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 See the GNU Lesser General Public License for more details.
16
17 You should have received a copy of
18 the GNU Lesser General Public License
19 along with @repository_name@.
20 If not, see <https://www.gnu.org/licenses/>.
21
22 ©Copyright @copyright_string@

```

3.22 build_and_checks_dependencies/lines_counts.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "lines_counts.sh" to
25 # "lines_counts.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=get_common_text_glob_patterns.lib.sh
31 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
32 # shellcheck source=lines_filters.lib.sh

```

```

33 source "$LFBFL_subdir/lines_filters.lib.sh"
34
35 all_code_lines() {
36     # Options:
37     #   --verbose
38     #   --work-directory=""
39     declare -i LFBFL_i_verbose=0
40     get_verbose_option "$@"
41     local LFBFL_work_directory=""
42     get_work_directory_option "$@"
43     declare -a LFBFL_return_traps_stack
44     init_return_trap
45     pushd_to_work_directory --trap-popd
46     can_continue_after_enhanced_pushd || return 1
47
48     enhanced_set_shell_option pipefail --trap-unset
49
50     get_COMMON_TEXT_FILES_GLOB_PATTERNS
51     local LFBFL_pattern
52     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}; do
53         if [[ LFBFL_i_verbose -eq 1 ]]; then
54             printf "all_code_lines: Iterating on pattern: %s.\n" \
55                 "${LFBFL_pattern}"
56         fi
57         find . -type f -name "${LFBFL_pattern}" -printf '%P\n' \
58             | tr '\n' '\0' \
59             | xargs -0 grep --invert-match --perl-regexp --with-filename \
60                 -- 'a(?!a)a'
61     done
62 }
63
64 all_self_code_lines() {
65     # Options:
66     #   --verbose
67     #   --work-directory=""
68     declare -i LFBFL_i_verbose=0
69     get_verbose_option "$@"
70     local LFBFL_work_directory=""
71     get_work_directory_option "$@"
72     declare -a LFBFL_return_traps_stack
73     init_return_trap
74     pushd_to_work_directory --trap-popd
75     can_continue_after_enhanced_pushd || return 1
76
77     enhanced_set_shell_option pipefail --trap-unset
78

```

```

79 declare -a LFBFL_common_options=("$@")
80 keep_options LFBFL_common_options --verbose
81
82 # below and another lines, see SC2145 false positive
83 all_code_lines "${LFBFL_common_options[@]}" \
84     | relevant_grep\
85     | not_license_grep\
86     | not_main_tex_grep\
87     | not_main_html_grep
88 }
89
90 # shellcheck disable=SC2120
91 all_self_empty_code_lines() {
92     # Options:
93     #   --verbose
94     #   --work-directory=""
95     declare -i LFBFL_i_verbose=0
96     get_verbose_option "$@"
97     local LFBFL_work_directory=""
98     get_work_directory_option "$@"
99     declare -a LFBFL_return_traps_stack
100     init_return_trap
101     pushd_to_work_directory --trap-popd
102     can_continue_after_enhanced_pushd || return 1
103
104     enhanced_set_shell_option pipefail --trap-unset
105
106     declare -a LFBFL_common_options=("$@")
107     keep_options LFBFL_common_options --verbose
108
109     all_self_code_lines "${LFBFL_common_options[@]}" \
110         | empty_lines_after_file_name
111 }
112
113 # shellcheck disable=SC2120
114 all_self_not_empty_code_lines() {
115     # Options:
116     #   --verbose
117     #   --work-directory=""
118     declare -i LFBFL_i_verbose=0
119     get_verbose_option "$@"
120     local LFBFL_work_directory=""
121     get_work_directory_option "$@"
122     declare -a LFBFL_return_traps_stack
123     init_return_trap
124     pushd_to_work_directory --trap-popd

```

```

125     can_continue_after_enhanced_pushd || return 1
126
127     enhanced_set_shell_option pipefail --trap-unset
128
129     declare -a LFBFL_common_options=("$@")
130     keep_options LFBFL_common_options --verbose
131
132     all_self_code_lines "${LFBFL_common_options[@]}" \
133         | not_empty_lines_after_file_name
134 }
135
136 code_lines_count_all() {
137     # Options:
138     #   --verbose
139     #   --work-directory=""
140     declare -i LFBFL_i_verbose=0
141     get_verbose_option "$@"
142     local LFBFL_work_directory=""
143     get_work_directory_option "$@"
144     declare -a LFBFL_return_traps_stack
145     init_return_trap
146     pushd_to_work_directory --trap-popd
147     can_continue_after_enhanced_pushd || return 1
148
149     enhanced_set_shell_option pipefail --trap-unset
150
151     declare -gi i_code_lines_count_all_result
152     i_code_lines_count_all_result=$(
153         all_self_code_lines\
154         | wc --lines
155     )
156     # The printf is for command line use.
157     # A script should use the result value instead.
158     # shellcheck disable=SC2248
159     printf "%s\n" ${i_code_lines_count_all_result}
160 }
161
162 code_lines_count_empty() {
163     # Options:
164     #   --verbose
165     #   --work-directory=""
166     declare -i LFBFL_i_verbose=0
167     get_verbose_option "$@"
168     local LFBFL_work_directory=""
169     get_work_directory_option "$@"
170     declare -a LFBFL_return_traps_stack

```

```

171     init_return_trap
172     pushd_to_work_directory --trap-popd
173     can_continue_after_enhanced_pushd || return 1
174
175     enhanced_set_shell_option pipefail --trap-unset
176
177     declare -gi i_code_lines_count_empty_result
178     # shellcheck disable=SC2119
179     i_code_lines_count_empty_result=$(
180         all_self_empty_code_lines\
181         | wc --lines
182     )
183     # The printf is for command line use.
184     # A script should use the result value instead.
185     # shellcheck disable=SC2248
186     printf "%s\n" "${i_code_lines_count_empty_result}"
187 }
188
189 code_lines_count_not_empty() {
190     # Options:
191     #   --verbose
192     #   --work-directory=""
193     declare -i LFBFL_i_verbose=0
194     get_verbose_option "$@"
195     local LFBFL_work_directory=""
196     get_work_directory_option "$@"
197     declare -a LFBFL_return_traps_stack
198     init_return_trap
199     pushd_to_work_directory --trap-popd
200     can_continue_after_enhanced_pushd || return 1
201
202     enhanced_set_shell_option pipefail --trap-unset
203
204     declare -gi i_code_lines_count_not_empty_result
205     # shellcheck disable=SC2119
206     i_code_lines_count_not_empty_result=$(
207         all_self_not_empty_code_lines\
208         | wc --lines
209     )
210     # The printf is for command line use.
211     # A script should use the result value instead.
212     # shellcheck disable=SC2248
213     printf "%s\n" "${i_code_lines_count_not_empty_result}"
214 }

```


3.23 build_and_checks_dependencies/lines_filters.lib.sh

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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11 # DevOrSysAdminScripts is distributed
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15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "lines_filters.sh" to
25 # "lines_filters.lib.sh".
26
27 # The goal of this file is to have some common line filters
28 # for bash processing and pipes.
29 # As is the case for other .sh files in this repository,
30 # it can be copied somewhere in your home directory
31 # and sourced in your .bashrc.
32
33 # shellcheck source=common_options.lib.sh
34 source "./build_and_checks_dependencies/common_options.lib.sh"
35
36 ll_wc() {
37     # current options to wc:
38     # -c --bytes
39     # -m --chars
40     # -l --lines
41     # --files0-from # Incompatible with cat,
42     #                  # another thing intentional by ?
43     #                  # I don't see the meaning of "files0".
44     # -L --max-line-length
```

```

45 # -w --words
46 # --help
47 # --version
48 # New option by Laurent Lyaudet (and probably other before):
49 # -n --no-filenames display only number(s)
50 # I think I did that the first time many years ago (Teliae ?).
51 # Since then, they added it as ignored... (-n)
52 enhanced_set_shell_option pipefail\
53     && trap 'enhanced_unset_shell_option pipefail' RETURN
54 declare -a ll_wc_var_args=()
55 declare -i ll_wc_var_i_number_only=0
56 declare -i ll_wc_var_i=0
57 for ll_wc_var_arg in "$@"; do
58     # printf "%s\n" "${ll_wc_var_arg}"
59     if [[ "${ll_wc_var_arg}" == "-n" ]]; then
60         ll_wc_var_i_number_only=1
61     elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
62         ll_wc_var_i_number_only=1
63     elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
64         ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
65         ll_wc_var_i=$((ll_wc_var_i + 1))
66     elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
67         ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
68         ll_wc_var_i=$((ll_wc_var_i + 1))
69     fi
70 done
71 # typeset -p ll_wc_var_args
72 # printf "%s\n" "${ll_wc_var_i_number_only}"
73 if [[ ll_wc_var_i_number_only -gt 0 ]]; then
74     # Too simple code, only the base use case is handled now.
75     # Convenient but incomplete.
76     # With more time,
77     # the simpler would be to submit a patch to GNU coreutils,
78     # unless blockade from assholes.
79     # Otherwise, it may be more simply handled in PHP or Python
80     # (argparse, etc.).
81     # You'll need to start by improving cat first,
82     # with --files0-from handling.
83     # wc does not display file_name when stream...
84     # hence cat of last arg "${!#}"
85     cat -- "${!#}"\
86         | wc "${ll_wc_var_args[@]}"
87 else
88     # printf "normal\n"
89     wc "${ll_wc_var_args[@]}"
90 fi

```

```

91 }
92
93 # I submitted a patch for wc :).
94 # printf "https://domain/list-path/2024-05/msg00013.html" | \
95 #   sed --expression='s|domain|lists.gnu.org|' \
96 #     --expression='s|list-path|archive/html/bug-coreutils|'
97 # I found another way to have a long URL fit on 70 characters.
98 # If it is merged, the following would be better:
99 # If --no-filenames option available,
100 # we redefine ll_wc command as an alias.
101 wc_help_text=$(wc --help)
102 if [[ "${wc_help_text}" == *" --no-filenames *" ]]; then
103     alias ll_wc='wc'
104 fi
105
106 # Trace avant le (2e ?) renommage
107 # de wget_sha512.sh à wget_sha512.lib.sh.
108 # $ ll_wc wget_sha512.sh
109 # 41 189 1224 wget_sha512.sh
110 # $ ll_wc -n wget_sha512.sh
111 # 41 189 1224
112 # $ ll_wc -l wget_sha512.sh
113 # 41 wget_sha512.sh
114 # $ ll_wc -l -n wget_sha512.sh
115 # 41
116
117 in_place_grep(){
118     # Same arguments and options as grep.
119     declare -r LFBFL_temp="${!#}.in_place_grep.temp"
120     grep "$@" \
121         > "${LFBFL_temp}"
122     # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
123     # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
124     # printf "%s\n" "${LFBFL_i_lines_before}"
125     # printf "%s\n" "${LFBFL_i_lines_after}"
126     # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
127     # Almost as fast (on large files) and correct in all cases :).
128     # Thanks Paul Eggert for the suggestion :).
129     # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
130     if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
131         rm -- "${LFBFL_temp}"
132         return
133     fi
134     mv -- "${LFBFL_temp}" "${!#}"
135 }
136

```

```

137 grep_fixed_string_with_anchor() {
138     # $1=file_path
139     # $2=some_fixed_string
140     # Options:
141     #     --enforce-line-starts-with-fixed-string
142     #     --enforce-line-ends-with-fixed-string
143     # Apart from $1 and $2 and the options above,
144     # all other arguments are considered grep_options like --quiet
145     # for example.
146     # But note that --fixed-strings will be added below,
147     # you don't need to supply it.
148     declare -a LFBFL_grep_options=()
149     declare -a LFBFL_grep_options2=()
150     declare -i LFBFL_i=0
151     declare -i LFBFL_i_enforce_line_starts=0
152     declare -i LFBFL_i_enforce_line_ends=0
153     declare -i LFBFL_i_quiet=0
154     local LFBFL_argument
155     for LFBFL_argument in "$@"; do
156         ((++LFBFL_i))
157         if [[ LFBFL_i -lt 3 ]]; then
158             continue
159         fi
160         if [[
161             "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"
162         ]]; then
163             LFBFL_i_enforce_line_starts=1
164             continue
165         fi
166         if [[
167             "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
168         ]]; then
169             LFBFL_i_enforce_line_ends=1
170             continue
171         fi
172         LFBFL_grep_options+=("${LFBFL_argument}")
173         if [[
174             "${LFBFL_argument}" == "-q"
175             || "${LFBFL_argument}" == "--quiet"
176         ]]; then
177             LFBFL_i_quiet=1
178             continue
179         fi
180         LFBFL_grep_options2+=("${LFBFL_argument}")
181     done
182

```

```

183 if [[
184     LFBFL_i_enforce_line_starts -eq 0
185     && LFBFL_i_enforce_line_ends -eq 0
186 ]]; then
187     grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
188     return
189 fi
190 declare -r LFBFL_s_lines=$(
191     grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
192         "$2" "$1"
193 )
194 declare -i LFBFL_i_result=1
195 if [[ -z "${LFBFL_s_lines}" ]]; then
196     # shellcheck disable=SC2248
197     return ${LFBFL_i_result}
198 fi
199 declare -a LFBFL_arr_lines
200 mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
201 readonly LFBFL_arr_lines
202
203 declare -i LFBFL_i_fixed_string_length=${#2}
204 local LFBFL_some_line
205 local LFBFL_line_part
206 for LFBFL_some_line in "${LFBFL_arr_lines[@]"; do
207     if [[ "${LFBFL_some_line}" == "$2" ]]; then
208         LFBFL_i_result=0
209         if [[ LFBFL_i_quiet -eq 1 ]]; then
210             break
211         fi
212         printf "%s\n" "${LFBFL_some_line}"
213         continue
214     fi
215     if [[
216         LFBFL_i_enforce_line_starts -eq 1
217         && LFBFL_i_enforce_line_ends -eq 1
218     ]]; then
219         continue
220     fi
221     if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
222         LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
223     fi
224     if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
225         LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
226     fi
227     if [[ "${LFBFL_line_part}" == "$2" ]]; then
228         LFBFL_i_result=0

```

```

229         if [[ LFBFL_i_quiet -eq 1 ]]; then
230             break
231         fi
232         printf "%s\n" "${LFBFL_some_line}"
233     fi
234 done
235 # shellcheck disable=SC2248
236 return ${LFBFL_i_result}
237 }
238
239 grep_variable() {
240     # $1=file_path
241     # $2=variable_name
242     # Options :
243     # --replace-line-returns-by=""
244     # --result-variable-prefix="LFBFL_" for example
245     declare -r LFBFL_regexp="(?!<^$2=).*$"
246     # printf "%s\n" "${LFBFL_regexp}"
247     local LFBFL_variable_value
248     local LFBFL_s_replace
249     get_some_option LFBFL_s_replace --replace-line-returns-by ' ' $'\n' 1 "$@"
250     if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
251         # sed -Ez -e...
252         LFBFL_variable_value=$(
253             grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1" \
254             | sed --null-data --regexp-extended \
255             --expression='s/\n/' "${LFBFL_s_replace}"'/Mg'
256         )
257         # We remove the trailing replace string,
258         # since the intent is only to replace the intermediate \n.
259         if [[ -n "${LFBFL_s_replace}" ]]; then
260             LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
261         fi
262     else
263         LFBFL_variable_value=$(
264             grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
265         )
266     fi
267     local LFBFL_prefix=""
268     get_some_option LFBFL_prefix --result-variable-prefix ' ' ' ' 1 "$@"
269     declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
270     # printf "%s : %s\n" "${LFBFL_variable_name}" "${LFBFL_variable_value}"
271     # The declare line may be dropped by a local or declare
272     # in outer scope.
273     # declare -g "${LFBFL_variable_name}"
274     # So the assignment must be done on another line.

```

```

275     printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
276     # printf "%s\n" "${!LFBFL_variable_name}"
277 }
278
279 empty_lines() {
280     grep '^$'
281 }
282
283 not_empty_lines() {
284     grep --invert-match '^$'
285 }
286
287 space_starting_lines() {
288     grep '^ '
289 }
290
291 not_space_starting_lines() {
292     grep --invert-match '^ '
293 }
294
295 # Space starting or empty lines = spampty
296 spampty_lines() {
297     grep --extended-regexp '^( |$)'
298 }
299
300 not_spampty_lines() {
301     grep '^[^ ]'
302     # grep --extended-regexp --invert-match '^( |$)'
303 }
304
305 all_space_lines() {
306     grep '^[ ]\+$'
307 }
308
309 not_all_space_lines() {
310     grep --invert-match '^[ ]\+$'
311 }
312
313 # All space or empty lines = aspmpty
314 aspmpty_lines() {
315     grep '^[ ]*$'
316 }
317
318 not_aspmpty_lines() {
319     grep '[^ ]'
320     # grep --invert-match '^[ ]*$'

```

```

321 }
322
323 empty_lines_after_file_name() {
324     grep '^[:]\+:$'
325 }
326
327 not_empty_lines_after_file_name() {
328     grep --invert-match '^[:]\+:$'
329 }
330
331 space_starting_lines_after_file_name() {
332     grep '^[:]\+:'
333 }
334
335 not_space_starting_lines_after_file_name() {
336     grep --invert-match '^[:]\+:'
337 }
338
339 spampty_lines_after_file_name() {
340     grep --extended-regexp '^[:]\+:( |$)'
341 }
342
343 not_spampty_lines_after_file_name() {
344     # The two variants results differ on lines without filename before.
345     # Since the intent of the function is only on lines "blabla.sh:whatever",
346     # use the fastest one. I didn't do extensive testing and just assumed
347     # that the "positive" one is faster as usual when querying database.
348     # But there is no index here ;) XD.
349     grep '^[:]\+:[^ ]'
350     # grep --extended-regexp --invert-match '^[:]\+:( |$)'
351 }
352
353 all_space_lines_after_file_name() {
354     grep '^[:]\+:[ ]\+$'
355 }
356
357 not_all_space_lines_after_file_name() {
358     grep --invert-match '^[:]\+:[ ]\+$'
359 }
360
361 aspmpty_lines_after_file_name() {
362     grep '^[:]\+:[ ]*$'
363 }
364
365 not_aspmpty_lines_after_file_name() {
366     grep --invert-match '^[:]\+:[ ]*$'

```



```

367 }
368
369 # If you need to test the previous 20 functions,
370 # look at file spampty_data.txt in miscellaneous.
371 # And give a try at something like:
372 # cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
373 # -E = --show-ends :). Nice command :).
374
375 not_JS_dependencies_find() {
376     grep --extended-regexp --invert-match\
377         --regexp="(^|/)node_modules/"\
378         --regexp="(^|/)package-lock\.json$"
379 }
380
381 not_JS_dependencies_grep() {
382     grep --extended-regexp --invert-match\
383         --regexp="^([^:]+)/?node_modules/"\
384         --regexp="^([^:]+)/?package-lock\.json:"
385 }
386
387 not_dependencies_find() {
388     not_JS_dependencies_find
389 }
390
391 not_dependencies_grep() {
392     not_JS_dependencies_grep
393 }
394
395 not_php_cache_find() {
396     grep --extended-regexp --invert-match\
397         --regexp="(^|/)\.phpdoc/cache/"
398 }
399
400 not_php_cache_grep() {
401     grep --extended-regexp --invert-match\
402         --regexp="^([^:]+)/?\.phpdoc/cache/"
403 }
404
405 not_python_cache_find() {
406     grep --extended-regexp --invert-match\
407         --regexp="(^|/).__pycache__/"\
408         --regexp="(^|/)\.mypy_cache/"\
409         --regexp="(^|/)\.ruff_cache/"
410 }
411
412 not_python_cache_grep() {

```

```

413     grep --extended-regexp --invert-match\
414         --regexp="^([^\:]+)/?__pycache__/"\
415         --regexp="^([^\:]+)/?.mypy_cache/"\
416         --regexp="^([^\:]+)/?.ruff_cache/"
417 }
418
419 not_cache_find() {
420     not_php_cache_find\
421     | not_python_cache_find
422 }
423
424 not_cache_grep() {
425     not_php_cache_grep\
426     | not_python_cache_grep
427 }
428
429 not_git_find() {
430     grep --extended-regexp --invert-match "^(^|/)\.git/"
431 }
432
433 not_git_grep() {
434     grep --extended-regexp --invert-match "^(^[\:]+)/?.git/"
435 }
436
437 not_archive_find() {
438     grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
439 }
440
441 not_archive_grep() {
442     grep --extended-regexp --invert-match\
443         --regexp="^[^\:]* (\.gz|\.rar|\.tar|\.tgz|\.whl) : "
444 }
445
446 not_license_find() {
447     grep --extended-regexp --invert-match\
448         --regexp=" (^|/) COPYING$ "\
449         --regexp=" (^|/) COPYING\.LESSER$ "
450 }
451
452 not_license_grep() {
453     grep --extended-regexp --invert-match\
454         --regexp="^ ([^\:]+)/?COPYING: "\
455         --regexp="^ ([^\:]+)/?COPYING\.LESSER: "
456 }
457
458 not_main_tex_find() {

```

```

459     local LFBFL_repository_name=""
460     grep_variable build_and_checks_variables/repository_data.txt\
461         repository_name\
462         --result-variable-prefix="LFBFL_"\
463         --replace-line-returns-by=""
464     grep --extended-regexp --invert-match\
465         --regexp="(^|/){LFBFL_repository_name}\.tex$"
466 }
467
468 not_main_tex_grep() {
469     local LFBFL_repository_name=""
470     grep_variable build_and_checks_variables/repository_data.txt\
471         repository_name\
472         --result-variable-prefix="LFBFL_"\
473         --replace-line-returns-by=""
474     grep --extended-regexp --invert-match\
475         --regexp="^([^\:]+)/?${LFBFL_repository_name}\.tex:"
476 }
477
478 not_main_html_find() {
479     local LFBFL_repository_name=""
480     grep_variable build_and_checks_variables/repository_data.txt\
481         repository_name\
482         --result-variable-prefix="LFBFL_"\
483         --replace-line-returns-by=""
484     grep --extended-regexp --invert-match\
485         --regexp="(^|/){LFBFL_repository_name}\.html$"
486 }
487
488 not_main_html_grep() {
489     local LFBFL_repository_name=""
490     grep_variable build_and_checks_variables/repository_data.txt\
491         repository_name\
492         --result-variable-prefix="LFBFL_"\
493         --replace-line-returns-by=""
494     grep --extended-regexp --invert-match\
495         --regexp="^([^\:]+)/?${LFBFL_repository_name}\.html:"
496 }
497
498 not_temp_file_find() {
499     grep --extended-regexp --invert-match\
500         --regexp="(^|/){build_and_checks_variables/temp/"
501 }
502
503 not_temp_file_grep() {
504     grep --extended-regexp --invert-match\

```

```

505     --regexp="^([^.:]+)/?build_and_checks_variables/temp/[^.:]*:"
506 }
507
508 not_unchecked_builds_find() {
509     grep --extended-regexp --invert-match\
510         --regexp="(^|/)\.phpdoc/build/"
511 }
512
513 not_unchecked_builds_grep() {
514     grep --extended-regexp --invert-match\
515         --regexp="^([^.:]+)/?.phpdoc/build/"
516 }
517
518 relevant_find() {
519     enhanced_set_shell_option pipefail\
520     && trap 'enhanced_unset_shell_option pipefail' RETURN
521     not_dependencies_find\
522     | not_cache_find\
523     | not_git_find\
524     | not_archive_find\
525     | not_temp_file_find\
526     | not_unchecked_builds_find
527 }
528
529 relevant_grep() {
530     enhanced_set_shell_option pipefail\
531     && trap 'enhanced_unset_shell_option pipefail' RETURN
532     not_dependencies_grep\
533     | not_cache_grep\
534     | not_git_grep\
535     | not_archive_grep\
536     | not_temp_file_grep\
537     | not_unchecked_builds_grep
538 }

```

3.24 build_and_checks_dependencies/lines_maps.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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6  # under the terms of the GNU General Public License
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8  # either version 3 of the License,
9  # or (at your option) any later version.

```

```

10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "lines_maps.sh" to "lines_maps.lib.sh".
25
26 # https://en.wikibooks.org/wiki/Unicode/Character\_reference/2000-2FFF
27 # Some invisible unicode characters that are a problem:
28 # 2000-200F 2028-202F 205F 2060-206F
29 # [
30 #   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
31 #   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
32 #   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
33 #   '205F',
34 #   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
35 #   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
36 # ]
37 replace_non_ascii_spaces() {
38     sed\
39         --expression='s/\xc2\xa0/ /g'\
40         --expression='s/\xe2\x80\x80/ /g'\
41         --expression='s/\xe2\x80\x81/ /g'\
42         --expression='s/\xe2\x80\x82/ /g'\
43         --expression='s/\xe2\x80\x83/ /g'\
44         --expression='s/\xe2\x80\x84/ /g'\
45         --expression='s/\xe2\x80\x85/ /g'\
46         --expression='s/\xe2\x80\x86/ /g'\
47         --expression='s/\xe2\x80\x87/ /g'\
48         --expression='s/\xe2\x80\x88/ /g'\
49         --expression='s/\xe2\x80\x89/ /g'\
50         --expression='s/\xe2\x80\x8a/ /g'\
51         --expression='s/\xe2\x80\x8b/ /g'\
52         --expression='s/\xe2\x80\x8c/ /g'\
53         --expression='s/\xe2\x80\x8d/ /g'\
54         --expression='s/\xe2\x80\x8e/ /g'\
55         --expression='s/\xe2\x80\x8f/ /g'\

```

```

56     --expression=' s/\xe2\x80\xa8/ /g' \
57     --expression=' s/\xe2\x80\xa9/ /g' \
58     --expression=' s/\xe2\x80\xaa/ /g' \
59     --expression=' s/\xe2\x80\xab/ /g' \
60     --expression=' s/\xe2\x80\xac/ /g' \
61     --expression=' s/\xe2\x80\xad/ /g' \
62     --expression=' s/\xe2\x80\xae/ /g' \
63     --expression=' s/\xe2\x80\xaf/ /g' \
64     --expression=' s/\xe2\x81\x9f/ /g' \
65     --expression=' s/\xe2\x81\xa0/ /g' \
66     --expression=' s/\xe2\x81\xa1/ /g' \
67     --expression=' s/\xe2\x81\xa2/ /g' \
68     --expression=' s/\xe2\x81\xa3/ /g' \
69     --expression=' s/\xe2\x81\xa4/ /g' \
70     --expression=' s/\xe2\x81\xa5/ /g' \
71     --expression=' s/\xe2\x81\xa6/ /g' \
72     --expression=' s/\xe2\x81\xa7/ /g' \
73     --expression=' s/\xe2\x81\xa8/ /g' \
74     --expression=' s/\xe2\x81\xa9/ /g' \
75     --expression=' s/\xe2\x81\xaa/ /g' \
76     --expression=' s/\xe2\x81\xab/ /g' \
77     --expression=' s/\xe2\x81\xac/ /g' \
78     --expression=' s/\xe2\x81\xad/ /g' \
79     --expression=' s/\xe2\x81\xae/ /g' \
80     --expression=' s/\xe2\x81\xaf/ /g' \
81 }

```

3.25 build_and_checks_dependencies/overwrite_if_not_equal.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #

```

```

18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "overwrite_if_not_equal.sh"
25 # to "overwrite_if_not_equal.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30
31 overwrite_if_not_equal() {
32     # $1=target_file_path
33     # $2=temp_file_path
34     # Options:
35     #   --keep-temp : keep the temporary file
36     #   --quiet : we even don't want the message of diff saying that
37     #               files are distinct
38     #   --tree-mode : when the two files are the output of tree command,
39     #               we ignore if all the differences are only on directories timestamps.
40     # Return 0 target was the same
41     #       1 target was different
42     #       2 target did not already exist
43     declare -i LFBFL_i_keep_temp
44     get_some_flag LFBFL_i_keep_temp --keep-temp 1 "$@"
45     declare -i LFBFL_i_quiet
46     get_some_flag LFBFL_i_quiet --quiet 1 "$@"
47     declare -i LFBFL_i_tree_mode
48     get_some_flag LFBFL_i_tree_mode --tree-mode 1 "$@"
49
50     if [[ ! -f "$1" ]]; then
51         if [[ LFBFL_i_keep_temp -eq 1 ]]; then
52             cp --preserve=mode,ownership,timestamps -- "$2" "$1"
53         else
54             mv -- "$2" "$1"
55         fi
56         return 2
57     fi
58     declare -i LFBFL_i_is_equal
59     if [[ LFBFL_i_tree_mode -eq 1 ]]; then
60         # Not setting pipefail since the result would still be incorrect.
61         # shellcheck disable=SC2312
62         diff -- "$1" "$2" \
63             | grep --extended-regexp "^(>|<)" \

```

```

64     | grep --invert-match "/"$
65     LFBFL_i_is_equal=${PIPESTATUS[2]}
66     if [[ LFBFL_i_is_equal -eq 1 ]]; then
67         grep --only-matching --regexp="^[^ ]*/$" -- "$1"\
68         > "overwrite_if_not_equal.file1.temp"
69         grep --only-matching --regexp="^[^ ]*/$" -- "$2"\
70         > "overwrite_if_not_equal.file2.temp"
71         if [[ LFBFL_i_quiet -eq 1 ]]; then
72             diff overwrite_if_not_equal.file1.temp\
73                 overwrite_if_not_equal.file2.temp\
74                 > /dev/null
75         else
76             diff overwrite_if_not_equal.file1.temp\
77                 overwrite_if_not_equal.file2.temp
78         fi
79         LFBFL_i_is_equal=1-$?
80         rm overwrite_if_not_equal.file1.temp\
81             overwrite_if_not_equal.file2.temp
82     fi
83 else
84     if [[ LFBFL_i_quiet -eq 1 ]]; then
85         diff --brief -- "$1" "$2" > /dev/null
86     else
87         diff --brief -- "$1" "$2"
88     fi
89     LFBFL_i_is_equal=1-$?
90 fi
91 if [[ LFBFL_i_is_equal -eq 1 ]]; then
92     if [[ LFBFL_i_keep_temp -eq 0 ]]; then
93         rm -- "$2"
94     fi
95     return 0
96 fi
97 if [[ LFBFL_i_keep_temp -eq 1 ]]; then
98     cp --preserve=mode,ownership,timestamps -- "$2" "$1"
99 else
100     mv -- "$2" "$1"
101 fi
102 return 1
103 }

```

3.26 build_and_checks_dependencies/phpmd_baseline.xslt

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <!--
3 This file is part of DevOrSysAdminScripts.

```



```

4
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23
24 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
25 -->
26 <xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
27     version="1.1">
28     <xsl:output method="xml"/>
29     <xsl:template match="phpmd-baseline">
30         <xsl:copy>
31             <xsl:text>
32 </xsl:text>
33             <xsl:apply-templates select="node()">
34                 <xsl:sort select="@file"/>
35                 <xsl:sort select="@rule"/>
36                 <xsl:sort select="@method"/>
37             </xsl:apply-templates>
38         </xsl:copy>
39     </xsl:template>
40     <xsl:template match="violation">
41         <xsl:text> </xsl:text>
42         <xsl:copy>
43             <xsl:copy-of select="@*" />
44         </xsl:copy>
45         <xsl:text>
46 </xsl:text>
47     </xsl:template>
48     <!--
49     Remove all indentation and line returns in text nodes,

```

```

50     because otherwise it will be output first after sorting above.
51     -->
52     <xsl:template match="text()"></xsl:template>
53 </xsl:stylesheet>

```

3.27 build_and_checks_dependencies/python_black_complement.lib.sh

```

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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_black_complement.sh"
25 # to "python_black_complement.lib.sh".
26
27 # shellcheck source=common_options.lib.sh
28 source "../build_and_checks_dependencies/common_options.lib.sh"
29
30 # It was rejected to enhance black with this currently.
31 # https://github.com/psf/black/issues/2370
32 # And moreover, at some point, I saw the reverse rule
33 # in black or pylint to add an empty line...
34 # This regexp may give false positives,
35 # but that's not the end of the world.
36 check_no_empty_line_after_python_function_docstring() {
37     # Options:
38     #     --verbose
39     #     --work-directory=""

```

```

40 declare -i LFBFL_i_verbose=0
41 get_verbose_option "$@"
42 local LFBFL_work_directory=""
43 get_work_directory_option "$@"
44 declare -a LFBFL_return_traps_stack
45 init_return_trap
46 pushd_to_work_directory --trap-popd
47 can_continue_after_enhanced_pushd || return 1
48
49 enhanced_set_bash_option globstar --trap-unset
50
51 [[ LFBFL_i_verbose -eq 1 ]]\
52   && printf "Checking empty lines after Python function docstrings\n"
53   pcre2grep --multiline --\
54     '$def ([^"]|"(?!"))*"'([^"]|"(?!"))*"'"\n\n(?!s*def)'\
55     **/*.py
56 }
57
58 python_black_complement() {
59   # Options:
60   #   --verbose
61   #   --work-directory=""
62   declare -i LFBFL_i_verbose=0
63   get_verbose_option "$@"
64   local LFBFL_work_directory=""
65   get_work_directory_option "$@"
66   declare -a LFBFL_return_traps_stack
67   init_return_trap
68   pushd_to_work_directory --trap-popd
69   can_continue_after_enhanced_pushd || return 1
70
71   declare -a LFBFL_common_options=("$@")
72   keep_options LFBFL_common_options --verbose
73
74   check_no_empty_line_after_python_function_docstring\
75     "${LFBFL_common_options[@]}"
76 }

```

3.28 build_and_checks_dependencies/python_isort_complement.lib.sh

```

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2 # This file is part of DevOrSysAdminScripts.
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```

```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_isort_complement.sh"
25 # to "python_isort_complement.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=lines_filters.lib.sh
31 source "${LFBFL_subdir}/lines_filters.lib.sh"
32 # shellcheck source=overwrite_if_not_equal.lib.sh
33 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
34
35 check_collections_abc_place() {
36     # Options:
37     #   --verbose
38     #   --work-directory=""
39     # This function should be useless in a few months:
40     # https://github.com/PyCQA/isort/issues/2250
41     declare -i LFBFL_i_verbose=0
42     get_verbose_option "$@"
43     local LFBFL_work_directory=""
44     get_work_directory_option "$@"
45     declare -a LFBFL_return_traps_stack
46     init_return_trap
47     pushd_to_work_directory --trap-popd
48     can_continue_after_enhanced_pushd || return 1
49
50     [[ LFBFL_i_verbose -eq 1 ]]\
51     && printf\
52         "Checking that import of _collections_abc is at the right place.\n"

```

```

53
54     enhanced_set_shell_option pipefail --trap-unset
55
56     declare -r LFBFL_temp=".check_collections_abc_place.temp"
57     local LFBFL_file_path
58     declare -r LFBFL_s_files_paths=$(
59         find . -type f -name "*.py" -printf '%P\n'\
60         | relevant_find
61     )
62     if [[ -z "${LFBFL_s_files_paths}" ]]; then
63         [[ LFBFL_i_verbose -eq 1 ]]\
64         && printf "check_collections_abc_place: No file found.\n"
65         return
66     fi
67     declare -a LFBFL_arr_files_paths
68     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
69     readonly LFBFL_arr_files_paths
70     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
71         sed --regexp-extended --null-data\
72             --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
73             -- "${LFBFL_file_path}"\
74             > "${LFBFL_file_path}${LFBFL_temp}"
75         overwrite_if_not_equal "${LFBFL_file_path}"\
76             "${LFBFL_file_path}${LFBFL_temp}"
77     done
78 }
79
80 python_isort_complement() {
81     # Options:
82     #     --verbose
83     #     --work-directory=""
84     declare -i LFBFL_i_verbose=0
85     get_verbose_option "$@"
86     local LFBFL_work_directory=""
87     get_work_directory_option "$@"
88     declare -a LFBFL_return_traps_stack
89     init_return_trap
90     pushd_to_work_directory --trap-popd
91     can_continue_after_enhanced_pushd || return 1
92
93     declare -a LFBFL_common_options=("$@")
94     keep_options LFBFL_common_options --verbose
95
96     check_collections_abc_place "${LFBFL_common_options[@]}"
97 }

```

3.29 build_and_checks_dependencies/repository_data.lib.sh

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
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23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.lib.sh
27 source "${LFBFL_subdir}/common_options.lib.sh"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30
31 # /\ You need to activate extglob before parsing this file. /\
32 # And I'm kind enough to activate it inside functions that need it below.
33
34 get_upgrade_venvs_time_interval_in_seconds() {
35     # $1=data_file_path
36     # Options:
37     #   --verbose
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     declare -a LFBFL_return_traps_stack
41     init_return_trap
42
43     enhanced_set_bash_option extglob --trap-unset
44 }
```

```

45 grep_variable "$1"\
46     upgrade_venvs_time_interval_in_seconds\
47     --result-variable-prefix="LFBFL_"
48
49 [[ LFBFL_i_verbose -eq 1 ]]\
50     && printf "You say \"%s\".\n\"\
51         "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
52
53 case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
54     +([0-9])) : ;;
55     wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
56     watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
57     *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
58 esac
59
60 [[ LFBFL_i_verbose -eq 1 ]]\
61     && printf "I say \"%s\".\n\"\
62         "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
63 }
64
65 grep_variable_with_multiple_files(){
66     # $1=data_file_path
67     # $2=variable_name
68     # Options :
69     # --result-variable-prefix="LFBFL_" for example
70     local LFBFL_prefix=""
71     get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
72     grep_variable "$1" "$2" --result-variable-prefix="${LFBFL_prefix}"
73     declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
74     declare -a LFBFL_arr
75     if [[ -z "${!LFBFL_variable_name}" ]]; then
76         return
77     fi
78     mapfile -t LFBFL_arr <<< "${!LFBFL_variable_name}"
79     local LFBFL_s=""
80     local LFBFL_file_path
81     local LFBFL_previous_file_path
82     for LFBFL_file_path in "${LFBFL_arr[@]}"; do
83         if [[
84             -n "${LFBFL_previous_file_path}"
85             && "${LFBFL_previous_file_path}" != */
86         ]]; then
87             LFBFL_s+=$'\n'
88         fi
89         LFBFL_s+="${LFBFL_file_path}"
90         LFBFL_previous_file_path="${LFBFL_file_path}"

```

```

91     done
92     printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_s}"
93 }

```

3.30 build_and_checks_dependencies/shell_checks_complement.lib.sh

```

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23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_black_complement.sh"
25 # to "python_black_complement.lib.sh".
26
27 # shellcheck source=common_options.lib.sh
28 source "../build_and_checks_dependencies/common_options.lib.sh"
29
30 check_no_size_of_array_first_element() {
31     # Options:
32     #   --verbose
33     #   --work-directory=""
34     declare -i LFBFL_i_verbose=0
35     get_verbose_option "$@"
36     local LFBFL_work_directory=""
37     get_work_directory_option "$@"
38     declare -a LFBFL_return_traps_stack
39     init_return_trap
40     pushd_to_work_directory --trap-popd

```



```

41     can_continue_after_enhanced_pushd || return 1
42
43     enhanced_set_bash_option globstar --trap-unset
44
45     local LFBFL_s="Checking that shell scripts do not"
46     LFBFL_s+=" contain wrong array size expansion.\n"
47     # shellcheck disable=SC2059
48     [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"
49
50     pcre2grep --multiline -- '$\\$\\#{#(|.*_)arr_(?!.*\\[@\\])}' **/*.sh
51 }
52
53 check_no_misplaced_then() {
54     # Options:
55     #   --verbose
56     #   --work-directory=""
57     #   --max-line-length
58     declare -i LFBFL_i_verbose=0
59     get_verbose_option "$@"
60     local LFBFL_work_directory=""
61     get_work_directory_option "$@"
62     declare -a LFBFL_return_traps_stack
63     init_return_trap
64     pushd_to_work_directory --trap-popd
65     can_continue_after_enhanced_pushd || return 1
66
67     declare -i LFBFL_i_max_line_length
68     get_max_line_length_option "$@"
69     declare -i LFBFL_i_max_previous_line_length
70     LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))
71
72     enhanced_set_bash_option globstar --trap-unset
73
74     [[ LFBFL_i_verbose -eq 1 ]]\
75     && printf\
76         "Checking that shell scripts do not contain misplaced 'then'.\n"
77
78     local LFBFL_regexp
79     LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
80     LFBFL_regexp+="(?!<=\\|\\);[^\n]{0,${LFBFL_i_max_previous_line_length}})"
81     LFBFL_regexp+="(?!<^\\s{0,${LFBFL_i_max_previous_line_length}}\\|\\|);)"
82     LFBFL_regexp+="\\n\\s*then"
83     # pcre2grep --multiline --
84     # '^[^\n]{1,70}(?!<=\\|\\);[^\n]{0,70})'
85     # '(?!<^\\s{0,70}\\|\\|);)\\n\\s*then'
86     pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh

```

```

87 }
88
89 check_no_negation_before_bash_test() {
90     # Options:
91     #     --verbose
92     #     --work-directory=""
93     declare -i LFBFL_i_verbose=0
94     get_verbose_option "$@"
95     local LFBFL_work_directory=""
96     get_work_directory_option "$@"
97     declare -a LFBFL_return_traps_stack
98     init_return_trap
99     pushd_to_work_directory --trap-popd
100     can_continue_after_enhanced_pushd || return 1
101
102     enhanced_set_bash_option globstar --trap-unset
103
104     [[ LFBFL_i_verbose -eq 1 ]]\
105         && printf "Checking that shell scripts do not contain ! before [.\n"
106
107     pcre2grep --multiline -- '$!\s*\[\[\[' '**/*.sh
108 }
109
110 shell_checks_complement() {
111     # Options:
112     #     --verbose
113     #     --work-directory=""
114     declare -i LFBFL_i_verbose=0
115     get_verbose_option "$@"
116     local LFBFL_work_directory=""
117     get_work_directory_option "$@"
118     declare -a LFBFL_return_traps_stack
119     init_return_trap
120     pushd_to_work_directory --trap-popd
121     can_continue_after_enhanced_pushd || return 1
122
123     declare -a LFBFL_common_options=("$@")
124     keep_options LFBFL_common_options --verbose --max-line-length
125
126     check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
127     check_no_misplaced_then "${LFBFL_common_options[@]}"
128     check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
129 }

```

3.31 build_and_checks_dependencies/split_score.exec.php

```
1  #!/usr/bin/php
2  <?php
3
4  /**
5   This file is part of DevOrSysAdminScripts.
6
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25
26  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27  2026-01-04T13:38+01:00: This file was renamed from
28  "split_line_at_most.exec.php"
29  to
30  "split_score.exec.php".
31  This file was renamed from
32  "call_split_score_after_before_or_before_after.exec.php" to
33  "split_line_at_most.exec.php".
34  This file was renamed from
35  "call_split_score_after_before_or_before_after.php" to
36  "call_split_score_after_before_or_before_after.exec.php".
37
38  @category Executable
39  @package DevOrSysAdminScripts
40  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
41  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
42  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
43  */
44
```

```

45 declare(strict_types=1);
46 declare(encoding='UTF-8');
47
48 require_once './build_and_checks_dependencies/split_score.lib.php';
49
50 if(PHP_SAPI !== 'cli'){
51     fwrite(
52         STDERR,
53         "split_score.exec.php must be run in a shell.\n",
54     );
55     die(1);
56 }
57
58 if(count($argv) !== 7){
59     fwrite(
60         STDERR,
61         "split_score.exec.php must be given exactly 6 arguments.\n",
62     );
63     die(1);
64 }
65
66 /*
67 php -r\
68 $'require_once \'./build_and_checks_dependencies/split_score.lib.php\';'\
69 $' echo(generate_split_score'\
70 $'(true, 70, [\'/\']) (\'/\', 60, true));'
71
72 php -r \
73 $'require_once \'./build_and_checks_dependencies/split_score.lib.php\';'\
74 $'echo(\SplitScore\generate_split_score(true, 70, [\'/\'])'\
75 $' (\'/\', 60, true));'
76
77 php -r \
78 $'require_once \'./build_and_checks_dependencies/split_score.lib.php\';'\
79 $' $o_split_score_closure ='\
80 $' new \SplitScore\SplitScoreClosure(true, 70, [\'/\']);'\
81 $'echo($o_split_score_closure(\'/\', 60, true));'
82 */
83
84 /*
85 // The first cast "(int) $func..." is to avoid false
86 // positive errors on possible XSS if output is unescaped.
87 echo(
88     (int) generate_split_score(
89         (bool) $argv[1],
90         (int) $argv[2],

```

```

91     explode(',', $argv[3]),
92     )($argv[4], (int) $argv[5], (bool) $argv[6])
93 );
94 /**/
95 // AHAHAH PHPCS...
96 /* AHAHAH PHPCS XD */
97
98 $o_split_score_closure = new \SplitScore\SplitScoreClosure(
99     (bool) $argv[1],
100     (int) $argv[2],
101     explode(',', $argv[3]),
102 );
103 echo($o_split_score_closure($argv[4], (int) $argv[5], (bool) $argv[6]));
104
105 ?>

```

3.32 build_and_checks_dependencies/split_score.lib.php

```

1  <?php
2
3  /**
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23  If not, see <https://www.gnu.org/licenses/>.
24
25  ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
26  2026-01-04T13:38+01:00: This file was renamed from
27  "split_line_at_most.lib.php"
28  to

```

```

29 "split_score.lib.php".
30 This file was renamed from "split_line_at_most.php" to
31 "split_line_at_most.lib.php".
32
33 @category Library
34 @package DevOrSysAdminScripts
35 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
36 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
37 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
38 */
39
40 declare(strict_types=1);
41 declare(encoding='UTF-8');
42
43
44
45 namespace SplitScore;
46
47 /**
48  Given the 3 following arguments, this function returns a closure that can
49  be used to compute the split score given the current string
50  (hopefully matching a delimiter), the current cut/split position,
51  and if that split position is after or before the current string.
52
53  @param bool          $b_larger_after
54                      Score is larger when splitting after,
55                      previous name of this parameter was $b_after_before.
56  @param int           $i_max_length
57                      The maximum length allowed (goal length).
58  @param array<string> $arr_delimiter_strings_domain
59                      An array of "delimiters".
60
61  @return \Closure {
62      The "split_score" closure to use repeatedly afterward.
63
64      @CLparam string $s_delimiter_string The string tested to be a delimiter.
65      @CLparam int    $i_cut_position     The offset of the cut in the string.
66      @CLparam bool   $b_is_cut_after     Is the offset taken after the string?
67
68      @CLreturn int The split score.
69  }
70 */
71 function generate_split_score(
72     bool $b_larger_after,
73     int $i_max_length,
74     array $arr_delimiter_strings_domain,

```

```

75 ) : \Closure {
76     return function (
77         string $s_delimiter_string,
78         int $i_cut_position,
79         bool $b_is_cut_after,
80     ) use (
81         $b_larger_after,
82         $i_max_length,
83         $arr_delimiter_strings_domain,
84     ) : int {
85         if(
86             !in_array(
87                 $s_delimiter_string,
88                 $arr_delimiter_strings_domain,
89                 true,
90             )
91         ){
92             // I do not handle regexps for the moment.
93             // I'm too brainfucked. I just want to finish this feature fast
94             // for HTML and Tex/PDF listings generation with HTML and Tex on
95             // at most 70 characters.
96             return 0;
97         }
98         if($b_is_cut_after === $b_larger_after){
99             // When $b_larger_after,
100             // always way better to cut after '/' than before it.
101             // When !$b_larger_after (aka larger_before),
102             // always way better to cut before '/' than after it.
103             return 1 + $i_max_length + $i_cut_position;
104         }
105         return 1 + $i_cut_position;
106     };
107 } //end generate_split_score()
108
109
110
111
112 /**
113  * Given the type system of PHP, a better way to have a closure computing
114  * the split score .
115  *
116  * @category Library
117  * @package DevOrSysAdminScripts
118  * @class SplitScoreClosure
119  * @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
120  * @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+

```

```

121 */
122 final class SplitScoreClosure {
123     /**
124      Score is larger when splitting after.
125
126      @var bool $b_larger_after
127      */
128     private bool $b_larger_after;
129
130     /**
131      The maximum length allowed (goal length).
132
133      @var int $i_max_length
134      */
135     private int $i_max_length;
136
137     /**
138      An array of "delimiters".
139
140      @var array<string> $arr_delimiter_strings_domain
141      */
142     private array $arr_delimiter_strings_domain;
143
144
145
146     /**
147      Given the 3 following arguments,
148      this constructor creates an object that will behave like a Closure
149      that can be used to compute the split score given the current string
150      (hopefully matching a delimiter), the current cut/split position,
151      and if that split position is after or before the current string.
152
153      @param bool          $b_larger_after
154                          Score is larger when splitting after,
155                          previous name of this parameter was $b_after_before.
156      @param int           $i_max_length
157                          The maximum length allowed (goal length).
158      @param array<string> $arr_delimiter_strings_domain
159                          An array of "delimiters".
160
161      @return void
162      */
163     public function __construct(
164         bool $b_larger_after,
165         int $i_max_length,
166         array $arr_delimiter_strings_domain,

```



```

167 ) {
168     $this->b_larger_after = $b_larger_after;
169     $this->i_max_length = $i_max_length;
170     $this->arr_delimiter_strings_domain = $arr_delimiter_strings_domain;
171 }//end __construct() SplitScoreClosure
172
173
174
175 /**
176  The "split_score" closure to use repeatedly afterward.
177
178  @param string $s_delimiter_string The string tested to be a delimiter.
179  @param int    $i_cut_position      The offset of the cut in the string.
180  @param bool   $b_is_cut_after      Is the offset taken after the string?
181
182  @return int The split score.
183  */
184 public function __invoke(
185     string $s_delimiter_string,
186     int $i_cut_position,
187     bool $b_is_cut_after,
188 ) : int {
189     if(
190         !in_array(
191             $s_delimiter_string,
192             $this->arr_delimiter_strings_domain,
193             true,
194         )
195     ){
196         return 0;
197     }
198     if($b_is_cut_after === $this->b_larger_after){
199         // When $b_larger_after,
200         // always way better to cut after '/' than before it.
201         // When !$b_larger_after (aka larger_before),
202         // always way better to cut before '/' than after it.
203         return 1 + $this->i_max_length + $i_cut_position;
204     }
205     return 1 + $i_cut_position;
206 }//end __invoke() SplitScoreClosure
207 }//end class SplitScoreClosure
208
209 ?>

```

3.33 build_and_checks_dependencies/strings_functions.lib.sh

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # I think I created a file strings_functions.sh or renamed to
25 # strings_functions.sh at some point, before committing.
26 # Hence, someone must have changed or reversed the name of this file
27 # to string_functions.sh at some point.
28 # This file was renamed from "string_functions.sh" to
29 # "strings_functions.lib.sh".
30
31 LFBFL_subdir="build_and_checks_dependencies"
32 # shellcheck source=get_common_text_glob_patterns.lib.sh
33 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
34 # shellcheck source=comparisons.lib.sh
35 source "${LFBFL_subdir}/comparisons.lib.sh"
36 # shellcheck source=too_long_code_lines.lib.sh
37 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
38
39 split_line_at() {
40     # $1=line
41     # $2=position : number of characters in split_line_at_result_beginning
42     declare -g split_line_at_result_beginning="${1:0:$2}"
43     declare -g split_line_at_result_end="${1:$2}"
44     # printf "%s\n" "$split_line_at_result_beginning"
```

```

45     # printf "%s\n" "$split_line_at_result_end"
46 }
47
48 #-----
49 #-----
50 # Properties for the score for a possible split/cut in the string between
51 # two characters.
52 #-----
53 # Some preliminary remarks:
54 # a) If the character before the split position is a forbidden previous
55 #     character, the score is -1.
56 #     (Usually, either we have no or one forbidden previous character that is
57 #     a backslash.)
58 # b) If the previous character is not forbidden,
59 #     and either the character before the position is among the delimiters,
60 #     or the character after the position is among the delimiters,
61 #     then the score is computed according to the delimiter found,
62 #     the position, and the fact that the position is after or before the
63 #     delimiter.
64 # c) If two consecutive delimiters are found in the string,
65 #     the maximum score between the one after the first delimiter
66 #     and before the second delimiter is taken for the score between the
67 #     two delimiters.
68 # d) Otherwise, by default,
69 #     the score for cutting/splitting at some position is 0.
70 # e) A substring is given by the goal length given for the resulting
71 #     strings/lines after split.
72 # f) When some scores are equal,
73 #     we split at the rightmost position with maximum score in the substring.
74 # g) Remark f) implies that below we do not need to consider strict
75 #     inequalities for the score for the result.
76 #     But if you look at split_line_at_most code,
77 #     you will see a micro-optimization where distinguishing < and <= could
78 #     have been interesting.
79 # h) The score function SF(position) which will dictate our choice for
80 #     split/cut position is in reality
81 #     SF1a(
82 #         position, previous_character, next_character, delimiters, forbiddens
83 #     )
84 #     but at the cost of expliciting its internal and adding verbosity,
85 #     we can see that the score really depends on the following arguments
86 #     SF1b(
87 #         position, matched_delimiter_before, matched_delimiter_after,
88 #         previous_character_is_forbidden
89 #     )
90 #     If it is needed,

```

```

91 # then matched_delimiter_before or matched_delimiter_after is NULL.
92 # Somehow previous_character can be seen as position.previous_character,
93 # and next_character can be seen as position.next_character.
94 # We can see that the previous or next character without
95 # knowing that it matches a delimiter is irrelevant.
96 # The fact that there is 1 or more delimiters may be interesting.
97 # The fact that there is 0 or more forbidden previous characters
98 # may be interesting, but we haven't integrated it in the split_score
99 # functions.
100 # SF1a when the previous character is not forbidden
101 # needs a score function SF2a with the following parameters
102 # SF2a(position, current_character, delimiter, is_after):
103 # SF1a(
104 #     position, previous_character, next_character, delimiters, forbiddens
105 # ) = max(
106 #     For all delimiter in delimiters,
107 #     SF2a(position, previous_character, delimiter, TRUE),
108 #     SF2a(position, next_character, delimiter, FALSE),
109 # )
110 # = max(
111 #     For all delimiter in delimiters,
112 #     For all couple C in {
113 #         (position.previous_character, TRUE),
114 #         (position.next_character, FALSE)
115 #     }
116 #     SF2a(position, C.1, delimiter, C.2),
117 # )
118 # Similarly,
119 # SF1b(
120 #     position, matched_delimiter_before, matched_delimiter_after,
121 #     previous_character_is_forbidden
122 # ) = max(
123 #     For all delimiter in delimiters,
124 #     SF2b(position, matched_delimiter_before, TRUE),
125 #     SF2b(position, matched_delimiter_after, FALSE),
126 # )
127 # where the score function SF2b has the following parameters
128 # SF2b(cut_position, matched_delimiter, is_after).
129 # We have also:
130 # SF1b(
131 #     position, matched_delimiter_before, matched_delimiter_after,
132 #     previous_character_is_forbidden
133 # ) = max(
134 #     For all delimiter in delimiters,
135 #     SF2c(position-1, matched_delimiter_before, TRUE),
136 #     SF2c(position, matched_delimiter_after, FALSE),

```

```

137 # )
138 # where the score function SF2c has the following parameters
139 # SF2c(character_position, matched_delimiter, is_after).
140 # current_character is only used to check if it is equal to the
141 # delimiter, otherwise the score is zero;
142 # but when expressing properties it makes sense to compare after
143 # and before the same character position.
144 # position, delimiter, and is_after may all three impact the score
145 # once the current character matches the delimiter.
146 # Position is an integer from 1 to goal length.
147 # Cutting at position 0 would be stupid,
148 # and moreover there would be no previous character.
149 # Since we ask for a split, the current length is above goal length,
150 # and there is a next character at the last considered position.
151 # So, in search of properties that may have an impact on the algorithm,
152 # we discard the characters to only keep matched delimiters.
153 # Let us use p1 and p2 for position1 and position2,
154 # mdb1 and mdb2 for matched_delimiter_before1, etc.,
155 # mda1 and mda2 for matched_delimiter_after1, etc.,
156 # pcif for previous_character_is_forbidden, etc.,
157 # md1 and md2 for matched_delimiter1, etc.,
158 # ial and ia2 for is_after1, etc.
159 #
160 # What we consider as properties are of 3 forms.
161 # The first form is like:
162 #   For all p1 in [1, goal_length], for all p2 in [1, goal_length],
163 #   for all mdb1 in delimiters union {NULL},
164 #   for all mdb2 in delimiters union {NULL},
165 #   for all mda1 in delimiters union {NULL},
166 #   for all mda2 in delimiters union {NULL},
167 #   RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
168 #   implies RelationS(
169 #       SF1b(p1, mdb1, mda1, false),
170 #       SF1b(p2, mdb2, mda2, false),
171 #   )
172 # There are many choices for RelationP, RelationM and RelationS.
173 # We can apply the without loss of generality argument only once when
174 # breaking symmetry between equivalent properties.
175 # Breaking symmetry on another relation that RelationS first
176 # is a challenge that we do not tackle.
177 # Indeed RelationS is the only relation where all variables appear.
178 # And breaking symmetry on RelationP would not allow to break symmetry
179 # on RelationM freely since some variables are correlated by the
180 # suffixes 1 or 2.
181 # But some choices for some relation may allow to break symmetry on
182 # another relation.

```

```

183 # Breaking symmetry on the result, there are only 2 choices for
184 # RelationS:
185 # RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
186 # RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
187 # But then there are 5+1 choices for RelationP:
188 # RP1 : p1 = p2,
189 # RP2 : p1 < p2,
190 # RP3 : p1 <= p2,
191 # RP4 : p1 > p2,
192 # RP5 : p1 >= p2,
193 # plus the choice that p1 and p2 are not necessarily related
194 # (empty RP or RPe).
195 # We can add RP6 : p1 = p2 - 1 and RP7 : p1 = p2 + 1, since it implies
196 # some equalities between matched delimiters.
197 # Note that p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
198 # logical relations compared to p1 < p2 or p2 < p1;
199 # it only yields subslices of other properties and open a slightly silly
200 # door on infinity.
201 # If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
202 # and maybe RP6.
203 # Note that RSe or empty RS would have no interest at all.
204 # There are many choices for RelationM:
205 # mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
206 # interesting at all since both scores are 0 or -1.
207 # mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
208 # implies that second score is always larger, that's not an additional
209 # property, that is a theorem.
210 # Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
211 # must (have the possibility to) be non-NULL.
212 # If we start listing the remaining relations, we have:
213 # 1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
214 #    AND mda2 is not NULL
215 #    RMA1A2
216 # 2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
217 #    AND mda2 is not NULL
218 #    AND mda1 = mda2
219 #    RMA1A2=
220 # 3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
221 #    AND mda2 is not NULL
222 #    AND mda1 != mda2 -> This case is not interesting (alone,
223 #    full analysis could combine it with some other case,
224 #    but it seems gratis combinatorial explosion without insight):
225 #    If we only know they are different, we cannot say anything on
226 #    properties related to delimiters that are different unless the score
227 #    doesn't depend on the delimiter, and that case is already handled by
228 #    1) that contains it. The same remark applies to some other choices.

```

```

229 # The only thing that could be said would be if we had an order on
230 # delimiters scores that are uniform for the position, or the score
231 # before some delimiter would always be =, >=, <=, etc. the score
232 # before or after another one. So it would only make sense with RPe,
233 # and mda1 <= mda2 for example, but that would trivially imply RS<=.
234 # It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
235 # ignores NULL value. You do not deduce or specify anything new here.
236 # 4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
237 # AND mda2 is NULL
238 # RMB1B2
239 # and so on. So if we use the short notation RM followed by A1, resp. A2,
240 # resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
241 # is not null, we have the following possibilities:
242 # RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
243 # RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
244 # with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
245 # and _4 (XX = YY = ZZ)
246 # for equalities on the 1st, 2nd or 3rd couple or between the three
247 # variables by transitivity,
248 # RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
249 # RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
250 # for one edge;
251 # RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
252 # RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
253 # RMALL_ALL square with diagonals.
254 # So we see that we listed at least 39 possibilities only for RM.
255 # We did not list yet the possibilities where the variables not present
256 # in some equality can be NULL or not NULL, etc.
257 # We see 9 of them : RMe (40), and the following 8:
258 # RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
259 # RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
260 # RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
261 # RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
262 # RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
263 # RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
264 # RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
265 # RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
266 # We don't do complete analysis at the moment.
267 # If we wanted to filter out some possibilities,
268 # we would keep only the RM possibilities, among the 38,
269 # that are symmetric between 1 and 2:
270 # RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
271 # RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
272 # RMALL_ALL. Still 10 possibilities.
273 # We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
274 # combinations (life is too short ;) );

```

```

275 # if we keep 48 RM possibilities, there is 624 combinations,
276 # but we did not analyze totally which RM relations are useless.
277 #
278 # The second form is like:
279 #   For all p1 in [1, goal_length], for all p2 in [1, goal_length],
280 #   for all md1 in delimiters union {NULL},
281 #   for all md2 in delimiters union {NULL},
282 #   for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
283 #   RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
284 #   implies RelationS(
285 #       SF2b(p1, md1, ia1)
286 #       SF2b(p2, md2, ia2)
287 #   )
288 # The third form is like:
289 #   For all p1 in [1, goal_length], for all p2 in [1, goal_length],
290 #   for all md1 in delimiters union {NULL},
291 #   for all md2 in delimiters union {NULL},
292 #   for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
293 #   RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
294 #   implies RelationS(
295 #       SF2c(p1, md1, ia1)
296 #       SF2c(p2, md2, ia2)
297 #   )
298 # This time the combinatorics is much more pleasant :).
299 # As above, we have RS= and RS<=, RPe, RP1 to RP7;
300 # For RelationMM, assume that the possible delimiters correspond to
301 # integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
302 # in [0,i], either 0 for NULL or the delimiter index.
303 # Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
304 # information we can use is k = l or k != l.
305 # If we allow that md1 != 0 and md2 = 0,
306 # then both RS= and RS<= are false.
307 # Thus RMM= is not possible.
308 # If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
309 # whilst RS= is false.
310 # Thus properties with RS<= can trivially be extended with
311 # RMM-"allow-md1-0", not useful to explicit this for the moment.
312 # If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
313 # nothing more.
314 # Thus we can focus on: md1 and md2 are not NULL,
315 # and are equal RMM= or not RMM!=. It seems that a property with RMM!=
316 # could be extended to a property RMM=or!=/RMMNotNULL.
317 # and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
318 # 3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
319 # or FALSE. Wrong ! Not pleasant ! XD
320 # So for RIA, we keep either: empty RIAe (RIA8),

```



```

321 #   ia1 = ia2 RIA= (RIA0 or RIA4),
322 #   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
323 #   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
324 #   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
325 #   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
326 #   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
327 #   even less pleasant XD : 11 possibilities.
328 #   In total, we have now 1 (RS=) times 5 times 2 times 11
329 #   + 1 (RS<=) times 8 times 2 times 11 = 286
330 #   combinations (life is even shorter ;) :) ).
331 #   We only listed below properties of the third form.
332 # i) By the previous remark, there is a good chance that we missed some
333 #   possible property.
334 #   We hope we did found the properties that had a simple meaning,
335 #   in particular related to the algorithm where either you
336 #   loop on the string forward and keep last position with maximum score,
337 #   or you loop backward and abort as soon as possible as soon you know
338 #   that you found some position at the end that must have maximum score.
339 # j) These properties are "boolean" flags (integers that are powers of 2).
340 #   We first put the exact integers,
341 #   then wondered about switching everything to exponentiation 2**i,
342 #   then we took the solution that minimizes the character length between
343 #   [1-9][0-9]* and 2**i, then reverted to initial before committing.
344 #   Most efficient interpreted code is what we did first,
345 #   if there is no opcode cache for constants.
346 #   Most safe code is when the language hides the details and handles
347 #   families of flags,
348 #   but beware if some flag goes into the database.
349 #   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
350 #   or $(2<<0)...
351 #   What I would love to see in many languages is a flag type, as a subtype
352 #   of integers, flag32, flag64 for C for example.
353 #   I think there is no such thing because sooner or later you play with
354 #   pseudo-flags that are combination of flags, hence integers.
355 #   But just for the declaration to let the type checker check that the
356 #   integer is a power of 2: flag32 MY_FLAG = 4;
357 #   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
358 #   Another fact is that practice makes perfect, so working with the powers
359 #   of 2 written out in base 10 gets you accustomed to them.
360 #   Safe code should be favored since (pre-)compilation optimizes it,
361 #   but for this project, I prefer to keep the integers in decimal and
362 #   check that it doubles when reading my listing, because I'm familiar
363 #   with that and the powers of 2 doesn't go beyond what I can check fast
364 #   enough in my brain.
365 # k) Some of the properties below are union of other properties,
366 #   and are here to simplify not only code but understanding.

```

```

367 #   split_score_properties_logical_closure() function below checks that
368 #   all flags that are consequences of already activated flags are set.
369 # 1) We use SSP_prefix for SPLIT_SCORE_PROPERTY_.
370 # Some of these remarks are repeated below in the corresponding functions,
371 # and some remarks are added just after the constants.
372 #-----
373 # Any score after a character that matches a delimiter
374 # is larger than or equal to
375 # any score before a character that matches a delimiter.
376 # RPPe RMMNotNull RIA!=T2 RS<=
377 declare -gir SSP_ALWAYS_LARGER_AFTER=1
378
379 # Any score before a character that matches a delimiter
380 # is larger than or equal to
381 # any score after a character that matches a delimiter.
382 # RPPe RMMNotNull RIA!=T1 RS<=
383 declare -gir SSP_ALWAYS_LARGER_BEFORE=2
384
385 # The score doesn't depend on the delimiter that is found.
386 # RPP1 RMMNotNull RIA= RS=
387 declare -gir SSP_DELIMITER_UNIFORM=4
388
389 # Whenever a character that matches a delimiter is found,
390 # the score after is equal to the score before.
391 # Or equivalently:
392 # For any delimiter, and for any character position,
393 # if the character at this position matches this delimiter,
394 # the score after the character
395 # is equal to
396 # the score before the character.
397 # RPP1 RMM= RIA!= RS=
398 declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
399 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=
400
401 # Whenever a character that matches a delimiter is found,
402 # the score after is larger than or equal to the score before.
403 # Or equivalently:
404 # For any delimiter, and for any character position,
405 # if the character at this position matches this delimiter,
406 # the score after the character
407 # is larger than or equal to
408 # the score before the character.
409 # RPP1 RMM= RIA!=T2 RS<=
410 declare -gir SSP_LARGER_AFTER=16
411 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=
412

```

```

413 # Whenever a character that matches a delimiter is found,
414 # the score before is larger than or equal to the score after.
415 # Or equivalently:
416 # For any delimiter, and for any character position,
417 # if the character at this position matches this delimiter,
418 # the score before the character
419 # is larger than or equal to
420 # the score after the character.
421 # RPP1 RMM= RIA!=T1 RS<=
422 declare -gir SSP_LARGER_BEFORE=32
423 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=
424
425 # The score after, resp. before, a character that matches a delimiter
426 # doesn't decrease when the position in the string increase.
427 # (RPP2 or RPP3) RMM= RIA= RS<=
428 declare -gir SSP_POSITION_NOT_DECREASING=64
429
430 # The score after a character that matches a delimiter doesn't decrease
431 # when the position in the string increases.
432 # (RPP2 or RPP3) RMM= RIA=T RS<=
433 declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128
434
435 # The score before a character that matches a delimiter doesn't decrease
436 # when the position in the string increases.
437 # (RPP2 or RPP3) RMM= RIA=F RS<=
438 declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256
439
440 # The score after, resp. before, a character that matches a delimiter
441 # doesn't depend on the position in the string.
442 # (RPP2 or RPP3) RMM= RIA= RS=
443 declare -gir SSP_POSITION_UNIFORM=512
444
445 # The score after a character that matches a delimiter doesn't depend
446 # on the position in the string.
447 # (RPP2 or RPP3) RMM= RIA=T RS=
448 declare -gir SSP_POSITION_UNIFORM_AFTER=1024
449
450 # The score before a character that matches a delimiter doesn't depend
451 # on the position in the string.
452 # (RPP2 or RPP3) RMM= RIA=F RS=
453 declare -gir SSP_POSITION_UNIFORM_BEFORE=2048
454
455 # For any delimiter,
456 # any score after a character that matches this delimiter
457 # is equal to
458 # any score before a character that matches this delimiter.

```

```

459 # RPpE RMM= RIA!= RS=
460 # <=> RPpE RMM= RIAe RS=
461 declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096
462
463 # For any delimiter,
464 # any score after a character that matches this delimiter
465 # is larger than or equal to
466 # any score before a character that matches this delimiter.
467 # RPpE RMM= RIA!=T2 RS<=
468 declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192
469
470 # For any delimiter,
471 # any score before a character that matches this delimiter
472 # is larger than or equal to
473 # any score after a character that matches this delimiter.
474 # RPpE RMM= RIA!=T1 RS<=
475 declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384
476
477 # For any character position,
478 # assuming that a character at this position matches some delimiter,
479 # the score after a character at this position that matches a delimiter
480 # is equal to
481 # the score before a character at this position that matches a delimiter.
482 # RPP1 RMMNotNull RIA!= RS=
483 declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768
484
485 # For any character position,
486 # assuming that a character at this position matches some delimiter,
487 # the score after a character at this position that matches a delimiter
488 # is larger than or equal to
489 # the score before a character at this position that matches a delimiter.
490 # RPP1 RMMNotNull RIA!=T2 RS<=
491 declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536
492
493 # For any character position,
494 # assuming that a character at this position matches some delimiter,
495 # the score before a character at this position that matches a delimiter
496 # is larger than or equal to
497 # the score after a character at this position that matches a delimiter.
498 # RPP1 RMMNotNull RIA!=T1 RS<=
499 declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072
500
501 # For all possible split/cut positions, the score is equal as long as
502 # at least one of the previous and next characters is a delimiter.
503 # Or equivalently:
504 # Any score after a character that matches a delimiter

```

```

505 # is equal to
506 # any score before a character that matches a delimiter.
507 # is_after doesn't change anything.
508 # So we cut on the last position adjacent to a delimiter without
509 # forbidden previous character (edge case, on last position).
510 # RPpe RMMNotNull RIAe RS=
511 declare -gir SSP_UNIFORM=262144
512 # declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1
513
514 #-----
515 # m) Note that a property telling that the score is always the same
516 # even if the current character isn't among delimiters is not needed,
517 # since, in that case, a null split score command will split at the goal
518 # length.
519 #-----
520 #-----
521
522 split_score_properties_logical_closure() {
523     # $1=split_score_properties
524     # Since this code is not performance critical,
525     # we followed lexicographic order on constants,
526     # instead of an order to minimize the operations.
527     # It is also safer to avoid forgetting some relation.
528     declare -i LFBFL_i_result=$(( $1 ))
529     declare -i LFBFL_i_result_old=0
530     declare -i LFBFL_i_all_flags=$((
531         SSP_ALWAYS_LARGER_AFTER
532         | SSP_ALWAYS_LARGER_BEFORE
533         | SSP_DELIMITER_UNIFORM
534         | SSP_EQUAL_AFTER_OR_BEFORE
535         | SSP_LARGER_AFTER
536         | SSP_LARGER_BEFORE
537         | SSP_POSITION_NOT DECREASING
538         | SSP_POSITION_NOT DECREASING_AFTER
539         | SSP_POSITION_NOT DECREASING_BEFORE
540         | SSP_POSITION_UNIFORM
541         | SSP_POSITION_UNIFORM_AFTER
542         | SSP_POSITION_UNIFORM_BEFORE
543         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
544         | SSP_SAME_DELIMITER_LARGER_AFTER
545         | SSP_SAME_DELIMITER_LARGER_BEFORE
546         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
547         | SSP_SAME_POSITION_LARGER_AFTER
548         | SSP_SAME_POSITION_LARGER_BEFORE
549         | SSP_UNIFORM
550     ))

```

```

551 until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
552     LFBFL_i_result_old=$((LFBFL_i_result))
553     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
554         LFBFL_i_result=$((
555             LFBFL_i_result
556             | SSP_LARGER_AFTER
557             | SSP_SAME_DELIMITER_LARGER_AFTER
558             | SSP_SAME_POSITION_LARGER_AFTER
559         ))
560     fi
561     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
562         LFBFL_i_result=$((
563             LFBFL_i_result
564             | SSP_LARGER_BEFORE
565             | SSP_SAME_DELIMITER_LARGER_BEFORE
566             | SSP_SAME_POSITION_LARGER_BEFORE
567         ))
568     fi
569     if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
570         LFBFL_i_result=$((
571             LFBFL_i_result
572             | SSP_POSITION_NOT_DECREASING_AFTER
573             | SSP_POSITION_NOT_DECREASING_BEFORE
574         ))
575     fi
576     if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
577         LFBFL_i_result=$((
578             LFBFL_i_result
579             | SSP_POSITION_UNIFORM_AFTER
580             | SSP_POSITION_UNIFORM_BEFORE
581             | SSP_POSITION_NOT_DECREASING
582             | SSP_POSITION_NOT_DECREASING_AFTER
583             | SSP_POSITION_NOT_DECREASING_BEFORE
584         ))
585     fi
586     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
587         LFBFL_i_result=$((
588             LFBFL_i_result
589             | SSP_POSITION_NOT_DECREASING_AFTER
590         ))
591     fi
592     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
593         LFBFL_i_result=$((
594             LFBFL_i_result
595             | SSP_POSITION_NOT_DECREASING_BEFORE
596         ))

```

```

597     fi
598     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
599         LFBFL_i_result=$((
600             LFBFL_i_result
601             | SSP_EQUAL_AFTER_OR_BEFORE
602             | SSP_LARGER_AFTER
603             | SSP_LARGER_BEFORE
604             | SSP_SAME_DELIMITER_LARGER_AFTER
605             | SSP_SAME_DELIMITER_LARGER_BEFORE
606         ))
607     fi
608     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
609         LFBFL_i_result=$((
610             LFBFL_i_result
611             | SSP_LARGER_AFTER
612         ))
613     fi
614     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
615         LFBFL_i_result=$((
616             LFBFL_i_result
617             | SSP_LARGER_BEFORE
618         ))
619     fi
620     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
621         LFBFL_i_result=$((
622             LFBFL_i_result
623             | SSP_DELIMITER_UNIFORM
624             | SSP_EQUAL_AFTER_OR_BEFORE
625             | SSP_LARGER_AFTER
626             | SSP_LARGER_BEFORE
627             | SSP_SAME_POSITION_LARGER_AFTER
628             | SSP_SAME_POSITION_LARGER_BEFORE
629         ))
630     fi
631     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
632         LFBFL_i_result=$((
633             LFBFL_i_result
634             | SSP_LARGER_AFTER
635         ))
636     fi
637     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
638         LFBFL_i_result=$((
639             LFBFL_i_result
640             | SSP_LARGER_BEFORE
641         ))
642     fi

```

```

643     if ((LFBFL_i_result & SSP_UNIFORM)); then
644         LFBFL_i_result=$((LFBFL_i_all_flags))
645         break
646     fi
647     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
648     && ((
649         LFBFL_i_result & (
650             SSP_ALWAYS_LARGER_BEFORE
651             | SSP_EQUAL_AFTER_OR_BEFORE
652             | SSP_LARGER_BEFORE
653             | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
654             | SSP_SAME_DELIMITER_LARGER_BEFORE
655             | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
656             | SSP_SAME_POSITION_LARGER_BEFORE
657         )
658     )); then
659         LFBFL_i_result=$((LFBFL_i_all_flags))
660         break
661     fi
662     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
663     && ((
664         LFBFL_i_result & (
665             SSP_EQUAL_AFTER_OR_BEFORE
666             | SSP_LARGER_AFTER
667             | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
668             | SSP_SAME_DELIMITER_LARGER_AFTER
669             | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
670             | SSP_SAME_POSITION_LARGER_AFTER
671         )
672     )); then
673         LFBFL_i_result=$((LFBFL_i_all_flags))
674         break
675     fi
676     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
677     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
678         LFBFL_i_result=$((
679             LFBFL_i_result
680             | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
681         ))
682     fi
683     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
684     && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
685         LFBFL_i_result=$((
686             LFBFL_i_result
687             | SSP_SAME_POSITION_LARGER_AFTER
688         ))

```



```

689     fi
690     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
691     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
692         LFBFL_i_result=$((
693             LFBFL_i_result
694             | SSP_SAME_POSITION_LARGER_BEFORE
695         ))
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
699         LFBFL_i_result=$((LFBFL_i_all_flags))
700         break
701     fi
702     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
703     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
704         LFBFL_i_result=$((
705             LFBFL_i_result
706             | SSP_ALWAYS_LARGER_AFTER
707         ))
708     fi
709     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
710     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
711         LFBFL_i_result=$((
712             LFBFL_i_result
713             | SSP_ALWAYS_LARGER_BEFORE
714         ))
715     fi
716     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
717     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
718         LFBFL_i_result=$((
719             LFBFL_i_result
720             | SSP_EQUAL_AFTER_OR_BEFORE
721         ))
722     fi
723     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
724     && ((
725         LFBFL_i_result & (
726             SSP_POSITION_UNIFORM
727             | SSP_POSITION_UNIFORM_AFTER
728         )
729     )); then
730         LFBFL_i_result=$((
731             LFBFL_i_result
732             | SSP_SAME_DELIMITER_LARGER_AFTER
733         ))
734     fi

```

```

735 if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
736 && ((
737     LFBFL_i_result & (
738         SSP_POSITION_UNIFORM
739         | SSP_POSITION_UNIFORM_BEFORE
740     )
741 )); then
742     LFBFL_i_result=$((
743         LFBFL_i_result
744         | SSP_SAME_DELIMITER_LARGER_BEFORE
745     ))
746 fi
747 if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
748 && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
749     LFBFL_i_result=$((
750         LFBFL_i_result
751         | SSP_POSITION_NOT_DECREASING
752     ))
753 fi
754 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
755 && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
756     LFBFL_i_result=$((
757         LFBFL_i_result
758         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
759     ))
760 fi
761 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
762 && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
763     LFBFL_i_result=$((LFBFL_i_all_flags))
764     break
765 fi
766 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
767 && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
768     LFBFL_i_result=$((
769         LFBFL_i_result
770         | SSP_ALWAYS_LARGER_AFTER
771     ))
772 fi
773 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
774 && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
775     LFBFL_i_result=$((
776         LFBFL_i_result
777         | SSP_ALWAYS_LARGER_BEFORE
778     ))
779 fi
780 if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\

```

```

781      && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
782          LFBFL_i_result=$((
783              LFBFL_i_result
784              | SSP_POSITION_UNIFORM
785          ))
786      fi
787      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
788      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
789          LFBFL_i_result=$((
790              LFBFL_i_result
791              | SSP_ALWAYS_LARGER_AFTER
792          ))
793      fi
794      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
795      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
796          LFBFL_i_result=$((
797              LFBFL_i_result
798              | SSP_ALWAYS_LARGER_BEFORE
799          ))
800      fi
801      if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
802      && ((
803          LFBFL_i_result & (
804              SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
805              | SSP_SAME_POSITION_LARGER_AFTER
806              | SSP_SAME_POSITION_LARGER_BEFORE
807          )
808      )); then
809          LFBFL_i_result=$((LFBFL_i_all_flags))
810          break
811      fi
812      if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
813      && ((
814          LFBFL_i_result & (
815              SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
816              | SSP_SAME_POSITION_LARGER_BEFORE
817          )
818      )); then
819          LFBFL_i_result=$((LFBFL_i_all_flags))
820          break
821      fi
822      if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
823      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
824          LFBFL_i_result=$((
825              LFBFL_i_result
826              | SSP_ALWAYS_LARGER_AFTER

```

```

827     ))
828   fi
829   if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
830   && ((
831     LFBFL_i_result & (
832       SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
833       | SSP_SAME_POSITION_LARGER_AFTER
834     )
835   )); then
836     LFBFL_i_result=$((LFBFL_i_all_flags))
837     break
838   fi
839   if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
840   && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
841     LFBFL_i_result=$((
842       LFBFL_i_result
843       | SSP_ALWAYS_LARGER_BEFORE
844     ))
845   fi
846   if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
847   && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
848     LFBFL_i_result=$((
849       LFBFL_i_result
850       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
851     ))
852   fi
853 done
854 declare -gi i_split_score_properties_logical_closure_result=$((
855   LFBFL_i_result
856 ))
857 }
858
859 split_score_simple() {
860   # $1=larger_after (The name was "after_before" previously,
861   #                 the meaning was not modified.
862   #                 Remark not repeated below.)
863   # $2=delimiters_strings_domain concatenated characters/delimiters
864   # $3=delimiter_string a single character
865   # $4=is_cut_after
866   declare -gi i_split_score_result
867   if [[ "$2" != *$3* ]]; then
868     i_split_score_result=0
869     return
870   fi
871   if [[ "$4" == "$1" ]]; then
872     i_split_score_result=2

```

```

873     return
874 fi
875 i_split_score_result=1
876 }
877
878 get_split_score_simple() {
879     # $1=larger_after
880     # $2=max_length
881     # $3=delimiters_strings_domain
882     declare -g get_split_score_result="split_score_simple $1 '$3'"
883     declare -gi i_get_split_score_result2
884     # i_get_split_score_result2=$((11-$1*4))
885     # if [[ $1 -eq 1 ]]; then
886     #     i_get_split_score_result2=7
887     # else
888     #     i_get_split_score_result2=11
889     # fi
890     i_get_split_score_result2=$((
891         SSP_DELIMITER_UNIFORM
892         + SSP_POSITION_UNIFORM
893         + SSP_LARGER_BEFORE
894         + (
895             $1 * (
896                 SSP_LARGER_AFTER
897                 - SSP_LARGER_BEFORE
898             )
899         )
900     ))
901     # shellcheck disable=SC2248
902     split_score_properties_logical_closure ${i_get_split_score_result2}
903     i_get_split_score_result2=$((
904         i_split_score_properties_logical_closure_result
905     ))
906 }
907
908 get_split_score_exec() {
909     declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
910     SPLIT_SCORE_EXEC+="/split_score.exec.php"
911 }
912
913 split_score() {
914     # $1=larger_after
915     # $2=max_length
916     # $3=delimiters_strings_domain concatenated characters/delimiters
917     # $4=delimiter_string a single character
918     # $5=cut_position

```

```

919 # $6=is_cut_after
920 get_split_score_exec
921 declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
922 declare -gi i_split_score_result
923 i_split_score_result=$(eval -- "${LFBFL_command}")
924 }
925
926 # This function is currently not called,
927 # hence we need this shellcheck disable.
928 # shellcheck disable=SC2034
929 get_split_score() {
930     # $1=larger_after
931     # $1=max_length
932     # $2=delimiters_strings_domain
933     declare -g get_split_score_result="split_score $1 $2 '$3'"
934     declare -gi i_get_split_score_result2
935     # i_get_split_score_result2=$((9-$1*4))
936     i_get_split_score_result2=$((
937         SSP_DELIMITER_UNIFORM
938         + SSP_LARGER_BEFORE
939         + (
940             $1 * (
941                 SSP_LARGER_AFTER
942                 - SSP_LARGER_BEFORE
943             )
944         )
945     ))
946     # shellcheck disable=SC2248
947     split_score_properties_logical_closure ${i_get_split_score_result2}
948     i_get_split_score_result2=$((
949         i_split_score_properties_logical_closure_result
950     ))
951 }
952
953 split_line_at_most() {
954     # $1=line
955     # $2=max_length of beginning result string
956     # $3=split_score_command
957     # negative score means do not consider as a possible split
958     # $4=split_score_command_properties :
959     # These properties are flags.
960     # If someday some property is not a flag,
961     # then create split_score_command_properties2.
962     # Old values to get a simple idea without reading everything at the
963     # beginning of file:
964     # - null or 0 no property;

```

```

965 # - 1 split score doesn't depend on which delimiter matches;
966 # - 2 split score doesn't depend on the current position;
967 # - 4 any split score after is always larger
968 #   than any split score before;
969 # - 8 any split score before is always larger
970 #   than any split score after;
971 # $5=forbidden_previous_character
972 # $6=split_score_command_delimiter_strings_domain
973 #   optimisation if small domain
974 # - this argument can be "null", then we will use $3 only instead,
975 #   in that case, only mono-character strings are considered,
976 # - this argument can be an array of characters,
977 #   optimize accordingly,
978 # - this argument can be an array of strings,
979 #   optimize accordingly,
980 # - this argument can be an array of regexps,
981 #   optimize accordingly.
982 #   Not sure an array of regexps makes more sense than a single
983 #   one. Maybe they could be both cases with corresponding
984 #   optimizations. Same for a single character or substring.
985 #
986 # Note that we're always looking for the last position to split
987 # among the positions with maximum score;
988 # if some position gets a score from 2 delimiters,
989 # its score is always the maximum of the two corresponding scores.
990 declare -g split_line_at_most_result_start
991 declare -g split_line_at_most_result_end
992 declare -A LFBFL_positions
993 # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
994 # For my use case in Bash scripts, I will need only an array of
995 # characters. See get_split_score().
996 if [[ -n "$6" ]]; then
997     printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
998 fi
999 if [[ $2 -ge ${#1} ]]; then
1000     split_line_at_most_result_start=$1
1001     split_line_at_most_result_end=""
1002     return
1003 fi
1004 # We know that ${#1} > $2, hence ${#1} - 1 >= $2
1005 # Edge cases:
1006 # There is always at least one character that is overlength,
1007 # but we need to look only at the score before that character that is
1008 # overlength, the score after that character is invalid,
1009 # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1010 # That first character that is overlength has index $2.

```

```

1011 # The cut position after that character has index $2 + 1.
1012 # Looping from character position going from 0 to $2, we need to drop
1013 # score before character 0 and drop score after character $2.
1014 # We do this with forbidden score -1.
1015 declare -ir LFBFL_i_max=$(( $2 ))
1016 declare -i LFBFL_i
1017 declare -i LFBFL_j
1018 local LFBFL_current_char
1019
1020 if [[ $2 -lt 2 ]]; then
1021     printf "split_line_at_most: max_length should be at least 2...\n"
1022     return
1023 fi
1024 if [[ -n "$5" && "$1" =~ ^"$5"*$ ]]; then
1025     for ((LFBFL_i = 2; LFBFL_i <= $2; LFBFL_i <= 1)); do
1026         :
1027     done
1028     ((LFBFL_i >>= 1))
1029     split_line_at_most_result_start="${1:0:LFBFL_i}"
1030     split_line_at_most_result_end="${1:LFBFL_i}"
1031     return
1032 fi
1033
1034 # At the beginning, each position has score 0.
1035 LFBFL_positions["0"]="-1"
1036 for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1037     LFBFL_positions["${LFBFL_i}"]="0"
1038 done
1039 LFBFL_positions["${LFBFL_i}"]="-1"
1040
1041 # If we have a forbidden previous character,
1042 # we mark the forbidden positions with -1.
1043 if [[ -n "$5" ]]; then
1044     for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1045         LFBFL_current_char="${1:${LFBFL_i}:1}"
1046         ((++LFBFL_i))
1047         if [[ "${LFBFL_current_char}" == "$5" ]]; then
1048             LFBFL_positions["${LFBFL_i}"]="-1"
1049         fi
1050     done
1051 fi
1052
1053 declare -r LFBFL_sort_command="sort --numeric-sort"
1054
1055 update_position_score() {
1056     # $1=position

```



```

1057     # Returns 0 if updated, 1 otherwise
1058     # printf "%s|%s\n" "${i_split_score_result}" "$1"
1059     if [[ i_split_score_result -ge 1 ]]; then
1060         # printf "valid result"
1061         if [[ ${LFBFL_positions["$1"]} != "-1" ]]; then
1062             LFBFL_positions["$1"]=$(
1063                 max "${LFBFL_sort_command}" \
1064                     "${LFBFL_positions["$1"]}" \
1065                     "${i_split_score_result}"
1066             )
1067             return 0
1068         fi
1069     fi
1070     return 1
1071 }
1072
1073 local LFBFL_split_score_command=$3
1074 declare -i LFBFL_split_score_command_properties=$4
1075
1076 normal_or_position_uniform_call() {
1077     # $1=cut_position
1078     # $2=is_cut_after
1079     if ((LFBFL_split_score_command_properties & SSP_POSITION_UNIFORM));
1080     then
1081         eval --\
1082             "${LFBFL_split_score_command} '${LFBFL_current_char}' $2"
1083     else
1084         eval --\
1085             "${LFBFL_split_score_command} '${LFBFL_current_char}' $1 $2"
1086     fi
1087 }
1088
1089 if (($4 & SSP_DELIMITER_UNIFORM)) \
1090 && (($4 & SSP_LARGER_AFTER)) \
1091 && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1092     for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1093         LFBFL_current_char="${1:${LFBFL_i}:1}"
1094         LFBFL_j=$((LFBFL_i + 1))
1095         # shellcheck disable=SC2248
1096         normal_or_position_uniform_call ${LFBFL_j} 1
1097         # shellcheck disable=SC2248
1098         if update_position_score ${LFBFL_j}
1099         then
1100             break
1101         fi
1102     done

```

```

1103     # shellcheck disable=SC2248
1104     normal_or_position_uniform_call ${LFBFL_i} 0
1105     # shellcheck disable=SC2248
1106     update_position_score ${LFBFL_i}
1107 done
1108 elif (($4 & SSP_DELIMITER_UNIFORM))\
1109 && (($4 & SSP_LARGER_BEFORE))\
1110 && (($4 & SSP_POSITION_NOT DECREASING_BEFORE)); then
1111     for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1112         LFBFL_current_char="${1:${LFBFL_i}:1}"
1113         # We still need to break in second if, after computing score after,
1114         # because SSP_LARGER_BEFORE implies large inequality instead of
1115         # strict inequality.
1116         # And since the algorithm is to take the largest position with
1117         # maximum score, there is a possibility that we'll break on a found
1118         # score before but for splitting on a score after.
1119         LFBFL_j=$((LFBFL_i + 1))
1120         # shellcheck disable=SC2248
1121         normal_or_position_uniform_call ${LFBFL_j} 1
1122         # shellcheck disable=SC2248
1123         update_position_score ${LFBFL_j}
1124
1125         # shellcheck disable=SC2248
1126         normal_or_position_uniform_call ${LFBFL_i} 0
1127         # shellcheck disable=SC2248
1128         if update_position_score ${LFBFL_i}
1129         then
1130             break
1131         fi
1132     done
1133 else
1134     for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1135         LFBFL_current_char="${1:${LFBFL_i}:1}"
1136         eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1137         # shellcheck disable=SC2248
1138         update_position_score ${LFBFL_i}
1139         ((++LFBFL_i))
1140         eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1141         # shellcheck disable=SC2248
1142         update_position_score ${LFBFL_i}
1143     done
1144 fi
1145 declare -ir LFBFL_i_max_score=$(
1146     max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1147 )
1148 local LFBFL_value

```

```

1149 declare -i LFBFL_i_best_position
1150 for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1151     if [[ LFBFL_i -eq 0 ]]; then
1152         printf $'/!\\split_line_at_most bug: no split position found./!\\n'
1153     fi
1154     local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1155     # printf "%s %s\n" "${LFBFL_i}" "${LFBFL_value}"
1156     if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1157         LFBFL_i_best_position=$((LFBFL_i))
1158         break
1159     fi
1160 done
1161 # shellcheck disable=SC2248
1162 split_line_at "$1" ${LFBFL_i_best_position}
1163 split_line_at_most_result_start="${split_line_at_result_beginning}"
1164 split_line_at_most_result_end="${split_line_at_result_end}"
1165 # printf "%s\n" "$split_line_at_most_result_start"
1166 # printf "%s\n" "$split_line_at_most_result_end"
1167 }
1168
1169 split_last_line() {
1170     # $1=new_lines
1171     # $2=prefix
1172     # $3=max_length
1173     # $4=suffix
1174     # $5=split_score_command
1175     # $6=split_score_command_properties
1176     # $7=forbidden_previous_character ('\' usually)
1177     # Options:
1178     # --verbose
1179     # In general, a prefix/suffix can be a required line prefix/suffix
1180     # for all final lines like a comment prefix,
1181     # or it can be a continuation line prefix/suffix applied only
1182     # when splitting a line in 2 lines,
1183     # or it can be a prefix/suffix for the string made of all the lines.
1184     # Clearly, there is no gain to couple handling a prefix/suffix for
1185     # the string made of all the lines in this function.
1186     # Since we add a prefix/suffix when splitting,
1187     # it is easy to see that we do not need to distinguish between
1188     # required line prefix/suffix and continuation line prefix/suffix.
1189     # The $2 and $4 arguments are given as concatenation of the
1190     # required and continuation line prefixes/suffixes as needed by the
1191     # user.
1192     # Thus lines arguments should be given already prefixed
1193     # and suffixed.
1194     # This is logic, since it would be cumbersome to handle prefixing

```

```

1195 # or not already splitted lines in this function
1196 # (see repeated_split_last_line).
1197 # The given max_length is the goal.
1198 # This function adapts the effective max position for the split by
1199 # taking into account the length of the suffix.
1200 declare -i LFBFL_i_verbose=0
1201 get_verbose_option "$@"
1202 declare -a LFBFL_return_traps_stack
1203 init_return_trap
1204
1205 enhanced_set_shell_option pipefail --trap-unset
1206
1207 declare -g split_last_line_result="$1"
1208 declare -ir LFBFL_i_length2=$((3 - ${#4}))
1209 get_overlength_regexp "$3"
1210
1211 # Testing if some line is too long.
1212 printf "%s" "$1"\
1213 | grep --quiet "${get_overlength_regexp_result}"
1214 if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1215     return
1216 fi
1217
1218 declare -r LFBFL_start=$(
1219     printf "%s" "$1"\
1220     | head --lines=-1
1221 )
1222 # printf "start: %s\n" "${LFBFL_start}"
1223 declare -r LFBFL_last_line=$(
1224     printf "%s" "$1"\
1225     | tail --lines=1
1226 )
1227 # printf "last_line: %s\n" "${LFBFL_last_line}"
1228 split_last_line_result=""
1229 if [[ -n "${LFBFL_start}" ]]; then
1230     split_last_line_result="${LFBFL_start}"
1231     # Last line-return of LFBFL_start seems dropped by $() and not ="${} ".
1232     # printf "%s %s\n" "${#split_last_line_result}"
1233     # "${split_last_line_result}"
1234     split_last_line_result+="${split_last_line_result}"
1235     # something="${split_last_line_result}"
1236     # printf "%s %s\n" "${#something}" "${something}"
1237 fi
1238 if [[ -n "$5" ]]; then
1239     # shellcheck disable=SC2248
1240     split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\

```

```

1241     "$5" "$6" "$7"
1242     split_last_line_result+="${split_line_at_most_result_start}"
1243     split_last_line_result+="$4"
1244     split_last_line_result+=$'\n'
1245     split_last_line_result+="$2${split_line_at_most_result_end}"
1246 else
1247     split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
1248     split_last_line_result+="$4"
1249     split_last_line_result+=$'\n'
1250     split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
1251 fi
1252 # printf "result: %s\n" "${split_last_line_result}"
1253 }
1254
1255 repeated_split_last_line() {
1256     # See split_last_line for the first arguments.
1257     # $8=final_suffix to handle when the end of the original string
1258     #     is the suffix.
1259     declare -g repeated_split_last_line_result="$1"
1260     local LFBFL_result="$1x"
1261     until
1262         [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
1263     do
1264         split_last_line "${repeated_split_last_line_result}" "$2" "$3\"
1265             "$4" "$5" "$6" "$7"
1266         LFBFL_result="${repeated_split_last_line_result}"
1267         repeated_split_last_line_result="${split_last_line_result}"
1268         # printf "%s\n" "${repeated_split_last_line_result}"
1269     done
1270     if [[ -n "$4" && -n "$8" ]]; then
1271         declare -ir LFBFL_i_offset=$((-${#4}))
1272         local LFBFL_lines_end
1273         LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
1274         if [[ "${LFBFL_lines_end}" == "$4" ]]; then
1275             repeated_split_last_line_result+="$8"
1276         fi
1277     fi
1278 }

```

3.34 build_and_checks_dependencies/too_long_code_lines.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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```

```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "too_long_code_lines.sh"
25 # to "too_long_code_lines.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=get_common_text_glob_patterns.lib.sh
31 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
32 # shellcheck source=lines_filters.lib.sh
33 source "${LFBFL_subdir}/lines_filters.lib.sh"
34
35 get_overlength_regexp() {
36     # $1=length
37     # Option:
38     # --length-is-overlength
39     declare -i LFBFL_i_overlength=$1
40     declare -i LFBFL_i_length_is_overlength
41     get_some_flag LFBFL_i_length_is_overlength --length-is-overlength 1 "$@"
42     if [[ LFBFL_i_length_is_overlength -eq 0 ]]; then
43         ((++LFBFL_i_overlength))
44     fi
45     readonly LFBFL_i_overlength
46     declare -g get_overlength_regexp_result=".\\${LFBFL_i_overlength}\\}"
47     # declare -g get_overlength_regexp_result="^.\\${LFBFL_i_overlength}\\}"
48     # declare -g get_overlength_regexp_result=".\\${LFBFL_i_overlength}\\}\\$"
49     # First one seems to be faster by a few percents when testing like this:
50     # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
51     # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done

```

```

52     # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
53 }
54
55 too_long_code_lines(){
56     # Options:
57     #     --verbose
58     #     --work-directory=""
59     #     --max-line-length
60     declare -i LFBFL_i_verbose=0
61     get_verbose_option "$@"
62     local LFBFL_work_directory=""
63     get_work_directory_option "$@"
64     declare -a LFBFL_return_traps_stack
65     init_return_trap
66     pushd_to_work_directory --trap-popd
67     can_continue_after_enhanced_pushd || return 1
68
69     declare -i LFBFL_i_max_line_length
70     get_max_line_length_option "$@"
71
72     # shellcheck disable=SC2248
73     get_overlength_regexp "${LFBFL_i_max_line_length}"
74
75     enhanced_set_shell_option pipefail --trap-unset
76
77     get_COMMON_TEXT_FILES_GLOB_PATTERNS
78     local LFBFL_pattern
79     local LFBFL_long_line
80     local LFBFL_file_path
81     local LFBFL_line
82     local LFBFL_extension
83     local LFBFL_base_name
84     local LFBFL_s_long_lines
85     declare -a LFBFL_arr_long_lines
86     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
87         if [[ LFBFL_i_verbose -eq 1 ]]; then
88             printf "too_long_code_lines: Iterating on pattern: %s.\n" \
89                 "${LFBFL_pattern}"
90         fi
91         # shellcheck disable=SC2248
92         LFBFL_s_long_lines=$(
93             find . -type f -name "${LFBFL_pattern}" -printf '%P\n' \
94             | relevant_find \
95             | not_license_find \
96             | tr '\n' '\0' \
97             | xargs -0 grep --line-number --with-filename \

```

```

98         --regexp="${get_overlength_regexp_result}" --
99     )
100     if [[ -z "${LFBFL_s_long_lines}" ]]; then
101         [[ LFBFL_i_verbose -eq 1 ]]\
102         && printf "too_long_code_lines: No file found with long lines.\n"
103         continue
104     fi
105     mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
106     for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
107         LFBFL_file_path=${LFBFL_long_line%:*} # Drop long ':*' suffix
108         # Remove file_path and line_number
109         LFBFL_line=${LFBFL_long_line#*:} # Drop short '*: ' prefix
110         LFBFL_line=${LFBFL_line#*:} # Drop short '*: ' prefix
111         # LFBFL_line="${LFBFL_line/%\\\/}" # Drop/Replace '\\' suffix by ''
112         LFBFL_extension=${LFBFL_file_path##*.*} # Drop long '*.*' prefix
113         LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
114         if [[ "${LFBFL_extension}" == "html" ]]; then
115             if [[ -f "${LFBFL_base_name}.md" ]]; then
116                 continue
117             fi
118             if grep --quiet --fixed-strings -- "${LFBFL_line}"\
119                 "./build_and_checks_variables/temp/files_listing.html.sub"
120             then
121                 continue
122             fi
123             printf "%s\n" "${LFBFL_long_line}"
124             elif [[ "${LFBFL_extension}" == "md" ]]; then
125                 if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
126                     printf "%s\n" "${LFBFL_long_line}"
127                 fi
128             elif [[ "${LFBFL_extension}" == "neon" ]]; then
129                 if [[ ! -f "${LFBFL_base_name}.neon.tpl" ]]; then
130                     printf "%s\n" "${LFBFL_long_line}"
131                 fi
132             else
133                 printf "%s\n" "${LFBFL_long_line}"
134             fi
135         done
136     done
137 }

```

3.35 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #

```



```

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23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "update_common_build_and_checks.sh"
25 # to "update_common_build_and_checks.exec.sh".
26
27 cd .. || exit
28 # shellcheck source=overwrite_if_not_equal.lib.sh
29 source "../build_and_checks_dependencies/overwrite_if_not_equal.lib.sh"
30 cd build_and_checks_dependencies || exit
31
32 update_common_build_and_checks() {
33     # Option:
34     # --verbose
35     declare -i LFBFL_i_verbose=0
36     if [[ "$*" == *--verbose* ]]; then
37         printf "%s %s\n" "$0" "$*"
38         LFBFL_i_verbose=1
39     fi
40     readonly LFBFL_i_verbose
41
42     if [[ ! -o pipefail ]]; then
43         [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
44         set -o pipefail
45         trap 'set +o pipefail' RETURN
46     fi
47
48     declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
49     cp -- "${LFBFL_common_file_name}.tpl" \

```

```

50     "./${LFBFL_common_file_name}.temp"
51
52 declare -ar LFBFL_files_names=(
53     "build_dependencies_notes.exec.php"
54     "build_dependencies_notes.lib.r.php"
55     "build_file_from_printable_file__standard.exec.sh"
56     "build_md_from_printable_md.exec.sh"
57     "check_shell_scripts_beginnings.lib.r.sh"
58     "check_shell_scripts_indentation.lib.r.sh"
59     "check_URLs.lib.r.sh"
60     "common_options.lib.r.sh"
61     "comparisons.lib.r.sh"
62     "create_PDF.exec.sh"
63     "generate_from_template.lib.r.sh"
64     "get_common_text_glob_patterns.lib.r.sh"
65     "grammar_and_spelling_check.lib.r.sh"
66     "licenses_templates/build_licenses_templates.exec.sh"
67     "licenses_templates/license_file_header_AGPLv3+.tpl"
68     "licenses_templates/license_file_header_GPLv3+.tpl"
69     "licenses_templates/license_file_header_LGPLv3+.tpl"
70     "lines_counts.lib.r.sh"
71     "lines_filters.lib.r.sh"
72     "lines_maps.lib.r.sh"
73     "overwrite_if_not_equal.lib.r.sh"
74     "phpmd_baseline.xslt"
75     "python_black_complement.lib.r.sh"
76     "python_isort_complement.lib.r.sh"
77     "repository_data.lib.r.sh"
78     "shell_checks_complement.lib.r.sh"
79     "split_score.exec.php"
80     "split_score.lib.r.php"
81     "strings_functions.lib.r.sh"
82     "too_long_code_lines.lib.r.sh"
83     "update_or_check_files_names_listing.exec.sh"
84 )
85
86 local LFBFL_file_name
87 local LFBFL_file_sha512
88 local LFBFL_file_sha512_all
89 local LFBFL_base_file_name
90 for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
91     LFBFL_file_sha512=$(
92         sha512sum "./${LFBFL_file_name}" \
93         | cut --delimiter=' ' --fields=1
94     )
95     LFBFL_file_sha512_all="LFBFL_correct_sha512="

```

```

96     LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
97     LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
98     LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
99     LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
100    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}' "
101    LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
102    sed --in-place\
103        "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
104        "./${LFBFL_common_file_name}.temp"
105    done
106
107    overwrite_if_not_equal "./${LFBFL_common_file_name}"\
108        "./${LFBFL_common_file_name}.temp"
109
110    declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
111    cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"
112
113    LFBFL_file_sha512=$(
114        sha512sum "./${LFBFL_common_file_name}"\
115        | cut --delimiter=' ' --fields=1
116    )
117    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
118    LFBFL_file_sha512_all+="    LFBFL_correct_sha512="
119    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
120    LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
121    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
122    LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
123    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}' "
124    sed --in-place\
125        "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
126        "../${LFBFL_main_file_name}.temp"
127
128    overwrite_if_not_equal "../${LFBFL_main_file_name}"\
129        "../${LFBFL_main_file_name}.temp"
130 }
131
132 update_common_build_and_checks "$@"

```

3.36 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License

```

```

7 # as published by the Free Software Foundation,
8 # either version 3 of the License,
9 # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 # 2025-12-30T21:33+01:00: This file was moved from
26 # "./build_and_checks_dependencies/listings"
27 # to
28 # "./build_and_checks_dependencies".
29 # This file was renamed from "update_or_check_files_names_listing.sh"
30 # to "update_or_check_files_names_listing.exec.sh".
31 # This file was renamed from "update_or_check_file_names_listing.sh"
32 # to "update_or_check_files_names_listing.sh"?
33
34 LFBFL_subdir="build_and_checks_dependencies"
35 # shellcheck source=common_options.lib.sh
36 source "${LFBFL_subdir}/common_options.lib.sh"
37 # shellcheck source=lines_filters.lib.sh
38 source "${LFBFL_subdir}/lines_filters.lib.sh"
39 # shellcheck source=strings_functions.lib.sh
40 source "${LFBFL_subdir}/strings_functions.lib.sh"
41
42 update_or_check_files_names_listing() {
43     # Options:
44     #   --verbose
45     #   --work-directory=""
46     declare -i LFBFL_i_verbose=0
47     get_verbose_option "$@"
48     local LFBFL_work_directory=""
49     get_work_directory_option "$@"
50     declare -a LFBFL_return_traps_stack
51     init_return_trap
52     pushd_to_work_directory --trap-popd

```

```

53 can_continue_after_enhanced_pushd || return 1
54
55 declare -i LFBFL_i_write
56 get_some_flag LFBFL_i_write --write 1 "$@"
57 declare -i LFBFL_i_append
58 get_some_flag LFBFL_i_append --append 1 "$@"
59
60 enhanced_set_shell_option pipefail --trap-unset
61
62 LFBFL_subdir2="build_and_checks_variables"
63 declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
64 declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"
65
66 declare -i LFBFL_i_max_line_length
67 grep_variable "${LFBFL_data_file_path}" max_line_length\
68     --result-variable-prefix="LFBFL_i_"
69
70 if [[ LFBFL_i_write -eq 1 ]]; then
71     : > "${LFBFL_listing}"
72 fi
73 # Remove line returns that are here to keep lines short.
74 sed --regexp-extended --null-data --expression='s/(.)\\n/\\1/Mg'\
75     -- "${LFBFL_listing}"\
76     > "${LFBFL_listing}.temp"
77 # shopt -s dotglob was needed at some point but I don't see why now.
78 # shellcheck disable=SC2248
79 get_split_score_simple 1 ${LFBFL_i_max_line_length} /
80 declare -r LFBFL_split_score_command="${get_split_score_result}"
81 declare -i LFBFL_i_split_score_command_properties
82 LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
83 readonly LFBFL_i_split_score_command_properties
84 local LFBFL_suffix=\\ # instead of '\\' to avoid shellcheck SC1003
85 readonly LFBFL_suffix
86 # There is always a $'\n' final suffix from >>.
87 # But when origin string ends with the suffix we must add something.
88 declare -r LFBFL_final_suffix=$'\n\\'
89
90 local LFBFL_s_files_paths
91 LFBFL_s_files_paths=$(
92     find . -type f -printf '%P\n'\
93     | relevant_find\
94     | sort
95 )
96 # printf "%s\n" "${LFBFL_s_files_paths}"
97 declare -a LFBFL_arr_files_paths
98 if [[ -n "${LFBFL_s_files_paths}" ]]; then

```

```

99     if [[ LFBFL_i_verbose -eq 1 ]]; then
100         printf\
101             "update_or_check_files_names_listing: Some files to process.\n"
102     fi
103     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
104 fi
105 # printf "%s\n" "${LFBFL_arr_files_paths[@]}"
106
107 local LFBFL_file_path
108 local LFBFL_base_file_name
109 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
110     # if [[ LFBFL_i_verbose -eq 1 ]]; then
111     #     printf\
112     #         "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
113     #         "${LFBFL_file_path}"
114     # fi
115     git check-ignore --quiet -- "${LFBFL_file_path}" && continue
116     LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
117     [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
118     || continue
119     if [[ "${LFBFL_base_file_name}" == *.md ]]; then
120         if [[ -f "${LFBFL_file_path}.tpl" ]]; then
121             continue
122         fi
123     fi
124     if [[ LFBFL_i_verbose -eq 1 ]]; then
125         printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
126             "${LFBFL_file_path}"
127     fi
128     # shellcheck disable=SC2248
129     repeated_split_last_line "${LFBFL_file_path}"\
130         ""\
131         "${LFBFL_i_max_line_length}"\
132         "${LFBFL_suffix}"\
133         "${LFBFL_split_score_command}"\
134         "${LFBFL_i_split_score_command_properties}"\
135         ""\
136         "${LFBFL_final_suffix}"
137     if [[ LFBFL_i_write -eq 1 ]]; then
138         # shellcheck disable=SC2001
139         printf "%s\n" "${repeated_split_last_line_result}"\
140             >> "${LFBFL_listing}"
141         continue
142     fi
143     if ! grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
144         "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;

```

```

145     then
146         printf "The file %s is not listed in %s.\n"\
147             "${LFBFL_file_path}"\
148             "${LFBFL_listing}"
149         if [[ LFBFL_i_append -eq 1 ]]; then
150             # shellcheck disable=SC2001
151             printf "%s\n" "${repeated_split_last_line_result}"\
152                 >> "${LFBFL_listing}"
153         fi
154     fi
155 done
156
157 LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
158 if [[ -n "${LFBFL_s_files_paths}" ]]; then
159     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
160 fi
161
162 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
163     [[ "${LFBFL_file_path}" != '/'/* ]] || continue
164     if [[ ! -f "${LFBFL_file_path}" ]]; then
165         printf "The non-file %s is listed in %s.\n"\
166             "${LFBFL_file_path}"\
167             "${LFBFL_listing}"
168     fi
169 done
170
171 rm -- "${LFBFL_listing}.temp"
172
173 if [[ LFBFL_i_verbose -eq 1 ]]; then
174     cat -- "${LFBFL_listing}"
175 fi
176 }
177
178 update_or_check_files_names_listing "$@"

```

3.37 build_and_checks_variables/.gitignore

```

1 temp/
2 upgrade_venvs_ts

```

3.38 build_and_checks_variables/bandit.ini

```

1 # FILE: .bandit
2 [bandit]

```

3.39 build_and_checks_variables/bandit_baseline.json

```
1 {"results":[]}
```

3.40 build_and_checks_variables/black.toml

```
1 [tool.black]
2 line-length = 75
```

3.41 build_and_checks_variables/dependencies_data.json

```
1 [
2   { "dependency_name": "Bash", "URLs": [
3     "https://www.gnu.org/software/bash/",
4     "https://tiswww.case.edu/php/chet/bash/bashtop.html",
5     "https://savannah.gnu.org/projects/bash/"
6   ] },
7   { "dependency_name": "GNU coreutils", "URLs": [
8     "https://www.gnu.org/software/coreutils/",
9     "https://savannah.gnu.org/projects/coreutils/"
10  ] },
11  { "dependency_name": "utils coreutils", "URLs": [
12    "https://utils.github.io/coreutils/",
13    "https://github.com/utils/coreutils"
14  ] },
15  { "dependency_name": "diff (GNU diffutils)", "URLs": [
16    "https://www.gnu.org/software/diffutils/",
17    "https://savannah.gnu.org/projects/diffutils/"
18  ] },
19  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
20    "https://www.gnu.org/software/findutils/",
21    "https://savannah.gnu.org/projects/findutils/"
22  ] },
23  { "dependency_name": "grep (GNU grep)", "URLs": [
24    "https://www.gnu.org/software/grep/",
25    "https://savannah.gnu.org/projects/grep/"
26  ] },
27  { "dependency_name": "sed (GNU sed)", "URLs": [
28    "https://www.gnu.org/software/sed/",
29    "https://savannah.gnu.org/projects/sed/"
30  ] },
31  { "dependency_name": "ShellCheck", "URLs": [
32    "https://www.shellcheck.net/",
33    "https://github.com/koalaman/shellcheck"
34  ] },
35  { "dependency_name": "GNU Wget", "URLs": [
```



```

36         "https://www.gnu.org/software/wget/",
37         "https://savannah.gnu.org/projects/wget/"
38     ] },
39     { "dependency_name": "more of util-linux", "URLs": [
40         "https://www.kernel.org/pub/linux/utils/util-linux/",
41         "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42         {
43             "URL": "https://github.com/util-linux/util-linux",
44             "comment": "official?"
45         }
46     ] },
47     { "dependency_name": "pandoc", "URLs": [
48         "https://pandoc.org/",
49         "https://github.com/jgm/pandoc"
50     ] },
51     { "dependency_name": "pdftex/pdflatex", "URLs": [
52         "https://www.tug.org/applications/pdftex/",
53         "https://tug.org/texlive/svn/"
54     ] },
55     { "dependency_name": "tree", "URLs": [
56         "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57         "https://gitlab.com/OldManProgrammer/unix-tree",
58         "https://github.com/Old-Man-Programmer/tree"
59     ] },
60     { "dependency_name": "git", "URLs": [
61         "https://git-scm.com/about",
62         "https://github.com/git/git"
63     ] },
64     { "dependency_name": "pylint", "URLs": [
65         "https://www.pylint.org/",
66         "https://pylint.readthedocs.io/en/stable/",
67         "https://github.com/pylint-dev/pylint",
68         "https://pypi.org/project/pylint/"
69     ] },
70     { "dependency_name": "notify-send", "URLs": [
71         "https://gitlab.gnome.org/GNOME/libnotify/",
72         "https://gnome.pages.gitlab.gnome.org/libnotify/"
73     ] },
74     { "dependency_name": "black", "URLs": [
75         "https://github.com/psf/black",
76         "https://black.readthedocs.io/en/stable/",
77         "https://pypi.org/project/black/"
78     ] },
79     { "dependency_name": "isort", "URLs": [
80         "https://pycqa.github.io/isort/",
81         "https://github.com/pycqa/isort/"

```

```

82         "https://pypi.org/project/isort/"
83     ] },
84     { "dependency_name": "mypy", "URLs": [
85         "https://mypy-lang.org/",
86         "https://mypy.readthedocs.io/en/stable/",
87         "https://github.com/python/mypy",
88         "https://pypi.org/project/mypy/"
89     ] },
90     { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91         "https://phpdoc.org/",
92         "https://docs.phpdoc.org/",
93         "https://github.com/phpdocumentor/phpdocumentor"
94     ] },
95     { "dependency_name": "pip", "URLs": [
96         "https://pip.pypa.io/en/stable/",
97         "https://github.com/pypa/pip",
98         {
99             "URL": "https://pypi.org/project/pip/",
100             "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "xmllint (libxml2)", "URLs": [
109         "https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home",
110         "https://gitlab.gnome.org/GNOME/libxml2"
111     ] },
112     { "dependency_name": "bandit", "URLs": [
113         "https://bandit.readthedocs.io/en/latest/",
114         "https://github.com/PyCQA/bandit",
115         "https://pypi.org/project/bandit/"
116     ] },
117     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
118         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
119         "https://packagist.org/packages/squizlabs/php_codesniffer"
120     ] },
121     { "dependency_name": "phpmd", "URLs": [
122         "https://phpmd.org/",
123         "https://github.com/phpmd/phpmd",
124         "https://packagist.org/packages/phpmd/phpmd"
125     ] },
126     { "dependency_name": "php", "URLs": [
127         "https://www.php.net/",

```

```

128         "https://github.com/php/php-src"
129     ] },
130     { "dependency_name": "tex", "URLs": [
131         "https://ctan.org/tex-archive/systems/knuth/dist/tex"
132     ] },
133     { "dependency_name": "pcregrep, pcre2grep", "URLs": [
134         "https://www.pcre.org/",
135         "https://pcre2project.github.io/pcre2/",
136         "https://github.com/PCRE2Project/pcre1",
137         "https://github.com/PCRE2Project/pcre2",
138         "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
139     ] },
140     { "dependency_name": "python3", "URLs": [
141         "https://www.python.org/",
142         "https://docs.python.org/3/"
143     ] }
144 ]

```

3.42 build_and_checks_variables/dependencies_notes.txt

```

1 Bash can be found at the following URLs:
2 https://www.gnu.org/software/bash/
3 https://tiswww.case.edu/php/chet/bash/bashtop.html
4 https://savannah.gnu.org/projects/bash/
5
6 GNU coreutils can be found at the following URLs:
7 https://www.gnu.org/software/coreutils/
8 https://savannah.gnu.org/projects/coreutils/
9
10 utils coreutils can be found at the following URLs:
11 https://utils.github.io/coreutils/
12 https://github.com/utils/coreutils
13
14 diff (GNU diffutils) can be found at the following URLs:
15 https://www.gnu.org/software/diffutils/
16 https://savannah.gnu.org/projects/diffutils/
17
18 find, xargs (GNU findutils) can be found at the following URLs:
19 https://www.gnu.org/software/findutils/
20 https://savannah.gnu.org/projects/findutils/
21
22 grep (GNU grep) can be found at the following URLs:
23 https://www.gnu.org/software/grep/
24 https://savannah.gnu.org/projects/grep/
25
26 sed (GNU sed) can be found at the following URLs:

```

27 <https://www.gnu.org/software/sed/>
28 <https://savannah.gnu.org/projects/sed/>
29
30 ShellCheck can be found at the following URLs:
31 <https://www.shellcheck.net/>
32 <https://github.com/koalaman/shellcheck>
33
34 GNU Wget can be found at the following URLs:
35 <https://www.gnu.org/software/wget/>
36 <https://savannah.gnu.org/projects/wget/>
37
38 more of util-linux can be found at the following URLs:
39 <https://www.kernel.org/pub/linux/utils/util-linux/>
40 <https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/>
41 <https://github.com/util-linux/util-linux> official?
42
43 pandoc can be found at the following URLs:
44 <https://pandoc.org/>
45 <https://github.com/jgm/pandoc>
46
47 pdftex/pdflatex can be found at the following URLs:
48 <https://www.tug.org/applications/pdftex/>
49 <https://tug.org/texlive/svn/>
50
51 tree can be found at the following URLs:
52 <https://oldmanprogrammer.net/source.php?dir=projects/tree>
53 <https://gitlab.com/OldManProgrammer/unix-tree>
54 <https://github.com/Old-Man-Programmer/tree>
55
56 git can be found at the following URLs:
57 <https://git-scm.com/about>
58 <https://github.com/git/git>
59
60 pylint can be found at the following URLs:
61 <https://www.pylint.org/>
62 <https://pylint.readthedocs.io/en/stable/>
63 <https://github.com/pylint-dev/pylint>
64 <https://pypi.org/project/pylint/>
65
66 notify-send can be found at the following URLs:
67 <https://gitlab.gnome.org/GNOME/libnotify/>
68 <https://gnome.pages.gitlab.gnome.org/libnotify/>
69
70 black can be found at the following URLs:
71 <https://github.com/psf/black>
72 <https://black.readthedocs.io/en/stable/>

73 <https://pypi.org/project/black/>
74
75 isort can be found at the following URLs:
76 <https://pycqa.github.io/isort/>
77 <https://github.com/pycqa/isort/>
78 <https://pypi.org/project/isort/>
79
80 mypy can be found at the following URLs:
81 <https://mypy-lang.org/>
82 <https://mypy.readthedocs.io/en/stable/>
83 <https://github.com/python/mypy>
84 <https://pypi.org/project/mypy/>
85
86 phpdoc (phpDocumentor) can be found at the following URLs:
87 <https://phpdoc.org/>
88 <https://docs.phpdoc.org/>
89 <https://github.com/phpdocumentor/phpdocumentor>
90
91 pip can be found at the following URLs:
92 <https://pip.pypa.io/en/stable/>
93 <https://github.com/pypa/pip>
94 <https://pypi.org/project/pip/> (egg and chicken ;))
95
96 ruff can be found at the following URLs:
97 <https://docs.astral.sh/ruff/>
98 <https://github.com/astrol-sh/ruff>
99 <https://pypi.org/project/ruff/>
100
101 xmllint (libxml2) can be found at the following URLs:
102 <https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home>
103 <https://gitlab.gnome.org/GNOME/libxml2>
104
105 bandit can be found at the following URLs:
106 <https://bandit.readthedocs.io/en/latest/>
107 <https://github.com/PyCQA/bandit>
108 <https://pypi.org/project/bandit/>
109
110 phpcs (PHP_CodeSniffer) can be found at the following URLs:
111 https://github.com/PHPCSStandards/PHP_CodeSniffer
112 https://packagist.org/packages/squizlabs/php_codesniffer
113
114 phpmd can be found at the following URLs:
115 <https://phpmd.org/>
116 <https://github.com/phpmd/phpmd>
117 <https://packagist.org/packages/phpmd/phpmd>
118

```
119 php can be found at the following URLs:
120 https://www.php.net/
121 https://github.com/php/php-src
122
123 tex can be found at the following URL:
124 https://ctan.org/tex-archive/systems/knuth/dist/tex
125
126 pcregrep, pcre2grep can be found at the following URLs:
127 https://www.pcre.org/
128 https://pcre2project.github.io/pcre2/
129 https://github.com/PCRE2Project/pcre1
130 https://github.com/PCRE2Project/pcre2
131 https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c
132
133 python3 can be found at the following URLs:
134 https://www.python.org/
135 https://docs.python.org/3/
136
137 If you see something that I missed regarding my dependencies,
138 please tell me/email me :).
```

3.43 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <!--
4 This file is part of DevOrSysAdminScripts.
5
6 DevOrSysAdminScripts is free software:
7 you can redistribute it and/or modify it
8 under the terms of the GNU Lesser General Public License
9 as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
```

```

24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 -->
27 <head>
28 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
29 <meta name="generator" content="DevOrSysAdminScripts">
30 <style>
31 .title{
32     margin: 2ex auto;
33     text-align: center;
34 }
35 .titlemain{
36     margin: 1ex 2ex 2ex 1ex;
37 }
38 .titlerest{
39     font-size: 1.17em;
40     margin: 0ex 2ex;
41 }
42 table{
43     border-collapse: collapse;
44 }
45 td{
46     padding: 0;
47 }
48 pre{
49     text-align: left;
50     margin-left: 0ex;
51     margin-right: auto;
52 }
53 pre.numbered_lines:before{
54     background: snow;
55     border-right: 1px black solid;
56     color: #010101;
57     content: attr(line-numbers);
58     float: left;
59     padding: 0 1ch;
60     text-align: right;
61 }
62 blockquote{
63     margin-left: 4ex;
64     margin-right: 4ex;
65     text-align: left;
66 }
67 ol.level1, ol.level2{
68     list-style-type: none;
69     /*

```

```

70     list-style-type: decimal;
71     Previous rule is shorter and more efficient if you need only level1.
72     But you don't have an accessible counter if you use decimal.
73     */
74 }
75 ol.level1{
76     counter-reset: level1;
77 }
78 ol.level2{
79     counter-reset: level2;
80 }
81 ol.level1 li:before{
82     content: counter(level1) ". ";
83     counter-increment: level1;
84 }
85 ol.level2 li:before{
86     content: counter(level1) "." counter(level2) " ";
87     counter-increment: level2;
88 }
89 </style>
90 <title>@repository_name@</title>
91 </head>
92 <body >
93 <table class="title">
94     <tr>
95         <td style="padding:1ex">
96             <h1 class="titlemain">@repository_name@</h1>
97             <h2 class="titlerest">
98                 @author_full_name@
99                 <br>
100                 <a href="@author_website@">
101                     <span style="font-family:monospace">
102                         @author_website@
103                     </span>
104                 </a>
105                 <br>
106                 <a href="mailto:@author_email@">
107                     <span style="font-family:monospace">
108                         @author_email@
109                     </span>
110                 </a>
111             </h2>
112         </td>
113     </tr>
114 </table>
115

```



```

116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
123 <p>
124 Code lines: @number_of_lines@
125 </p>
126
127
128 <h2>Contents</h2>
129
130 <ol class="level1">
131   <li><a href="#section1">Files trees</a></li>
132   <li><a href="#section2">Notes on licenses and dependencies</a></li>
133   <li><a href="#section3">Listing of files</a><br>
134     <ol class="level2">
135       @files_lis@
136     </ol>
137   </li>
138 </ol>
139
140
141 <h2 id="section1">1 Files trees</h2>
142 <pre class="verbatim">
143 @current_tree_light@
144 </pre>
145 <pre class="verbatim">
146 @current_tree@
147 </pre>
148
149
150 <h2 id="section2">2 Notes on licenses and dependencies</h2>
151
152 <p>
153 This repository is dual licensed.
154 All the Bash scripts source code is covered by GPLv3+.
155 All the PHP scripts source code is covered by LGPLv3+.
156 All the Python scripts source code is covered by LGPLv3+.
157 </p>
158
159 <p>
160 The text of the licenses is available at:
161 <a href="https://www.gnu.org/licenses/"><span>

```

```

162     style="font-family:monospace"
163 >https://www.gnu.org/licenses/</span></a>.
164 The git repository of this source code is also available at:
165 <a href="@repository_web_url@"><span
166     style="font-family:monospace"
167 >@repository_web_url@</span></a>.
168 Clone it with this URL:
169 <span style="font-family:monospace"
170 >@repository_git_url@</span>.
171 </p>
172
173 <div>
174 <p>
175 Here is a listing of the dependencies of the code in this repository:
176 </p>
177 <ul>
178 <li> The following dependencies are GPLv3+ licensed:<br>
179 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
180 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
181 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
182 <li> The following dependencies are GPLv2+ licensed:<br>
183 more, pandoc, pdftex, pdflatex, tree;
184 <li> The following dependencies are GPLv2 licensed:<br>
185 git, pylint;
186 <li> The following dependencies are LGPLv2.1+ licensed:<br>
187 notify-send;
188 <li> The following dependencies are MIT licensed:<br>
189 black, isort, mypy, phpdoc, pip, ruff, xmllint;
190 <li> The following dependencies are Apache-2.0 licensed:<br>
191 bandit;
192 <li> The following dependencies are BSD-3-Clause licensed:<br>
193 phpcs, phpmd;
194 <li> php is licensed under PHP License v3.01;
195 <li> tex is licensed under Knuth license;
196 <li> pcregrep, pcre2grep has a custom license from Cambridge
197 University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
198 <li> python3 is licensed under PSF License Version 2 and the Zero-Clause
199 BSD license.
200 </ul>
201 </div>
202
203 <p>
204 Remark: Some commands listed under GPLv3+ are POSIX
205 and can have different implementations under another license;
206 for example, utils coreutils is MIT licensed,
207 whilst GNU coreutils is GPLv3+ licensed.

```

```

208 </p>
209
210 @dependencies_notes@
211
212
213 <h2 id="section3">3 Listing of files</h2>
214
215 @files_listing_HTMLPreInput@
216
217 <p>
218 @acknowledgments@
219 </p>
220
221 <script>
222 "use strict";
223 const pres = Array.from(
224   document.getElementsByClassName('numbered_lines')
225 );
226 pres.forEach(
227   (pre) => {
228     const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
229     let numbers = "";
230     for(let i = 1; i <= nb; ++i){
231       numbers += i.toString();
232       if(i < nb){ numbers += "\n "; }
233     }
234     pre.setAttribute('line-numbers', numbers);
235   }
236 );
237 </script>
238 </html>

```

3.44 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```

1  % This file is part of DevOrSysAdminScripts.
2  %
3  % DevOrSysAdminScripts is free software:
4  % you can redistribute it and/or modify it
5  % under the terms of the GNU Lesser General Public License
6  % as published by the Free Software Foundation,
7  % either version 3 of the License,
8  % or (at your option) any later version.
9  %
10 % DevOrSysAdminScripts is distributed
11 % in the hope that it will be useful,
12 % but WITHOUT ANY WARRANTY;

```

```

13 % without even the implied warranty of
14 % MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 % See the GNU Lesser General Public License for more details.
16 %
17 % You should have received a copy of
18 % the GNU Lesser General Public License
19 % along with DevOrSysAdminScripts.
20 % If not, see <https://www.gnu.org/licenses/>.
21 %
22 % ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
23
24 % 2025-12-29T17:08+01:00: This file was moved from
25 % "./latex"
26 % to
27 % "./build_and_checks_variables".
28
29 \documentclass{article}
30
31 \usepackage[utf8]{inputenc}
32 \usepackage[T1]{fontenc}
33 \usepackage{subfigure}
34 \usepackage{amsmath}
35 \usepackage{amssymb}
36 \usepackage{amsthm}
37 \usepackage[pdftex]{hyperref}
38 \usepackage{tikz}
39 \usepackage{caption}
40 \usepackage[round]{natbib}
41 \usepackage{fancyhdr}
42 \usepackage{amsfonts}
43 \usepackage{times}
44 \usepackage{ifpdf}
45 \usepackage{latexsym}
46 \usepackage{graphicx}
47 \usepackage{enumerate}
48 \usepackage{pmbboxdraw}
49 \usepackage{fancyvrb}
50 \usepackage[french,english]{babel}
51
52 % *** les environnements ***
53 %\theoremstyle{break}
54 \newtheorem{definition}{Definition}[section]
55 \newtheorem{proposition}[definition]{Proposition}
56 \newtheorem{theorem}[definition]{Theorem}
57 \newtheorem{lemma}[definition]{Lemma}
58 \newtheorem{corollary}[definition]{Corollary}

```

```

59 \newtheorem{remark}[definition]{Remark}
60 \newtheorem{openproblem}[definition]{Open problem}
61
62 \begin{document}
63
64 \author{
65     @author_full_name@\\
66     \url{@author_website@}\\
67     \href{mailto:@author_email@}{@author_email@}
68 }
69 \title{@repository_name@}
70
71 \maketitle
72 \begin{abstract}
73 @abstract@
74 \end{abstract}
75
76 Current version: @current_date@
77
78 Current number of commits: @number_of_commits@
79
80 Current git SHA1: @current_git_SHA1@
81
82 Code lines: @number_of_lines@
83
84 \tableofcontents
85 \newpage
86
87 \section{Files trees}
88 \label{section:trees}
89
90 \begin{verbatim}
91 @current_tree_light@
92 \end{verbatim}
93
94 \begin{verbatim}
95 @current_tree@
96 \end{verbatim}
97
98 \section{Notes on licenses and dependencies}
99 \label{section:notes_on}
100
101 This repository is dual licensed.
102 All the Bash scripts source code is covered by GPLv3+.
103 All the PHP scripts source code is covered by LGPLv3+.
104 All the Python scripts source code is covered by LGPLv3+.

```

```

105
106 The text of the licenses is available at:
107 \url{https://www.gnu.org/licenses/}.
108 The git repository of this source code is also available at:
109 \url{@repository_web_url@}.
110 Clone it with this URL:
111 \url{@repository_git_url@}.
112
113 Here is a listing of the dependencies of the code in this repository:
114 \begin{itemize}
115 \item The following dependencies are GPLv3+ licensed:\newline
116 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
117 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
118 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
119 \item The following dependencies are GPLv2+ licensed:\newline
120 more, pandoc, pdftex, pdflatex, tree;
121 \item The following dependencies are GPLv2 licensed:\newline
122 git, pylint;
123 \item The following dependencies are LGPLv2.1+ licensed:\newline
124 notify-send;
125 \item The following dependencies are MIT licensed:\newline
126 black, isort, mypy, phpdoc, pip, ruff, xmllint;
127 \item The following dependencies are Apache-2.0 licensed:\newline
128 bandit;
129 \item The following dependencies are BSD-3-Clause licensed:\newline
130 phpcs, phpmo;
131 \item php is licensed under PHP License v3.01;
132 \item tex is licensed under Knuth license;
133 \item pcregrep, pcre2grep has a custom license from Cambridge
134 University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
135 \item python3 is licensed under PSF License Version 2 and the Zero-Clause
136 BSD license.
137 \end{itemize}
138
139 Remark: Some commands listed under GPLv3+ are POSIX
140 and can have different implementations under another license;
141 for example, utils coreutils is MIT licensed,
142 whilst GNU coreutils is GPLv3+ licensed.
143
144 @dependencies_notes@
145
146 \section{Listing of files}
147 \label{section:listing}
148
149 @files_listing_VerbatimInput@
150

```

```

151 @acknowledgments@
152
153 \end{document}

```

3.45 build_and_checks_variables/files_names_listing.txt

```

1 // This file is part of DevOrSysAdminScripts.
2 //
3 // DevOrSysAdminScripts is free software:
4 // you can redistribute it and/or modify it under the terms
5 // of the GNU Lesser General Public License
6 // as published by the Free Software Foundation,
7 // either version 3 of the License,
8 // or (at your option) any later version.
9 //
10 // DevOrSysAdminScripts is distributed in the hope
11 // that it will be useful,
12 // but WITHOUT ANY WARRANTY;
13 // without even the implied warranty of
14 // MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 // See the GNU Lesser General Public License for more details.
16 //
17 // You should have received a copy of
18 // the GNU Lesser General Public License
19 // along with DevOrSysAdminScripts.
20 // If not, see <https://www.gnu.org/licenses/>.
21 //
22 // ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
23 // 2024-10-31T21:37+01:00: This file was moved from
24 // "../build_and_checks_dependencies/listings"
25 // to
26 // "../build_and_checks_variables".
27 archive_libre_office_documents.exec.php
28 // Blasé20240517Résultat.txt
29 // Blasé20240517.sh
30 // Blasé20240518_2.txt
31 // Blasé20240518_3.txt
32 build_and_checks_dependencies/build_and_checks.exec.sh.tpl
33 build_and_checks_dependencies/build_dependencies_notes.exec.php
34 build_and_checks_dependencies/build_dependencies_notes.lib.php
35 build_and_checks_dependencies/\
36 build_file_from_printable_file__standard.exec.sh
37 build_and_checks_dependencies/build_md_from_printable_md.exec.sh
38 build_and_checks_dependencies/check_shell_scripts_beginnings.lib.sh
39 build_and_checks_dependencies/check_shell_scripts_indentation.lib.sh
40 build_and_checks_dependencies/check_URLs.lib.sh

```

```

41 build_and_checks_dependencies/common_build_and_checks.exec.sh
42 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
43 build_and_checks_dependencies/common_options.lib.sh
44 build_and_checks_dependencies/comparisons.lib.sh
45 build_and_checks_dependencies/create_PDF.exec.sh
46 build_and_checks_dependencies/generate_from_template.lib.sh
47 build_and_checks_dependencies/get_common_text_glob_patterns.lib.sh
48 build_and_checks_dependencies/grammar_and_spelling_check.lib.sh
49 build_and_checks_dependencies/licenses_templates/\
50 build_licenses_templates.exec.sh
51 build_and_checks_dependencies/licenses_templates/\
52 license_file_header_AGPLv3+.tpl
53 build_and_checks_dependencies/licenses_templates/\
54 license_file_header_GPLv3+.tpl
55 build_and_checks_dependencies/licenses_templates/\
56 license_file_header_LGPLv3+.tpl
57 build_and_checks_dependencies/lines_counts.lib.sh
58 build_and_checks_dependencies/lines_filters.lib.sh
59 build_and_checks_dependencies/lines_maps.lib.sh
60 // build_and_checks_dependencies/listings/files_listing.tex.tpl
61 build_and_checks_dependencies/overwrite_if_not_equal.lib.sh
62 build_and_checks_dependencies/phpmd_baseline.xslt
63 build_and_checks_dependencies/python_black_complement.lib.sh
64 build_and_checks_dependencies/python_isort_complement.lib.sh
65 build_and_checks_dependencies/repository_data.lib.sh
66 build_and_checks_dependencies/shell_checks_complement.lib.sh
67 build_and_checks_dependencies/split_score.exec.php
68 build_and_checks_dependencies/split_score.lib.php
69 build_and_checks_dependencies/strings_functions.lib.sh
70 build_and_checks_dependencies/too_long_code_lines.lib.sh
71 build_and_checks_dependencies/update_common_build_and_checks.exec.sh
72 build_and_checks_dependencies/\
73 update_or_check_files_names_listing.exec.sh
74 build_and_checks_variables/.gitignore
75 build_and_checks_variables/bandit.ini
76 build_and_checks_variables/bandit_baseline.json
77 build_and_checks_variables/black.toml
78 build_and_checks_variables/dependencies_data.json
79 build_and_checks_variables/dependencies_notes.txt
80 build_and_checks_variables/DevOrSysAdminScripts.html.tpl
81 // build_and_checks_variables/DevOrSysAdminScripts.tex
82 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
83 build_and_checks_variables/files_names_listing.txt
84 build_and_checks_variables/licenses_notes.txt
85 build_and_checks_variables/phpcs_ruleset.xml
86 build_and_checks_variables/phpmd_baseline.xml

```



```

87 // build_and_checks_variables/phpstan_baseline.neon
88 build_and_checks_variables/phpstan_baseline.neon.tpl
89 // build_and_checks_variables/phpstan_config.neon
90 build_and_checks_variables/phpstan_config.neon.tpl
91 build_and_checks_variables/pylintrc.toml
92 build_and_checks_variables/repository_data.txt
93 build_and_checks_variables/repository_data.txt.tpl
94 build_and_checks_variables/ruff.toml
95 build_and_checks_variables/shellcheck.ini
96 build_and_checks_variables/shellcheck_notes.txt
97 // build_and_checks_variables/upgrade_venvs_ts
98 build_and_checks.exec.sh
99 COPYING
100 COPYING.LESSER
101 // DevOrSysAdminScripts.html
102 // DevOrSysAdminScripts.pdf
103 // EncoreBlasé20240517-20240518.txt
104 files.lib.php
105 git_helpers/git_checks.lib.sh
106 git_helpers/post-commit
107 git_helpers/post_commit_renaming_comment.lib.sh
108 git_helpers/pre-commit
109 IMAP_transfert.exec.py
110 LiensUtiles.txt
111 // Uncomment some files in miscellaneous to do some testing.
112 // miscellaneous/\
113 a_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
114 e_that_it_works.txt
115 miscellaneous/funny.sh
116 miscellaneous/grep_benchmark.sh
117 miscellaneous/spamty_data.txt
118 miscellaneous/string_escaping.html
119 miscellaneous/string_escaping.xhtml
120 // miscellaneous/\
121 yet_another_long_long_long_long_long_long_long_long_filename_so_that_\
122 you_can_see_that_it_works.txt
123 // README.html
124 // README.md
125 README.md.tpl
126 string_escaping.lib.php
127 unicode.lib.php
128 unicode_exceptions.lib.php
129 upgrade_build_and_checks.exec.sh
130 wget_sha512.lib.sh

```

3.46 build_and_checks_variables/licenses_notes.txt

```
1 This repository is dual licensed.
2 All the Bash scripts source code is covered by GPLv3+.
3 All the PHP scripts source code is covered by LGPLv3+.
4 All the Python scripts source code is covered by LGPLv3+.
5
6 Here is a listing of the dependencies of the code in this repository:
7 - The following dependencies are GPLv3+ licensed:
8 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
9 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
10 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
11 - The following dependencies are GPLv2+ licensed:
12 more, pandoc, pdftex, pdflatex, tree;
13 - The following dependencies are GPLv2 licensed:
14 git, pylint;
15 - The following dependencies are LGPLv2.1+ licensed:
16 notify-send;
17 - The following dependencies are MIT licensed:
18 black, isort, mypy, phpdoc, pip, ruff, xmllint;
19 - The following dependencies are Apache-2.0 licensed:
20 bandit;
21 - The following dependencies are BSD-3-Clause licensed:
22 phpcs, phpmid;
23 - php is licensed under PHP License v3.01;
24 - tex is licensed under Knuth license;
25 - pcregrep, pcre2grep has a custom license from Cambridge University,
26 but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
27 - python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
28 license.
29
30 Remark: Some commands listed under GPLv3+ are POSIX
31 and can have different implementations under another license;
32 for example, utils coreutils is MIT licensed,
33 whilst GNU coreutils is GPLv3+ licensed.
```

3.47 build_and_checks_variables/phpcs_ruleset.xml

```
1 <?xml version="1.0"?>
2 <ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
3   <description>A custom coding standard</description>
4   <config name="tab_width" value="2"/>
5
6   <file>../</file>
7
8   <arg name="extensions" value="php"/>
```

```

9      <arg name="report" value="summary"/>
10     <arg name="colors"/>
11     <arg value="sp"/>
12
13     <!--
14     I added all the standards, then excluded all the rules that gave me
15     false positives or that seemed inappropriate when looking at the
16     documentation.
17     Some are bugs, some are conflicting with my own coding style.
18     You may delete an exclude and see what pops up to maybe understand
19     my choices.
20     -->
21
22     <rule ref="Generic">
23         <!-- You have to allow long or short or both syntaxes. -->
24         <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
25         <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
26         <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
27         <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
28         <!-- Back to my habits at Teliae with always a closing tag. -->
29         <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
30         <!--
31         Not standard in Unix/Linux world to keep files not ending with \n.
32         This is always an option of the code editors.
33         Since PHP doesn't output the final \n after ?>, it is ok.
34         -->
35         <exclude name="Generic.Files.EndFileNoNewline.Found"/>
36         <exclude name="Generic.Files.ExecutableFile.Executable"/>
37         <exclude name="Generic.Files.OneClassPerFile.MultipleFound"/>
38         <exclude name="Generic.Files.OneObjectStructurePerFile.MultipleFound"/>
39         <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
40         <exclude
41             name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
42             />
43         <exclude
44             name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
45             />
46         <exclude
47             name="Generic.NamingConventions.CamelCapsFunctionName.ScopeNotCamelCaps"
48             />
49         <!-- Applies only to true, false and null. -->
50         <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
51         <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
52         <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
53         <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
54     </rule>

```

```

55 <rule ref="PEAR">
56   <exclude name="PEAR.Classes.ClassDeclaration.OpenBraceNewLine"/>
57   <exclude name="PEAR.Commenting.ClassComment.MissingLinkTag"/>
58   <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
59   <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
60   <!--
61   Look at the comment before generate_split_score.
62   PEAR and Squiz require to align the parameter names parts in the same
63   comment, and also the parameter comments parts in the same comment,
64   but with a long parameter name like $a_delimiter_strings_domain
65   and a line length limit at 75 characters, parameter comments parts
66   have only 24 characters left per line, which is ridiculous.
67   Our choice to put the parameter comments parts starting on another line
68   with same "indent" as parameter names parts is way better in those
69   cases. Alignment has always been a trap with short lines.
70   Since phpDocumentor handles the parameter comments parts correctly,
71   it is enough for me.
72   Worse is the handling of "nested doc block comment" for the Closure
73   returned by generate_split_score().
74   Our solution makes at least the source code readable.
75   Maybe one day the dev tools will handle it correctly.
76   -->
77   <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
78   <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
79   <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
80   <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
81   <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
82   <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
83   <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
84   <exclude
85     name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
86   />
87   <exclude
88     name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
89   />
90   <exclude
91     name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
92   />
93   <!--
94   Don't laugh when looking at the false positives of the two exclude
95   below ;) XD.
96   They don't check for multiline function declarations apparently.
97   -->
98   <exclude
99     name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
100  />

```

```

101     <exclude
102         name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
103     />
104     <exclude
105         name="PEAR.Functions.FunctionDeclaration.SpaceBeforeOpenBrace"
106     />
107     <exclude
108         name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
109     />
110     <exclude
111         name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
112     />
113     <exclude
114         name="PEAR.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
115     />
116     <!--
117     You have to choose between the following exclude or another one
118     for PSR2.
119     -->
120     <exclude
121         name="PEAR.NamingConventions.ValidVariableName.PrivateNoUnderscore"
122     />
123 </rule>
124 <rule ref="PSR1">
125     <exclude name="PSR1.Classes.ClassDeclaration.MultipleClasses"/>
126     <exclude
127         name="PSR1.Methods.CamelCapsMethodName.NotCamelCaps"
128     />
129 </rule>
130 <rule ref="PSR2">
131     <exclude name="PSR2.Classes.ClassDeclaration.OpenBraceNewLine"/>
132     <!-- Back to my habits at Teliae with always a closing tag. -->
133     <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
134     <exclude
135 name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
136     />
137 </rule>
138 <rule ref="PSR12">
139     <exclude
140 name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
141     />
142     <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
143     <!--
144     3 false positives, see:
145     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
146     -->

```

```

147 <exclude
148     name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
149 />
150 <exclude
151     name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
152 />
153 <exclude
154     name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
155 />
156 <!-- and even a fourth and fifth false positives -->
157 <exclude
158     name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
159 />
160 <exclude
161     name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
162 />
163 <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
164 <exclude
165     name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
166 />
167 <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
168 <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
169 <!-- I agree except for string concatenation. -->
170 <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
171 <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
172 </rule>
173 <rule ref="Squiz">
174     <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceAfterBracket"/>
175     <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceBeforeBracket"/>
176     <exclude name="Squiz.Arrays.ArrayDeclaration.DoubleArrowNotAligned"/>
177     <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
178     <exclude name="Squiz.Classes.ClassDeclaration.MultipleClasses"/>
179     <exclude
180         name="Squiz.Classes.ClassDeclaration.NewlinesAfterCloseBrace"
181     />
182     <exclude name="Squiz.Classes.ClassDeclaration.OpenBraceNewLine"/>
183     <exclude name="Squiz.Classes.ClassDeclaration.SpaceBeforeCloseBrace"/>
184     <exclude name="Squiz.Classes.ClassFileName.NoMatch"/>
185     <exclude name="Squiz.Classes.ValidClassName.NotPascalCase"/>
186     <!--
187     It doesn't know the //*/ trick...
188     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
189     -->
190     <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
191     <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
192     <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>

```

```

193 <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
194 <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
195 <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
196 <exclude name="Squiz.Commenting.BlockComment.SingleLine"/>
197 <!-- Problem for doc block comment before namespace -->
198 <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
199 <exclude name="Squiz.Commenting.ClassComment.TagNotAllowed"/>
200 <exclude name="Squiz.Commenting.ClosingDeclarationComment.Incorrect"/>
201 <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
202 <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
203 <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
204 <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
205 <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
206 <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
207 <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
208 <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
209 <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
210 <exclude name="Squiz.Commenting.FunctionComment.EmptyThrows"/>
211 <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
212 <!-- asks for boolean instead of bool which is incorrect -->
213 <exclude
214     name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
215 />
216 <exclude name="Squiz.Commenting.FunctionComment.IncorrectTypeHint"/>
217 <!-- asks for boolean instead of bool which is incorrect -->
218 <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
219 <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
220 <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
221 <exclude
222     name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
223 />
224 <exclude
225     name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
226 />
227 <exclude name="Squiz.Commenting.FunctionCommentThrowTag.WrongNumber"/>
228 <!-- Problem for doc block comment before namespace -->
229 <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
230 <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
231 <!-- It doesn't know the //*/ trick... -->
232 <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
233 <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
234 <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
235 <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
236 <!-- asks for boolean instead of bool which is incorrect -->
237 <exclude name="Squiz.Commenting.VariableComment.IncorrectVarType"/>
238 <exclude

```

```

239         name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
240     />
241     <exclude
242 name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
243     />
244     <exclude
245         name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
246     />
247     <exclude
248         name="Squiz.ControlStructures.ForEachLoopDeclaration.SpacingBeforeAs"
249     />
250     <exclude
251         name="Squiz.ControlStructures.InlineIfDeclaration.NotSingleLine"
252     />
253     <exclude name="Squiz.Files.FileExtension.ClassFound"/>
254     <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
255     <exclude name="Squiz.Functions.GlobalFunction.Found"/>
256     <exclude
257         name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
258     />
259     <exclude
260 name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
261     />
262     <exclude
263 name="Squiz.Functions.MultiLineFunctionDeclaration.SpaceBeforeOpenBrace"
264     />
265     <exclude
266         name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
267     />
268     <exclude
269         name="Squiz.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
270     />
271     <exclude
272         name="Squiz.NamingConventions.ValidVariableName.MemberNotCamelCaps"
273     />
274     <exclude
275         name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
276     />
277     <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
278     <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
279     <exclude name="Squiz.PHP.CommentedOutCode"/>
280     <exclude name="Squiz.PHP.DisallowBooleanStatement.Found"/>
281     <exclude
282         name="Squiz.PHP.DisallowComparisonAssignment.AssignedComparison"
283     />
284     <exclude name="Squiz.PHP.DisallowInlineIf.Found"/>

```



```

285     <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
286     <exclude
287         name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
288     />
289     <exclude name="Squiz.WhiteSpace.FunctionSpacing"/>
290     <exclude name="Squiz.WhiteSpace.MemberVarSpacing.FirstIncorrect"/>
291     <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
292     <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
293     <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
294
295 </rule>
296 <rule ref="Zend">
297     <!-- Back to my habits at Teliae with always a closing tag. -->
298     <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
299     <exclude
300         name="Zend.NamingConventions.ValidVariableName.MemberVarNotCamelCaps"
301     />
302     <exclude
303         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
304     />
305     <exclude
306         name="Zend.NamingConventions.ValidVariableName.PrivateNoUnderscore"
307     />
308 </rule>
309
310 <rule ref="Generic.Arrays.ArrayIndent">
311     <properties>
312         <property name="indent" value="2" />
313     </properties>
314 </rule>
315 <rule ref="Generic.Files.LineLength">
316     <properties>
317         <property name="lineLimit" value="75"/>
318         <property name="absoluteLineLimit" value="75"/>
319     </properties>
320 </rule>
321 <rule ref="Generic.Formatting.SpaceAfterNot">
322     <properties>
323         <property name="spacing" value="0" />
324     </properties>
325 </rule>
326 <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
327     <properties>
328         <property name="checkFunctions" value="true"/>
329         <property name="checkClosures" value="true"/>
330     </properties>

```

```

331 </rule>
332 <rule ref="Generic.Metrics.CyclomaticComplexity">
333   <properties>
334     <property name="complexity" value="40" />
335     <property name="absoluteComplexity" value="80" />
336   </properties>
337 </rule>
338 <rule ref="Generic.Metrics.NestingLevel">
339   <properties>
340     <property name="nestingLevel" value="7" />
341     <property name="absoluteNestingLevel" value="10" />
342   </properties>
343 </rule>
344 <rule ref="Generic.Strings.UnnecessaryStringConcat">
345   <properties>
346     <property name="allowMultiline" value="true" />
347   </properties>
348 </rule>
349 <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
350   <properties>
351     <property name="ignoreNewlines" value="true" />
352   </properties>
353 </rule>
354 <rule ref="Generic.WhiteSpace.ScopeIndent">
355   <properties>
356     <property name="ignoreIndentationTokens" type="array">
357       <element value="T_COMMENT"/>
358       <element value="T_DOC_COMMENT_OPEN_TAG"/>
359     </property>
360     <property name="indent" value="2"/>
361   </properties>
362 </rule>
363
364 <rule ref="PEAR.Functions.FunctionCallSignature">
365   <properties>
366     <property name="indent" value="2"/>
367   </properties>
368 </rule>
369 <rule ref="PEAR.Functions.FunctionDeclaration">
370   <properties>
371     <property name="indent" value="2"/>
372   </properties>
373 </rule>
374 <rule ref="PEAR.WhiteSpace.ScopeIndent">
375   <properties>
376     <property name="ignoreIndentationTokens" type="array">

```

```

377         <element value="T_COMMENT"/>
378         <element value="T_DOC_COMMENT_OPEN_TAG"/>
379     </property>
380     <property name="indent" value="2"/>
381 </properties>
382 </rule>
383
384 <rule ref="PSR2.Methods.FunctionCallSignature">
385     <properties>
386         <property name="indent" value="2"/>
387     </properties>
388 </rule>
389
390 <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
391     <properties>
392         <property name="indent" value="2"/>
393     </properties>
394 </rule>
395
396 <rule ref="Squiz.Classes.ClassDeclaration">
397     <properties>
398         <property name="indent" value="2" />
399     </properties>
400 </rule>
401 <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
402     <properties>
403         <property name="indent" value="2"/>
404     </properties>
405 </rule>
406 <rule ref="Squiz.Strings.ConcatenationSpacing">
407     <properties>
408         <property name="ignoreNewlines" value="true" />
409     </properties>
410 </rule>
411 <rule ref="Squiz.WhiteSpace.FunctionSpacing">
412     <properties>
413         <property name="spacing" value="3" />
414         <property name="spacingBeforeFirst" value="0" />
415         <property name="spacingAfterLast" value="0" />
416     </properties>
417 </rule>
418 </ruleset>

```

3.48 build_and_checks_variables/phpmd_baseline.xml

```
1 <?xml version="1.0"?>
2 <phpmd-baseline>
3   <violation
4     rule="PHPMD\Rule\CleanCode\ElseExpression"
5     file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
6   />
7   <violation
8     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
9     file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
10  />
11  <violation
12    rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
13    file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
14  />
15  <violation
16    rule="PHPMD\Rule\CyclomaticComplexity"
17    file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
18  />
19  <violation
20    rule="PHPMD\Rule\Design\LongMethod"
21    file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
22  />
23  <violation
24    rule="PHPMD\Rule\Design\NpathComplexity"
25    file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
26  />
27  <violation
28    rule="PHPMD\Rule\Naming\LongVariable"
29    file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
30  />
31  <violation
32    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
33    file="build_and_checks_dependencies/split_score.lib.php"
34  />
35  <violation
36    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
37    file="build_and_checks_dependencies/split_score.lib.php"
38    method="__construct"
39  />
40  <violation
41    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
42    file="build_and_checks_dependencies/split_score.lib.php"
43    method="__invoke"
44  />
```

```

45 <violation
46     rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
47     file="build_and_checks_dependencies/split_score.lib.php"
48 />
49 <violation
50     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
51     file="build_and_checks_dependencies/split_score.lib.php"
52 />
53 <violation
54     rule="PHPMD\Rule\Naming\LongVariable"
55     file="build_and_checks_dependencies/split_score.lib.php"
56 />
57 <violation
58     rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
59     file="files.lib.php"
60 />
61 <violation
62     rule="PHPMD\Rule\CleanCode\ElseExpression"
63     file="files.lib.php"
64 />
65 <violation
66     rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
67     file="files.lib.php"
68     method="add_file_data"
69 />
70 <violation
71     rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
72     file="files.lib.php"
73     method="add_file_data_for_file_path"
74 />
75 <violation
76     rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
77     file="files.lib.php"
78     method="get_s_content"
79 />
80 <violation
81     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
82     file="files.lib.php"
83 />
84 <violation
85     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
86     file="files.lib.php"
87     method="__construct"
88 />
89 <violation
90     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"

```

```

91     file="files.lib.php"
92     method="add_file_data"
93 />
94 <violation
95     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
96     file="files.lib.php"
97     method="add_file_data_for_file_path"
98 />
99 <violation
100     rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
101     file="files.lib.php"
102 />
103 <violation
104     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
105     file="files.lib.php"
106 />
107 <violation
108     rule="PHPMD\Rule\CyclomaticComplexity"
109     file="files.lib.php"
110 />
111 <violation
112     rule="PHPMD\Rule\Design\LongMethod"
113     file="files.lib.php"
114 />
115 <violation
116     rule="PHPMD\Rule\Design\NpathComplexity"
117     file="files.lib.php"
118 />
119 <violation
120     rule="PHPMD\Rule\Naming\LongVariable"
121     file="files.lib.php"
122 />
123 <violation
124     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
125     file="string_escaping.lib.php"
126 />
127 <violation
128     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
129     file="string_escaping.lib.php"
130 />
131 <violation
132     rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
133     file="unicode.lib.php"
134 />
135 <violation
136     rule="PHPMD\Rule\CleanCode\ElseExpression"

```

```

137     file="unicode.lib.php"
138 />
139 <!-- https://github.com/phpmd/phpmd/issues/910 -->
140 <!-- <violation -->
141 <!-- rule="PHPMD\Rule\CleanCode\MissingImport" -->
142 <!-- file="unicode.lib.php" -->
143 <!-- /> -->
144 <violation
145     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
146     file="unicode.lib.php"
147 />
148 <violation
149     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
150     file="unicode.lib.php"
151 />
152 <violation
153     rule="PHPMD\Rule\CyclomaticComplexity"
154     file="unicode.lib.php"
155 />
156 <violation
157     rule="PHPMD\Rule\Design\LongMethod"
158     file="unicode.lib.php"
159 />
160 <violation
161     rule="PHPMD\Rule\Design\LongParameterList"
162     file="unicode.lib.php"
163 />
164 <violation
165     rule="PHPMD\Rule\Design\NpathComplexity"
166     file="unicode.lib.php"
167 />
168 <violation
169     rule="PHPMD\Rule\Naming\LongVariable"
170     file="unicode.lib.php"
171 />
172 <violation
173     rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
174     file="unicode_exceptions.lib.php"
175     method="get_data"
176 />
177 <violation
178     rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
179     file="unicode_exceptions.lib.php"
180     method="__construct"
181 />
182 <violation

```

```

183     rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
184     file="unicode_exceptions.lib.php"
185   />
186   <violation
187     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
188     file="unicode_exceptions.lib.php"
189   />
190 </phpmd-baseline>

```

3.49 build_and_checks_variables/phpstan_baseline.neon.tpl

```

1 parameters:
2   ignoreErrors:
3     # https://github.com/phpstan/phpstan/issues/14585
4     -
5       rawMessage: Variable $argv might not be defined.
6       identifier: variable.undefined
7       count: 1
8       path: ../archive_libre_office_documents.exec.php
9
10    -
11      rawMessage: Variable $argv might not be defined.
12      identifier: variable.undefined
13      count: 1
14      path: ../build_and_checks_dependencies/build_dependencies_notes.exec.php
15
16    -
17      rawMessage: Variable $argv might not be defined.
18      identifier: variable.undefined
19      count: 1
20      path: ../build_and_checks_dependencies/split_score.exec.php
21
22    -
23      rawMessage: 'Comparison operation "<" between int<0, 1114111> \
24                and 0 is always false.'
25      identifier: smaller.alwaysFalse
26      count: 1
27      path: ../unicode.lib.php
28
29    -
30      rawMessage: 'Comparison operation ">" between int<-283305, 283305> \
31                and 1114111 is always false.'
32      identifier: greater.alwaysFalse
33      count: 1
34      path: ../unicode.lib.php
35

```



```

36 -
37   rawMessage: 'Parameter #1 $i_unicode_code_point of function \
38               DOSAS_unicode\unicode_code_point_to_UTF8 expects \
39               int<0, 1114111>, int<-1114095, 1114111> given.'
40   identifier: argument.type
41   count: 1
42   path: ../unicode.lib.php
43
44 -
45   rawMessage: 'Parameter #1 $i_unicode_code_point of function \
46               DOSAS_unicode\unicode_code_point_to_UTF8 expects \
47               int<0, 1114111>, int<-283305, 283305> given.'
48   identifier: argument.type
49   count: 1
50   path: ../unicode.lib.php
51
52 -
53   rawMessage: 'Parameter #2 $i_current_octet of function \
54               DOSAS_unicode\get_message_and_data_array expects int, \
55               int<0, 191>|int<194, 244>|int<246, 254>|null given.'
56   identifier: argument.type
57   count: 1
58   path: ../unicode.lib.php

```

3.50 build_and_checks_variables/phpstan_config.neon.tpl

```

1  includes:
2    - phpstan_baseline.neon
3    - /home/laurent/.config/composer/vendor/\
4      phpstan/phpstan/conf/bleedingEdge.neon
5
6  parameters:
7    level: max
8
9    paths:
10      - ..
11
12    checkTooWideReturnTypesInProtectedAndPublicMethods: true

```

3.51 build_and_checks_variables/pylintrc.toml

```

1  [tool.pylint.main]
2  exit-zero = true
3  ignore = ["node_modules"]
4  jobs = 0
5  [tool.pylint.basic]

```

```

6 # We use uppercase acronyms like HTML, IMAP, etc.
7 module-rgx = "^[_a-zA-Z0-9+]+$"
8 function-rgx = "^[_a-zA-Z0-9+]+$"
9 [tool.pylint."messages control"]
10 # Code like R0912 is better to order disables by category.
11 disable = [
12     "C1804", # use-implicit-booleaness-not-comparison-to-string
13     "I0011", # locally-disabled
14     "I0020", # suppressed-message
15     "R0912", # too-many-branches
16     "R0913", # too-many-arguments
17     "R0914", # too-many-locals
18     "R0915", # too-many-statements
19     "R0917", # too-many-positional-arguments
20 ]
21 enable = "all"
22 [tool.pylint.reports]
23 output-format = "colorized"

```

3.52 build_and_checks_variables/repository_data.txt

```

1 Look at grep_variable to understand the structure of this file.
2 Basically, it has almost no structure apart from some lines
3 starting with a "variable_name=".
4 Anything else is comments.
5
6 repository_name=DevOrSysAdminScripts
7 repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
8 repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git
9
10 abstract=Some "small" useful scripts
11 abstract=for enhancing development or system-administrators tasks
12
13 license=GPLv3+
14 license2=LGPLv3+
15
16 author_email=Laurent.Lyaudet@gmail.com
17 author_full_name=Laurent Frédéric Bernard François Lyaudet
18 author_last_name=Lyaudet
19 author_name=Laurent Lyaudet
20 author_website=https://lyaudet.eu/laurent/
21
22 acknowledgments=Merci Dieu !
23 acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !
24
25 Great day! Now I allow myself lines of 75 characters instead of 70 :).

```

```

26 I checked how many characters fit in the PDF file generated by tex with
27 default paper size,
28 when using verbatim environment with line numbers for source code listing.
29 With a screen capture in GIMP,
30 I measured 197px between two-digits line numbers and the left border of
31 the page.
32 (It is less than 197px for three-digits or more line numbers.)
33 The same distance from the right border yields 75 characters allowed.
34 Since when printing the HTML file, the number of allowed characters in
35 <pre> tags is way larger, 75 is the right limit with the tools I use.
36
37 max_line_length=75
38
39 latex_command=pdflatex
40
41 grammar_and_spelling_checker_command=
42
43
44 Why one venv for each Python dependency ?
45 Because no language of my knowledge did it right to handle all
46 possible dependency conflicts.
47 So in case of such a conflict, one specialized venv can get you out
48 of any problem.
49 The best solution would have been to allow install of different
50 versions of the same package, and have a mechanism:
51 - default behavior, use the latest available version of a package among
52 those that satisfy the declared dependency constraints of the package
53 that has this dependency;
54 it requires that all dependency constraints are declared;
55 - custom behavior, check a dependency overload configuration file
56 with tree structure: if calling python package is A version 1.1.1
57 with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
58 you could have along that another rule saying that when called
59 directly B 2.2.2 should use C 4.4.4;
60 and even more complicated:
61 if calling python package is A version 1.1.1 with dependency B and D
62 both themselves with dependency C, use B 2.2.2, D 0.1.0,
63 and C 3.3.3 for B, and C 4.4.4 for D;
64 once you've seen that it can never be more complicated than that once
65 the same version number guarantees that the code is the same,
66 you create a config language "à la CSS" for cascading dependencies ;)
67 XD fit into JSON the dumbest possible way and you stop bothering
68 about dependency conflicts for the rest of history of informatics.
69 {
70     "A": {
71         "1.1.1": {

```

```

72     "B": {
73         "2.2.2": {
74             "C": {
75                 // Dumb, you did not found it by yourself? Failed exam.
76                 "3.3.3" : {}
77             }
78         }
79     },
80     "D": {
81         "0.1.0": {
82             "C": {
83                 // Dumb, you did not found it by yourself? Failed exam.
84                 "4.4.4" : {}
85             }
86         }
87     }
88 },
89 "B": {
90     "2.2.2": {
91         "C": {
92             // Dumb, you did not found it by yourself? Failed exam.
93             "4.4.4" : {}
94         }
95     }
96 }
97 }
98 }
99 Wonderful someone rediscovered nested structures XD
100 Seriously so many engineers or computer scientists are dumb to cry.
101
102 bandit_venv=../venv_bandit
103 black_venv=../venv_black
104 isort_venv=../venv_isort
105 mypy_venv=../venv_mypy
106 pylint_venv=../venv_pylint
107 ruff_venv=../venv_ruff
108
109
110 upgrade_venvs can be auto or ask, any other value is interpreted
111 as ask.
112
113 upgrade_venvs=ask
114
115
116 time interval can be 0 to always upgrade or ask to upgrade,
117 or a number at most bash max int

```

```

118 https://superuser.com/questions/1030122/
119 what-is-the-maximum-value-of-a-numeric-bash-shell-variable
120 it should be enough to simulate never "upgrade or ask for upgrade"
121 unless you claim to live many billions years ;)
122 If it is wat, this is a random int XD (see RANDOM).
123 Note that the probability that it upgrades your dependencies is not
124 1/2, it increases over time, and is 1 after slightly
125 less than one day.
126 If it is watyouwant?, see SRANDOM XD.
127
128 upgrade_venvs_time_interval_in_seconds=wat
129
130
131 For the Nu HTML, CSS, and SVG checker.
132
133 vnu_jar_path=./jar_venv/vnu.jar
134
135 Not fully implemented yet, it will only warn that vnu jar was not
136 updated since some time.
137 "wat" works, but "watyouwant?" joke was not repeated ;).
138
139 upgrade_vnu_jar_time_interval_in_seconds=604800
140
141 use_vnu_for_xhtml=false
142
143 For xmllint_files_without_DTD and files_without_license_header,
144 each line is a file path, except if the line ends with a /,
145 then it is concatenated with the following line(s) to obtain a path.
146
147 xmllint_files_without_DTD=./build_and_checks_variables/phpcs_ruleset.xml
148 xmllint_files_without_DTD=./build_and_checks_variables/phpmd_baseline.xml
149
150 If you use Ubuntu/Debian, install w3c-sgml-lib instead of =true.
151 See https://gitlab.gnome.org/GNOME/libxml2/-/work\_items/1119
152
153 xmllint_use_downloaded_DTDs=false
154
155 files_without_license_header=build_and_checks_variables/phpcs_ruleset.xml
156 files_without_license_header=build_and_checks_variables/
157 files_without_license_header=phpmd_baseline.xml
158 files_without_license_header=README.html
159 files_without_license_header=miscellaneous/string_escaping.html
160 files_without_license_header=build_and_checks_variables/ruff.toml
161 files_without_license_header=build_and_checks_variables/black.toml
162 files_without_license_header=build_and_checks_variables/pylintrc.toml

```

3.53 build_and_checks_variables/repository_data.txt.tpl

```
1 Look at grep_variable to understand the structure of this file.
2 Basically, it has almost no structure apart from some lines
3 starting with a "variable_name=".
4 Anything else is comments.
5 The goal of this template is to list all possible variables.
6
7 repository_name=
8 repository_web_url=
9 repository_git_url=
10
11 abstract=
12 abstract=
13
14 license=
15 license2=
16
17 author_email=
18 author_full_name=
19 author_last_name=
20 author_name=
21 author_website=
22
23 acknowledgments=
24 acknowledgments=
25
26 max_line_length=
27
28 latex_command=
29
30 grammar_and_spelling_checker_command=
31
32 bandit_venv=
33 black_venv=
34 isort_venv=
35 mypy_venv=
36 pylint_venv=
37 ruff_venv=
38
39 upgrade_venvs=
40
41 upgrade_venvs_time_interval_in_seconds=
42
43 vnu_jar_path=
44
```

```

45 upgrade_vnu_jar_time_interval_in_seconds=
46
47 use_vnu_for_xhtml=
48
49 npm_lint_directories=
50 npm_lint_directories=
51
52 xmllint_files_without_DTD=
53 xmllint_files_without_DTD=
54
55 xmllint_use_downloaded_DTDs=
56
57 files_without_license_header=
58 files_without_license_header=

```

3.54 build_and_checks_variables/ruff.toml

```

1  [lint]
2  select = [
3      "ALL",
4      "D211", # blank-line-before-class no...before-class-docstring!
5  ]
6  # ignore = ["D203", "D212", "D404"]
7  ignore = [
8      "C901", # complex-structure AKA give-me-toy-code
9      "D203", # (want-)?incorrect-blank-line-before-class
10     "D205", # missing-blank-line-after-summary So so cute|true ;) XD.
11     "D212", # multi-line-summary-first-line
12     "D404", # docstring-starts-with-this Shame! XD
13
14     # raw-string-in-exception may be useful for a real project
15     # with "dictionnaires" of errors made of constants.
16     "EM101",
17
18     # commented-out-code Of course git log is always faster...
19     # Note that if you're tired or someone else in your team is,
20     # it may catch some commit that lacked some review,
21     # or automated tests: "Last manual test succeeded, ah but when I
22     # debugged at some point I commented a part that was needed in
23     # another case."...
24     "ERA001",
25
26     "N802", # invalid-function-name IMAP, HTML, etc.
27     "PLR0912", # too-many-branches AKA ...
28     "PLR0913", # too-many-arguments AKA ...
29     "PLR0915", # too-many-statements AKA ...

```

```

30 "PLR0917", # too-many-positional-arguments AKA ...
31 # Not enough reading when young makes brain slow to read and
32 # understand lengthy functions.
33 # If the project is finished, you may play with AKA rules.
34 ]
35 [lint.per-file-ignores]
36 "build_and_checks_dependencies/licenses_templates/*.py" = [
37     "INP001" # implicit-namespace-package
38 ]
39 "IMAP_transfert.exec.py" = [
40     "T201", # print Use log instead in full applications.
41     "TRY003", # raise-vanilla-args
42 ]

```

3.55 build_and_checks_variables/shellcheck.ini

```

1 enable=all
2 check-sourced=true
3 external-sources=true
4 source-path="."
5 source-path="SCRIPTDIR"
6 source-path="./build_and_checks_dependencies"
7 source-path="./git_helpers"

```

3.56 build_and_checks_variables/shellcheck_notes.txt

```

1 A listing of the current suppressed warnings/errors from shellcheck,
2 and also some of the warnings that were suppressed in some previous
3 state of the git repository (in that case, we use X suffix, see SC2162X):
4 - SC1090 : Can't follow non-constant source.
5         Use a directive to specify location
6         # shellcheck source=bla/bli.sh
7         Only remaining cases are in common_build_and_checks
8         when sourcing some Python venv
9         source venv/bin/active
10 - SC1091 : Not following (file not found most of the time).
11         Only remaining cases are in common_build_and_checks
12         when sourcing some Python venv
13         source venv/bin/active
14 - SC1111 : Use of slanted double quotes (valid inside single quotes)
15         False positive, but may catch some problem.
16 - SC2001 : Replace some sed with bash builtin ${a//b/c}
17         False positive when c contains \n.
18 - SC2016 : Expressions don't expand in single quotes,
19         use double quotes for that.
20 - SC2034 : Unused variable

```



```

21         False positive for globals that are declared in functions
22         that are not currently called.
23         Somehow it can be used to detect dead code.
24 - SC2059 : Don't use variables in the printf format string.
25         Use printf '...%s...' "$foo".
26         False positive when using a variable to build the format string.
27         Usually because the format string is very long.
28 - SC2086 : Double quote to prevent globbing and word splitting.
29         False positive when returning integer variable
30         via an indirection like return ${!foo}.
31         See SC2248 below and the corresponding issue on GitHub.
32 - SC2119 : Use foo "$@" if function's $1 should mean script's $1.
33         False positive when passing arguments would yield
34         a wrong result.
35 - SC2120 : foo references arguments, but none are ever passed.
36         False positives when some arguments are meant for debug in CLI.
37 - SC2125 : Brace expansions and globs are literal.
38         False positive when assigning a variable declared with -i.
39         https://github.com/koalaman/shellcheck/issues/914
40 - SC2145 : Argument mixes string and array.
41         False positive when using "$@" line return and pipes.
42         https://github.com/koalaman/shellcheck/issues/3192
43         Example:
44             # shellcheck disable=SC2145
45             all_self_code_lines "$@"\
46             | empty_lines_after_file_name
47 - SC2154 : var is referenced but not assigned.
48         Very useful but false positive when I play with
49         variable variable names.
50 - SC2162X: read without -r will mangle backslashes.
51         False positive when read is used without variable:
52         https://github.com/koalaman/shellcheck/issues/623
53 - SC2248 : Prefer double quoting
54         False positive on returning integer values.
55         https://github.com/koalaman/shellcheck/issues/3405
56 - SC2309 : -eq treats this as a variable.
57         Use = to compare as string (or expand explicitly with $var).
58         Very useful but false positive when I play with
59         variable variable names.
60 - SC2312 : Some command is not invoked separately and thus its return
61         value is masked. Frequent with pipes.
62 - SCNNNNX: Use printf instead of echo.
63         https://github.com/koalaman/shellcheck/issues/3419
64
65 The command to check if there is anything else:
66 grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$\\"

```

```

67 -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
68 -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$"\
69 -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$"\
70 -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"

```

3.57 build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "build_and_checks.exec.sh.tpl" was renamed from
25 # "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
26 # The file "build_and_checks.exec.sh" generated from the file
27 # "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
28 # was renamed from
29 # "build_and_checks.sh" to "build_and_checks.exec.sh".
30
31 build_and_checks() {
32     # $1=work_directory
33     # Options:
34     # --check-download=y|keep (=keep to keep the downloaded files)
35     # --verbose
36     local LFBFL_work_directory="."
37     if [[ -n "$1" ]]; then
38         LFBFL_work_directory="$1"
39     fi

```

```

40  readonly LFBFL_work_directory
41
42  local LFBFL_verbose=""
43  if [[ "$*" == *--verbose* ]]; then
44      printf "%s %s\n" "$0" "$*"
45      LFBFL_verbose="--verbose"
46  fi
47  readonly LFBFL_verbose
48
49  local LFBFL_check_download=""
50  if [[ "$*" == *--check-download=keep* ]]; then
51      LFBFL_check_download="--check-download=keep"
52  elif [[ "$*" == *--check-download=y* ]]; then
53      LFBFL_check_download="--check-download=y"
54  fi
55  readonly LFBFL_check_download
56
57  source ./wget_sha512.lib.sh
58
59  LFBFL_subdir="build_and_checks_dependencies"
60  mkdir --parents "${LFBFL_subdir}/licenses_templates"
61  local LFBFL_variables_directory
62  LFBFL_variables_directory="${LFBFL_work_directory}/"
63  LFBFL_variables_directory+="build_and_checks_variables"
64  readonly LFBFL_variables_directory
65  mkdir --parents -- "${LFBFL_variables_directory}/temp"
66  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67      printf "temp/\nupgrade_venvs_ts\n" \
68          > "${LFBFL_variables_directory}/.gitignore"
69  fi
70
71  local LFBFL_dependencies_URL
72  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74  readonly LFBFL_dependencies_URL
75
76  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77  local LFBFL_script_download_URL
78  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80  readonly LFBFL_script_download_URL
81  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82  local LFBFL_correct_sha512
83  LFBFL_correct_sha512='b1e12643d3320e7b9be2c7172c8a334b14a19d7ef28ab'
84  LFBFL_correct_sha512+='fa30a15e20daa01b9662bff6658c4dc6d59b19ad841a'
85  LFBFL_correct_sha512+='6311c2f7d367983ff9f67a6f27dc63bec5a0aae'

```

```

86 wget_sha512 "${LFBFL_file_path}"\
87     "${LFBFL_script_download_URL}"\
88     "${LFBFL_correct_sha512}"\
89     "${LFBFL_verbose}"\
90     "${LFBFL_check_download}"
91 chmod +x "${LFBFL_file_path}"
92
93 "${LFBFL_file_path}" "${LFBFL_work_directory}"\
94     "${LFBFL_dependencies_URL}"\
95     "${LFBFL_verbose}"\
96     "${LFBFL_check_download}"
97 }
98
99 if [[ "$*" == *--fixed-point-build* ]]; then
100     printf -- "--fixed-point-build\n"
101     source "./build_and_checks_dependencies/lines_filters.lib.sh"
102     LFBFL_work_directory="."
103     if [[ -n "$1" ]]; then
104         LFBFL_work_directory="$1"
105     fi
106     LFBFL_variables_directory="${LFBFL_work_directory}/"
107     LFBFL_variables_directory+="build_and_checks_variables"
108     LFBFL_data_file_name="${LFBFL_variables_directory}/"
109     LFBFL_data_file_name+="repository_data.txt"
110     LFBFL_repository_name=""
111     grep_variable "${LFBFL_data_file_name}" repository_name\
112         --result-variable-prefix="LFBFL_"\
113         --replace-line-returns-by=""
114     # Touching the 3 following files first let us iterate
115     # upgrade_build_and_checks 2 times instead of 3 to reach
116     # a fixed point.
117     touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
118     touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
119     touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
120     # The two iterations of build_and_checks needs to complete
121     # during the same minute to have a fixed point.
122     build_and_checks "$@"
123     # build_and_checks "$@" second iteration below
124     # Then calling this script without --fixed-point-build should not
125     # yield any new superficial modifications.
126 fi
127
128 build_and_checks "$@"

```

3.58 COPYING

1 GNU GENERAL PUBLIC LICENSE
2 Version 3, 29 June 2007
3
4 Copyright (C) 2007 Free Software Foundation, Inc. <<https://fsf.org/>>
5 Everyone is permitted to copy and distribute verbatim copies
6 of this license document, but changing it is not allowed.
7
8 Preamble
9
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12
13 The licenses for most software and other practical works are designed
14 to take away your freedom to share and change the works. By contrast,
15 the GNU General Public License is intended to guarantee your freedom to
16 share and change all versions of a program--to make sure it remains free
17 software for all its users. We, the Free Software Foundation, use the
18 GNU General Public License for most of our software; it applies also to
19 any other work released this way by its authors. You can apply it to
20 your programs, too.
21
22 When we speak of free software, we are referring to freedom, not
23 price. Our General Public Licenses are designed to make sure that you
24 have the freedom to distribute copies of free software (and charge for
25 them if you wish), that you receive source code or can get it if you
26 want it, that you can change the software or use pieces of it in new
27 free programs, and that you know you can do these things.
28
29 To protect your rights, we need to prevent others from denying you
30 these rights or asking you to surrender the rights. Therefore, you have
31 certain responsibilities if you distribute copies of the software, or if
32 you modify it: responsibilities to respect the freedom of others.
33
34 For example, if you distribute copies of such a program, whether
35 gratis or for a fee, you must pass on to the recipients the same
36 freedoms that you received. You must make sure that they, too, receive
37 or can get the source code. And you must show them these terms so they
38 know their rights.
39
40 Developers that use the GNU GPL protect your rights with two steps:
41 (1) assert copyright on the software, and (2) offer you this License
42 giving you legal permission to copy, distribute and/or modify it.
43
44 For the developers' and authors' protection, the GPL clearly explains

45 that there is no warranty for this free software. For both users' and
46 authors' sake, the GPL requires that modified versions be marked as
47 changed, so that their problems will not be attributed erroneously to
48 authors of previous versions.

49
50 Some devices are designed to deny users access to install or run
51 modified versions of the software inside them, although the manufacturer
52 can do so. This is fundamentally incompatible with the aim of
53 protecting users' freedom to change the software. The systematic
54 pattern of such abuse occurs in the area of products for individuals to
55 use, which is precisely where it is most unacceptable. Therefore, we
56 have designed this version of the GPL to prohibit the practice for those
57 products. If such problems arise substantially in other domains, we
58 stand ready to extend this provision to those domains in future versions
59 of the GPL, as needed to protect the freedom of users.

60
61 Finally, every program is threatened constantly by software patents.
62 States should not allow patents to restrict development and use of
63 software on general-purpose computers, but in those that do, we wish to
64 avoid the special danger that patents applied to a free program could
65 make it effectively proprietary. To prevent this, the GPL assures that
66 patents cannot be used to render the program non-free.

67
68 The precise terms and conditions for copying, distribution and
69 modification follow.

70 71 TERMS AND CONDITIONS

72 73 0. Definitions.

74
75 "This License" refers to version 3 of the GNU General Public License.

76
77 "Copyright" also means copyright-like laws that apply to other kinds of
78 works, such as semiconductor masks.

79
80 "The Program" refers to any copyrightable work licensed under this
81 License. Each licensee is addressed as "you". "Licensees" and
82 "recipients" may be individuals or organizations.

83
84 To "modify" a work means to copy from or adapt all or part of the work
85 in a fashion requiring copyright permission, other than the making of an
86 exact copy. The resulting work is called a "modified version" of the
87 earlier work or a work "based on" the earlier work.

88
89 A "covered work" means either the unmodified Program or a work based
90 on the Program.

91
92 To "propagate" a work means to do anything with it that, without
93 permission, would make you directly or secondarily liable for
94 infringement under applicable copyright law, except executing it on a
95 computer or modifying a private copy. Propagation includes copying,
96 distribution (with or without modification), making available to the
97 public, and in some countries other activities as well.
98
99 To "convey" a work means any kind of propagation that enables other
100 parties to make or receive copies. Mere interaction with a user through
101 a computer network, with no transfer of a copy, is not conveying.
102
103 An interactive user interface displays "Appropriate Legal Notices"
104 to the extent that it includes a convenient and prominently visible
105 feature that (1) displays an appropriate copyright notice, and (2)
106 tells the user that there is no warranty for the work (except to the
107 extent that warranties are provided), that licensees may convey the
108 work under this License, and how to view a copy of this License. If
109 the interface presents a list of user commands or options, such as a
110 menu, a prominent item in the list meets this criterion.
111
112 1. Source Code.
113
114 The "source code" for a work means the preferred form of the work
115 for making modifications to it. "Object code" means any non-source
116 form of a work.
117
118 A "Standard Interface" means an interface that either is an official
119 standard defined by a recognized standards body, or, in the case of
120 interfaces specified for a particular programming language, one that
121 is widely used among developers working in that language.
122
123 The "System Libraries" of an executable work include anything, other
124 than the work as a whole, that (a) is included in the normal form of
125 packaging a Major Component, but which is not part of that Major
126 Component, and (b) serves only to enable use of the work with that
127 Major Component, or to implement a Standard Interface for which an
128 implementation is available to the public in source code form. A
129 "Major Component", in this context, means a major essential component
130 (kernel, window system, and so on) of the specific operating system
131 (if any) on which the executable work runs, or a compiler used to
132 produce the work, or an object code interpreter used to run it.
133
134 The "Corresponding Source" for a work in object code form means all
135 the source code needed to generate, install, and (for an executable
136 work) run the object code and to modify the work, including scripts to

137 control those activities. However, it does not include the work's
138 System Libraries, or general-purpose tools or generally available free
139 programs which are used unmodified in performing those activities but
140 which are not part of the work. For example, Corresponding Source
141 includes interface definition files associated with source files for
142 the work, and the source code for shared libraries and dynamically
143 linked subprograms that the work is specifically designed to require,
144 such as by intimate data communication or control flow between those
145 subprograms and other parts of the work.

146
147 The Corresponding Source need not include anything that users
148 can regenerate automatically from other parts of the Corresponding
149 Source.

150
151 The Corresponding Source for a work in source code form is that
152 same work.

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157 copyright on the Program, and are irrevocable provided the stated
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159 permission to run the unmodified Program. The output from running a
160 covered work is covered by this License only if the output, given its
161 content, constitutes a covered work. This License acknowledges your
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166 in force. You may convey covered works to others for the sole purpose
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168 with facilities for running those works, provided that you comply with
169 the terms of this License in conveying all material for which you do
170 not control copyright. Those thus making or running the covered works
171 for you must do so exclusively on your behalf, under your direction
172 and control, on terms that prohibit them from making any copies of
173 your copyrighted material outside their relationship with you.

174
175 Conveying under any other circumstances is permitted solely under
176 the conditions stated below. Sublicensing is not allowed; section 10
177 makes it unnecessary.

178 179 3. Protecting Users' Legal Rights From Anti-Circumvention Law.

180
181 No covered work shall be deemed part of an effective technological
182 measure under any applicable law fulfilling obligations under article

183 11 of the WIPO copyright treaty adopted on 20 December 1996, or
184 similar laws prohibiting or restricting circumvention of such
185 measures.

186
187 When you convey a covered work, you waive any legal power to forbid
188 circumvention of technological measures to the extent such circumvention
189 is effected by exercising rights under this License with respect to
190 the covered work, and you disclaim any intention to limit operation or
191 modification of the work as a means of enforcing, against the work's
192 users, your or third parties' legal rights to forbid circumvention of
193 technological measures.

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197 You may convey verbatim copies of the Program's source code as you
198 receive it, in any medium, provided that you conspicuously and
199 appropriately publish on each copy an appropriate copyright notice;
200 keep intact all notices stating that this License and any
201 non-permissive terms added in accord with section 7 apply to the code;
202 keep intact all notices of the absence of any warranty; and give all
203 recipients a copy of this License along with the Program.

204
205 You may charge any price or no price for each copy that you convey,
206 and you may offer support or warranty protection for a fee.

207 208 5. Conveying Modified Source Versions.

209
210 You may convey a work based on the Program, or the modifications to
211 produce it from the Program, in the form of source code under the
212 terms of section 4, provided that you also meet all of these conditions:

213
214 a) The work must carry prominent notices stating that you modified
215 it, and giving a relevant date.

216
217 b) The work must carry prominent notices stating that it is
218 released under this License and any conditions added under section
219 7. This requirement modifies the requirement in section 4 to
220 "keep intact all notices".

221
222 c) You must license the entire work, as a whole, under this
223 License to anyone who comes into possession of a copy. This
224 License will therefore apply, along with any applicable section 7
225 additional terms, to the whole of the work, and all its parts,
226 regardless of how they are packaged. This License gives no
227 permission to license the work in any other way, but it does not
228 invalidate such permission if you have separately received it.

229
230 d) If the work has interactive user interfaces, each must display
231 Appropriate Legal Notices; however, if the Program has interactive
232 interfaces that do not display Appropriate Legal Notices, your
233 work need not make them do so.

234
235 A compilation of a covered work with other separate and independent
236 works, which are not by their nature extensions of the covered work,
237 and which are not combined with it such as to form a larger program,
238 in or on a volume of a storage or distribution medium, is called an
239 "aggregate" if the compilation and its resulting copyright are not
240 used to limit the access or legal rights of the compilation's users
241 beyond what the individual works permit. Inclusion of a covered work
242 in an aggregate does not cause this License to apply to the other
243 parts of the aggregate.

244 245 6. Conveying Non-Source Forms.

246
247 You may convey a covered work in object code form under the terms
248 of sections 4 and 5, provided that you also convey the
249 machine-readable Corresponding Source under the terms of this License,
250 in one of these ways:

251
252 a) Convey the object code in, or embodied in, a physical product
253 (including a physical distribution medium), accompanied by the
254 Corresponding Source fixed on a durable physical medium
255 customarily used for software interchange.

256
257 b) Convey the object code in, or embodied in, a physical product
258 (including a physical distribution medium), accompanied by a
259 written offer, valid for at least three years and valid for as
260 long as you offer spare parts or customer support for that product
261 model, to give anyone who possesses the object code either (1) a
262 copy of the Corresponding Source for all the software in the
263 product that is covered by this License, on a durable physical
264 medium customarily used for software interchange, for a price no
265 more than your reasonable cost of physically performing this
266 conveying of source, or (2) access to copy the
267 Corresponding Source from a network server at no charge.

268
269 c) Convey individual copies of the object code with a copy of the
270 written offer to provide the Corresponding Source. This
271 alternative is allowed only occasionally and noncommercially, and
272 only if you received the object code with such an offer, in accord
273 with subsection 6b.

274

275 d) Convey the object code by offering access from a designated
276 place (gratis or for a charge), and offer equivalent access to the
277 Corresponding Source in the same way through the same place at no
278 further charge. You need not require recipients to copy the
279 Corresponding Source along with the object code. If the place to
280 copy the object code is a network server, the Corresponding Source
281 may be on a different server (operated by you or a third party)
282 that supports equivalent copying facilities, provided you maintain
283 clear directions next to the object code saying where to find the
284 Corresponding Source. Regardless of what server hosts the
285 Corresponding Source, you remain obligated to ensure that it is
286 available for as long as needed to satisfy these requirements.

287
288 e) Convey the object code using peer-to-peer transmission, provided
289 you inform other peers where the object code and Corresponding
290 Source of the work are being offered to the general public at no
291 charge under subsection 6d.

292
293 A separable portion of the object code, whose source code is excluded
294 from the Corresponding Source as a System Library, need not be
295 included in conveying the object code work.

296
297 A "User Product" is either (1) a "consumer product", which means any
298 tangible personal property which is normally used for personal, family,
299 or household purposes, or (2) anything designed or sold for incorporation
300 into a dwelling. In determining whether a product is a consumer product,
301 doubtful cases shall be resolved in favor of coverage. For a particular
302 product received by a particular user, "normally used" refers to a
303 typical or common use of that class of product, regardless of the status
304 of the particular user or of the way in which the particular user
305 actually uses, or expects or is expected to use, the product. A product
306 is a consumer product regardless of whether the product has substantial
307 commercial, industrial or non-consumer uses, unless such uses represent
308 the only significant mode of use of the product.

309
310 "Installation Information" for a User Product means any methods,
311 procedures, authorization keys, or other information required to install
312 and execute modified versions of a covered work in that User Product from
313 a modified version of its Corresponding Source. The information must
314 suffice to ensure that the continued functioning of the modified object
315 code is in no case prevented or interfered with solely because
316 modification has been made.

317
318 If you convey an object code work under this section in, or with, or
319 specifically for use in, a User Product, and the conveying occurs as
320 part of a transaction in which the right of possession and use of the

321 User Product is transferred to the recipient in perpetuity or for a
322 fixed term (regardless of how the transaction is characterized), the
323 Corresponding Source conveyed under this section must be accompanied
324 by the Installation Information. But this requirement does not apply
325 if neither you nor any third party retains the ability to install
326 modified object code on the User Product (for example, the work has
327 been installed in ROM).

328
329 The requirement to provide Installation Information does not include a
330 requirement to continue to provide support service, warranty, or updates
331 for a work that has been modified or installed by the recipient, or for
332 the User Product in which it has been modified or installed. Access to a
333 network may be denied when the modification itself materially and
334 adversely affects the operation of the network or violates the rules and
335 protocols for communication across the network.

336
337 Corresponding Source conveyed, and Installation Information provided,
338 in accord with this section must be in a format that is publicly
339 documented (and with an implementation available to the public in
340 source code form), and must require no special password or key for
341 unpacking, reading or copying.

342
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344
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400 additional terms that apply to those files, or a notice indicating
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404 form of a separately written license, or stated as exceptions;
405 the above requirements apply either way.

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447

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611
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624

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651
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658 under certain conditions; type 'show c' for details.

659
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21
22 An "Application" is any work that makes use of an interface provided
23 by the Library, but which is not otherwise based on the Library.
24 Defining a subclass of a class defined by the Library is deemed a mode
25 of using an interface provided by the Library.
26
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28 Application with the Library. The particular version of the Library
29 with which the Combined Work was made is also called the "Linked
30 Version".
31
32 The "Minimal Corresponding Source" for a Combined Work means the
33 Corresponding Source for the Combined Work, excluding any source code
34 for portions of the Combined Work that, considered in isolation, are
35 based on the Application, and not on the Linked Version.
36
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38 object code and/or source code for the Application, including any data
39 and utility programs needed for reproducing the Combined Work from the
40 Application, but excluding the System Libraries of the Combined Work.
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64

65 The object code form of an Application may incorporate material from
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67 code under terms of your choice, provided that, if the incorporated
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84 engineering for debugging such modifications, if you also do each of
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94 c) For a Combined Work that displays copyright notices during
95 execution, include the copyright notice for the Library among
96 these notices, as well as a reference directing the user to the
97 copies of the GNU GPL and this license document.
98

99 d) Do one of the following:
100

101 0) Convey the Minimal Corresponding Source under the terms of this
102 License, and the Corresponding Application Code in a form
103 suitable for, and under terms that permit, the user to

104 recombine or relink the Application with a modified version of
105 the Linked Version to produce a modified Combined Work, in the
106 manner specified by section 6 of the GNU GPL for conveying
107 Corresponding Source.

108

109 1) Use a suitable shared library mechanism for linking with the
110 Library. A suitable mechanism is one that (a) uses at run time
111 a copy of the Library already present on the user's computer
112 system, and (b) will operate properly with a modified version
113 of the Library that is interface-compatible with the Linked
114 Version.

115

116 e) Provide Installation Information, but only if you would otherwise
117 be required to provide such information under section 6 of the
118 GNU GPL, and only to the extent that such information is
119 necessary to install and execute a modified version of the
120 Combined Work produced by recombining or relinking the
121 Application with a modified version of the Linked Version. (If
122 you use option 4d0, the Installation Information must accompany
123 the Minimal Corresponding Source and Corresponding Application
124 Code. If you use option 4d1, you must provide the Installation
125 Information in the manner specified by section 6 of the GNU GPL
126 for conveying Corresponding Source.)

127

128 5. Combined Libraries.

129

130 You may place library facilities that are a work based on the
131 Library side by side in a single library together with other library
132 facilities that are not Applications and are not covered by this
133 License, and convey such a combined library under terms of your
134 choice, if you do both of the following:

135

136 a) Accompany the combined library with a copy of the same work based
137 on the Library, uncombined with any other library facilities,
138 conveyed under the terms of this License.

139

140 b) Give prominent notice with the combined library that part of it
141 is a work based on the Library, and explaining where to find the
142 accompanying uncombined form of the same work.

143

144 6. Revised Versions of the GNU Lesser General Public License.

145

146 The Free Software Foundation may publish revised and/or new versions
147 of the GNU Lesser General Public License from time to time. Such new
148 versions will be similar in spirit to the present version, but may
149 differ in detail to address new problems or concerns.

150
151 Each version is given a distinguishing version number. If the
152 Library as you received it specifies that a certain numbered version
153 of the GNU Lesser General Public License "or any later version"
154 applies to it, you have the option of following the terms and
155 conditions either of that published version or of any later version
156 published by the Free Software Foundation. If the Library as you
157 received it does not specify a version number of the GNU Lesser
158 General Public License, you may choose any version of the GNU Lesser
159 General Public License ever published by the Free Software Foundation.
160
161 If the Library as you received it specifies that a proxy can decide
162 whether future versions of the GNU Lesser General Public License shall
163 apply, that proxy's public statement of acceptance of any version is
164 permanent authorization for you to choose that version for the
165 Library.

3.60 files.libc.php

```
1 <?php
2
3 /**
4 This file is part of DevOrSysAdminScripts.
5
6 DevOrSysAdminScripts is free software:
7 you can redistribute it and/or modify it
8 under the terms of the GNU Lesser General Public License
9 as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
26
27 @category Library
```

```

28 @package DevOrSysAdminScripts
29 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
30 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
31 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
32 */
33
34 declare(strict_types=1);
35 declare(encoding='UTF-8');
36
37
38
39 /**
40 The namespace DOSASfiles contain additional classes and functions
41 to deal with files.
42
43 @package DevOrSysAdminScripts
44 @subpackage DOSASfiles
45 */
46 namespace DOSASfiles {
47
48     use Exception;
49
50
51
52
53     /**
54     Kind of a "dataclass" to keep the informations related with a file,
55     slightly enhanced with:
56     - a constructor that extract the distinct parts of the filepath;
57     - a method/getter handling the file content loading.
58
59     @category Library
60     @package DevOrSysAdminScripts
61     @subpackage DOSASfiles
62     @class FileData
63     @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
64     @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
65     */
66     final class FileData {
67         /**
68         The hash function to use for file contents.
69
70         @var string $s_hash_function
71         */
72         public static string $s_hash_function = 'md5';
73

```



```

74      /**
75       * The filepath of the file.
76       * https://www.youtube.com/watch?v=2NR1Z1ZME5U
77       * Why?
78       * https://github.com/squizlabs/PHP\_CodeSniffer/issues/258
79       *
80       * @var string $s_filepath
81       */
82      public string $s_filepath;
83
84      /**
85       * The dirname of the filepath.
86       *
87       * @var string $s_dirname
88       */
89      public string $s_dirname;
90
91      /**
92       * The basename of the filepath.
93       *
94       * @var string $s_basename
95       */
96      public string $s_basename;
97
98      /**
99       * The filename of the basename.
100      *
101      * @var string $s_filename
102      */
103      public string $s_filename;
104
105      /**
106       * The extension of the basename.
107       *
108       * @var string $s_extension
109       */
110      public string $s_extension;
111
112      /**
113       * The filesubpath starting from a reference directory.
114       *
115       * @var string $s_path_from_reference_directory
116       */
117      public string $s_path_from_reference_directory;
118
119      /**

```

```

120     The size of the content of the file.
121
122     @var int $i_size
123     */
124     public int $i_size;
125
126     /**
127     The hash of the content of the file.
128
129     @var string $s_hash
130     */
131     public string $s_hash;
132
133     /**
134     The content of the file as a string.
135
136     @var string $s_content
137     */
138     public $s_content;
139
140     /**
141     Statuses relative to another directory : 1.
142
143     Does the file exists with same relative path and same content
144     At The Same Time (ATST)?
145
146     @var bool $b_exists_with_same__relative_path_and_content
147     */
148     public bool $b_exists_with_same__relative_path_and_content; // ATST
149
150     /**
151     Statuses relative to another directory : 2.
152
153     Does the file exists with same relative path?
154
155     @var bool $b_exists_with_same_relative_path
156     */
157     public bool $b_exists_with_same_relative_path;
158
159     /**
160     Statuses relative to another directory : 3.
161
162     Does the file exists with same content?
163
164     @var bool $b_exists_with_same_content
165     */

```

```

166     public bool $b_exists_with_same_content;
167
168     /**
169     Statuses relative to another directory : 4.
170
171     Does the file exists with suffixed relative path?
172     Only computed if some regular expression for valid suffixes
173     is supplied.
174     An empty suffix is always considered valid,
175     but it is handled if your regexp doesn't match the empty string.
176
177     @var bool $b_exists_with_suffixed_relative_path
178     */
179     public bool $b_exists_with_suffixed_relative_path;
180
181     /**
182     Statuses relative to another directory : 5.
183
184     Does the file exists with suffixed relative path and same content
185     At The Same Time (ATST)?
186     Only computed if some regular expression for valid suffixes
187     is supplied.
188     An empty suffix is always considered valid,
189     but it is handled if your regexp doesn't match the empty string.
190
191     @var bool $b_exists_with_suffixed_relative_path_and
192     */
193     public bool $b_exists_with_suffixed_relative_path_and; // ATST
194
195
196
197     /**
198     Constructor for FileData.
199
200     @param string $s_filepath The filepath of the file.
201     @param string $s_reference_directory_path
202         The directorypath of the reference directory.
203
204     @return void
205     */
206     public function __construct(
207         string $s_filepath,
208         string $s_reference_directory_path,
209     ) {
210         $this->s_filepath = $s_filepath;
211         $path_parts = pathinfo($this->s_filepath);

```

```

212     $this->s_dirname = (
213         isset($path_parts['dirname']) ? $path_parts['dirname'] : ''
214     );
215     $this->s_basename = $path_parts['basename'];
216     $this->s_filename = $path_parts['filename'];
217     if(!isset($path_parts['extension'])) {
218         $this->s_extension = '';
219     }
220     else {
221         $this->s_extension = $path_parts['extension'];
222         if($this->s_filename === '') {
223             // Hidden file
224             $this->s_filename = $this->s_basename;
225             $this->s_extension = '';
226         }
227     }
228     $this->s_path_from_reference_directory = substr(
229         $this->s_filepath,
230         strlen($s_reference_directory_path),
231     );
232     // To enforce a load attempt.
233     $this->i_size = 1;
234     $this->s_hash = '';
235     $this->s_content = '';
236     $this->get_s_content();
237     $this->b_exists_with_same__relative_path_and_content = false;
238     $this->b_exists_with_same_relative_path = false;
239     $this->b_exists_with_same_content = false;
240     $this->b_exists_with_suffixed_relative_path = false;
241     $this->b_exists_with_suffixed_relative_path_and = false;
242 } //end __construct() FileData
243
244
245
246 /**
247  Returns the content of the file as a string after fetching it from
248  the filesystem.
249
250  @throws \Exception When the content of the file cannot be retrieved
251          from the filesystem.
252
253  @return string
254  */
255 public function get_s_content() : string {
256     if($this->i_size !== strlen($this->s_content)) {
257         $s_content = file_get_contents($this->s_filepath);

```

```

258         if($s_content === false){
259             throw new Exception(
260                 'Unexpected failure when fetching content of '
261                 .$this->s_filepath
262             );
263         }
264         $this->s_content = $s_content;
265         if($this->s_hash === ''){
266             $this->i_size = strlen($this->s_content);
267             $this->s_hash = md5($this->s_content);
268         }
269     }
270     return $this->s_content;
271 } //end get_s_content()
272 } //end class FileData
273
274
275
276
277 /**
278  * Kind of a "dataclass" to keep the informations related with the files
279  * under a directory.
280
281  * @category Library
282  * @package DevOrSysAdminScripts
283  * @subpackage DOSASfiles
284  * @class DataOfFilesUnderDirectory
285  * @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
286  * @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
287  */
288 final class DataOfFilesUnderDirectory {
289     /**
290      * The path of the reference directory.
291
292      * @var string $s_reference_directory_path
293      */
294     public string $s_reference_directory_path;
295
296     /**
297      * The data of the files under the reference directory
298      * put in order by absolute path.
299
300      * @var array<string, FileData> $arr_files_by_path
301      */
302     public array $arr_files_by_path;
303

```

```

304     /**
305     The data of the files under the reference directory
306     put in order by path from reference directory.
307
308     @var array<string, FileData> $arr_files_by_path_from
309     */
310     public array $arr_files_by_path_from;
311
312     /**
313     The data of the files under the reference directory
314     put in order by size, hash, and order of addition to this object.
315
316     @var array<int, array<string, array<int, FileData>>> $arr_files_by_size
317     */
318     public array $arr_files_by_size;
319
320
321
322     /**
323     Constructor for DataOfFilesUnderDirectory.
324
325     @param string $s_reference_directory_path
326             The directorypath of the reference directory.
327
328     @return void
329     */
330     public function __construct(string $s_reference_directory_path) {
331         $this->s_reference_directory_path = $s_reference_directory_path;
332         $this->arr_files_by_path = [];
333         $this->arr_files_by_path_from = [];
334         $this->arr_files_by_size = [];
335     } //end __construct() DataOfFilesUnderDirectory
336
337
338
339     /**
340     Adds a FileData object in the arrays used for fast access by
341     different keys.
342
343     @param FileData $o_file_data The FileData object to add.
344
345     @return void
346     */
347     public function add_file_data(FileData $o_file_data) : void {
348         $this->arr_files_by_path[$o_file_data->s_filepath] = $o_file_data;
349         $this->arr_files_by_path_from[

```

```

350         $o_file_data->s_path_from_reference_directory
351     ] = $o_file_data;
352     if(!isset($this->arr_files_by_size[$o_file_data->i_size])){
353         $this->arr_files_by_size[$o_file_data->i_size] = [];
354     }
355     if(
356         !isset(
357             $this->arr_files_by_size[$o_file_data->i_size][
358                 $o_file_data->s_hash
359             ]
360         )
361     ){
362         $this->arr_files_by_size[$o_file_data->i_size][
363             $o_file_data->s_hash
364         ] = [];
365     }
366     $this->arr_files_by_size[$o_file_data->i_size][
367         $o_file_data->s_hash
368     ] []= $o_file_data;
369 }//end add_file_data()
370
371
372
373 /**
374  Create a FileData object for the given FilePath.
375  And adds this FileData object in the arrays used for fast access by
376  different keys.
377
378  @param string $s_filepath The filepath required.
379
380  @return void
381  */
382 public function add_file_data_for_file_path(
383     string $s_filepath
384 ) : void {
385     $o_file_data = new FileData(
386         $s_filepath,
387         $this->s_reference_directory_path,
388     );
389     $o_file_data->s_content = '';
390     $this->add_file_data($o_file_data);
391 }//end add_file_data_for_file_path()
392 }//end class DataOfFilesUnderDirectory
393
394
395

```

```

396
397  /**
398  Loads the data of files under a given directory,
399  but doesn't keep the contents loaded.
400  Recusivity is not implemented yet.
401
402  @param string $s_dirpath The directory to search in.
403
404  @return \DOSASfiles\DataOfFilesUnderDirectory
405  */
406  function load_files_data_under_directory(
407      string $s_dirpath,
408  ) : DataOfFilesUnderDirectory {
409      $o_data_of_files_under_directory = new DataOfFilesUnderDirectory(
410          $s_dirpath
411      );
412      $arr_s_paths = glob($s_dirpath.'*');
413      if($arr_s_paths === false){
414          return $o_data_of_files_under_directory;
415      }
416      foreach($arr_s_paths as $s_filepath){
417          $o_data_of_files_under_directory->add_file_data_for_file_path(
418              $s_filepath
419          );
420      }//end foreach($arr_s_paths as $s_filepath)
421      return $o_data_of_files_under_directory;
422  }//end load_files_data_under_directory()
423
424
425
426  /**
427  Loads the data of files under two given directories,
428  but doesn't keep the contents loaded.
429  And then compares the files under both directories.
430  Recusivity is not implemented yet.
431
432  @param string $s_source_dirpath The first directory to search in.
433  @param string $s_dest_dirpath The second directory to search in.
434  @param bool   $b_compute_also_reverse_status
435              If you want that the status of files under dest_dirpath
436              must also be computed.
437  @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
438
439  @return array{
440      'source_files_data': \DOSASfiles\DataOfFilesUnderDirectory,
441      'dest_files_data': \DOSASfiles\DataOfFilesUnderDirectory

```



```

442     }
443     */
444     function compare_files_under_directories(
445         string $s_source_dirpath,
446         string $s_dest_dirpath,
447         bool $b_compute_also_reverse_status = false,
448         string $s_valid_suffixes_regexp = '',
449     ) : array {
450         $o_files_data_source = load_files_data_under_directory(
451             $s_source_dirpath
452         );
453         $o_files_data_dest = load_files_data_under_directory(
454             $s_dest_dirpath
455         );
456         foreach($o_files_data_source->arr_files_by_path_from as $o_file_data){
457             if(
458                 isset(
459                     $o_files_data_dest->arr_files_by_path_from[
460                         $o_file_data->s_path_from_reference_directory
461                     ]
462                 )
463             ){
464                 $o_file_data->b_exists_with_same_relative_path = true;
465                 $o_file_data2 = (
466                     $o_files_data_dest->arr_files_by_path_from[
467                         $o_file_data->s_path_from_reference_directory
468                     ]
469                 );
470                 $o_file_data2->b_exists_with_same_relative_path = true;
471                 $o_file_data->b_exists_with_suffixed_relative_path = true;
472                 $o_file_data2->b_exists_with_suffixed_relative_path = true;
473                 if(
474                     $o_file_data2->i_size === $o_file_data->i_size
475                     && $o_file_data2->s_hash === $o_file_data->s_hash
476                     && $o_file_data2->get_s_content()
477                     === $o_file_data->get_s_content()
478                 ){
479                     $o_file_data->b_exists_with_same__relative_path_and_content = (
480                         true
481                     );
482                     $o_file_data->b_exists_with_same_content = true;
483                     $o_file_data->b_exists_with_suffixed_relative_path_and = (
484                         true
485                     );
486                     $o_file_data2->b_exists_with_same__relative_path_and_content = (
487                         true

```

```

488         );
489         $o_file_data2->b_exists_with_same_content = true;
490         $o_file_data2->b_exists_with_suffixed_relative_path_and = (
491             true
492         );
493         $o_file_data->s_content = '';
494         $o_file_data2->s_content = '';
495         continue;
496     }
497 } //end if(isset($o_files_data_dest->arr_files_by_path_from[...]))
498 if(
499     isset($o_files_data_dest->arr_files_by_size[$o_file_data->i_size])
500     && isset(
501         $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
502             $o_file_data->s_hash
503         ]
504     )
505 ){
506     foreach(
507         $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
508             $o_file_data->s_hash
509         ]
510         as $o_file_data2
511     ){
512         if(
513             $o_file_data2->get_s_content()
514             === $o_file_data->get_s_content()
515         ){
516             $o_file_data->b_exists_with_same_content = true;
517             $o_file_data2->b_exists_with_same_content = true;
518             $o_file_data->s_content = '';
519             $o_file_data2->s_content = '';
520             if($s_valid_suffixes_regexp !== ''){
521                 if(
522                     preg_match(
523                         '/'
524                         .preg_quote(
525                             $o_file_data->s_path_from_reference_directory,
526                             '/'
527                         )
528                         .$s_valid_suffixes_regexp
529                         .'/',
530                         $o_file_data2->s_path_from_reference_directory,
531                     )
532                 ){
533                     $o_file_data->b_exists_with_suffixed_relative_path = true;

```

```

534         $o_file_data->b_exists_with_suffixed_relative_path_and = (
535             true
536         );
537     }
538     if(
539         $b_compute_also_reverse_status
540         && preg_match(
541             '/',
542             .preg_quote(
543                 $o_file_data2->s_path_from_reference_directory,
544                 '/',
545             )
546             . $s_valid_suffixes_regexp
547             . '/',
548             $o_file_data->s_path_from_reference_directory,
549         )
550     ){
551         $o_file_data2->b_exists_with_suffixed_relative_path = true;
552         $o_file_data2->b_exists_with_suffixed_relative_path_and = (
553             true
554         );
555     }
556     } //end if($s_valid_suffixes_regexp !== '')
557 } //end if(file_data2->content == file_data->content)
558 } //end foreach(file with same size and same hash)
559 } //end if(there is some file with same size and same hash)
560 } //end foreach($arr_files_data_source[...] as $o_file_data)
561
562 if($b_compute_also_reverse_status){
563     foreach($o_files_data_dest->arr_files_by_path_from as $o_file_data){
564         if($o_file_data->b_exists_with_suffixed_relative_path){
565             continue;
566         }
567         $arr_s_paths = glob(
568             $s_source_dirpath.$o_file_data->s_basename.'*'
569         );
570         if($arr_s_paths === false){
571             continue;
572         }
573         foreach($arr_s_paths as $s_filepath){
574             if(!isset($o_files_data_source->arr_files_by_path[$s_filepath])){
575                 // A newly created file that wasn't previously loaded
576                 // is ignored.
577                 continue;
578             }
579             $o_file_data2 = (

```

```

580         $o_files_data_source->arr_files_by_path[$s_filepath]
581     );
582     if(
583         preg_match(
584             '/',
585             .preg_quote(
586                 $o_file_data->s_path_from_reference_directory,
587                 '/',
588             )
589             . $s_valid_suffixes_regexp
590             . '/',
591             $o_file_data2->s_path_from_reference_directory,
592         )
593     ){
594         $o_file_data->b_exists_with_suffixed_relative_path = true;
595     }
596     }//end foreach($arr_s_paths as $s_filepath)
597 }//end foreach($arr_files_data_dest[...] as $o_file_data)
598 }//end if($b_compute_also_reverse_status)
599
600 return [
601     'source_files_data' => $o_files_data_source,
602     'dest_files_data' => $o_files_data_dest,
603 ];
604 }//end compare_files_under_directories()
605
606
607
608 /**
609  Archive some directory files into another directory.
610
611  @param string $s_source_dirpath The source directory to search in.
612  @param string $s_dest_dirpath The archive directory to move files in.
613  @param bool   $b_archive_if_missing_relative_filepath
614               Archive if no destination file is found with a relative
615               path that is identical.
616  @param bool   $b_archive_if_missing_content
617               Archive if no destination file is found with same content.
618  @param bool   $b_archive_if_missing_same_relative_path_content
619               Archive if no destination file is found with a relative
620               path that is identical and with same content.
621  @param bool   $b_archive_if_missing_suffixed_relative_path
622               Archive if no destination file is found with a relative
623               path that has a valid suffix
624               after the source relative path.
625  @param bool   $b_archive_if_missing_suffixed_relative_path_content

```

```

626             Archive if no destination file is found with a relative
627             path that has a valid suffix
628             after the source relative path
629             and with same content.
630 @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
631
632 @throws \Exception When $s_valid_suffixes_regexp is required
633             and not supplied.
634
635 @return void
636 */
637 function archive_directory_into_another(
638     string $s_source_dirpath,
639     string $s_dest_dirpath,
640     bool $b_archive_if_missing_relative_filepath,
641     bool $b_archive_if_missing_content,
642     bool $b_archive_if_missing_same_relative_path_content,
643     bool $b_archive_if_missing_suffixed_relative_path,
644     bool $b_archive_if_missing_suffixed_relative_path_content,
645     string $s_valid_suffixes_regexp = '',
646 ) : void {
647     if(
648         (
649             $b_archive_if_missing_suffixed_relative_path
650             || $b_archive_if_missing_suffixed_relative_path_content
651         )
652         && $s_valid_suffixes_regexp === ''
653     ){
654         throw new Exception(
655             'archive_directory_into_another is missing an'
656             .' $s_valid_suffixes_regexp argument.'
657         );
658     }
659     $arr_files_data = compare_files_under_directories(
660         $s_source_dirpath,
661         $s_dest_dirpath,
662         false,
663         $s_valid_suffixes_regexp,
664     );
665     foreach(
666         $arr_files_data['source_files_data']->arr_files_by_path_from
667         as $o_file_data
668     ){
669         if(
670             (
671                 $b_archive_if_missing_relative_filepath

```

```

672         && !$o_file_data->b_exists_with_same_relative_path
673     )
674     ||
675     (
676         $b_archive_if_missing_content
677         && !$o_file_data->b_exists_with_same_content
678     )
679     ||
680     (
681         $b_archive_if_missing_same_relative_path_content
682         && !$o_file_data->b_exists_with_same__relative_path_and_content
683     )
684     ||
685     (
686         $b_archive_if_missing_suffixed_relative_path
687         && !$o_file_data->b_exists_with_suffixed_relative_path
688     )
689     ||
690     (
691         $b_archive_if_missing_suffixed_relative_path_content
692         && !$o_file_data->b_exists_with_suffixed_relative_path_and
693     )
694 ) {
695     shell_exec(
696         'cp --backup=numbered'
697         ." '$".preg_replace("/'/", "'\"'\"'", $o_file_data->s_filepath)
698         ." "
699         ." '$".preg_replace("/'/", "'\"'\"'", $s_dest_dirpath)."'"
700     );
701 }
702 } //end foreach($arr_files_data['source_files_data'] as $o_file_data)
703 } //end archive_directory_into_another()
704 } //end namespace DOSASfiles
705
706
707
708
709 /*
710 $ php
711 <?php
712 require_once 'files.lib.php';
713 var_dump(
714     DOSASfiles\compare_files_under_directories(
715         '/home/laurent/.config/libreoffice/4/user/backup/',
716         '/home/laurent/Documents/LibreOfficeArchives/',
717     )

```

```

718 );
719 DOSASfiles\archive_directory_into_another(
720     '/home/laurent/.config/libreoffice/4/user/backup/',
721     '/home/laurent/Documents/LibreOfficeArchives/',
722     false,
723     true,
724     false,
725     false,
726     false,
727     '',
728 );
729 */
730 ?>

```

3.61 git_helpers/git_checks.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
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14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:57+02:00: This file was renamed from
25 # "git_checks.lib.sh"
26 # to
27 # "git_checks.lib.sh".
28 # 2024-05-31T01:57+02:00: This file was moved from
29 # "."
30 # to

```

```

31 # "./git_helpers".
32 # 2024-05-31T01:49+02:00: This file was renamed from
33 # "git_checks.lib.sh"
34 # to
35 # "git_checks.lib.sh".
36 # 2024-05-31T01:49+02:00: This file was moved from
37 # "."
38 # to
39 # "./git_helpers".
40 # This file was renamed from "git_checks.sh" to "git_checks.lib.sh".
41
42 check_files() {
43     # Checks used on Django projects.
44     # Use 'git commit --no-verify' after check.
45     declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46     declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations."
47     local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48     LFBFL_send_body_2+=" dès la création du modèle."
49     readonly LFBFL_send_body_2
50     declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51     declare -r LFBFL_send_body_3=\
52 "Les nouveaux preloadings sont interdits : -> #Prefetch()."
53     declare -i LFBFL_i_error=0
54     git diff --cached --name-only\
55     > /tmp/DOSAS_django_git_check1.temp
56     if grep models /tmp/DOSAS_django_git_check1.temp;
57     then
58         notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59         notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60         LFBFL_i_error=1
61     fi
62     if grep serializer /tmp/DOSAS_django_git_check1.temp;
63     then
64         notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65         LFBFL_i_error=1
66     fi
67     rm /tmp/DOSAS_django_git_check1.temp
68     # shellcheck disable=SC2248
69     return ${LFBFL_i_error}
70 }
71
72
73 check_no_abusive_trailing_comma() {
74     # Useful for Python code, although it regularly yields
75     # false positives in Python code
76     # (but it is easy to distinguish by hand).

```



```

77 # Use 'git commit --no-verify' after check.
78 declare -r LFBFL_send_summary_1="ATTENTION"
79 local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80 LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81 readonly LFBFL_send_body_1
82 declare -i LFBFL_i_error=0
83 git diff --cached\
84 > /tmp/DOSAS_django_git_check2.temp
85 if grep --perl-regexp ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86 then
87     notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88     LFBFL_i_error=1
89 fi
90 rm /tmp/DOSAS_django_git_check2.temp
91 # shellcheck disable=SC2248
92 return ${LFBFL_i_error}
93 }
94
95
96 check_black_code_formatting(){
97     # attention ça ne marche que sur les fichiers "stagés" avec git add
98     declare -r LFBFL_s_files_paths=$(
99         git diff --cached --name-only\
100         | grep '\.py$'
101     )
102     printf "%s\n" "${LFBFL_s_files_paths}"
103     declare -a LFBFL_arr_files_paths
104     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
105     readonly LFBFL_arr_files_paths
106     declare -i LFBFL_i_error=0
107     local LFBFL_file_path
108     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
109         printf "Black will check formatting on file %s.\n" "${LFBFL_file_path}"
110         if ! black --check --diff -- "${LFBFL_file_path}";
111         then
112             LFBFL_i_error=1
113         fi
114     done
115     # shellcheck disable=SC2248
116     return ${LFBFL_i_error}
117 }

```

3.62 git_helpers/post-commit

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.

```

```

3  #
4  # DevOrSysAdminScripts is free software:
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22 #
23 # ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:49+02:00: This file was renamed from
25 # "post-commit"
26 # to
27 # "post-commit".
28 # 2024-05-31T01:49+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32
33 LFBFL_some_directory="../DevOrSysAdminScripts"
34 declare -i LFBFL_i_pushd_result3
35 pushd "${LFBFL_some_directory}" || {
36     LFBFL_i_pushd_result3=$?
37     printf "post-commit pushd %s no such directory\n"\
38         "${LFBFL_some_directory}"
39     # shellcheck disable=SC2248
40     exit ${LFBFL_i_pushd_result3}
41 }
42 # shellcheck source=post_commit_renaming_comment.lib.sh
43 source "./git_helpers/post_commit_renaming_comment.lib.sh"
44 # shellcheck source=lines_filters.lib.sh
45 source "./build_and_checks_dependencies/lines_filters.lib.sh"
46 declare -i LFBFL_i_popd_result3
47 popd || {
48     LFBFL_i_popd_result3=$?

```

```

49     printf "post-commit no popd\n"
50     # shellcheck disable=SC2248
51     exit ${LFBFL_i_popd_result3}
52 }
53
54 LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
55 LFBFL_max_comment_line_length=""
56 if [[ -f "${LFBFL_data_file_name}" ]]; then
57     LFBFL_max_line_length=""
58     grep_variable "${LFBFL_data_file_name}" "max_line_length"\
59     --result-variable-prefix="LFBFL_"
60     LFBFL_max_comment_line_length="--max-comment-line-length="
61     LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
62 fi
63
64 commit_a_file_renamed_comment --verbose --log-directory-change\
65     "${LFBFL_max_comment_line_length}"
66 exit 0

```

3.63 git_helpers/post_commit_renaming_comment.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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18 # You should have received a copy of
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:49+02:00: This file was renamed from
25 # "post_commit_renaming_comment.lib.sh"

```

```

26 # to
27 # "post_commit_renaming_comment.lib.sh".
28 # 2024-05-31T01:49+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32 # This file was renamed from "post_commit_renaming_comment"
33 # to "post_commit_renaming_comment.lib.sh".
34 # (Before first commit it seems. But I had the old name in listing.)
35
36 LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
37 LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
38 LFBFL_some_directory="${LFBFL_this_file_directory}/../"
39 declare -i LFBFL_i_pushd_result2
40 pushd -- "${LFBFL_some_directory}" || {
41     LFBFL_i_pushd_result2=$?
42     printf\
43         "post_commit_renaming_comment.lib.sh pushd %s no such directory\n"\
44         "${LFBFL_some_directory}"
45     # shellcheck disable=SC2248
46     exit ${LFBFL_i_pushd_result2}
47 }
48 # shellcheck source=common_options.lib.sh
49 source ../build_and_checks_dependencies/common_options.lib.sh"
50 # shellcheck source=strings_functions.lib.sh
51 source ../build_and_checks_dependencies/strings_functions.lib.sh"
52 declare -i LFBFL_i_popd_result2
53 popd || {
54     LFBFL_i_popd_result2=$?
55     printf "post_commit_renaming_comment.lib.sh no popd\n"
56     # shellcheck disable=SC2248
57     exit ${LFBFL_i_popd_result2}
58 }
59
60 # Never forget that the worse security is the illusion of security
61 # (Kevin Mitnick). So, do not expect to have an accurate trackability
62 # of code history just because you use this post-commit hook or
63 # something else more evolved. This is useful but not a guarantee.
64 # This function will only handle the files that were only renamed in the
65 # preceding commit.
66 # It assumes that `diff` is the current default difference utility
67 # parametered with `git`. This point could be enhanced.
68 commit_a_file_renamed_comment() {
69     # Options:
70     # --verbose
71     # --log-directory-change

```

```

72 # --max-comment-line-length
73 declare -i LFBFL_i_verbose=0
74 get_verbose_option "$@"
75 declare -i LFBFL_i_log_directory_change
76 get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
77 declare -i LFBFL_i_max_comment_line_length
78 get_some_option LFBFL_i_max_comment_line_length\
79 --max-comment-line-length 70 70 1 "$@"
80 if [[ LFBFL_i_verbose -eq 1 ]]; then
81     LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
82     LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
83     # shellcheck disable=SC2059,SC2248
84     printf "${LFBFL_s_format}"\
85         "${LFBFL_i_log_directory_change}"\
86         "${LFBFL_i_max_comment_line_length}"
87 fi
88 if [[ LFBFL_i_max_comment_line_length -lt 10 ]]; then
89     declare -r LFBFL_send_summary="Max comment line length below 10."
90     notify-send "${LFBFL_send_summary}" "${LFBFL_send_summary}"
91     if [[ LFBFL_i_verbose -eq 1 ]]; then
92         printf "Max comment line length below 10, aborting.\n"
93     fi
94     return
95 fi
96 local LFBFL_s_format
97 #-----
98
99 enhanced_set_shell_option pipefail\
100     && trap 'enhanced_unset_shell_option pipefail' RETURN
101
102 declare -r LFBFL_renaming_lines=$(
103     git log --max-count=1 --patch\
104     | grep --after-context=1 '^diff --git'
105 )
106 if [[ LFBFL_i_verbose -eq 1 ]]; then
107     printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
108 fi
109
110 declare -a LFBFL_renaming_lines_array
111 mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
112 readonly LFBFL_renaming_lines_array
113
114 declare -r LFBFL_timestamp=$(date --iso-8601="minute")
115 if [[ LFBFL_i_verbose -eq 1 ]]; then
116     printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
117 fi

```

```

118 local LFBFL_split_score_command1
119 declare -i LFBFL_i_split_score_command_properties1
120 # shellcheck disable=SC2248
121 get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
122 LFBFL_split_score_command1="${get_split_score_result}"
123 LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
124 readonly LFBFL_split_score_command1
125 readonly LFBFL_i_split_score_command_properties1
126 local LFBFL_split_score_command2
127 declare -i LFBFL_i_split_score_command_properties2
128 # shellcheck disable=SC2248
129 get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
130 LFBFL_split_score_command2="${get_split_score_result}"
131 LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
132 readonly LFBFL_split_score_command2
133 readonly LFBFL_i_split_score_command_properties2
134 declare -i LFBFL_i_renaming_happened=0
135 declare -i LFBFL_i
136 declare -i LFBFL_j
137 declare -r LFBFL_special_file_name="files_names_listing.txt"
138 local LFBFL_diff_line
139 local LFBFL_similarity_line
140 local LFBFL_old_file_path
141 local LFBFL_old_file_name
142 local LFBFL_old_file_directory
143 local LFBFL_new_file_path
144 local LFBFL_new_file_name
145 local LFBFL_new_file_directory
146 local LFBFL_useful_file_name
147 local LFBFL_extension
148 local LFBFL_comment_prefix
149 local LFBFL_comment_prefix2
150 local LFBFL_old_file_name2
151 local LFBFL_old_file_directory2
152 local LFBFL_new_file_name2
153 local LFBFL_new_file_directory2
154 local LFBFL_new_comment
155 declare -i LFBFL_i_copyright_line_number
156 declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
157 declare -i LFBFL_i_line_count
158 declare -i LFBFL_i_lines_after
159 local LFBFL_temp_file_path
160 for ((LFBFL_i = 0; LFBFL_i < ${#LFBFL_renaming_lines_array[@]};)); do
161     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
162     LFBFL_i=$((LFBFL_i + 1))
163     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"

```

```

164 LFBFL_i=$((LFBFL_i + 1))
165 # We need to increment 3 times since grep adds "--" lines between.
166 LFBFL_i=$((LFBFL_i + 1))
167 if [[ LFBFL_i_verbose -eq 1 ]]; then
168     printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n" \
169         "${LFBFL_diff_line}" \
170         "${LFBFL_similarity_line}"
171 fi
172 if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
173     continue
174 fi
175
176 # Extract old file name from diff line. -----
177 LFBFL_old_file_path=$(
178     printf "%s" "${LFBFL_diff_line}" \
179     | cut --delimiter=' ' --fields=3
180 )
181 # It starts by "a/".
182 LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
183 LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
184 LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
185 # Extract new file name from diff line. -----
186 LFBFL_new_file_path=$(
187     printf "%s" "${LFBFL_diff_line}" \
188     | cut --delimiter=' ' --fields=4
189 )
190 # It starts by "b/".
191 LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
192 LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
193 LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")
194 if [[ LFBFL_i_verbose -eq 1 ]]; then
195     LFBFL_s_format="LFBFL_old_file_path: %s\n"
196     LFBFL_s_format+="LFBFL_old_file_name: %s\n"
197     LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
198     LFBFL_s_format+="LFBFL_new_file_path: %s\n"
199     LFBFL_s_format+="LFBFL_new_file_name: %s\n"
200     LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
201     # shellcheck disable=SC2059
202     printf "${LFBFL_s_format}" \
203         "${LFBFL_old_file_path}" \
204         "${LFBFL_old_file_name}" \
205         "${LFBFL_old_file_directory}" \
206         "${LFBFL_new_file_path}" \
207         "${LFBFL_new_file_name}" \
208         "${LFBFL_new_file_directory}"
209 fi

```

```

210 # Extract "LFBFL_extension" with smart guessing for .tpl -----
211 # and set prefix for line with ©Copyright. -----
212 LFBFL_useful_file_name="{LFBFL_new_file_name}"
213 LFBFL_extension="{LFBFL_useful_file_name##*}"
214 if [[ LFBFL_i_verbose -eq 1 ]]; then
215     printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n" \
216         "${LFBFL_useful_file_name}" \
217         "${LFBFL_extension}"
218 fi
219 for ((LFBFL_j = 0; LFBFL_j < 3; ++LFBFL_j)); do
220     if [[ "${LFBFL_extension}" != "tpl" ]]; then
221         break
222     fi
223     LFBFL_useful_file_name="{LFBFL_useful_file_name:0:-4}"
224     LFBFL_extension="{LFBFL_useful_file_name##*}"
225     if [[ LFBFL_i_verbose -eq 1 ]]; then
226         printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n" \
227             "${LFBFL_useful_file_name}" \
228             "${LFBFL_extension}"
229     fi
230 done
231 LFBFL_comment_prefix=""
232 if [[
233     "${LFBFL_extension}" == "sh"
234     || "${LFBFL_extension}" == "toml"
235     || "${LFBFL_extension}" == "yaml"
236 ]]; then
237     LFBFL_comment_prefix="# "
238 fi
239 if [[ "${LFBFL_extension}" == "sql" ]]; then
240     LFBFL_comment_prefix="-- "
241 fi
242 if [[ "${LFBFL_extension}" == "tex" ]]; then
243     LFBFL_comment_prefix="% "
244 fi
245 if [[
246     "${LFBFL_useful_file_name}" == "${LFBFL_special_file_name}"
247     || "${LFBFL_extension}" == "rs"
248     || "${LFBFL_extension}" == "rlib"
249 ]]; then
250     LFBFL_comment_prefix="// "
251 fi
252 if [[ LFBFL_i_verbose -eq 1 ]]; then
253     printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
254 fi
255 fi

```



```

256 #-----
257 # Split if necessary old_file_name, new_file_name, etc. to have
258 # comment lines below max_comment_line_length.
259 LFBFL_comment_prefix2="${LFBFL_comment_prefix}"' "'
260 # -----
261 LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
262 LFBFL_old_file_name2+="${LFBFL_old_file_name}"
263 # shellcheck disable=SC2248
264 repeated_split_last_line "${LFBFL_old_file_name2}"\
265   "${LFBFL_comment_prefix2}"\
266   "${LFBFL_i_max_comment_line_length}"\
267   "' '\
268   "${LFBFL_split_score_command1}"\
269   "${LFBFL_i_split_score_command_properties1}"
270 LFBFL_old_file_name2="${repeated_split_last_line_result}"
271 # -----
272 LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
273 LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
274 # shellcheck disable=SC2248
275 repeated_split_last_line "${LFBFL_old_file_directory2}"\
276   "${LFBFL_comment_prefix2}"\
277   "${LFBFL_i_max_comment_line_length}"\
278   "' '\
279   "${LFBFL_split_score_command2}"\
280   "${LFBFL_i_split_score_command_properties2}"
281 LFBFL_old_file_directory2="${repeated_split_last_line_result}"
282 # -----
283 LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
284 LFBFL_new_file_name2+="${LFBFL_new_file_name}"
285 # shellcheck disable=SC2248
286 repeated_split_last_line "${LFBFL_new_file_name2}"\
287   "${LFBFL_comment_prefix2}"\
288   "${LFBFL_i_max_comment_line_length}"\
289   "' '\
290   "${LFBFL_split_score_command1}"\
291   "${LFBFL_i_split_score_command_properties1}"
292 LFBFL_new_file_name2="${repeated_split_last_line_result}"
293 # -----
294 LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
295 LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
296 # shellcheck disable=SC2248
297 repeated_split_last_line "${LFBFL_new_file_directory2}"\
298   "${LFBFL_comment_prefix2}"\
299   "${LFBFL_i_max_comment_line_length}"\
300   "' '\
301   "${LFBFL_split_score_command2}"\

```

```

302     ${LFBFL_i_split_score_command_properties2}
303 LFBFL_new_file_directory2="${repeated_split_last_line_result}"
304 # Create new comments strings. -----
305 LFBFL_new_comment=""
306 if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
307     LFBFL_new_comment="${LFBFL_comment_prefix}"
308     LFBFL_new_comment+="${LFBFL_timestamp}"
309     LFBFL_new_comment+=": This file was renamed from"
310     LFBFL_new_comment+=$'\n'
311     LFBFL_new_comment+="${LFBFL_old_file_name2}"' "'
312     LFBFL_new_comment+=$'\n'
313     LFBFL_new_comment+="${LFBFL_comment_prefix}to"
314     LFBFL_new_comment+=$'\n'
315     LFBFL_new_comment+="${LFBFL_new_file_name2}"' ". '
316 fi
317 if [[
318     LFBFL_i_log_directory_change -eq 1
319     && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
320 ]];
321 then
322     if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
323         LFBFL_new_comment+=$'\n'
324     fi
325     LFBFL_new_comment+="${LFBFL_comment_prefix}"
326     LFBFL_new_comment+="${LFBFL_timestamp}"
327     LFBFL_new_comment+=": This file was moved from"
328     LFBFL_new_comment+=$'\n'
329     LFBFL_new_comment+="${LFBFL_old_file_directory2}"' "'
330     LFBFL_new_comment+=$'\n'
331     LFBFL_new_comment+="${LFBFL_comment_prefix}to"
332     LFBFL_new_comment+=$'\n'
333     LFBFL_new_comment+="${LFBFL_new_file_directory2}"' ". '
334 fi
335 if [[ LFBFL_i_verbose -eq 1 ]]; then
336     printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
337 fi
338 # Find line with ©Copyright. -----
339 LFBFL_i_copyright_line_number=$(
340     grep --line-number -- '©Copyright' "${LFBFL_new_file_path}"\
341     | head --lines=1\
342     | cut --delimiter=':' --fields=1
343 )
344 if [[ LFBFL_i_verbose -eq 1 ]]; then
345     printf "LFBFL_i_copyright_line_number: %s\n"\
346         "${LFBFL_i_copyright_line_number}"
347 fi

```

```

348 if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
349     notify-send "${LFBFL_send_summary_2}"\
350     "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
351     if [[ LFBFL_i_verbose -eq 1 ]]; then
352         printf "No copyright line. Skipping file.\n"
353     fi
354     continue
355 fi
356 # Get total number of lines. -----
357 LFBFL_i_line_count=$(
358     wc --lines\
359     < "${LFBFL_new_file_path}"
360 )
361 if [[ LFBFL_i_verbose -eq 1 ]]; then
362     printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
363 fi
364 # Compute number of lines after. -----
365 LFBFL_i_lines_after=$((
366     LFBFL_i_line_count - LFBFL_i_copyright_line_number
367 ))
368 if [[ LFBFL_i_verbose -eq 1 ]]; then
369     printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
370 fi
371 # Update file. -----
372 LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
373 {
374     head --lines="${LFBFL_i_copyright_line_number}"\
375     -- "${LFBFL_new_file_path}"
376     printf "%s\n" "${LFBFL_new_comment}"
377     tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
378 } >> "${LFBFL_temp_file_path}"
379 mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
380 # Add file to stage. -----
381 git add -- "${LFBFL_new_file_path}"
382 LFBFL_i_renaming_happened=1
383 done
384 readonly LFBFL_i_renaming_happened
385 if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
386     # Commit -----
387     local LFBFL_message="commit_a_file_renamed_comment()"
388     LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
389     readonly LFBFL_message
390     printf "Commit message:\n%s\n" "${LFBFL_message}"
391     git commit --no-verify --message="${LFBFL_message}"
392 fi
393 }

```

3.64 git_helpers/pre-commit

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it under the terms
6  # of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed in the hope
12 # that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:57+02:00: This file was renamed from
25 # "pre-commit"
26 # to
27 # "pre-commit".
28 # 2024-05-31T01:57+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32
33 LFBFL_some_directory="../../DevOrSysAdminScripts"
34 declare -i LFBFL_i_pushd_result3
35 pushd "${LFBFL_some_directory}" || {
36     LFBFL_i_pushd_result3=$?
37     printf "pre-commit pushd %s no such directory\n"\
38         "${LFBFL_some_directory}"
39     # shellcheck disable=SC2248
40     exit ${LFBFL_i_pushd_result3}
41 }
42 # shellcheck source=git_checks.lib.sh
43 source "./git_helpers/git_checks.lib.sh"
44 declare -i LFBFL_i_popd_result3
```

```

45 popd || {
46     LFBFL_i_popd_result3=$?
47     printf "pre-commit no popd\n"
48     # shellcheck disable=SC2248
49     exit ${LFBFL_i_popd_result3}
50 }
51
52 declare -gi LFBFL_i_git_hook_error=0
53 if ! check_files; then
54     LFBFL_i_git_hook_error=1
55 fi
56 if ! check_no_abusive_trailing_comma; then
57     LFBFL_i_git_hook_error=1
58 fi
59 if ! check_black_code_formatting; then
60     LFBFL_i_git_hook_error=1
61 fi
62 readonly LFBFL_i_git_hook_error
63 # shellcheck disable=SC2248
64 exit ${LFBFL_i_git_hook_error}

```

3.65 IMAP_transfert.exec.py

```

1  """
2  This file is part of DevOrSysAdminScripts.
3
4  DevOrSysAdminScripts is free software:
5  you can redistribute it and/or modify it
6  under the terms of the GNU Lesser General Public License
7  as published by the Free Software Foundation,
8  either version 3 of the License,
9  or (at your option) any later version.
10
11  DevOrSysAdminScripts is distributed
12  in the hope that it will be useful,
13  but WITHOUT ANY WARRANTY;
14  without even the implied warranty of
15  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16  See the GNU Lesser General Public License for more details.
17
18  You should have received a copy of
19  the GNU Lesser General Public License
20  along with DevOrSysAdminScripts.
21  If not, see <https://www.gnu.org/licenses/>.
22
23  ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet

```

```

24 2026-05-12T12:06+02:00: This file was renamed from
25 "IMAP_transfert.py"
26 to
27 "IMAP_transfert.exec.py".
28
29 Pour transférer d'un compte IMAP à un autre
30 Penser à autoriser les apps moins sûres sur Gmail si nécessaire
31 """
32
33 import argparse
34 import email
35 import imaplib
36
37
38 class IMAPTransfertBadArgumentError(Exception):
39     """
40     A class to avoid a linter warning,
41     that needs an Error suffix to avoid another linter warning,
42     that needs a docstring to avoid another linter warning.
43     If you don't see its intent, try badminton.
44     """
45
46
47 class SurpriseError(Exception):
48     """
49     A class to avoid a linter warning,
50     that needs an Error suffix to avoid another linter warning,
51     that needs a docstring to avoid another linter warning.
52     If you don't see its intent after seeing its use, try volleyball.
53     But "Cheer up!", now you don't need "pass" XD.
54     """
55
56
57 def IMAP_transfert(
58     from_login: str,
59     from_password: str,
60     from_host: str,
61     from_port: int,
62     to_login: str,
63     to_password: str,
64     to_host: str,
65     to_port: int,
66     *,
67     to_new_folder: str = "",
68     verbose: bool = False,
69 ) -> None:

```

```

70     """
71     Transfert all emails from some IMAP account
72     to another IMAP account.
73     """
74     empty = ""
75     if from_login == "":
76         raise IMAPTransfertBadArgumentError("From login is empty.")
77     # if from_password == "":
78     if from_password == empty:
79         # https://github.com/PyCQA/bandit/issues/714
80         # Dumb
81         raise IMAPTransfertBadArgumentError("From password is empty.")
82     if from_host == "":
83         raise IMAPTransfertBadArgumentError("From host is empty.")
84     if from_port <= 0:
85         raise IMAPTransfertBadArgumentError(
86             "From port is not positive.",
87         )
88     if to_login == "":
89         raise IMAPTransfertBadArgumentError("To login is empty.")
90     if to_password == empty:
91         # And Dumber The infinite collection XD
92         raise IMAPTransfertBadArgumentError("To password is empty.")
93     if to_host == "":
94         raise IMAPTransfertBadArgumentError("To host is empty.")
95     if to_port <= 0:
96         raise IMAPTransfertBadArgumentError(
97             "To port is not positive.",
98         )
99     # pylint: disable=pointless-string-statement
100    """
101    Maybe one day Python will ship something like that in its standard
102    library:
103    https://www.php.net/manual/fr/function.filter-var.php
104    But now urlparse doesn't do any validation.
105    You may want to check that the hosts or other fields are not
106    "anything".
107    """
108
109    # pylint: disable=unreachable
110    raise SurpriseError(
111        "I haven't tested this version of my script... Ooooooh!\n"
112        "But I linted it without this exception ;). Aaaaaaaah!\n"
113        "XD XD XD\n"
114        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
115    )

```

```

116 # """
117 # I haven't tested any version with an empty to_new_folder...
118
119 if verbose:
120     print('Connecting to "from account".')
121 from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
122 from_account.login(from_login, from_password)
123
124 if verbose:
125     print('Connecting to "to account".')
126 to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
127 to_account.login(to_login, to_password)
128
129 if to_new_folder != "":
130     if verbose:
131         print('Creating new folder in "to account".')
132     to_account.create(to_new_folder)
133
134 if verbose:
135     print('Fetching all emails ids in "from account".')
136 from_account.select()
137 result, ids = from_account.uid("search", "all")
138 if verbose:
139     print(f'from_account.uid("search") result:{result} ids:{ids}')
140 ids = ids[0].decode().split()
141 if verbose:
142     print('Fetching and transferring each email in "from account".')
143 for my_id in ids:
144     result, raw_message = from_account.uid(
145         "fetch",
146         my_id,
147         "(RFC822)",
148     )
149     if verbose:
150         print(f'from_account.uid("fetch", {my_id}) result:{result}')
151     message = raw_message[0][1]
152     my_email = email.message_from_string(message.decode("utf-8"))
153     flags = imaplib.ParseFlags(raw_message[0][0])
154     # flags = " ".join(flags) mypy is unhappy but IMAP flags are
155     # ASCII strings, we give mypy what it wants.
156     flags_as_strs = [flag.decode("ascii") for flag in flags]
157     flags_str = " ".join(flags_as_strs)
158     # date = imaplib.Internaldate2tuple(raw_message[0][0])
159     date = '"" + my_email["date"] + ""'
160     if verbose:
161         print(

```



```

162         f'from_account.uid("fetch", {my_id})'
163         f" flags:{flags} date:{date}",
164     )
165     to_account.append(to_new_folder, flags_str, date, message)
166
167
168 def main() -> None:
169     """
170     Parse command-line arguments and
171     transfert all emails from some IMAP account
172     to another IMAP account.
173     """
174     parser = argparse.ArgumentParser(
175         description=(
176             "Transfert all emails from some IMAP account"
177             " to another IMAP account"
178         ),
179     )
180     parser.add_argument(
181         "from_login",
182         type=str,
183         help="Email account login of user one.",
184     )
185     parser.add_argument(
186         "from_password",
187         type=str,
188         help="Email account password of user one.",
189     )
190     parser.add_argument(
191         "to_login",
192         type=str,
193         help="Email account login of user two.",
194     )
195     parser.add_argument(
196         "to_password",
197         type=str,
198         help="Email account password of user two.",
199     )
200     parser.add_argument(
201         "--to_new_folder",
202         type=str,
203         default="",
204         help="Transfert emails into a new folder of this name.",
205     )
206     parser.add_argument(
207         "--from_host",

```

```

208         type=str,
209         default="imap.gmail.com",
210         help="Host of first email account",
211     )
212     parser.add_argument(
213         "--from_port",
214         type=int,
215         default=993,
216         help="Port on host of first email account",
217     )
218     parser.add_argument(
219         "--to_host",
220         type=str,
221         default="imap.gmail.com",
222         help="Host of second email account",
223     )
224     parser.add_argument(
225         "--to_port",
226         type=int,
227         default=993,
228         help="Port on host of second email account",
229     )
230     parser.add_argument(
231         "-v",
232         "--verbose",
233         action="store_true",
234         help="Print details of operations",
235     )
236
237     args = parser.parse_args()
238
239     if args.verbose:
240         print(
241             "-----\n"
242             "IMAP transfert:\n"
243             "-----\n"
244             "From account:\n"
245             f" - Login: {args.from_login}\n"
246             f" - Password: {args.from_password}\n"
247             f" - Host: {args.from_host}\n"
248             f" - Port: {args.from_port}\n"
249             "To account:\n"
250             f" - Login: {args.to_login}\n"
251             f" - Password: {args.to_password}\n"
252             f" - Host: {args.to_host}\n"
253             f" - Port: {args.to_port}\n"

```

```

254         "-----\n",
255     )
256     if args.to_new_folder is not None:
257         print(
258             f"Emails will go in folder {args.to_new_folder}"
259             " after it is created.",
260         )
261     try:
262         IMAP_transfert(
263             args.from_login,
264             args.from_password,
265             args.from_host,
266             args.from_port,
267             args.to_login,
268             args.to_password,
269             args.to_host,
270             args.to_port,
271             to_new_folder=args.to_new_folder,
272             verbose=args.verbose,
273         )
274     except (IMAPTransfertBadArgumentError, SurpriseError) as e:
275         print(e)
276
277
278 if __name__ == "__main__":
279     main()

```

3.66 LiensUtiles.txt

```

1  https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
2  -configurez-le-service-de-migration-de-donn%C3%A9es
3  -----
4  https://stackoverflow.com/a/78413334/5796086
5  https://stackoverflow.com/questions/21612980/\
6  why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
7  https://web.archive.org/web/20240501105801/https://stackoverflow.com\
8  /questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
9  78413334#78413334
10 -----
11 https://news.ycombinator.com/item?id=22214690
12 https://github.com/koalaman/shellcheck/issues/2005
13 https://opensource.stackexchange.com/questions/6271/\
14 calling-gpld-programs-from-a-shell-script
15 -----
16 https://pubs.opengroup.org/onlinepubs/9699919799/
17 When looking `man sed` GNU sed says to use POSIX -E instead of

```

```

18 -r or --regexp-extended for portability.
19 But the link above says that still now only -e, -f, -n are POSIX options
20 for sed.
21 For this reason we replaced all short option names by long option names
22 when possible, even for sed.

```

3.67 miscellaneous/funny.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
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13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24
25 unprotected_funny() {
26     # Works in a script but not in shell if job control is active.
27     shopt -s lastpipe
28     declare -i LFBFL_i_result=0
29     local LFBFL_line
30     printf "hahaha\n" | while read -r LFBFL_line; do
31         if [[ "${LFBFL_line}" == "hahaha" ]]; then
32             LFBFL_i_result=1
33         fi
34     done
35     [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
36 }
37
38 protected_funny() {

```

```

39 # Always works regarding the echoes.
40 printf "hahaha\n" | while read -r LFBFL_line; do
41     unprotected_funny
42 done
43 }

```

3.68 miscellaneous/grep_benchmark.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
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13 # but WITHOUT ANY WARRANTY;
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15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
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18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 some_diff(){
26     for ((i=0; i < 1000; ++i)); do
27         # shellcheck disable=SC2312
28         diff --suppress-common-lines\
29             build_and_checks_variables/phpmd_baseline.xml\
30             build_and_checks_variables/temp/phpmd_baseline.xml\
31             | grep --invert-match\
32                 --regexp='<!-- '\
33                 --regexp='^[0-9]\+d[0-9]\+$'\
34                 --regexp='^[0-9]\+, [0-9]\+d[0-9]\+$'
35     done
36 }
37
38 some_diff2(){

```

```

39     for ((i=0; i < 1000; ++i)); do
40         # shellcheck disable=SC2312
41         diff --suppress-common-lines\
42             build_and_checks_variables/phpmd_baseline.xml\
43             build_and_checks_variables/temp/phpmd_baseline.xml\
44             | grep --invert-match\
45                 --regexp='<!-- '\
46                 --regexp='^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$'
47     done
48 }
49
50 some_diff3(){
51     for ((i=0; i < 1000; ++i)); do
52         # shellcheck disable=SC2312
53         diff --suppress-common-lines\
54             build_and_checks_variables/phpmd_baseline.xml\
55             build_and_checks_variables/temp/phpmd_baseline.xml\
56             | grep --invert-match\
57                 --regexp='\(<!-- \|^[0-9]\+d[0-9]\+$\|^ [0-9]\+, [0-9]\+d[0-9]\+$\)'
58     done
59 }
60
61 some_diff4(){
62     for ((i=0; i < 1000; ++i)); do
63         # shellcheck disable=SC2312
64         diff --suppress-common-lines\
65             build_and_checks_variables/phpmd_baseline.xml\
66             build_and_checks_variables/temp/phpmd_baseline.xml\
67             | grep --invert-match\
68                 --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
69     done
70 }
71
72 # $ time some_diff
73 # real      0m1,663s
74 # user      0m0,795s
75 # sys       0m1,975s
76 # $ time some_diff2
77 # real      0m1,597s
78 # user      0m0,705s
79 # sys       0m2,009s
80 # $ time some_diff3
81 # real      0m1,619s
82 # user      0m0,758s
83 # sys       0m1,969s
84 # $ time some_diff4

```

```

85 # real          0m1,574s
86 # user          0m0,750s
87 # sys           0m1,932s

```

3.69 miscellaneous/spamty_data.txt

```

1
2
3
4 a
5 a
6 empty.txt:
7 one_space.txt:
8 all_space.txt:
9 normal.txt:a
10 normal_again.txt: a

```

3.70 miscellaneous/string_escaping.html

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="utf-8">
5   <title>Truc</title>
6 </head>
7 <body>
8 <header id="title-block-header">
9 <h1 class="title">Truc</h1>
10 </header>
11 &lt;abc> &amp; &lt;de>
12 <p title="<&>'
13   abcd\\ machin">
14 This is some text
15 </p>
16
17 <pre>
18 "&lt;&amp;>'&quot;
19 abcd\\ machin"
20 This is some text
21 </pre>
22
23 <script>
24 let a = 'abcd & < > \\ \n  \"\' \u{00003c}/script>';
25 alert(a);
26 let b = "abcd & < > \\ \n  \"\' ";
27 alert(b);

```

```

28 </script>
29
30 </body>
31 </html>

```

3.71 miscellaneous/string_escaping.xhtml

```

1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
2   "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
3 <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
4 <head>
5 <meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
6   <title>Truc</title>
7 </head>
8 <body>
9 <h1 class="title">Truc</h1>
10 <lt;abc> & <de>
11
12 <p title="& >'&#10;
13   abcd\ machin">
14   This is some text
15 </p>
16
17 <pre>
18   "& >'&quot;
19   abcd\ machin"
20   This is some text
21 </pre>
22
23 <script type="text/javascript">
24   let a = 'abcd & < > \\ \n  \"'\"';
25   alert(a);
26   let b = "abcd & < > \\ \n  \"'\"";
27   alert(b);
28 </script>
29
30 </body>
31 </html>

```

3.72 README.md.tpl

```

1 # DevOrSysAdminScripts
2
3 [![CodeFactor-badge]] [CodeFactor-package-page]
4 [![Codacy-badge]] [Codacy-package-page]
5 ![GitHub-top-language-badge]

```



```

6  ![GitHub-code-size-in-bytes-badge]
7  [![Security:bandit:badge]][Bandit-GitHub.com]
8  [![Code-style:black:badge]][Black-GitHub.com]
9  [![Imports:isort:badge]][Isort-GitHub.io]
10 [![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
11 [![Linting:pylint:badge]][Pylint-GitHub.com]
12 [![Linting:ruff:badge]][Ruff-GitHub.com]
13
14 |                                     **Some "small" useful scripts**                                     |
15 | :-----: |
16 | **for enhancing development or system-administrators tasks** |
17
18 Currently, the biggest part of this repository is in
19 a suite of shell scripts about build and checks that
20 I use in some of my other repositories.
21
22 I tried to follow some conventions for my bash code:
23
24 - I tried to encapsulate most of the code in functions.
25 - Unless used as return values, variables in functions are defined
26   as local variables (`local` keyword or `declare`)
27   with prefix "LFBFL_".
28 - When used as return values, variables in functions are defined
29   as global variables (`declare -g`) with prefix
30   "@function_name@_result_".
31 - Whenever a variable is no more modified after some point,
32   add the keyword `readonly` or use `declare -r`.
33 - Whenever a variable will only contain integer values (or boolean
34   values as 0 or 1), use `declare -i`.
35 - I named bash files with suffixes ".exec.sh" when the script can
36   be runned
37   (executable, abbreviation stops before a vowel).
38 - I named bash files with suffixes ".lib.sh" when the script contains
39   only functions definitions
40   (library, abbreviation stops before a vowel).
41 - But the files "pre-commit" and "post-commit" were not renamed,
42   since it is not possible to give them other names to use them as
43   pre-commit/post-commit hooks in `git`.
44 - When some ".exec.sh" code is encapsulated into functions,
45   either these functions can be reused and go in some ".lib.sh" file,
46   either they are truly specific to this script and they are kept
47   in ".exec.sh" file.
48 - When using `source` builtin, although any path/filename with a slash
49   will only be searched for from the current directory (tested),
50   we add the prefix "./", to make it explicit.
51 - We use mapfile instead of read -r most of the time, but with the

```

```

52     following exceptions: real use of read -r to read the command line of the
53     shell, comments (of course), and a file funny.sh where it is kind of
54     required to use read -r.
55 - We replaced all echo calls by printf calls to follow
56   "The Open Group Base Specifications" since at least 2 decades:
57   "New applications are encouraged to use printf instead of echo.",
58   <https://pubs.opengroup.org/onlinepubs/000095399/>.
59 - We used `'_?i_'` or `'_?j_'`, etc. in integer variables names in Bash
60   scripts.
61
62 [CodeFactor-badge]: https://www.codefactor.io/repository/github/\
63 llyaudet/DevOrSysAdminScripts/badge
64
65 [CodeFactor-package-page]: https://www.codefactor.io/repository/\
66 github/llyaudet/DevOrSysAdminScripts
67
68 [Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
69 3dd6ef85e8db4460991d1ebc404b68dd
70
71 [Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
72 DevOrSysAdminScripts/dashboard?utm\_source=gh&utm\_medium=referral\
73 &utm\_content=&utm\_campaign=Badge\_grade
74
75 [GitHub-top-language-badge]: https://img.shields.io/github/\
76 languages/top/llyaudet/DevOrSysAdminScripts
77
78 [GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
79 languages/code-size/llyaudet/DevOrSysAdminScripts
80
81 [Security:bandit:badge]: https://img.shields.io/badge/\
82 security-bandit-yellow.svg
83
84 [Bandit-GitHub.com]: https://github.com/PyCQA/bandit
85
86 [Code-style:black:badge]: https://img.shields.io/badge/\
87 code%20style-black-000000.svg
88
89 [Black-GitHub.com]: https://github.com/psf/black
90
91 [Imports:isort:badge]: https://img.shields.io/badge/\
92 %20imports-isort-%231674b1?style=flat&labelColor=ef8336
93
94 [Isort-GitHub.io]: https://pycqa.github.io/isort/
95
96 [Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
97 mypy\_badge.svg

```

```

98
99 [Typecheck-mypy-lang.org]: https://mypy-lang.org/
100
101 [Linting:pylint:badge]: https://img.shields.io/badge/\
102 linting-pylint-yellowgreen
103
104 [Pylint-GitHub.com]: https://github.com/pylint-dev/pylint
105
106 [Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
107 https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
108 badge/v2.json
109
110 [Ruff-GitHub.com]: https://github.com/astral-sh/ruff

```

3.73 string_escaping.lib.php

```

1 <?php
2
3 /**
4  This file is part of DevOrSysAdminScripts.
5
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
26
27 @category Library
28 @package DevOrSysAdminScripts
29 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
30 @copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet

```

```

31 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
32 */
33
34 declare(strict_types=1);
35 declare(encoding='UTF-8');
36
37
38
39 /**
40 The namespace string_escaping is splitted into subnamespaces for each
41 language.
42 This subnamespace is for HTML (surprised?)...
43
44 @package DevOrSysAdminScripts
45 @subpackage string_escaping\HTML
46 */
47 namespace string_escaping\HTML {
48     /**
49      When you want that the string appears as is in HTML text block.
50
51      @param string $s_string The string you want to escape.
52
53      @return string The escaped string.
54     */
55     function escape_text(string $s_string) : string {
56         return str_replace(
57             [
58                 '&',
59                 '<',
60             ],
61             [
62                 '&',
63                 '<',
64             ],
65             $s_string,
66         );
67     } //end escape_text()
68
69
70
71     /**
72      When you want that the string appears as is in an HTML tag attribute
73      that starts and ends with double quotes: my_attribute="something".
74      Example: a title attribute.
75      /*! in double quotes /*! my_attribute='something' is not supported.
76

```

```

77  @param string $s_string The string you want to escape.
78
79  @return string The escaped string.
80  */
81  function escape_attribute(string $s_string) : string {
82      return str_replace(
83          ['"',
84           '"'],
85          $s_string,
86      );
87  }//end escape_attribute()
88
89
90
91  /**
92   When you want that the string appears verbatim inside HTML <pre> tags.
93
94   @param string $s_string The string you want to escape.
95
96   @return string The escaped string.
97   */
98   function escape_pre(string $s_string) : string {
99       return str_replace(
100          [
101              '&',
102              '<',
103          ],
104          [
105              '&amp;',
106              '&lt;',
107          ],
108          $s_string,
109      );
110  }//end escape_pre()
111  }//end namespace string_escaping\HTML
112
113
114
115
116  /**
117   The namespace string_escaping is splitted into subnamespaces for each
118   language.
119   This subnamespace is for XHTML (still surprised?)...
120
121   @package DevOrSysAdminScripts
122   @subpackage string_escaping\XHTML

```

```

123 */
124 namespace string_escaping\XHTML {
125     /**
126     When you want that the string appears as is in XHTML text block.
127
128     @param string $s_string The string you want to escape.
129
130     @return string The escaped string.
131     */
132     function escape_text(string $s_string) : string {
133         return str_replace(
134             [
135                 '&',
136                 '<',
137             ],
138             [
139                 '&amp;',
140                 '&lt;',
141             ],
142             $s_string,
143         );
144     } //end escape_text()
145
146
147
148     /**
149     When you want that the string appears as is in an XHTML tag attribute
150     that starts and ends with double quotes: my_attribute="something".
151     Example: a title attribute.
152     /*!\ in double quotes /*!\ my_attribute='something' is not supported.
153
154     @param string $s_string The string you want to escape.
155
156     @return string The escaped string.
157     */
158     function escape_attribute(string $s_string) : string {
159         return str_replace(
160             [
161                 '&',
162                 '<',
163                 '"',
164                 "\n",
165             ],
166             [
167                 '&amp;',
168                 '&lt;',

```

```

169         '&quot;','
170         '&#10;','
171     ],
172     $s_string,
173 );
174 }//end escape_attribute()
175
176
177
178 /**
179  When you want that the string appears verbatim inside XHTML <pre> tags.
180
181  @param string $s_string The string you want to escape.
182
183  @return string The escaped string.
184  */
185  function escape_pre(string $s_string) : string {
186      return str_replace(
187          [
188              '&',
189              '<',
190          ],
191          [
192              '&amp;',
193              '&lt;',
194          ],
195          $s_string,
196      );
197  }//end escape_pre()
198 }//end namespace string_escaping\XHTML
199
200
201
202
203 /**
204  The namespace string_escaping is splitted into subnamespaces for each
205  language.
206  This subnamespace is for JS (you're hopeless?)...
207
208  @package DevOrSysAdminScripts
209  @subpackage string_escaping\JS
210  */
211  namespace string_escaping\JS {
212
213      use function string_escaping\XHTML\escape_text as XHTML_escape_text;
214

```

```

215
216
217  /**
218  When you want that the string can be put in a JS string definition.
219
220  @param string $s_string The string you want to escape.
221
222  @return string The escaped string.
223  */
224  function escape(string $s_string) : string {
225      return str_replace(
226          [
227              '\\',
228              '\n',
229              '"',
230              "'",
231          ],
232          [
233              '\\\\',
234              '\n',
235              '\\\\',
236              '\\\\',
237          ],
238          $s_string,
239      );
240  } //end escape()
241
242
243
244  /**
245  When you want that the string can be put in a JS string definition
246  to be included inside an HTML file <script> tag.
247
248  @param string $s_string The string you want to escape.
249
250  @return string The escaped string.
251  */
252  function escape_for_string_definition_inside_HTML(string $s_string)
253  : string {
254      return str_replace(
255          [ '</script>' ],
256          [ '\u{00003c}/script>' ],
257          escape($s_string),
258      );
259  } //end escape_for_string_definition_inside_HTML()
260

```



```

261
262
263  /**
264   When you want that the string can be put in a JS string definition
265   to be included inside an XHTML file <script> tag.
266
267   @param string $s_string The string you want to escape.
268
269   @return string The escaped string.
270   */
271   function escape_for_string_definition_inside_XHTML(string $s_string)
272   : string {
273       return XHTML_escape_text(escape($s_string));
274   } //end escape_for_string_definition_inside_XHTML()
275 } //end namespace string_escaping\JS
276
277
278
279
280  /**
281   The namespace string_escaping is splitted into subnamespaces for each
282   language.
283   This subnamespace is for TeX.
284
285   @package DevOrSysAdminScripts
286   @subpackage string_escaping\TeX
287   */
288   namespace string_escaping\TeX {
289       /**
290        When you want that the string can be written to a TeX file inside a
291        part where TeX is in text mode.
292
293        @param string $s_string The string you want to escape.
294
295        @return string The escaped string.
296        */
297        function escape_text(string $s_string) : string {
298            // https://tex.stackexchange.com/a/34586/270015
299            return str_replace(
300                [
301                    '\\',
302                    '&',
303                    '%',
304                    '$',
305                    '#',
306                    '_'

```

```

307         '{',
308         '}',
309         '~',
310         '^',
311     ],
312     [
313         '\\textbackslash',
314         '\\&',
315         '\\%',
316         '\\$',
317         '\\#',
318         '\\_',
319         '\\{',
320         '\\}',
321         '\\textasciitilde',
322         '\\textasciicircum',
323     ],
324     $s_string,
325 );
326 } //end escape_text()
327 } //end namespace string_escaping\\TeX
328
329 /*
330 $ php
331 <?php
332 require_once 'string_escaping.lib.php';
333 var_dump(string_escaping\\HTML\\escape_text("<&\\\"\\n'truc\\>"));
334 var_dump(string_escaping\\HTML\\escape_attribute("<&\\\"\\n'truc\\>"));
335 var_dump(string_escaping\\HTML\\escape_pre("<&\\\"\\n'truc\\>"));
336 var_dump(string_escaping\\XHTML\\escape_text("<&\\\"\\n'truc\\>"));
337 var_dump(string_escaping\\XHTML\\escape_attribute("<&\\\"\\n'truc\\>"));
338 var_dump(string_escaping\\XHTML\\escape_pre("<&\\\"\\n'truc\\>"));
339 var_dump(string_escaping\\JS\\escape("abcd & < > \\ \\n \\\"'"));
340 var_dump(string_escaping\\JS\\escape_for_string_definition_inside_HTML(
341     "abcd & < > \\ \\n \\\"' </script>"
342 ));
343 var_dump(string_escaping\\JS\\escape_for_string_definition_inside_XHTML(
344     "abcd & < > \\ \\n \\\"'"
345 ));
346 */
347 ?>

```

3.74 unicode.lib.php

```

1 <?php
2

```

```

3  /**
4  This file is part of DevOrSysAdminScripts.
5
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 This file was renamed from "unicode.php" to "unicode.lib.php".
27
28 @category Library
29 @package DevOrSysAdminScripts
30 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33 */
34
35 declare(strict_types=1);
36 declare(encoding='UTF-8');
37
38 namespace DOSAS_unicode;
39
40 use Exception;
41
42 /*
43 Max value for Unicode is 10FFFF != 16*255*255 = 16*65025 = 1040400... XD
44 Max value for Unicode is 10FFFF = (16*256+255)*256+255
45   = (2560+1536+255)*256+255 = (4096+255)*256+255 = 4351*256+255
46   = 4351*1000/4 + 4351*6 + 255
47   = 4351000 2175500 1087750
48   = 1087750 + 26106 + 255 = 1114111

```

```

49  so there is 1114112 possible values in the "uncarved" range.
50  244,143,191,191 for UTF-8 yields max code point is ((4*64+15)*64+63)*64+63
51    = ((256+15)*64+63)*64+63 = (271*64+63)*64+63 = (16260+1084+63)*64+63
52    = 17407*64+63 = 1044420 + 69628 + 63 = 1114111
53  */
54  const MAX_UNICODE_CODE_POINT = 1114111;
55
56
57
58  /**
59  Converts an int to the byte-string corresponding to the UTF-8 encoding
60  of the Unicode code point of same integer value.
61  Note that not all integers between 0 and 1114111 are valid code points
62  for UTF-8.
63
64  @param int<0, 1114111> $i_unicode_code_point The code point.
65
66  @throws \Exception When the code point is outside of Unicode range.
67
68  @return string The corresponding UTF-8 character.
69  */
70  function unicode_code_point_to_UTF8(int $i_unicode_code_point) : string {
71      // var_dump($i_unicode_code_point);
72      if($i_unicode_code_point < 0){
73          throw new Exception('A unicode code point must not be negative.');
```

```

95         + $i_last_byte_significant_bits
96     );
97     return $s_result;
98     */
99     return (
100         chr($i_first_byte_base_value + ($i_unicode_code_point >> 6))
101         .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
102     );
103 }//end if($i_unicode_code_point < 2048)
104 // 1110xxxx 10xxxxxx 10xxxxxx
105 if($i_unicode_code_point < 65536){ // 2**16
106     $i_first_byte_base_value = 224;
107     /*
108     $arr_arr_data_per_byte_reverse = [
109         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
110         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
111         ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
112     ];
113     $i_rest = $i_unicode_code_point;
114     $s_result = '';
115     foreach($arr_arr_data_per_byte_reverse as $arr_data){
116         // _Pragma unroll
117         // J'aurais pu "roller" les 4 cas des ifs.
118         // Mais pour les perfs, il faudrait tout unroller vu le nombre
119         // d'itérations. Et j'aimerais bien comparer du code unrollé
120         // aux 2 niveaux à la main avec celui unrollé par le compilateur
121         // sur les 2 niveaux. Je préjuge/présume qu'il unroll
122         // l'intérieur d'abord (pour ne pas refaire le même boulot)
123         // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
124         // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
125         $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
126         $i_rest = intdiv(
127             $i_rest - $i_significant_bits,
128             2**$arr_data['i_bits'],
129         );
130         // .= or =.
131         // Below could have been a case of "=".
132         $s_result = (
133             chr($i_significant_bits + $arr_data['i_base_value'])
134             . $s_result
135         );
136     }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
137     return $s_result;
138     */
139     return (
140         chr($i_first_byte_base_value + ($i_unicode_code_point >> 12))

```

```

141     .chr(
142         $i_continuation_base_value
143         + (($i_unicode_code_point >> 6) & 63)
144     )
145     .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
146 );
147 }//end if($i_unicode_code_point < 65536)
148 // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
149 // if($i_unicode_code_point < 2097152){ // 2**21
150 if($i_unicode_code_point < MAX_UNICODE_CODE_POINT){
151     $i_first_byte_base_value = 240;
152     /*
153     $arr_arr_data_per_byte_reverse = [
154         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
155         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
156         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
157         ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
158     ];
159     $i_rest = $i_unicode_code_point;
160     $s_result = '';
161     foreach($arr_arr_data_per_byte_reverse as $arr_data){
162         // _Pragma unroll
163         $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
164         $i_rest = intdiv(
165             $i_rest - $i_significant_bits,
166             2**$arr_data['i_bits'],
167         );
168         // .= or =.
169         // Below could have been a case of "=".
170         $s_result = (
171             chr($i_significant_bits + $arr_data['i_base_value'])
172             . $s_result
173         );
174     }
175     return $s_result;
176     */
177     return (
178         chr($i_first_byte_base_value + ($i_unicode_code_point >> 18))
179         .chr(
180             $i_continuation_base_value
181             + (($i_unicode_code_point >> 12) & 63)
182         )
183         .chr(
184             $i_continuation_base_value
185             + (($i_unicode_code_point >> 6) & 63)
186     )

```

```

187         .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
188     );
189 }//end if($i_unicode_code_point < MAX_UNICODE_CODE_POINT)
190 // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
191 throw new Exception(
192     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
193     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
194     ." 3 octets au lieu de 4 :face-palm:."
195     ." Du coup, ça peut donner quelques \"surprises\"."
196     ." Sinon, y'a aussi ASCII dans MySQL."
197     ." Et comme ça joue sur la mémoire et les performances,"
198     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
199     ." :) "
200 );
201 // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
202 }//end unicode_code_point_to_UTF8()
203
204
205
206 /**
207  Converts a string of hexadecimal digits to the byte-string corresponding
208  to the UTF-8 encoding of the Unicode code point of same integer value
209  that is denoted by the hexadecimal digits.
210
211  @param string $s_code_point_in_hexadecimal_notation The code point.
212
213  @throws \Exception When the code point is outside of Unicode range.
214
215  @return string The corresponding UTF-8 character.
216  */
217 function hexa_code_point_to_UTF8(
218     string $s_code_point_in_hexadecimal_notation
219 ) : string {
220     $s_code_point_in_hexadecimal_notation = strtoupper(
221         $s_code_point_in_hexadecimal_notation
222     );
223     $i_unicode_code_point = 0;
224     $i_ord_0 = ord('0');
225     $i_ord_A = ord('A');
226     $i_digit_offset = -$i_ord_0;
227     $i_letter_offset = 10 - $i_ord_A;
228     for(
229         $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
230         $i < $i_max;
231         ++$i
232     ) {

```

```

233     $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
234     $i_unicode_code_point *= 16;
235     if(ctype_digit($s_hexa_digit)){
236         $i_unicode_code_point += (
237             ord($s_hexa_digit) + $i_digit_offset
238         );
239         continue;
240     }
241     if(ctype_xdigit($s_hexa_digit)){
242         $i_unicode_code_point += (
243             ord($s_hexa_digit) + $i_letter_offset
244         );
245         continue;
246     }
247     throw new Exception('Not an hexadecimal digit.');
```

```

248 }
249 if($i_unicode_code_point > MAX_UNICODE_CODE_POINT){
250     throw new Exception(
251         'Hexadecimal integer is greater than UTF-8 maximum code point.'
252     );
253 }
254 return unicode_code_point_to_UTF8($i_unicode_code_point);
255 }//end hexa_code_point_to_UTF8()
256
257
258
259 /**
260  Converts a string of decimal digits to the byte-string corresponding
261  to the UTF-8 encoding of the Unicode code point of same integer value
262  that is denoted by the decimal digits.
263
264  @param string $s_code_point_in_decimal_notation The code point.
265
266  @throws \Exception When the code point is outside of Unicode range.
267
268  @return string The corresponding UTF-8 character.
269  */
270 function decimal_code_point_to_UTF8(
271     string $s_code_point_in_decimal_notation
272 ) : string {
273     $i_unicode_code_point = 0;
274     $i_ord_0 = ord('0');
275     $i_digit_offset = -$i_ord_0;
276     for(
277         $i = 0, $i_max = strlen($s_code_point_in_decimal_notation);
278         $i < $i_max;
```



```

279         ++$i
280     ) {
281         $s_digit = $s_code_point_in_decimal_notation[$i];
282         $i_unicode_code_point *= 10;
283         if (ctype_digit($s_digit)) {
284             $i_unicode_code_point += (ord($s_digit) + $i_digit_offset);
285             continue;
286         }
287         throw new Exception('Not a decimal digit.');
```

288 }

```

289     if ($i_unicode_code_point > MAX_UNICODE_CODE_POINT) {
290         throw new Exception(
291             'Decimal integer is greater than UTF-8 maximum code point.'
292         );
293     }
294     return unicode_code_point_to_UTF8($i_unicode_code_point);
295 } //end decimal_code_point_to_UTF8()
296
297
298
299 /**
300  This function throws an exception with extended debug informations
301  if the input string is not valid ASCII.
302  Otherwise, it returns true.
303
304  @param string $s_string The input string.
305
306  @throws \DOSAS_unicode\InvalidEncodingException
307          When the input string is not valid ASCII.
308
309  @return true
310  */
311 function check_string_is_valid_ASCII(string $s_string) : bool {
312     //Fast-path
313     if (mb_check_encoding($s_string, 'ASCII')) {
314         return true;
315     }
316
317     $i_offset_in_octets_from_string_start = 0;
318     $i_current_line_number = 1;
319     $i_offset_in_octets_from_line_start = -1;
320     for ($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i) {
321         $i_current_octet = ord($s_string[$i]);
322         $i_offset_in_octets_from_string_start = $i;
323         ++$i_offset_in_octets_from_line_start;
324     }

```

```

325     if($i_current_octet > 127){
326         throw new InvalidEncodingException(
327             'Non-ASCII character found on line '
328             .$i_current_line_number
329             .'; the octet/character '
330             .($i_offset_in_octets_from_line_start + 1)
331             .' has value '
332             .$i_current_octet
333             .' which is not ASCII.'
334             .' (Sequential position without line splitting:'
335             .' This is at octet/character '
336             .($i_offset_in_octets_from_string_start + 1)
337             .'.)',
338             [
339                 'current_octet' => $i_current_octet,
340                 'offset_from_string_start'
341                     => $i_offset_in_octets_from_string_start,
342                 'line' => $i_current_line_number,
343                 'offset_from_line_start'
344                     => $i_offset_in_octets_from_line_start,
345             ],
346         );
347     }//end if($i_current_octet > 127)
348
349     // if($s_string[$i] === "\n"){
350     if($i_current_octet === 10){
351         ++$i_current_line_number;
352         $i_offset_in_octets_from_line_start = -1;
353     }
354     }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
355     return true;
356 }//end check_string_is_valid_ASCII()
357
358
359
360 /**
361  This function throws an exception with extended debug informations
362  if the content of the file with given file_path is not valid ASCII.
363  Otherwise, it returns true.
364
365  @param string $s_file_path The input file_path.
366
367  @throws \Exception When the file is not found.
368  @throws \DOSAS_unicode\InvalidEncodingException
369      When the input string is not valid ASCII.
370

```

```

371 @return true
372 */
373 function check_file_is_valid_ASCII(string $s_file_path) : bool {
374     $s_string = file_get_contents($s_file_path);
375     if($s_string === false){
376         throw new Exception('File '.$s_file_path.' not found.');
```

```

377     }
```

```

378     return check_string_is_valid_ASCII($s_string);
```

```

379 }//end check_file_is_valid_ASCII()
```

```

380
```

```

381
```

```

382
```

```

383 /**
```

```

384 This function returns the message string and data array for
```

```

385 DOSAS_unicode\InvalidEncodingException inside check_string_is_valid_UTF8().
```

```

386
```

```

387 @param string $s_custom_message The custom part of the message.
```

```

388 @param int $i_current_octet The current octet in the string.
```

```

389 @param int $i_continuation_octet_needed The current number of continuation
390                                         octets needed.
```

```

391 @param int $i_offset_in_octets_from_string_start
```

```

392 The offset in octets from string start.
```

```

393 @param int $i_offset_in_characters_from_string_start
```

```

394 The offset in characters from string start.
```

```

395 @param int $i_character_start_position_from_string_start
```

```

396 The offset in octets from string start at which the current
397 character starts.
```

```

398 @param int $i_current_line_number The current line number.
```

```

399 @param int $i_offset_in_octets_from_line_start
```

```

400 The offset in octets from line start.
```

```

401 @param int $i_offset_in_characters_from_line_start
```

```

402 The offset in characters from line start.
```

```

403 @param int $i_character_start_position_from_line_start
```

```

404 The offset in octets from line start at which the current
405 character starts.
```

```

406 @param int $i_current_continuation_octet_minimum
```

```

407 The minimum value of the second continuation octet as specified
408 by the grammar.
```

```

409 @param int $i_current_continuation_octet_maximum
```

```

410 The maximum value of the second continuation octet as specified
411 by the grammar.
```

```

412
```

```

413 @return array{'message': string, 'data': array<mixed>}
```

```

414 */
```

```

415 function get_message_and_data_array(
```

```

416     string $s_custom_message,
```

```

417     int $i_current_octet,
418     int $i_continuation_octet_needed,
419     int $i_offset_in_octets_from_string_start,
420     int $i_offset_in_characters_from_string_start,
421     int $i_character_start_position_from_string_start,
422     int $i_current_line_number,
423     int $i_offset_in_octets_from_line_start,
424     int $i_offset_in_characters_from_line_start,
425     int $i_character_start_position_from_line_start,
426     int $i_current_continuation_octet_minimum,
427     int $i_current_continuation_octet_maximum,
428 ) : array {
429     return [
430         'message' => (
431             'Non-UTF8 character found on line '
432             .$i_current_line_number
433             .'; the octet '
434             .($i_offset_in_octets_from_line_start + 1)
435             .', part of the character '
436             .($i_offset_in_characters_from_line_start + 1)
437             .', has value '
438             .$i_current_octet
439             .$s_custom_message
440             .' This character starts at octet '
441             .($i_character_start_position_from_line_start + 1)
442             .' of the current line.'
443             .' (Sequential positions without line splitting:'
444             .' This is at character '
445             .($i_offset_in_characters_from_string_start + 1)
446             .' and octet '
447             .($i_offset_in_octets_from_string_start + 1)
448             .'.'
449             .' This character starts at octet '
450             .($i_character_start_position_from_string_start + 1)
451             .'.)'
452         ),
453         'data' => [
454             'current_octet' => $i_current_octet,
455             'continuation_octet_needed' => $i_continuation_octet_needed,
456             'offset_in_octets_from_string_start'
457             => $i_offset_in_octets_from_string_start,
458             'offset_in_characters_from_string_start'
459             => $i_offset_in_characters_from_string_start,
460             'character_start_position_from_string_start'
461             => $i_character_start_position_from_string_start,
462             'line' => $i_current_line_number,

```

```

463         'offset_in_octets_from_line_start'
464         => $i_offset_in_octets_from_line_start,
465         'offset_in_characters_from_line_start'
466         => $i_offset_in_characters_from_line_start,
467         'character_start_position_from_line_start'
468         => $i_character_start_position_from_line_start,
469         'current_continuation_octet_minimum'
470         => $i_current_continuation_octet_minimum,
471         'current_continuation_octet_maximum'
472         => $i_current_continuation_octet_maximum,
473     ],
474 ];
475 } //end get_message_and_data_array()
476
477
478
479 /**
480 Checks that the version of PHP allows using the fast-path inside
481 check_string_is_valid_UTF8.
482
483 @return bool
484 */
485 function get_use_fast_path() : bool {
486     // Before PHP 5.4, Unicode support has bugs,
487     // both in mb_check_encoding and preg_match.
488     // See https://github.com/php/php-src/issues/22279
489     return PHP_VERSION_ID >= 50400;
490 } //end get_use_fast_path()
491
492
493
494 /**
495 This function throws an exception with extended debug informations
496 if the input string is not valid UTF-8.
497 Otherwise, it returns true.
498
499 See https://datatracker.ietf.org/doc/html/rfc3629
500 See https://github.com/Seldaek/jsonlint/pull/98
501 for why it may be a good idea to have the last two parameters.
502 Note that a very clever compiler would merge both booleans into
503 a single integer parameter using binary flags.
504 If you want to use this function with fast-path and fallback on slow-path
505 in case of error, then set both booleans to true.
506 If, for testing/benchmarking purpose, you want to compare the results
507 of fast-path and slow-path, call this function twice with a single boolean
508 set, and store/compare the results.

```

```

509
510 @param string $s_string The input string.
511 @param bool $b_fast_path Use the fast-path or not to check quickly that
512                          there is no error.
513 @param bool $b_slow_path Use the slow-path or not to check that
514                          there is no error or give detailed error message.
515
516 @throws \Exception When called with $b_fast_path set to true but PHP
517                    version is incompatible.
518 @throws \DOSAS_unicode\InvalidEncodingException
519         When the input string is not valid UTF-8.
520
521 @return bool
522 */
523 function check_string_is_valid_UTF8(
524     string $s_string,
525     bool $b_fast_path = true,
526     bool $b_slow_path = true,
527 ) : bool {
528     // Fast-path
529     // But before PHP 5.4 Unicode support has bugs.
530     // See https://github.com/php/php-src/issues/22279
531     if($b_fast_path && !get_use_fast_path()){
532         throw new Exception(
533             'Fast-path is not available with version '.PHP_VERSION_ID.' of PHP.'
534         );
535     }
536     if($b_fast_path){
537         if(function_exists('mb_check_encoding')){
538             if(mb_check_encoding($s_string, 'UTF-8')){
539                 return true;
540             }
541         }
542         else{
543             if(preg_match('//u', $s_string) === 1){
544                 return true;
545             }
546         }
547     }
548     else{
549         if(!$b_slow_path){
550             throw new Exception(
551                 "Don't call check_string_is_valid_UTF8 without at least one of"
552                 .' fast-path or slow-path required.'
553             );
554         }
555     }

```

```

555     }
556
557     if (!$b_slow_path) {
558         return false;
559     }
560
561     $i_current_octet = null;
562     $i_continuation_octet_needed = 0;
563     $i_offset_in_octets_from_string_start = 0;
564     $i_offset_in_characters_from_string_start = -1;
565     $i_character_start_position_from_string_start = 0;
566     $i_current_line_number = 1;
567     $i_offset_in_octets_from_line_start = -1;
568     $i_offset_in_characters_from_line_start = -1;
569     $i_character_start_position_from_line_start = -1;
570
571     // For the second octet of a character, hence first continuation octet,
572     // further restriction may apply.
573     $I_CONTINUATION_OCTET_MINIMUM = 128;
574     $I_CONTINUATION_OCTET_MAXIMUM = 191;
575     $i_current_continuation_octet_minimum = $I_CONTINUATION_OCTET_MINIMUM;
576     $i_current_continuation_octet_maximum = $I_CONTINUATION_OCTET_MAXIMUM;
577     $s_message_for_continuation_octet_above_maximum = null;
578
579     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
580         $i_current_octet = ord($s_string[$i]);
581         $i_offset_in_octets_from_string_start = $i;
582         ++$i_offset_in_octets_from_line_start;
583
584         /*
585         The octet values C0, C1, F5 to FF never appear.
586         C0 = 12*16          = 192 = 11000000
587         C1 = 12*16 + 1     = 193 = 11000001
588         F5 = 15*16 + 5     = 245 = 11110101
589         FF = 15*16 + 15   = 255 = 11111111
590         */
591         if(
592             $i_current_octet === 192
593             || $i_current_octet === 193
594             || $i_current_octet === 245
595             || $i_current_octet === 255
596         ){
597             [
598                 'message' => $s_message,
599                 'data' => $arr_data,
600             ] = get_message_and_data_array(

```

```

601         ' which is one of the four forbidden values (C0, C1, F5, FF).',
602         $i_current_octet,
603         $i_continuation_octet_needed,
604         $i_offset_in_octets_from_string_start,
605         $i_offset_in_characters_from_string_start,
606         $i_character_start_position_from_string_start,
607         $i_current_line_number,
608         $i_offset_in_octets_from_line_start,
609         $i_offset_in_characters_from_line_start,
610         $i_character_start_position_from_line_start,
611         $i_current_continuation_octet_minimum,
612         $i_current_continuation_octet_maximum,
613     );
614     throw new InvalidEncodingException($s_message, $arr_data);
615 }
616
617 /*
618 UTF8-octets = *( UTF8-char )
619 UTF8-char   = UTF8-1 / UTF8-2 / UTF8-3 / UTF8-4
620 UTF8-1      = %x00-7F
621 UTF8-2      = %xC2-DF UTF8-tail
622     Notice that values C0 and C1 are forbidden, hence %xC2
623 UTF8-3      = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
624               %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
625     Notice that E0 = 224 adds a restriction on second octet.
626     Notice that ED = 237 adds a restriction on second octet.
627 UTF8-4      = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
628               %xF4 %x80-8F 2( UTF8-tail )
629     Notice that F0 = 240 adds a restriction on second octet.
630     Notice that F4 = 244 adds a restriction on second octet.
631 UTF8-tail   = %x80-BF
632 */
633 if($i_continuation_octet_needed > 0){
634     if(
635         $i_current_octet < $i_current_continuation_octet_minimum
636         || $i_current_octet > $i_current_continuation_octet_maximum
637     ){
638         [
639             'message' => $s_message,
640             'data' => $arr_data,
641         ] = get_message_and_data_array(
642             $s_message_for_continuation_octet_above_maximum !== null
643             && $i_current_octet > $i_current_continuation_octet_maximum
644             ? $s_message_for_continuation_octet_above_maximum
645             : ' which is not a continuation octet.',
646             $i_current_octet,

```



```

647         $i_continuation_octet_needed,
648         $i_offset_in_octets_from_string_start,
649         $i_offset_in_characters_from_string_start,
650         $i_character_start_position_from_string_start,
651         $i_current_line_number,
652         $i_offset_in_octets_from_line_start,
653         $i_offset_in_characters_from_line_start,
654         $i_character_start_position_from_line_start,
655         $i_current_continuation_octet_minimum,
656         $i_current_continuation_octet_maximum,
657     );
658     throw new InvalidEncodingException($s_message, $arr_data);
659 } //end if($i_current_octet < 128 || $i_current_octet >= 192)
660 --$i_continuation_octet_needed;
661 $i_current_continuation_octet_minimum = (
662     $I_CONTINUATION_OCTET_MINIMUM
663 );
664 $i_current_continuation_octet_maximum = (
665     $I_CONTINUATION_OCTET_MAXIMUM
666 );
667     continue;
668 } //end if($i_continuation_octet_needed > 0)
669
670 ++$i_offset_in_characters_from_string_start;
671 ++$i_offset_in_characters_from_line_start;
672 $i_character_start_position_from_string_start = (
673     $i_offset_in_octets_from_string_start
674 );
675 $i_character_start_position_from_line_start = (
676     $i_offset_in_octets_from_line_start
677 );
678
679 if($i_current_octet < 128){ // 0xxxxxxx ASCII
680     // if($s_string[$i] === "\n"){
681         if($i_current_octet === 10){
682             ++$i_current_line_number;
683             $i_offset_in_octets_from_line_start = -1;
684             $i_offset_in_characters_from_line_start = -1;
685         }
686         continue;
687     }
688
689 if(/*$i_current_octet >= 128 &&*/ $i_current_octet < 192){
690     [
691         'message' => $s_message,
692         'data' => $arr_data,

```

```

693     ] = get_message_and_data_array(
694         ' which is a continuation octet.',
695         $i_current_octet,
696         $i_continuation_octet_needed,
697         $i_offset_in_octets_from_string_start,
698         $i_offset_in_characters_from_string_start,
699         $i_character_start_position_from_string_start,
700         $i_current_line_number,
701         $i_offset_in_octets_from_line_start,
702         $i_offset_in_characters_from_line_start,
703         $i_character_start_position_from_line_start,
704         $i_current_continuation_octet_minimum,
705         $i_current_continuation_octet_maximum,
706     );
707     throw new InvalidEncodingException($s_message, $arr_data);
708 } //end if($i_current_octet >= 128 && $i_current_octet < 192)
709
710 if(/*$i_current_octet >= 192 &&*/ $i_current_octet < 224){
711     // 110xxxxx 10xxxxxx
712     $i_continuation_octet_needed = 1;
713     continue;
714 }
715
716 if(/*$i_current_octet >= 224 &&*/ $i_current_octet < 240){
717     // 1110xxxx 10xxxxxx 10xxxxxx
718     $i_continuation_octet_needed = 2;
719     /*
720     UTF8-3 = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
721             %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
722     Notice that E0 = 224 adds a restriction on second octet.
723     Notice that ED = 237 adds a restriction on second octet.
724
725     The definition of UTF-8 prohibits encoding character numbers between
726     U+D800 and U+DFFF.
727     D8 = 13*16 + 8   = 216 = 11010100
728     DF = 13*16 + 15  = 223 = 11010101
729     216*256 = 55296      = 1101 1000 0000 0000 = D800
730     to
731     223*256 + 255 = 57343 = 1101 1111 1111 1111 = DFFF
732
733     1110xxxx
734     11101101 = 237 = ED
735
736     237,160,128
737     237,191,191
738     Thus, the whole "continuation range" is forbidden if start

```

```

739     is 237 and second octet, first continuation octet, is >= 160.
740     */
741     if($i_current_octet === 224){
742         $i_current_continuation_octet_minimum = 160;
743         $i_current_continuation_octet_maximum = 191;    // Normal value
744     }
745     if($i_current_octet === 237){
746         $i_current_continuation_octet_minimum = 128;    // Normal value
747         $i_current_continuation_octet_maximum = 159;
748         $s_message_for_continuation_octet_above_maximum = (
749             ' which is into the forbidden range of surrogate pairs.'
750         );
751     }
752     continue;
753 } //end if (/*$i_current_octet >= 224 &&*/ $i_current_octet < 240)
754
755 if(/*$i_current_octet >= 240 &&*/ $i_current_octet < /*248*/ 245){
756     // 11110xxx 10xxxxxxx 10xxxxxxx 10xxxxxxx
757     $i_continuation_octet_needed = 3;
758     /*
759     UTF8-4 = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
760             %xF4 %x80-8F 2( UTF8-tail )
761     Notice that F0 = 240 adds a restriction on second octet.
762     Notice that F4 = 244 adds a restriction on second octet.
763     */
764     if($i_current_octet === 240){
765         $i_current_continuation_octet_minimum = 144;
766         $i_current_continuation_octet_maximum = 191;    // Normal value
767     }
768     if($i_current_octet === 244){
769         $i_current_continuation_octet_minimum = 128;    // Normal value
770         $i_current_continuation_octet_maximum = 143;
771     }
772     continue;
773 }
774
775 [
776     'message' => $s_message,
777     'data' => $arr_data,
778 ] = get_message_and_data_array(
779     ' which is invalid.',
780     $i_current_octet,
781     $i_continuation_octet_needed,
782     $i_offset_in_octets_from_string_start,
783     $i_offset_in_characters_from_string_start,
784     $i_character_start_position_from_string_start,

```

```

785         $i_current_line_number,
786         $i_offset_in_octets_from_line_start,
787         $i_offset_in_characters_from_line_start,
788         $i_character_start_position_from_line_start,
789         $i_current_continuation_octet_minimum,
790         $i_current_continuation_octet_maximum,
791     );
792     throw new InvalidEncodingException($s_message, $arr_data);
793 }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
794
795 if($i_continuation_octet_needed > 0){
796     [
797         'message' => $s_message,
798         'data' => $arr_data,
799     ] = get_message_and_data_array(
800         '; end of string was found instead of a continuation octet.',
801         $i_current_octet,
802         $i_continuation_octet_needed,
803         $i_offset_in_octets_from_string_start,
804         $i_offset_in_characters_from_string_start,
805         $i_character_start_position_from_string_start,
806         $i_current_line_number,
807         $i_offset_in_octets_from_line_start,
808         $i_offset_in_characters_from_line_start,
809         $i_character_start_position_from_line_start,
810         $i_current_continuation_octet_minimum,
811         $i_current_continuation_octet_maximum,
812     );
813     throw new InvalidEncodingException($s_message, $arr_data);
814 }
815
816 if($b_fast_path){
817     throw new InvalidEncodingException(
818         "Fast-path detected an error that this function couldn't found."
819         .'Please create an issue at '
820         .' "https://github.com/LLyaudet/DevOrSysAdminScripts/issues".',
821         [],
822     );
823 }
824
825 return true;
826 }//end check_string_is_valid_UTF8()
827
828
829
830 / **

```

```

831 This function throws an exception with extended debug informations
832 if the content of the file with given file_path is not valid UTF-8.
833 Otherwise, it returns true.
834
835 @param string $s_file_path The input file_path.
836 @param bool $b_fast_path Use the fast-path or not to check quickly that
837     there is no error.
838
839 @throws \Exception When the file is not found OR when called with
840     $b_fast_path set to true but PHP version is
841     incompatible.
842 @throws \DOSAS_unicode\InvalidEncodingException
843     When the input string is not valid UTF-8.
844
845 @return bool
846 */
847 function check_file_is_valid_UTF8(
848     string $s_file_path,
849     bool $b_fast_path = true,
850 ) : bool {
851     $s_string = file_get_contents($s_file_path);
852     if($s_string === false){
853         throw new Exception('File '.$s_file_path.' not found.');
```

```

854     }
855     return check_string_is_valid_UTF8($s_string, $b_fast_path);
856 }//end check_file_is_valid_UTF8()
857
858
859
860 /*
861 ?>
862 <?php
863 require_once "./unicode_exceptions.lib.php";
864 require_once "./unicode.lib.php";
865 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("002B"));
866 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("00E6"));
867 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("1400"));
868 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("10111"));
869 // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
870 // Some invisible unicode characters that are a problem:
871 // 2000-200F 2028-202F 205F 2060-206F
872 $arr = [
873     '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
874     '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
875     '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
876     '205F',
```

```

877     '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
878     '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
879 ];
880 $arr_s = [];
881 $arr_s2 = [];
882 $arr_s3 = [];
883 echo("sed\\\n");
884 foreach($arr as $s_hexa){
885     $s = DOSAS_unicode\hexa_code_point_to_UTF8($s_hexa);
886     $arr_s []= $s;
887     if(strlen($s) === 2){
888         $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
889     }
890     if(strlen($s) === 3){
891         $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
892             .'\x'.bin2hex($s[2]);
893     }
894     $arr_s2 []= $s2;
895     $s3 = " --expression='s/" . $s2 . "/ g'\\";
896     $arr_s3 []= $s3;
897     echo($s3 . "\n");
898 }
899
900 ?>
901 <?php
902 require_once "./unicode_exceptions.lib.php";
903 require_once "./unicode.lib.php";
904 // var_dump(DOSAS_unicode\check_file_is_valid_UTF8("validutf8.json"));
905 // var_dump(DOSAS_unicode\check_file_is_valid_UTF8("nonvalidutf8.json"));
906 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé"));
907 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", true, false));
908 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", false, true));
909 var_dump(
910     DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), true, false)
911 );
912 if(true){
913     var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233)));
914 }
915 else{
916     var_dump(
917         DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), false, true)
918     );
919 }
920 //*/
921 ?>

```

3.75 unicode_exceptions.lib.php

```
1  <?php
2
3  /**
4   This file is part of DevOrSysAdminScripts.
5
6   DevOrSysAdminScripts is free software:
7   you can redistribute it and/or modify it
8   under the terms of the GNU Lesser General Public License
9   as published by the Free Software Foundation,
10  either version 3 of the License,
11  or (at your option) any later version.
12
13  DevOrSysAdminScripts is distributed
14  in the hope that it will be useful,
15  but WITHOUT ANY WARRANTY;
16  without even the implied warranty of
17  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18  See the GNU Lesser General Public License for more details.
19
20  You should have received a copy of
21  the GNU Lesser General Public License
22  along with DevOrSysAdminScripts.
23  If not, see <https://www.gnu.org/licenses/>.
24
25  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26  This file was renamed from "unicode.php" to "unicode.lib.php".
27
28  @category Library
29  @package DevOrSysAdminScripts
30  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33  */
34
35  declare(strict_types=1);
36  declare(encoding='UTF-8');
37
38  namespace DOSAS_unicode;
39
40  /**
41   A class to enhance standard PHP Exception with an array of data.
42
43   @category Library
44   @package DevOrSysAdminScripts
```

```

45  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
46  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
47  */
48  class WithDataArrayException extends \Exception {
49      /**
50       A data array to store details related to an exception.
51
52       @var array<mixed> $arr_data
53       */
54       protected $arr_data;
55
56
57
58       /**
59       Constructor for WithDataArrayException.
60
61       @param string $message
62       @param array<mixed> $arr_data
63
64       @return void
65       */
66       public function __construct(string $message, array $arr_data = []) {
67           $this->arr_data = $arr_data;
68           parent::__construct($message);
69       } //end __construct() WithDataArrayException
70
71
72
73       /**
74       Getter to obtain the data array with details of the exception.
75
76       @return array<mixed>
77       */
78       public function get_data() : array {
79           return $this->arr_data;
80       } //end get_data()
81   } //end class WithDataArrayException
82
83
84
85
86   /**
87   A class to specialize WithDataArrayException for use with invalid
88   encoding detection (ASCII and UTF-8 validation) in unicode.lib.php.
89
90   @category Library

```



```

91 @package DevOrSysAdminScripts
92 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
93 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
94 */
95 class InvalidEncodingException extends WithDataArrayException {
96 }//end class InvalidEncodingException
97
98 ?>

```

3.76 upgrade_build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "upgrade_build_and_checks.sh"
25 # to "upgrade_build_and_checks.exec.sh".
26
27 upgrade_build_and_checks() {
28     # Options:
29     #   --add-test-files
30     #   --check-download=y|keep (=keep to keep the downloaded files)
31     #   --fixed-point-build
32     #   --verbose
33     local LFBFL_verbose=""
34     if [[ "$*" == *--verbose* ]]; then
35         printf "%s %s\n" "$0" "$*"

```

```

36     LFBFL_verbose="--verbose"
37 fi
38 readonly LFBFL_verbose
39 local LFBFL_fixed_point_build=""
40 if [[ "$*" == *--fixed-point-build* ]]; then
41     LFBFL_fixed_point_build="--fixed-point-build"
42 fi
43 readonly LFBFL_fixed_point_build
44 local LFBFL_check_download=""
45 if [[ "$*" == *--check-download=keep* ]]; then
46     LFBFL_check_download="--check-download=keep"
47 elif [[ "$*" == *--check-download=y* ]]; then
48     LFBFL_check_download="--check-download=y"
49 fi
50 readonly LFBFL_check_download
51 declare -i LFBFL_add_test_files=0
52 if [[ "$*" == *--add-test-files* ]]; then
53     LFBFL_add_test_files=1
54 fi
55 readonly LFBFL_add_test_files
56 declare -i LFBFL_remove_test_files=0
57 if [[ "$*" == *--remove-test-files* ]]; then
58     LFBFL_remove_test_files=1
59 fi
60 readonly LFBFL_remove_test_files
61 (
62     cd build_and_checks_dependencies/\
63     && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
64 )
65
66 declare -i LFBFL_i
67 local LFBFL_file_path="miscellaneous/123456789012345678901234567890"
68 declare -i LFBFL_digit
69 if [[ LFBFL_add_test_files -eq 1 ]]; then
70     mkdir --parents -- --a/
71     {
72         printf "<?php\nfunction my_function(;;){\n\n}\n"
73         # shellcheck disable=SC2016
74         printf '$my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
75         printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
76     } > --a.php
77     cp -- --a.php --a/
78     {
79         printf "from typing import Iterable\n"
80         printf "from _collections_abc import dict_items\n"
81         printf "\n"

```

```

82     printf "from python_none_objects import NoneIterable\n"
83     printf "\n"
84     printf "def my_function(my_argument : int) -> None:\n"
85     printf "    print(my_argument)\n"
86     printf "\n"
87     printf "my_function(\"abcd\")\n"
88     printf "\n"
89     printf "def my_function2(my_argument : int) -> None:\n"
90     printf "    \"\"\"Some comment.\"\"\"\n"
91     printf "    print(my_argument)\n"
92     printf "\n"
93     printf "def my_function3(my_argument : int) -> None:\n"
94     printf "    \"\"\"Some comment.\"\"\"\n"
95     printf "\n"
96     printf "    print(my_argument)\n"
97     printf "\n"
98     printf "def my_function4(my_argument : int):\n"
99     printf "    \"\"\"Some comment.\"\"\"\n"
100    printf "    def my_sub_function():\n"
101    printf "        print(my_argument)\n"
102    printf "    return my_sub_function\n"
103    printf "\n"
104    printf "def my_function5(my_argument : int):\n"
105    printf "    \"\"\"Some comment.\"\"\"\n"
106    printf "\n"
107    printf "    def my_sub_function():\n"
108    printf "        print(my_argument)\n"
109    printf "    return my_sub_function\n"
110    printf "\n"
111 } > --a.py
112 cat -- --a.py
113 cp -- --a.py --a/
114 {
115     printf "def my_function(;;):\n\n\n"
116     printf 'my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'\n'
117     printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
118 } > --a2.py
119 cat -- --a2.py
120 cp -- --a2.py --a/
121 {
122     # Error 0 is missing shell header.
123     printf "my_function(){\n"
124     printf "    # Error 1: Indentation is not correct.\n"
125     printf "    sed -e 's/a/b/g' \\\n"
126     printf "        -e 's/a/b/g' \"\$1\" # Indentation is correct.\n"
127     printf "    my_variable=\"%s\"\n\n"

```

```

128     # shellcheck disable=SC2016
129     printf '    printf "${my_variable}" # Error 2: shellcheck warning\n'
130     local LFBFL_ttp="ttp"
131     local LFBFL_URL="h${LFBFL_ttp}://example.com"
132     printf '    printf "%s" # Error 3: http -> https\n' "${LFBFL_URL}"
133     # shellcheck disable=SC2016
134     printf '    printf "${#some_'
135     printf 'arr_1}" # Error 4: no_size_of_array_first_element\n'
136     # shellcheck disable=SC2016
137     printf '    printf "${#_'
138     printf 'arr_1}" # Error 5: no_size_of_array_first_element\n'
139     printf '    if !'
140     # shellcheck disable=SC2016
141     printf '    [[ $1 -eq 1 ]]; # Error 6: no_negation_before_bash_test\n'
142     printf '    then # Error 7: misplaced then\n'
143     printf '        my_variable="a"\n'
144     printf '    fi\n'
145     printf "\n"
146     # shellcheck disable=SC2016
147     printf '    if [[ $1 -eq 1 ]];\n'
148     printf '    then # Error 8: another case of misplaced then\n'
149     printf '        my_variable="a"\n'
150     printf '    fi\n'
151     printf '    my_variable="aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
152     printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa" # Error 9 line too long.\n'
153     printf '}\n'
154 } > --a.sh
155 cp -- --a.sh --a/
156
157 # We add a file with space in it for tests.
158 touch "a b.sh"
159
160 # The following files cannot be committed,
161 # because the backslash in them breaks CodeFactor.io.
162 for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
163     LFBFL_digit=$((LFBFL_i % 10))
164     LFBFL_file_path+="${LFBFL_digit}"
165     touch "${LFBFL_file_path}"
166     touch "${LFBFL_file_path}\"
167 done
168 fi
169 ./build_and_checks.exec.sh ".\"
170     "${LFBFL_verbose}"\"
171     "${LFBFL_fixed_point_build}"\"
172     "${LFBFL_check_download}"\"
173 if [[ LFBFL_add_test_files -eq 1 ]]; then

```

```

174     cat -- --a.py
175     cat -- --a2.py
176 fi
177 if [[ LFBFL_remove_test_files -eq 1 ]]; then
178     rm --force --recursive -- --a/
179     rm --force -- --a.php --a.py --a2.py --a.sh
180     rm "a b.sh"
181     for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
182         LFBFL_digit=$((LFBFL_i % 10))
183         LFBFL_file_path+="{LFBFL_digit}"
184         rm --force "${LFBFL_file_path}"
185         rm --force "${LFBFL_file_path}\"
186     done
187 fi
188 }
189
190 upgrade_build_and_checks "$@"

```

3.77 wget_sha512.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
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19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "wget_sha512.sh"
25 # to "wget_sha512.lib.sh".
26

```

```

27 wget_sha512() {
28     # $1=file_path
29     # $2=download_URL
30     # $3=correct_sha512
31     # Options:
32     #   --check-download=y|keep (=keep to keep the downloaded file)
33     #   --verbose
34     local LFBFL_verbose=""
35     if [[ "$*" == *--verbose* ]]; then
36         printf "%s wget_sha512 %s\n" "$0" "$*"
37         LFBFL_verbose="--verbose"
38     fi
39     readonly LFBFL_verbose
40     local LFBFL_check_download=""
41     if [[ "$*" == *--check-download=keep* ]]; then
42         LFBFL_check_download="keep"
43     elif [[ "$*" == *--check-download=y* ]]; then
44         LFBFL_check_download="y"
45     fi
46     readonly LFBFL_check_download
47     declare -i LFBFL_i_error=0
48
49     if [[ ! -f "$1" ]]; then
50         wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
51             -- "$2"
52     fi
53     declare -r LFBFL_present_sha512=$(
54         sha512sum -- "$1"\
55         | cut --fields=1 --delimiter=' '
56     )
57     if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
58         printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
59             "$1" "$3" "${LFBFL_present_sha512}"
60         LFBFL_i_error=1
61     fi
62     if [[ -n "${LFBFL_check_download}" ]]; then
63         wget ${LFBFL_verbose:+"${LFBFL_verbose}"}\
64             --output-document="$1.wget_sha512.temp"\
65             -- "$2"
66         declare -r LFBFL_present_sha512_2=$(
67             sha512sum -- "$1.wget_sha512.temp"\
68             | cut --fields=1 --delimiter=' '
69         )
70         if [[ "${LFBFL_check_download}" != "keep" ]]; then
71             rm -- "$1.wget_sha512.temp"
72         fi

```

```

73     if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
74         printf\
75             "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
76             "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
77         LFBFL_i_error=1
78     fi
79 fi
80 # shellcheck disable=SC2248
81 return ${LFBFL_i_error}
82 }

```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !